

MATTHEW BRENNAN

ATTENTION FACTORY

The Story of TikTok & China's ByteDance



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& China's ByteDance**

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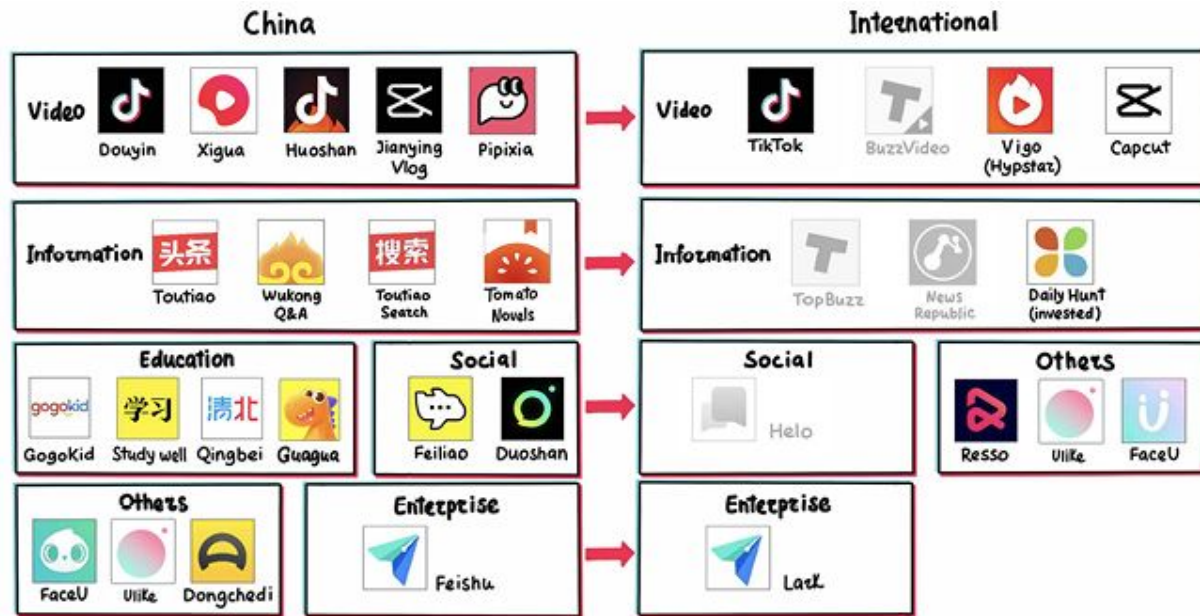
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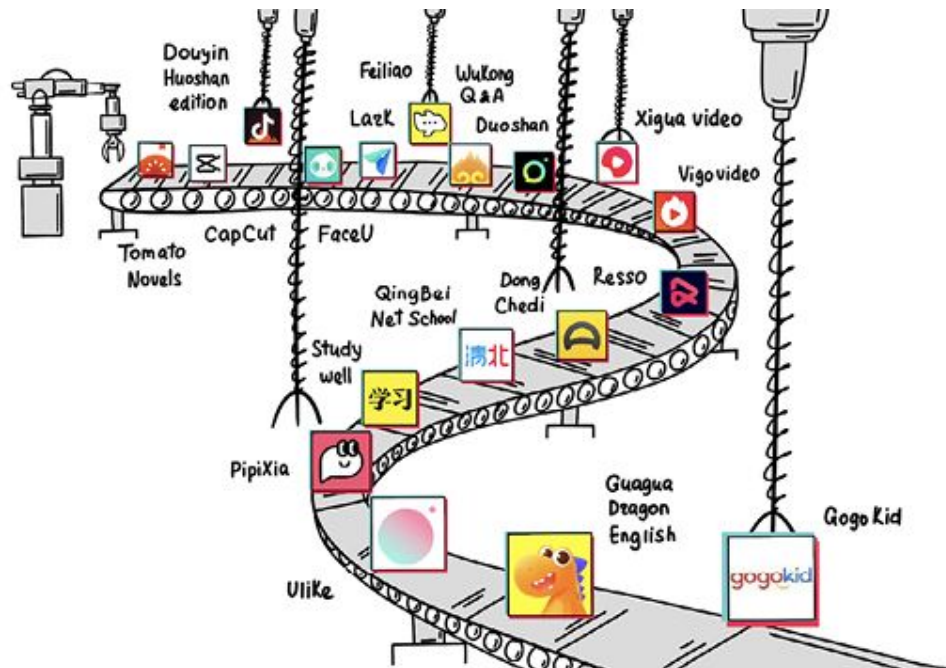
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RESOURCES

ByteDance's Key Mobile Applications



Greyed out icons indicate the app has recently shut down. For an extensive list of ByteDance's product suite refer to the back of this book.



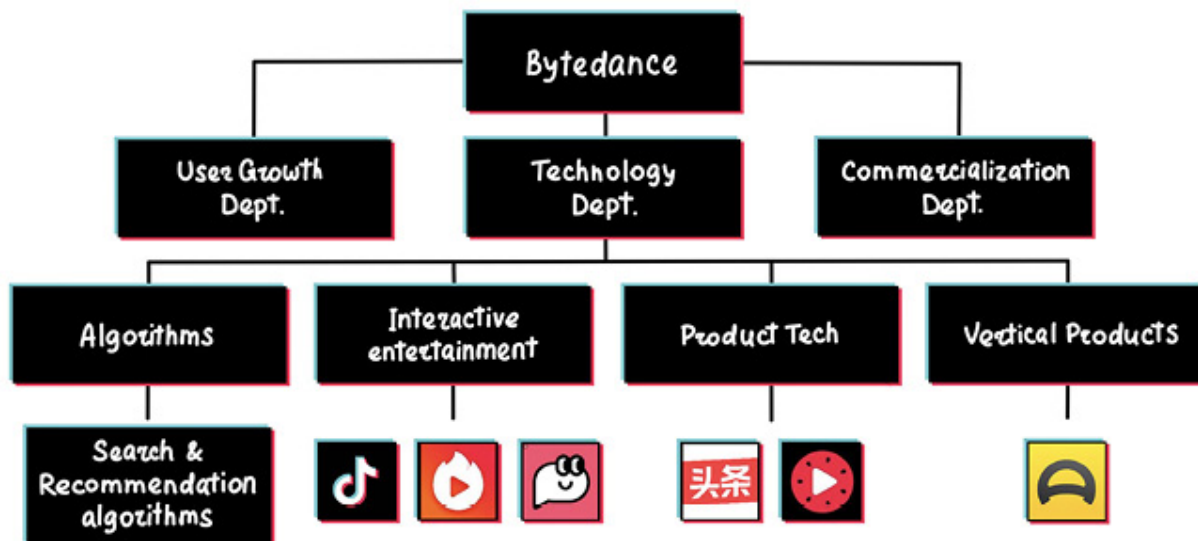
Key People at ByteDance



Higher Management (mid-2020)

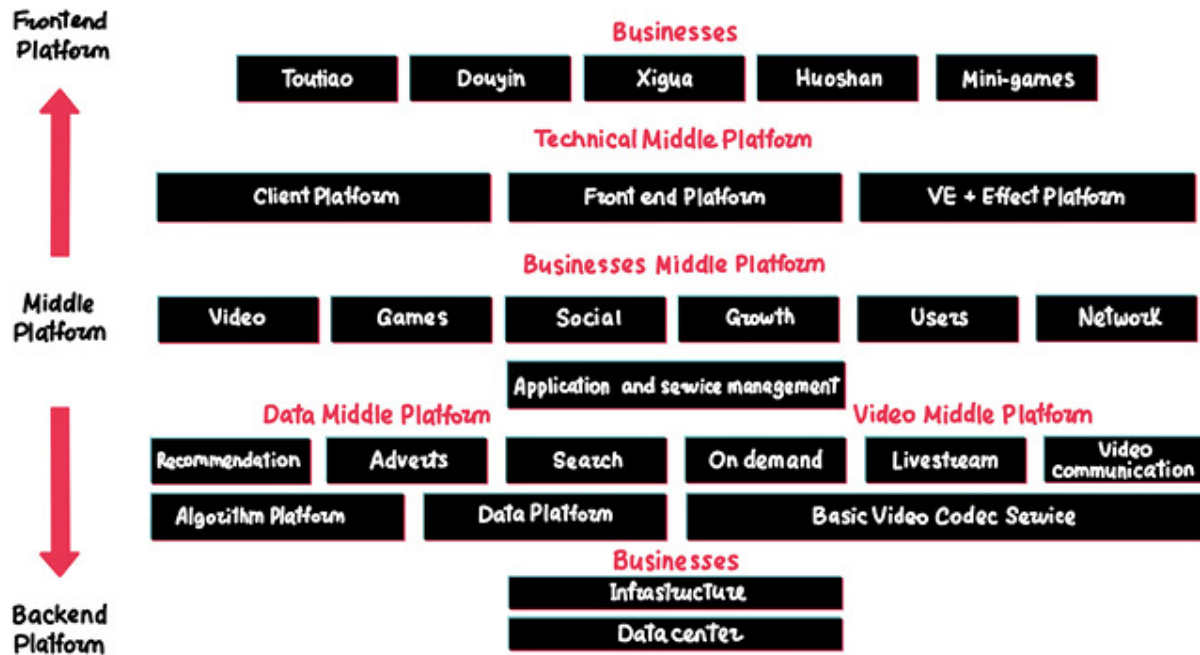


ByteDance Company Structure (Overview)

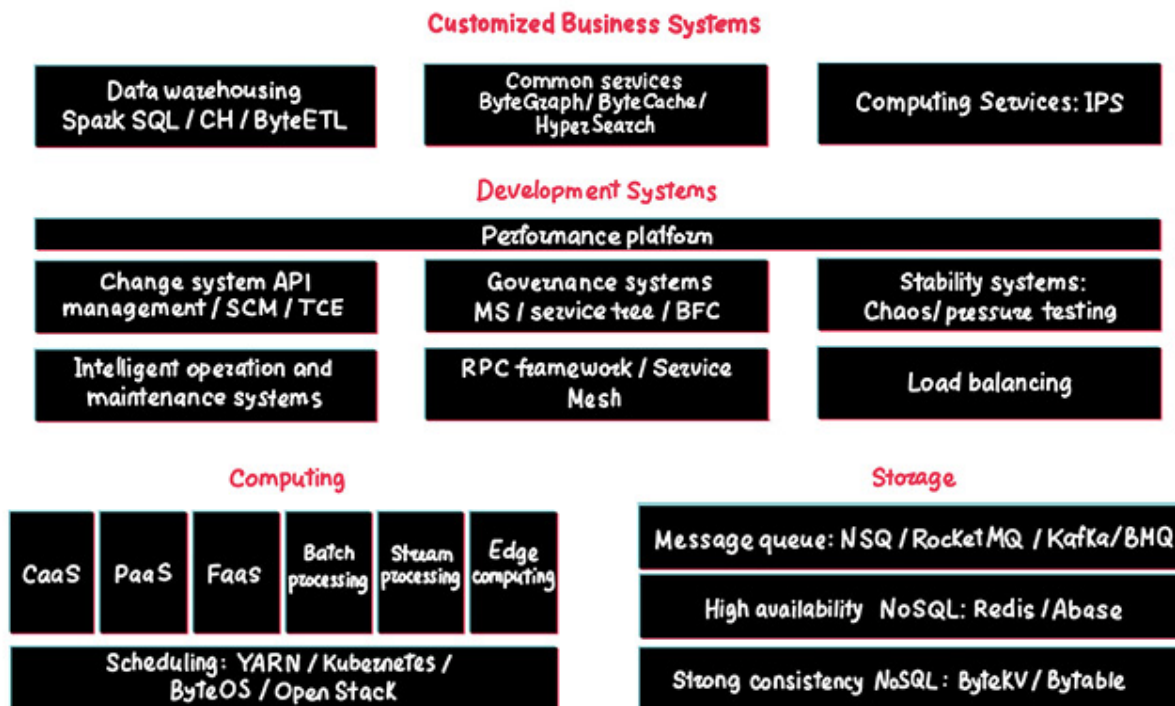


There are three core functional departments at ByteDance: user growth, technology, and commercialization, focused on user acquisition, product development, and monetization respectively.

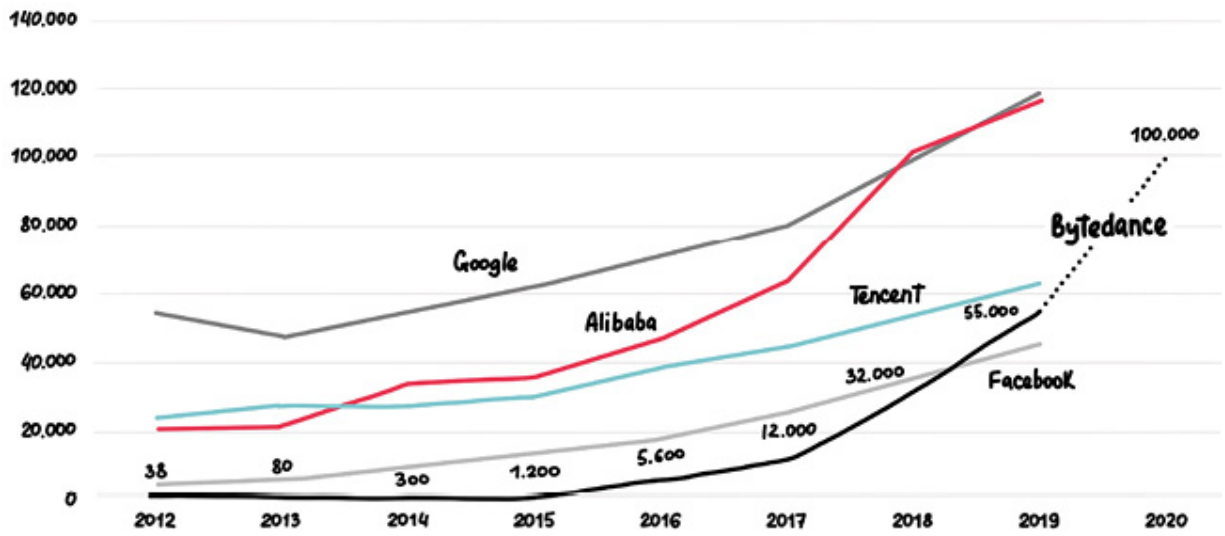
ByteDance Technical Systems (Overview)



ByteDance Technology System Infrastructure

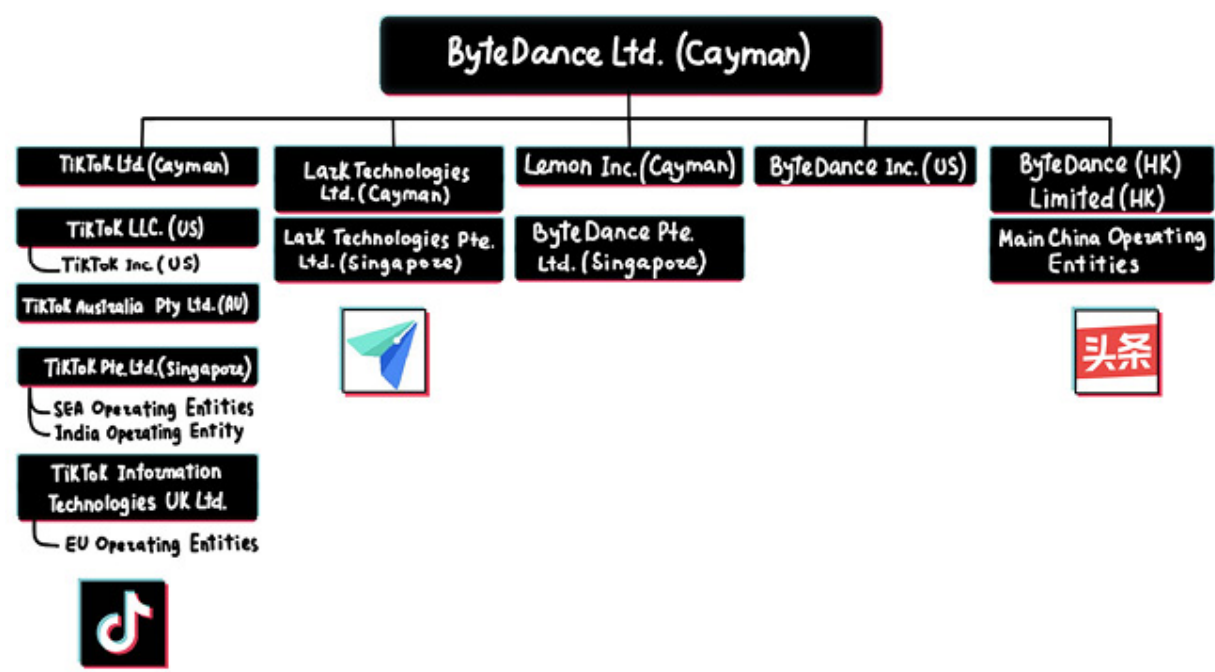


ByteDance Number of Global Employees (estimated, pre-India ban)



Sources: company financial reports, SinaTech, Reuters, 36Kr, TMT post, Huxiuwang

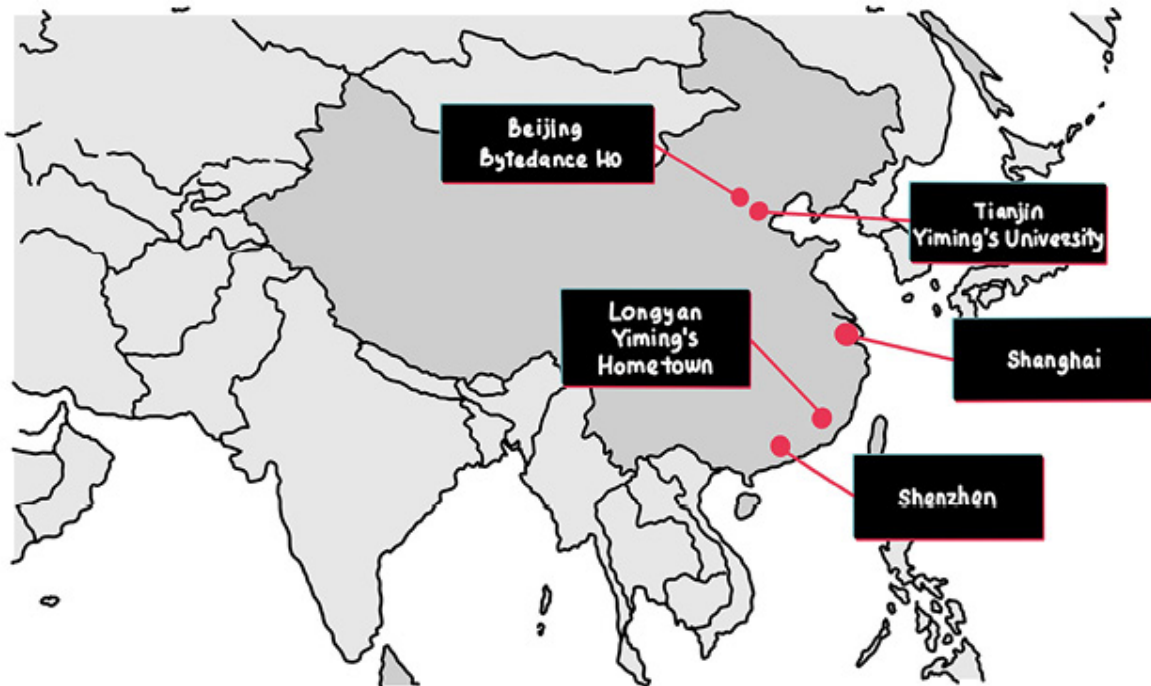
ByteDance Corporate Legal Structure



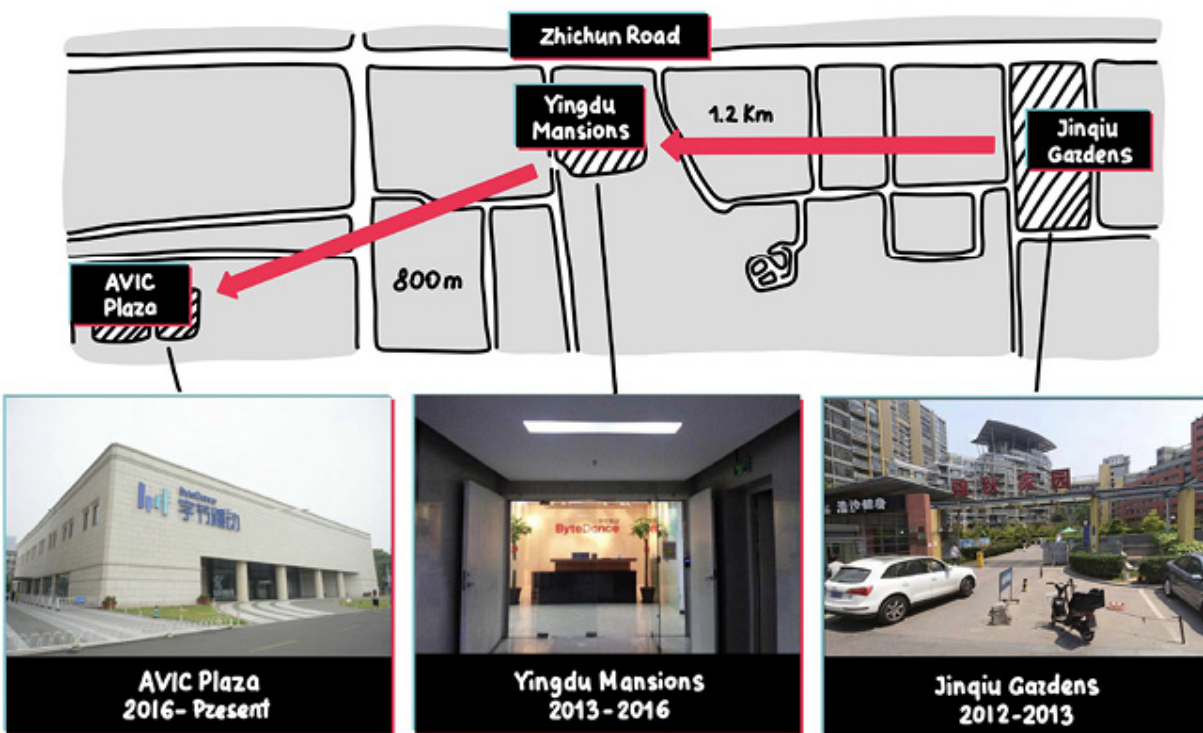
Investment rounds

Date	Round	Amount, USD	Lead Investors	Valuation, USD
2012 April	Angel	3 million	SIG Asia	-
2012 July	Series A	1 million	SIG Asia	-
2013 May	Series B	10 million	Yuri Milner (Apoletto), SIG Asia Source Code Capital	60 million
2014 June	Series C	100 million	Sequoia China, Weibo	500 million+
2016 Dec	Series D	1 billion	Sequoia China, China Construction Bank International	11 billion
2017 Sept	Series E	2 billion	General Atlantic	22 billion
2018 Nov	Pre- IPO	2.5-4 billion	General Atlantic, KKR, Primavera Capital Group, SoftBank	75 billion
2020 Mar	N/A	Undisclosed	Tiger Global Management	100 billion

Chinese Cities Mentioned in This Book



ByteDance Head Office Locations, North West Beijing



PREFACE

THE ORIGINAL Chinese version of TikTok, called Douyin, was the kind of app that you'd hear of once and then forget about. It was a late bloomer in a category widely considered to be only of second-tier importance. It was yet another uninspired clone destined to enjoy a fleeting moment of fame before withering away unceremoniously into internet obscurity and shut down without a second thought. It was supposed to be discarded and forgotten like thousands of apps before it in China's fast-paced, cutthroat internet industry.

But beyond all expectations, Douyin and its international variant, TikTok, skyrocketed to unimaginable heights of success. These two apps became a global phenomenon, exceeding even the wildest dreams of the founding team. So what happened?

Being based in China, I saw firsthand the rise of Douyin in late 2017 and the impact it was having on those around me. People started watching short videos while waiting for the subway and shooting short videos on the street. Friends discussed their favorite accounts, and soon after, new celebrities were formed. Songs became hyper-popular overnight, all due to Douyin. Eighteen months later, this same story started playing out across the globe, as interest in TikTok spiked to all-time highs.

The nine-month journey of writing this book has been a strange one. From private conversations dating back to 2018, I can tell you ByteDance staff were extremely aware of the escalating political risks of being a Chinese company operating in the U.S. and other Western markets. Even so, I don't think anyone could have imagined how dramatically things would play out.

Discussion around the company has become heavily politicized as of late, as most are well aware. Yet I did not choose to write this book to add to the already limitless number of articles, hot-takes, and deep-dives dissecting the geopolitical rivalries and "tech cold-wars" in which TikTok finds itself entangled.

Instead, this book asks the questions—Why TikTok? Why ByteDance? Why short video?

Inside you will find a detailed, accurate, and hopefully somewhat entertaining account of the rise of ByteDance's business fortunes as a private company, including the competitive dynamics with rival internet service providers. This book is fundamentally an exploration of the short video format and machine recommendation as the most important internet trends of the past three years.

My primary motivation for writing this book was to shine a light on the extensive knowledge gap on ByteDance between China and the rest of the world. Many anecdotes included in this book are widely known in China but still making their way into English for the first time. In addition, through original research, I have corrected some inaccuracies, dispelled a few myths, and sprinkled in a healthy dose of original analysis and insight regarding the subject, which I hope readers will find value in.

ByteDance has been one of the most globally impactful technology companies in recent years. My sincere hope is that this work can bring value to the broader discussion and understanding around TikTok, ByteDance, short video as a content format, and more in-depth insight into Chinese internet firms in general.

I consider myself neither pro nor anti-ByteDance and, as such, have attempted to depict the company objectively, acknowledging both its negative and positive aspects. I hope to have captured the story of ByteDance and TikTok so far but know that it is just the beginning.

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DISCLAIMER

THIS BOOK is **not** endorsed by ByteDance. I (the author) have no association or business relationship with ByteDance or any of its affiliate companies. Indeed, this work was not sponsored or supported by any organization. Nor do I have any political affiliation or agenda in creating this work.

Opinions, interpretations, and theoretical frameworks expressed in this book represent my own views, except in cases when explicitly attributed otherwise.

As of writing, ByteDance is a privately held company under no obligation to release audited financial information or user numbers. Quantitative data points in this book may deviate from documentation released later, if or when the company does go public.

This book is the result of extensive research, most of which was conducted in the Chinese language. Sources number in the many hundreds and include:

- Conversations with ByteDance former and current employees
- Media articles and essays, the most important of which are listed under the “Works cited” section of this book
- Video footage of presentations and in-depth interviews
- Analyst reports and academic papers
- Relevant social media accounts
- Extensive personal use of ByteDance’s product suite, both Chinese and international
- The base foundation of knowledge regarding the Chinese mobile internet industry gained through years on the ground in China covering rival Tencent

Link rot is a serious issue with Chinese internet content. Source links provided may cease to work as the content is moved or taken down. I have

done my best to verify and cross-check information, erring on the side of caution when sources conflicted.

Note on the Use of Names



The three most important executives at ByteDance all share the same surname—Zhang. ¹ In order to mitigate this confusing occurrence, I have leaned towards using given names where possible throughout the book at the risk of seeming overly familiar or informal at times, for which I apologize. For what it's worth, the use of given names is in keeping with the company's culture. The founder, Zhang Yiming, once sent out a company-wide email forbidding the use of job titles when addressing others and insisted that staff refer to him simply as Yiming.

Exchange rates

Monetary values are expressed either in United States dollars or Chinese yuan, otherwise known as Renminbi RMB. The exchange rate between these two currencies is set as RMB 1 = USD \$0.14 unless expressed otherwise.

1 Surnames in Chinese come before given names. For example with the name Zhang Lidong, Zhang is the surname, Lidong is the given name.

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ACKNOWLEDGMENTS

2020 HAS BEEN a tough year for all of us and a time that will be hard to forget for all the wrong reasons. Firstly, I wish to acknowledge the loving support of my family, which has kept me sane throughout the process of research and writing.

I would like to thank Rita Liao for being my sounding board and guide throughout the lengthy journey. The value of an editor willing to challenge assumptions and question supportively cannot be understated. I have nothing but praise for her efforts in keeping me on track when I could not see the woods for the trees. This work has benefited immensely from her contributions and would not be the same book without her.

A big thank you to those who read early versions of the manuscript. In particular, Ed Sander's eye for detail and intelligent questioning was truly invaluable.

I also wish to note the inspiration, advice, and support of John Artman, Pascal Coppens, Elijah Whaley, Fabian Bern, Bernard Leong, Jeffrey Towson, and Kit Harford at various points along the journey leading to this work.

Over the many years of becoming engrossed in the Chinese internet, I have significantly benefited from the thoughts and writings of others. I would like to acknowledge the influence of those who have enriched my thinking through their works, including Pan Luan, Jason Ng, Tracey Xiang, Rio Nook, Huang Hai, Sheji Ho, Jordan Schneider, and Dan Grover.

Finally, a hat tip to the individuals who made significant contributions but have asked to remain anonymous. You know who you are, and thank you.

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This book is dedicated to my daughter, Millie.

You are amazing.

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PART ONE

THE BACK END

Algorithmic Recommendation



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PROLOGUE

GAMING THE GATEKEEPERS



“You think becoming an online influencer is simply a matter of making interesting videos? Come on, don’t be so naïve!”²
-The manager

A FAINT HUM emanated from the room ahead, and for some reason the damp stairwell smelled of plastic packaging. “This is it,” the manager said through a cloud of cigarette smoke as the metal door creaked open. “We keep them all in the basement.” The two clients hesitantly stepped into the room, and everyone was silent as they absorbed the scene.

Before them lay several thousand smartphones arranged neatly in metallic racks.³ The screens were busy, animated, flickering with activity. A maze of wires interlaced themselves across the concrete floor of the large windowless room.

Curious, one of the clients walked over and picked up a phone, the wire still connected. He held it in his hand, fascinated—the device operated by itself: switching between apps, scrolling through newsfeeds, selecting videos. The pace was slow, with frequent pauses designed to mimic natural human behavior. The effect was both mesmerizing and deeply unnerving. The silent glow of countless automated screens, a ghostly orchestra. Its invisible conductor was a single PC, with central control system software⁴ to which each phone was connected by a wire.

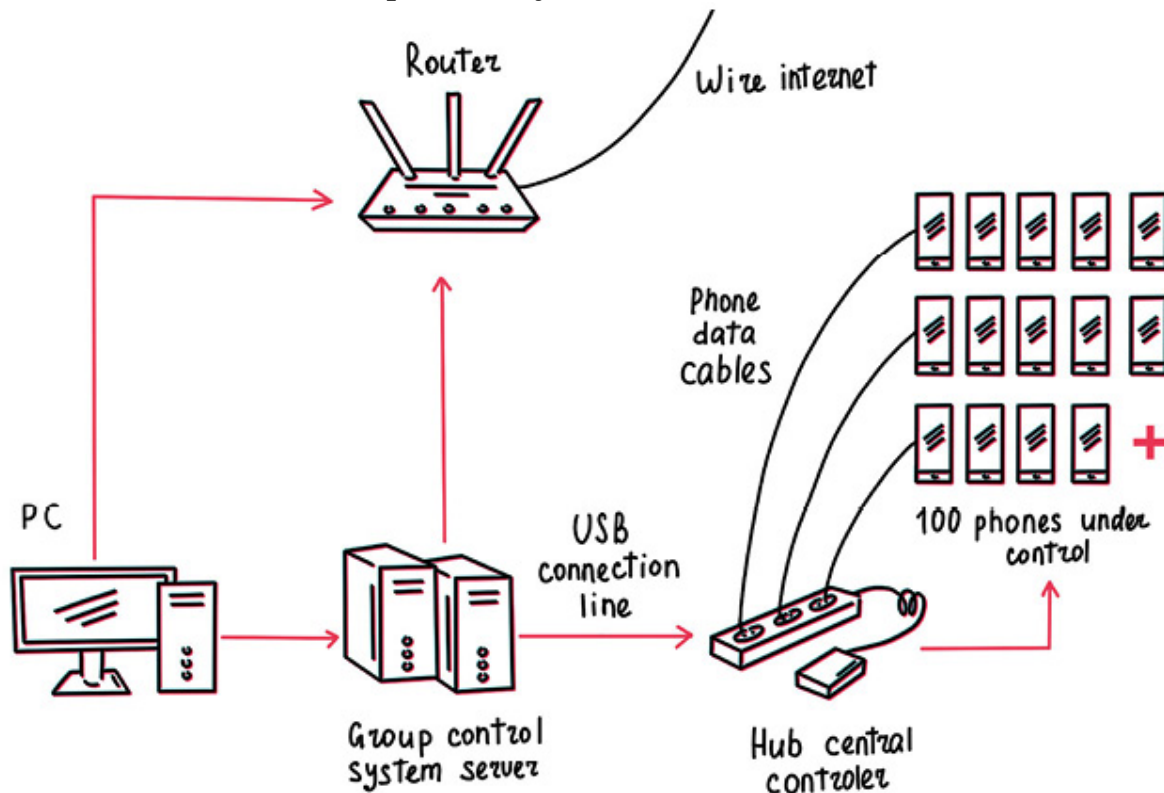
The manager walked over to the client holding the phone and slapped him on the back. “You think becoming an online influencer is simply a matter of making interesting videos?” he asked, laughing wholeheartedly at his own question. “Come on, don’t be so naïve! Now, which services are you interested in?” The two clients looked at each other, intimidated and unsure about who should speak. “Mainly Douyin,” replied one somewhat timidly, referring to the Chinese version of TikTok. The manager chuckled. “You’re not the only ones,” he said, offering the pair a wolfish smile as he reached into his pocket. “We got half these phones swiping through Douyin all day, helping guys like you.” He pulled out a crumpled soft pack of Kent cigarettes and offered them to the clients. “Smoke?”

This basement operation was a typical “click farm.” Hundreds of these underground businesses operated across China, playing a vital role in the modern online attention economy.

Skirting around bot-traffic detection systems and occasionally having accounts shutdown was an occupational hazard. Software that virtually emulated phones could be easily detected. Despite the equipment and

electricity costs, it was much more reliable to use physical devices; cheap Android smartphones controlled via USB cables with software simulating a real human's taps and swipes.

Helping others trick the algorithms and manipulate online attention was a lucrative business, and there were always willing customers. Fake likes, zombie followers, simulated livestream audiences, automated comments, rigged competition voting, mass reporting of content to trigger takedowns – the list of “services” was practically endless.



Above: a typical click farm equipment set up⁵

The manager leaned his head back slightly and blew a cloud of cigarette smoke up toward the concrete ceiling: “We sell set packages. With Douyin, 350 yuan (\$51) gets you 10k likes, 100k views, 5k shares, and 50 comments.” The clients nodded their heads in agreement; it was a reasonable price.

“It’s the shares that matter,” continued the manager. Again, the two clients looked at each other, unsure if they should say anything. Finally, one of them broke the silence. “We heard the ‘watch completion rate’ was most important.”

The manager took a long drag on his cigarette before dropping the butt to the floor and grinding it slowly into the concrete. After a long uncomfortable pause, he looked up and locked eyes with the client who had spoken.

“Every seller in China is looking to game this system now. You’ve got a lot to learn.”

How to game the TikTok algorithm 101

Millions upon millions of videos are uploaded to TikTok every day, the vast majority of which receive only modest views. The size of each video’s audience is decided predominantly by the system’s ever-changing and mysterious algorithms, and the key to gaming the system is understanding how these algorithms work.

The moment a video is uploaded to TikTok, the clip and its text description are queued up to go through an automated audit. Computer vision is used to analyze and identify elements within the clip, which are then tagged and categorized with keywords. Videos suspected of violating the platform’s content guidelines are flagged for human review. The audit cross-checks the footage against a massive archive for duplicate content. This system is designed to prevent plagiarism, as well as the practice of downloading popular videos, removing the watermark, and reuploading them to a new account. Videos identified as duplicate have their visibility significantly reduced.

After the screening process, the video is released to a small pool of a few hundred active users. Metrics such as the number of complete views,⁶ likes, comments, average play length, and shares will be analyzed to gauge the video’s popularity within its vertical category. Those that perform well pass through to the next level, where the video is then exposed to thousands of active users. Again, more metrics will be evaluated, with the top-performing videos passing on to the next level, where they gain exposure to an even larger audience. As the video moves up to higher tiers, it will gain exposure to potentially millions of users.

The process isn’t entirely run by an algorithm. At the higher levels, a person on the content moderation team will watch the video and follow a set of strict guidelines to confirm that it does not violate the platform’s

terms of service or have any copyright issues. There are cases of videos reaching up to a million views only to suddenly be taken down once they hit the human review process.

On a platform like TikTok, with such massive amounts of attentive users, there was no shortage of bad actors trying to find loopholes and shortcuts to game the system. By closely monitoring an account, it was possible to use activity from click farms to artificially inflate the metrics used by the algorithm to judge a new video's popularity, significantly increasing the chances of the video being boosted to the higher levels.

Rather than placing their eggs in one basket and boosting the metrics for just one account, unscrupulous marketers further increased their odds of success by operating tens and even hundreds of similar accounts. Longer videos were cut up and edited into shorter content sections using various filters and effects to evade the duplicate content detection systems.

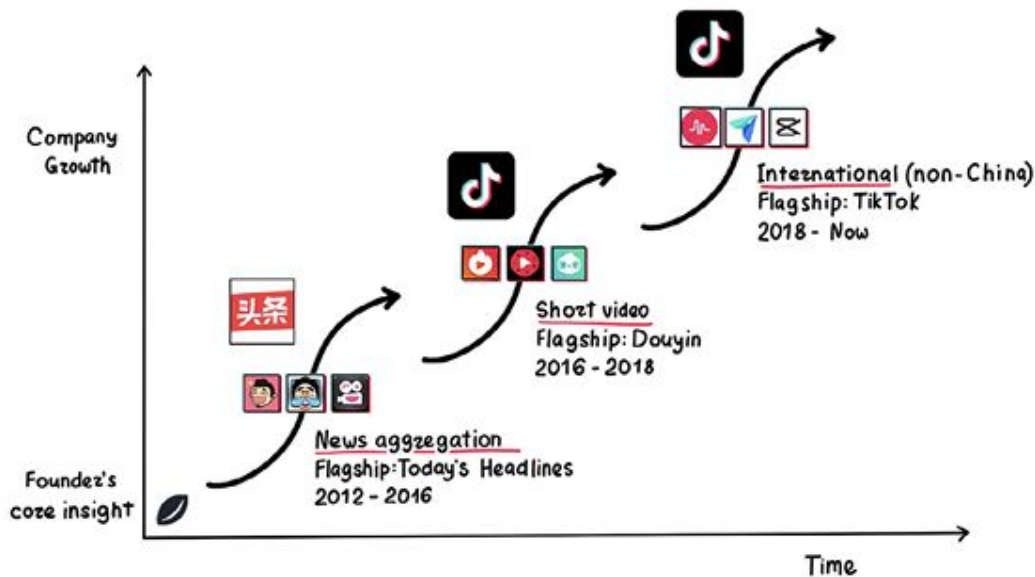
It was a never-ending game of cat and mouse as the platform became stricter and stricter in employing all kinds of methods to detect and counter nefarious conduct. Yet each time the platform managed to close down one loophole, another exploit would soon be found.

A set of established rules emerged for operating mass numbers of accounts at scale while avoiding detection. New accounts needed to be “cultivated” before the platform would judge them trustworthy. This involved mimicking the expected behavior of a typical user. Immediately batch uploading pre-edited videos on a new account was a sure-fire way to get “shadow-banned,”² rendering the account useless. The best practice was to passively watch videos for at least seven days before posting anything and record the first few videos within the app using the phone's camera. Using a separate phone number and SIM card for each account was a must. Never log in through multiple devices—that was a dead giveaway that the account was being operated professionally. To deceive machines, a person needed to have a scrupulous attention to detail.

These practices boiled down to a single goal: trick the app's recommendation systems. Recommendation—the use of machine learning to infer people's preferences for content based on their behavior alone—

was at the very heart of TikTok. It was the key to understanding the success of the app and its parent company, ByteDance.

ByteDance had been the earliest Chinese internet company to go “all in” on the then-nascent technology and commit to the daunting task of building a recommendation engine, challenging the status quo of human curation. This early bet paid off in spades. The foundations of TikTok’s success were laid many years before the app itself was built, and it was no coincidence that ByteDance was the company to make it.



Above: ByteDance’s three growth phrases, news aggregation from 2012 to 2016, short video from 2016 to 2018, and international from 2018 onwards

Today ByteDance is a sprawling corporate leviathan. Much like other large internet conglomerates, it has expanded into a myriad of online services consisting of gaming, education, enterprise productivity, payments, and much more. This is made possible by three businesses: Toutiao, Douyin, and TikTok, which are, at the time of the book’s writing in 2020, the core drivers of the company’s valuation and fast growth.

The rise of these businesses reveals three distinct growth stages driven by breakthrough flagship products that enabled the company to reach hundreds of millions of new users. Each stage featured a family of supporting applications, both self-developed and acquired, which extensively diversified the company’s offerings and product categories.

Each phase builds directly upon the products and technology developed in the previous one. TikTok's success was predicated on replicating its Chinese sister app Douyin's technology, product experience, and promotional playbook. In turn, Douyin's success relied on the recommendation engine, operational expertise, and cash reserves developed through Toutiao, the company's news aggregation app that made it a household name in China. Last but not least, the success of Toutiao can be traced back to the prescience and decisiveness of the company's founder, Zhang Yiming.

By 2011, seasoned entrepreneur Yiming recognized that smartphones would profoundly change how humans received and interacted with information. Although no one could have foreseen such at the time, this seed of thought, planted at the very founding of the company, eventually manifested itself through the global success of TikTok. To quote Steve Jobs, "You can't connect the dots looking forward; you can only connect them looking backwards."

To understand TikTok and the company behind it, ByteDance, we need to start at the beginning—in a small village in southeast China, Yiming's hometown.

2 Image: click farm equipment. Zombie phones stacked in racks and operated remotely using automated software.

3 <https://www.youtube.com/watch?v=AnpetUlz19A&list=UUXNHI7mRQIUUpLi6rOwOxg&index=63>

4 <https://youtu.be/2sUt-9-2Pxo?t=26>

5 Image adapted from source article: <http://www.woshipm.com/it/2484849.html>

6 A complete view is defined as watching the entire video from start to finish

7 A “shadow-banned” video would be deprived of reaching any audience, yet still viewable from the account page. To the account holder it may appear as if the video was just very unpopular.

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CHAPTER 1

YIMING

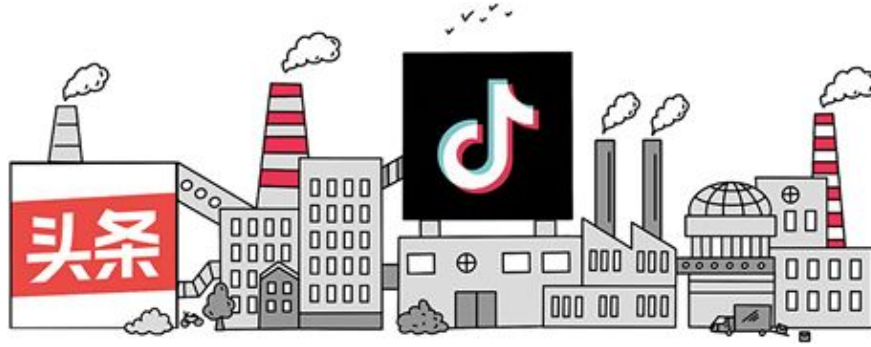


*“By many quantitative metrics, Zhang Yiming
is the top entrepreneur in the world.”⁸
-Time Magazine, 2019*

⁸ Image: Zhang Yiming at Musical.ly head offices in Shanghai

Quote source: <https://time.com/collection/100-most-influential-people-2019/5567716/zhang-yiming/>

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Chapter Timeline

- 1983 – Zhang Yiming is born in Longyan, Fujian province, China
- 2001 – Begins undergraduate study at Nankai University, Tianjin
- 2005 – Upon graduation, starts own company, fails within six months
- 2006 – Joins online travel search engine Kuxun as the first engineer
- 2007 – The first iPhone comes to market
- 2008 – Brief stint at Microsoft Beijing
- 2008 – Joins Twitter clone Fanfou
- 2009 – Becomes CEO of online real estate portal, 99Fang.com
- 2011 – WeChat's breakout marks the start of China's "smartphone era"
- 2012 – Decides to leave 99Fang.com and start ByteDance

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“I STARTED READING VERY YOUNG,” ⁹ Yiming recalled.

“Around the second year of kindergarten, my dad was ordering weekly children’s reading magazines for me.” And so began a lifelong habit—the love of consuming information. By fourth grade, Yiming had started to read several simple novels, biographies, newspapers, and periodicals. While other children watched cartoons, Yiming preferred science fiction and mythology.

During junior high school, he claimed to have read twenty to thirty newspapers a week. From local tabloids to national dailies, he devoured every article without missing a single word. Reflecting back in a later article, Yiming felt frustrated by the lack of options, as he could only read what was physically available in front of him. “I often think if I had a Kindle and an iPad when I was a kid, if I could access Wikipedia and YouTube, I would be so much smarter than I am now.”

His parents encouraged his reading. Their choice of name—Yiming—embodied the very hopes they held for their child’s future. The name alludes to a Chinese idiom ¹⁰ used to describe someone of low profile who suddenly rises to great success: a perfect choice given Yiming’s later life trajectory.

Yiming’s early life in rural China was a humble beginning. Still, it was already clear he was highly studious, held a voracious appetite for consuming information, and had a strong preference for self-improvement—all attributes that proved critical in his future career.

While in elementary school, Yiming’s dream was to become a scientist, a desire no doubt inspired by his parents’ scientific leanings. Yiming’s mother was a nurse, and his father was a member of the local city science and technology commission, who often discussed the latest technologies at home.

Yiming’s father, Zhang Hanping, would later resign from his position to try his luck in the booming “workshop of the world,” the Pearl River Delta, 300 miles southwest. He set up an electronics factory in the gritty industrial city of Dongguan, close to Shenzhen and Hong Kong. This business’s income ensured that Yiming would be well taken care of despite being born

and raised in a modest countryside environment. While not wealthy, his family was commonly described as being “comfortable.”

Kongfu,¹¹ the family hometown, was a village within the Longyan district of western Fujian—a coastal province that lies across from the island of Taiwan. The area is traditionally known for its entrepreneurial merchants, mountain ranges, and tea culture. Kongfu people, including Yiming’s family, are Hakka Chinese. They speak a distinct dialect, difficult for most other Chinese to understand, and even have their own Hakka-style cuisine, something Yiming sorely missed during his time away.

A college nobody

Yiming had once famously said that in launching a business, he is creating two types of products: first, the consumer-facing platforms, and second, the actual company itself. There is arguably a third product the entrepreneur has been carving out over the years: himself.

Indeed, Yiming runs his life as if it was a software product undergoing constant tests and iterations to reach optimization. To that end, he has created a set of formulas that help him make life choices.¹² One early example concerns how he came to choose his college, arguably the most critical decision in a young Chinese student’s life. The process would typically involve long discussions and consultations with parents, followed by careful evaluation and research. Instead, Yiming considered which outcomes were important to him personally and boiled everything down to four conditions. Based on these four conditions, he filtered through hundreds of options, deducing the conclusion from his premises in minutes.

“First, it has to be a well-known, comprehensive university,” the entrepreneur explained to a roomful of current Nankai University students when visiting his alma mater in 2019.¹³ That was because such an institution would enjoy a more balanced gender ratio than pure science-focused universities, making it easier for Yiming to find a girlfriend. Yiming indeed did find his first girlfriend at Nankai, eventually marrying her.

“Second, it has to be close to the ocean. Third, it should be far away from my hometown, and fourth, it has to snow in the winter,” the entrepreneur

continued. He grew up in the southern coastal province of Fujian, where it rarely snows.

The only school that met all these criteria was Nankai, an hour's drive from Beijing. That was where he would spend his four years of university.

This anecdote provides a revealing insight into Yiming's decision-making process. Based on a given set of conditions, he is able to simplify complex decisions and hold conviction in a single "optimal outcome" that is often not what others would expect. Nankai was a top twenty university in China, and by no means a poor choice, yet most Chinese students would aspire to more famous institutions. However, for young Yiming, winter snow, finding a girlfriend, the freedom of being away from family, and proximity to appetizing ocean seafood were all that mattered.

In college, he initially wanted to study biology, a subject that at the time was deemed to be "the bellwether of the 21st century." Unfortunately, his scores weren't high enough. He opted instead to enroll in electrical engineering, but before long had switched to software engineering. Electrical engineering, he reasoned, provided "few chances to apply textbook theory to real-life applications" whereas the cycle of computer programming was far shorter and the results could be seen faster.¹⁴

A person of few words and a child-like face that endures to this day, Yiming was hardly noticeable on campus. He performed well academically but never blended into his college's mainstream social scene. Nor did he have grandiose plans like other peers with good grades. "I didn't join any of the student clubs or study the GRE ¹⁵ to go abroad. I had none of those aspirations," he recalled.

Soon after the first semester began, he set up a desktop computer in the dorm room and practiced programming, familiarizing himself with the emerging internet industry. Unlike his peers, Yiming rarely played cards, indulged in computer games, or watched movies. Titling himself as a "moral champion," ¹⁶ he set about conducting personal improvement tasks, splitting his time after class between programming, reading, and repairing computers—the three pursuits that he later credited for helping him cultivate patience, knowledge, and friendship.

“Endurance, the ability to be alone, making judgments based on long-term considerations without being blinded by short-term factors, as well as having the patience for your plans and efforts to materialize—these are all very important in building your own startup,” he affirmed.

Despite having narrowly focused interests and a rigid schedule, Yiming set aside time for more spontaneous moments to bond with people. But even those events were meant to achieve optimal results. In addition to coding, as mentioned, Yiming often repaired computers, which was an efficient way to meet new people. Most of the requests for help came from female classmates, including his future wife.¹⁷

Yiming married his first girlfriend, whom he first came to know through the university’s online bulletin board system (BBS).¹⁸ There, he had burnished his reputation for having the best technical skills in fixing computers for his peers. Yiming later explained the rationale behind marrying his first partner as such: “If there are 20,000 people in the world who are suitable for me, I only have to find one of those 20,000. That’s an approximate optimal solution within an acceptable range.”¹⁹ How romantic.

Such quotes make it easy for people to characterize Yiming as being an unemotional robot. This depiction doesn’t always hold true when we consider his serial entrepreneurship ventures and bold, ambitious bets in the world of business. However, it is worth noting the juxtaposition between Yiming’s conservative personal life and his entrepreneurial career’s daring ambition. Many years later, Yiming would insist that ByteDance, contrary to public impression, is a “romantic” company, his definition of romance being “turning imagination into reality—to face reality and change it.”

Despite being goal-oriented and reclusive in college, Yiming still cherished friendship and camaraderie, which boded well for his future recruitment efforts. Yiming’s part-time pursuit of fixing computers together with another side gig in which he built websites for businesses, brought him a monthly income of two to three thousand yuan (roughly \$285 to \$430). This was no trivial amount for a Chinese student in the early 2000’s, and Yiming would spend some of his money on treating a few of his close classmates to barbecued meat skewers after staying up late together finishing assignments.

A handful of his Nankai friends, the most notable being his university dormitory roommate Liang Rubo, joined Yiming at ByteDance years later. Studying computers and programming together, the two forged a deep friendship, sharing a PC and playing badminton together every weekend.

Yiming held fond memories of consuming meat skewers with Rubo and his other college friends. So much so that when ByteDance moved into a new larger headquarters in Beijing, he got psyched about the opportunity to set up a barbeque coal rack on the company's rooftop to bring back his favorite late-night snack. "That reminds me of my joyous days at Nankai."²⁰ Which is interesting because Yiming isn't known for being someone who gets easily excited; he is consistently described as being incredibly calm and rational.²¹

During his four years of college, Yiming's grades were consistently good but never remarkable. A rare highlight being the time when he won second prize in a competition with circuit board automation software he wrote in his senior year. None of his classmates or teachers had considered that this inconspicuous outsider—an even slightly dull classmate—would later become one of the most accomplished entrepreneurs in the world, and the brain behind some of the apps that would come to dominate hundreds of millions of people's digital lives.²²

Entering the workforce

Having known Yiming for many years, Rubo described his friend's central conception of life as a "craving for excellence" or, to put it another way, a struggle to "escape the gravity of mediocrity." This goes some way to explaining why, upon graduation, Yiming immediately dived into entrepreneurship, co-founding a business with two schoolmates to develop collaborative office system software.

Yet the project quickly failed. The concept was too advanced for the China market at that time. No one bought the software, due in part to the team's inability to communicate its benefits effectively. Yiming was not discouraged. Unlike most college graduates in China, he seemingly did not value stability that much. Instead, Yiming was interested in taking bets on promising new fields. He rarely displayed much interest in material goods or personal wealth. Of course, it greatly helped that his father's electronics

factory's income afforded him a certain degree of social status, economic security, and freedom to pursue his interests.

Looking for a new challenge, Yiming posted his software engineering credentials and contact information to an online forum to find a job. Before long, another Nankai alumnus contacted him about joining a travel industry startup called Kuxun. Yiming went to Beijing for an interview with the fledgling online flight and hotel search engine site and impressed the interviewer by suggesting a useful technical improvement. He was immediately hired as employee number five. Within months, Yiming had quickly become the backbone of the company. "I was an ordinary engineer at the beginning, but by the second year, I was in charge of about forty to fifty people responsible for back-end technology and other tasks related to products," he later revealed in a speech.²³

"Because I was young at the time, I could work non-stop night and day and pull all-nighters. If I did go home early from work, I usually would read books and study until one or two at night. That period feels very fulfilling; for those two years, I was studying day and night." He took the opportunity to learn everything he could and even tagged along with the sales team to visit clients. Yiming cited this experience as invaluable years later when it came to setting up ByteDance's first advertising sales teams.

Later at ByteDance, Yiming would look to hire candidates in the mold of his younger self—young, driven, and dedicated. The entire Chinese internet industry was built around leveraging such inexperienced fresh graduates. China's private sector economy was fast-paced, dynamic, and in many areas, intensely competitive. The lucrative and rapidly expanding internet services sector amplified these characteristics taking them to a whole other level.

Business was conducted like a brutal guerilla war in which developers, engineers, and operations staff worked themselves to death under grueling schedules in which speed of execution was everything. Companies looked to raid universities to recruit young, unmarried, aspiring "front line soldiers," employees willing to be chewed up and spat out of the system by their mid-thirties in exchange for generous compensation or the chance to strike it rich with an IPO.²⁴

Despite its modest success, Yiming's new employer, Kuxun, holds a somewhat legendary status within China's internet industry. At its peak, the company employed just 170 people, yet over thirty of those staff members went on to become active entrepreneurs starting their own internet companies.²⁵ Yiming would become the most prominent member of the "Kuxun entrepreneurship gang," a term that echoed Silicon Valley's famous "PayPal mafia."

The Kuxun co-founder Chen Hua had ambitions of competing with search giant Baidu in the race to become "the Google of China." Yet Baidu, by this time, was already invincible, having defended its market share from Google, who later voluntarily²⁶ left the market, gifting its former rival a near-monopoly in Chinese language search. Yiming had just graduated from Nankai University around the time Baidu went public on the Nasdaq. By March 2011, Baidu's market value had reached \$44.5 billion, exceeding rival Tencent to become China's most valuable internet company.

Chen Hua realized that the window of opportunity to compete with Baidu in "general search" had passed. He turned to the smaller yet still sizable market of vertical search for the travel industry, covering bookings of flights, hotels, and tours. Regardless, Chen Hua's big ambition of one day challenging the invincible Baidu had instilled itself in many people at Kuxun, especially young Yiming. Unexpected by many, it was Yiming who would later realize Chen Hua's dream.

Unfortunately, the founders left Kuxun, and the company fell into a period of chaotic mismanagement just after it had gained traction in the market. Yiming realized it was time to move on, and in 2008 he joined the Microsoft Research Asia Institute (MSRA)²⁷ in Beijing with the idea of learning lessons from one of the most prestigious technology companies in the world.

Yet what he found was completely different from his expectations. Looking back on his time at Microsoft, Yiming described it as the most boring year in his career. Switching to a slow-moving big company environment filled with rules and regulations was a culture shock. He found himself idle for roughly half the day and broke the boredom through his favorite habit, admitting, "I read a lot of books."²⁸ During this time, he voraciously consumed informative autobiographies. Favorite titles include

Steven Covey's productivity classic *7 Habits of Highly Effective People* and the business bible *Winning* by legendary American CEO Jack Welch. Microsoft rewarded following processes and focusing on details. Yiming found himself frustrated by what he saw as inefficient work methods. "I don't think it is suitable for people who have special ideas and motivation... I still prefer a challenging and creative life," was his take.

During his time at Microsoft, Yiming bought his first apartment in Beijing—a big deal for anyone his age. Real estate is taken very seriously, particularly for young men in China, due to cultural expectations of homeownership before marriage. Owning their first property in the capital would be a lifetime milestone for any Chinese, with similar pricing to New York or London real estate and requiring a mortgage loan to be paid back over decades. The standard approach would include consulting with friends and work colleagues, intense deliberations with family relations, approaching multiple real estate companies, and making many visits in person to various housing compounds.

Yiming's method was, of course, different. He built a software program to crawl the web and collect all the available data online about Beijing's housing market, placing it into a database. After generating multiple excel spreadsheets, he crunched the numbers, ranked all the options, and finally came up with his single optimal answer. He took this to a real estate agent with the simple instruction, "get me an apartment in this compound."

After one year, the chosen apartment had more than doubled in price. Of course, the whole market was rising at that time, but the compound Yiming chose increased the most of any within the surrounding area. This anecdote again highlights Yiming's disregard for traditional approaches when making high stakes decisions and skepticism of the status quo. He simplified a time consuming and complex decision, opted instead for an efficient and logical comparison of data to produce a single optimal solution. The apartment he bought was later sold to raise money to start ByteDance, a move that unambiguously demonstrated his strong commitment to early team members and investors.

The Twitter of China

During his time at Kuxun, Yiming had gotten to know another young entrepreneur, Wang Xing, a small-time hustler who would later become one of the most influential figures in China's business world. Wang Xing and Yiming shared the same hometown, Longyan, a thousand miles from their work in Beijing. Wang Xing's father was also a factory owner. Wang Xing, by this time, was 29, with Yiming just four years his junior at 25. Wang Xing had also been a computer engineering student. The success of Facebook had inspired him to drop out of his Ph. D. program at the University of Delaware, returning home to become an internet entrepreneur. Yiming and Wang Xing were kindred spirits, two computer geeks far away from home, hustling their way through the wild, topsy turvy world of Chinese internet startups.

Wang Xing persuaded Yiming to join his latest venture, Fanfou, a clone of Twitter built for the China market. Yiming was like a fish in water, back in his entrepreneurial element after time away at big tech Microsoft. He served as a technical partner responsible for Fanfou's search features, trending topics, and social analytics. The site developed fast and, at the time, was considered one of the brightest stars of the Chinese internet.

Yiming later reflected on his time working at Fanfou, saying one of the big realizations for him was how social networking and obtaining information were two separate things. On a platform like Fanfou, or Twitter, users were both communicating with friends, staying in touch, and sharing thoughts. At the same time, they were also consuming information such as breaking news or articles of interest. These two activities were easy to conflate but represented two different needs. Clarity around this distinction later helped Yiming in setting ByteDance's early direction.

Fanfou's progress came to an abrupt halt in July 2009. Riots erupted in Ürümqi, the capital of China's far west province of Xinjiang, following protests by local Muslim Uighurs. The Chinese authorities said nearly 200 people were killed, mostly ethnic Han Chinese, though Uighur groups claimed the toll was higher.²⁹ To tighten control of online discourse, Beijing launched an unprecedented online crackdown.

The internet was shut down for nine months across Xinjiang, with thousands of websites being banned and blocked across the country, including Twitter, Fanfou, and Facebook. Fanfou was eventually permitted

to reopen after one and a half years. By that time, a new local player, Sina Weibo, had already come in and run away with the microblogging market, earning the “Twitter of China” crown.



Above: a side by side comparison of Twitter and Fanfou user interfaces.

Censorship is a risk that all media companies operating in China must confront; the key to survival is understanding and anticipating the authorities’ playbook. The overnight shutdown of Fanfou was Yiming’s first time experiencing the immense disruption falling foul of Beijing’s internet regulators could bring. It would not be his last.

It wasn’t long before Wang Xing’s focus shifted to his next venture Meituan, which would eventually become one of China’s largest and most successful internet companies providing food delivery services and local restaurant listings. Today Wang Xing and Yiming are two of China’s most respected CEOs, and their friendship endures.

Yiming becomes a CEO

With Fanfou’s website shut down, Yiming was left with little to do and waited two months before leaving. That was when venture capitalist Joan Wang pounced on the opportunity and persuaded Yiming to join a new venture. Joan was the managing director of SIG China Investment, a Chinese venture investment arm of the U.S. financial giant SIG. She knew Yiming well as she had led SIG’s investment s in Kuxun and became impressed by his capabilities.

Years earlier, Joan's first impression of Yiming had initially left her doubtful. "Seeing him at first glance, he felt like a high school student. I felt nervous. How could Kuxun use such a young boy to lead the company's important business line? But right after he talked to the entire board of directors... his vision, his understanding and control of technology immediately impressed us all."³⁰

The idea she had for him came from Kuxun's side business line, a search section focused on real estate information, which showed signs of being more profitable than the core travel and hotels search business. Joan saw there was an opportunity to build out a dedicated real estate search portal website, somewhat similar to Zillow in the U.S. market.

They named the site 99Fang.com, *fang* meaning "house" in Chinese. Yiming managed a small team, including his former roommate from Nankai University, Rubo. The company wasn't a runaway success, but it did well by many measures, and Yiming proved capable of leading a team. By December 2011, 99Fang had three hundred thousand daily visitors becoming the third-largest real estate information website in China.

99Fang was already Yiming's fourth entrepreneurial venture in six years. The exposure to such various working environments, technologies, and products greatly influenced his later work at ByteDance. Yiming had worked across multiple verticals, from enterprise software to travel and real estate. He had soaked up practical technical knowledge around operating content portals, search engines, and social networks. He'd experienced wildly disparate company cultures and management styles, from risk-averse Microsoft to working alongside some of the boldest and most ambitious startup founders. And at 99Fang, he had learned how to lead others.

When dealing with staff management issues, Yiming had a reputation for being very gentle. When unsatisfied with a staff member's performance, he would look to tackle the situation with mild reasoning and sincere encouragement, an approach that employees have described as having a charm of its own. Many, including Rubo, commented that Yiming considers anger a useless emotion and a form of mental laziness. Instead, he strived for an ideal state "between mild joy and mild depression."

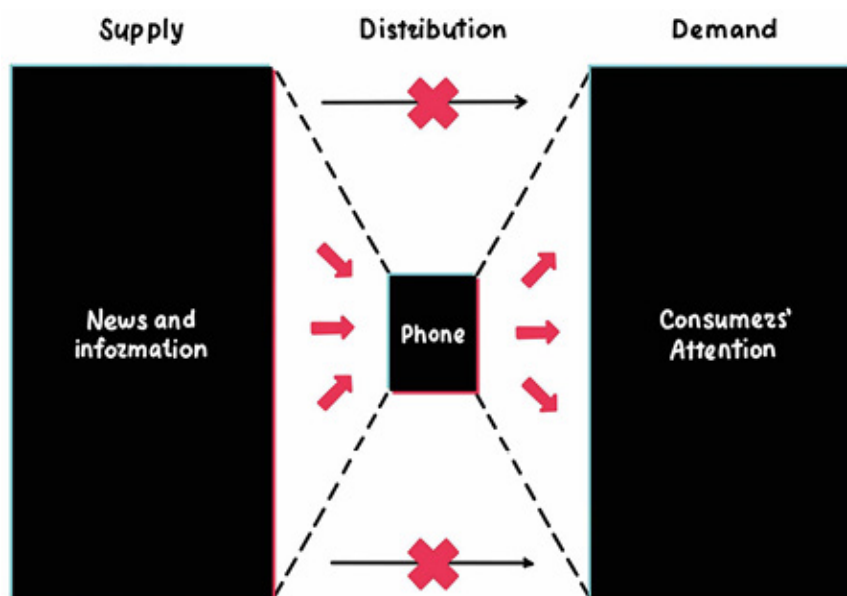
The mobile app era arrives

The early 2010s saw a seismic shift in China's internet industry, a rapidly changing landscape that would inspire ByteDance's most successful apps. The launch of the iPhone 4S in late 2011 was a landmark. It proved to be a blockbuster with local consumers queuing round the block to get their hands on one. That year was a breakthrough for smartphones with some 110 million units being shipped within China, exceeding the numbers from several previous years combined.³¹

Before smartphones, urban Chinese used feature phones, which were impractical for accessing all but a minimal selection of expensive internet services. The devices were held back by tiny low-resolution screens and poor user experience. Up until this point, desktop PCs had remained the default device for accessing the internet through web browsers and installed software programs displaying on large screens in fixed indoor locations such as offices or internet bars.

The arrival of the smartphone revolutionized access to information for most Chinese to a far greater extent than in Western countries. Traditional forms of media in China such as publication of books, newspapers, radio, and television, were heavily regulated and, in most cases, directly controlled by the state. In comparison, the internet was a "wild west" of information dissemination, evolving with complexity and speed such that regulation could not keep pace. Privately-owned internet media organizations had evolved, and online influencers were learning how to build large followings.

Smartphones were on course to massively expand the size of the market for services and information accessed via the internet. These new phones were dramatically different from PCs, and the screens were tiny compared with big PC monitors. Phones were carried around in consumer's pockets, which meant accessing online information in any place, any time was now a reality. China quickly became a smartphone obsessed nation. Most had never owned a desktop or laptop device, but across China's hundreds of large cities soon, almost every working-age adult had a smartphone.



Above: the phone as a dominant medium of news and information distribution acts as a choke point in reaching consumers.

Yiming realized he was witnessing the beginnings of a tectonic shift in the way people were obtaining information. At the beginning of 2011, commuters on the Beijing subway around him were still getting news through print newspapers; over just a few months, most had switched to reading on their smartphones.

“This is a revolution in information dissemination. Mobile phones will likely replace the newspaper to become the main medium for delivering information. Because of the relationship between humans and mobile phones, the demand for personalized recommendation will also certainly increase,” was Yiming’s assessment.³²

His observation was accurate, not just for news and information. The small screen in everyone’s pocket would become the most valuable and hotly contested real estate in China; a new choke point had emerged in gaining people’s attention. Other internet industry leaders were also becoming acutely aware of the profound change happening around them. WeChat founder Allen Zhang went as far as to claim: “The phone is an extension of our bodies.”

Yiming doubled down on mobile development. Over six months, the 99Fang team launched five mobile apps targeting separate subcategories of the real estate market, such as second-hand properties and rentals.³³

Together the family of apps gained 1.5 million mobile users, over a hundred thousand of whom were active daily, catapulting 99Fang to dominance within the niche category of Chinese real estate apps.

The process of buying a house entailed, for most people, a long decision-making cycle. Both buyers and sellers had to continually stay up to date with the latest news, such as new housing developments or local government policy changes. To meet this demand, 99Fang created an app plainly entitled “Real Estate Information,” which aggregated articles from the major news portal platforms and popular real estate websites.



Above: The 99Fang real estate news and article aggregation app.³⁴

The technology behind the service was basic, but it fulfilled a real need and proved popular. Aggregation of information from thousands of sources into a single feed on a smartphone had great potential. Technology could be used to extract, filter, and surface the most relevant information much faster than any human could hope to achieve. Yiming saw they had hit upon something with great potential and that building this kind of app just for the real estate sector was thinking too small.

In hindsight, it is clear that 99Fang’s “Real Estate Information” app was the direct predecessor of ByteDance’s early flagship app Toutiao (Today’s Headlines). The layouts, product logic, and even the newspaper logos are similar. The strategy of building a suite of apps that supported each other would also be taken over and used to significant effect at ByteDance.

Yiming resolved to resign from his position as CEO of 99Fang. He could no longer restrict himself to just the real estate sector. The rise of the mobile internet was a massive once in a lifetime opportunity for entrepreneurs like him. He felt compelled to try something bigger, something that could touch everyone.

⁹ <https://kknews.cc/tech/oplx28o.html>

¹⁰ (Yīmíngjīngrén)

¹¹ <http://cunwu.cuncun8.com/index.php?ctl=village&geoCode=76597251>

¹² <https://new.qq.com/omn/20190529/20190529A0QZQV.html>

¹³ https://www.sohu.com/na/349348869_766689?

¹⁴ <https://www.tmtpost.com/3145145.htm>

¹⁵ The Graduate Record Examinations, a standardized test required by many graduate schools in the U.S. and Canada.

¹⁶ <https://tech.sina.com.cn/i/2018-08-02/doc-ihhehtqf3594606.shtml>

¹⁷ <http://www.iceo.com.cn/com2013/2016/1129/302109.shtml>

¹⁸ It was not uncommon for early Chinese internet users to meet their spouses online. Tencent CEO Pony Ma famously met his wife through his company's messaging software QQ.

¹⁹ <https://new.qq.com/omn/20191226/20191226A0AB5D00.html>

²⁰ https://www.sohu.com/na/349348869_766689?

²¹ <http://goodyomo.com/archives/155>

²² <http://www.fjydyz.net/plus/view.php?aid=5329>

²³ <https://www.scmp.com/magazines/style/news-trends/article/3023093/how-did-tiktoks-zhang-yiming-become-one-chinas-richest>

²⁴ Medium average age of a ByteDance China employee is just 26.

Source: https://medium.com/@ming_ma/how-to-work-with-people-who-are-10-years-younger-than-you-71cd378b30e

²⁵ <http://www.startup-partner.com/3654.html>

²⁶ The voluntary withdrawal was in reaction to what the company described in a blog post as a “highly sophisticated and targeted attack on our corporate infrastructure originating from China that resulted in the theft of intellectual property”

<https://googleblog.blogspot.com/2010/01/new-approach-to-china.html>

²⁷ The Microsoft Asia Research Institute is somewhat of a West Point for China's top AI talent, having trained many executives and influential CEOs at leading Chinese AI startups. Many of which have described their time at MSRA in more favorable terms than Yiming.

²⁸ https://www.ixigua.com/pseries/6805466361402229262_6805110223536128515/

²⁹ <http://news.bbc.co.uk/2/hi/asia-pacific/8135203.stm>

³⁰ <https://www.huxiu.com/article/144262.htm>

³¹ http://usa.chinadaily.com.cn/business/2012-04/14/content_15047719.htm

³² https://www.sohu.com/na/349348869_766689?

³³ <https://baike.baidu.com/item/%E4%B9%9D%E4%B9%9D%E6%88%BF>

³⁴ Images source: <http://www.pc6.com/az/79802.html>

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CHAPTER 2

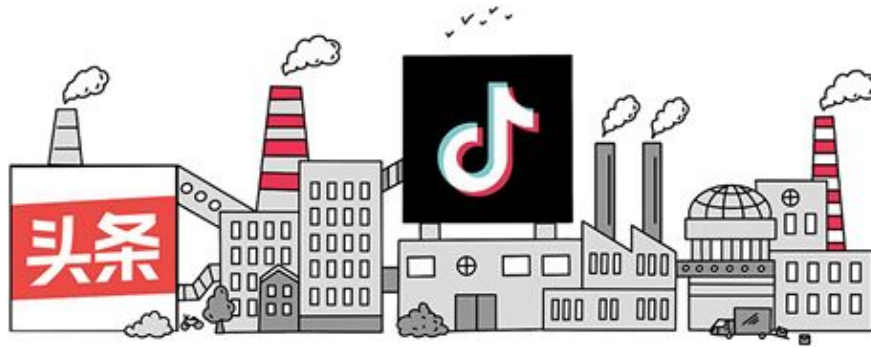
BYTEDANCE THE EARLY YEARS



*“They all thought, how can I possibly
associate myself with something so lowbrow?
There are even engineers who won’t join them.”
-Zhang Yiming, recalling the early days of ByteDance*

Image: ByteDance staff in 2013 at their original converted apartment offices

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Chapter Timeline

2012 Jan – ByteDance’s first app, “Hilarious Goofy Pics,” is released

2012 Mar – ByteDance is officially founded

2012 May – Launch of second app Neihan Duanzi (Implied Jokes)

2012 July – SIG China leads ByteDance’s Series A investment round

2012 Aug – Flagship app Toutiao (Today’s Headlines) goes online

2012 Sept – First personalized recommendation system went live

2012 Oct – Toutiao acquires 10 million users after 90 days of operation

2013 May – Yuri Milner leads ByteDance’s Series B investment round

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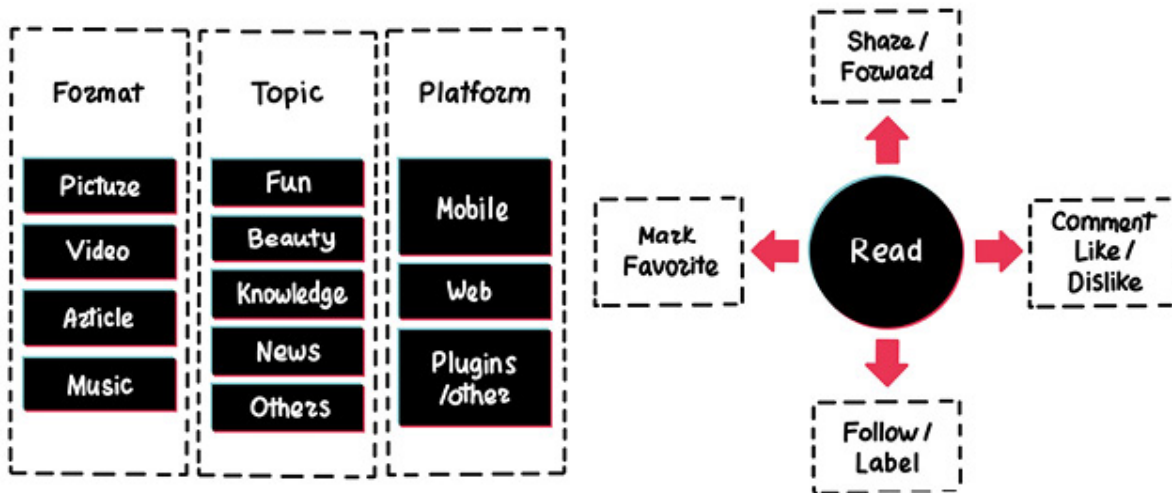
I T WAS BITTERLY COLD THAT DAY. The café had left the lights off because there were so few guests. Outside lay the grey endless concrete and polluted winter skies of Beijing. When Joan arrived, she spotted the slim, spectacled figure of Yiming sitting in a corner wearing a black jacket—he was reading.

While the entire country had been celebrating Chinese New Year with their families, Yiming had spent the festival absorbed in his ideas for building apps. As soon as the week-long holiday had ended, he reached out to Joan, the investor who had initially persuaded him to do 99Fang. They arranged to meet in a café not far from the company offices in Zhongguancun, a district of Beijing known as the home to some of China’s most powerful tech companies, and the closest equivalent China has to Silicon Valley.

Three months earlier, he had told Joan he was looking to “do something” outside of real estate to catch the new mobile internet wave. He now had a vision for what that “something” could be. The two talked.

Halfway through their discussions, Yiming grabbed a napkin from the table and started sketching out his ideas. Joan didn’t take much persuading. Venture capital was a game of investing in people, and she trusted Yiming. He was even willing to sell his apartment to help fund this new venture. Joan agreed to make an angel investment of \$80,000, with an additional guarantee to lead the startup’s next round.

Yiming was able to secure further funding from his personal network; two friends Liu Jun and Zhou Zijing, added amounts to make up a total of two million yuan. Joan is, without a doubt, the most important investor in ByteDance’s history.³⁵ Although she could have hardly known so at the time, her meeting in the café that day with Yiming was a career-defining moment, the kind of deal that every venture capitalist dreams of. When ByteDance eventually goes public, SIG stands to make multiple billion-dollar returns, exceeding even those of Benchmark Capital’s legendary early investment in Uber, which was worth a whopping \$6.8 billion upon IPO. In 2013 SIG owned roughly 12% of the company and is reported to have remained the largest institutional investor.³⁶ As of writing, ByteDance, including TikTok, is valued at over \$100 billion.



Above: what Yiming sketched on the napkin as remembered by Joan Wang.³⁷

SIG

In its home market, American financial services firm Susquehanna International Group (SIG) is essentially a hedge fund with investment banks and a research business focused on the public markets. However, SIG China, the regional arm where Joan worked, acts independently as a venture capital firm investing in private unlisted companies.³⁸ Entering China in 2005, SIG China initially made joint investments across varied industries, even delving into areas such as mining and industrial dye companies.

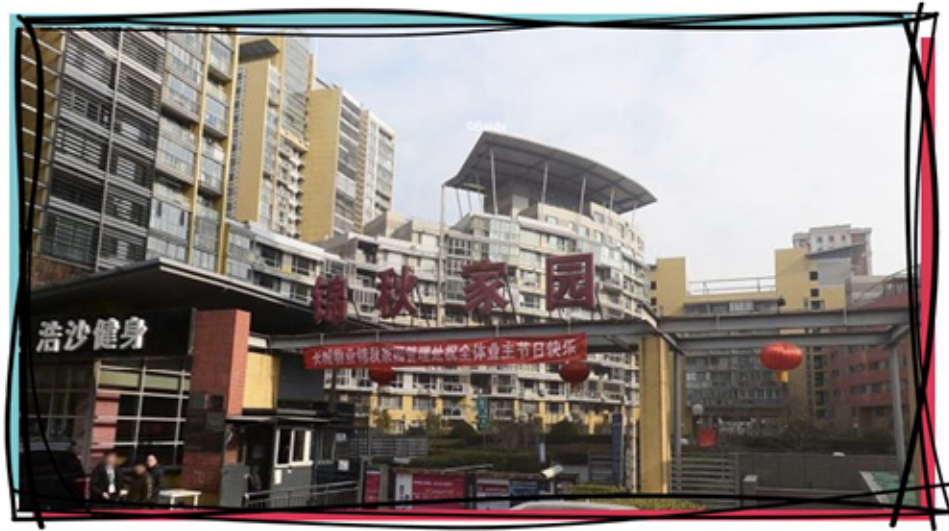
Joan, who grew up in China, studied electrical engineering at the State University of New York and went on to work in the IT and telecom industry for thirteen years before joining SIG as a partner in 2006 based out of Beijing. One of her first investments, where she also came to know Yiming, was Kuxun.

ByteDance was the company's first angel investment project. "This [early investment] allowed me to participate in the whole development from zero to one. I am proud in my heart of the company," gushed Joan, recalling her experiences.³⁹

Jinqiu Gardens

ByteDance's first offices were somewhat modest with the entire operations being based out of two converted apartments. For the company's first year,

home was Floor 6, Section D, Building 4, of the Jinqiu Gardens housing compound. Ten minutes' drive from the Zhongguancun technology hub in northwest Beijing and surrounded by major research universities such as the Beijing Institute of Technology and Tsinghua University. Yet otherwise, Jinqiu Gardens had no distinguishing features. It was yet another housing compound among Beijing's seemingly never-ending expanse of monotonous residential buildings.

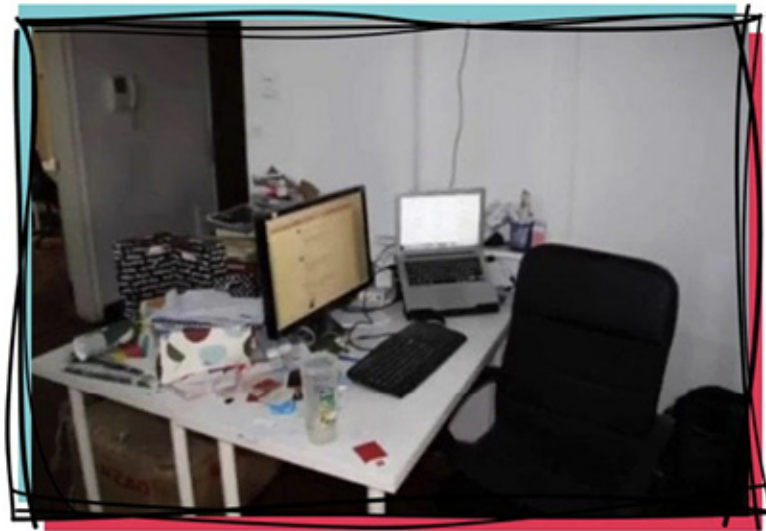


Above: the Jinqiu Gardens main gate⁴⁰

The team set up shop in a large four-bedroom, two-bathroom apartment and filled it with Ikea office furniture. Rent was 20,000 yuan a month (approx. \$3,170 at the time). This was a popular strategy for early-stage startups before they gained traction and secured enough investment to transition to more formal premises. It was China's equivalent to American startups' basement and garage offices.

Yiming took hiring seriously, personally conducting hundreds of interviews for technical positions in the earliest days. After several rounds of rigorous screening, a core team was formed. He had also taken with him several key people from his old company, 99Fang, including former university roommate Rubo. Yiming felt little remorse for switching roles and leaving the old company behind, "Entrepreneurship is like gambling. Success is an improbable event. Would you feel guilty for losing?" was his rationalization.

Jinqiu Gardens fostered a homely environment for the founding team. A chef cooked meals for everyone out of the apartment's kitchen, which saved both time and money for the thirty-plus staff. The only drawback being the distracting smell wafting around just before lunch. Employees bonded over cigarette breaks on the balcony. The company rejected formal office wear—shorts and T-shirts were the standard on hot summer days. Yiming surprised a reporter once during ByteDance's early years when he arrived for a formal media interview with a lunch bowl in hand and wearing sandals. Occasionally, new hires started work only to quit a few days later. "They probably thought this company was too small. Nothing was standardized," Yiming later reflected.



Above: Yiming's workspace at the Jinqiu Gardens offices.⁴¹

Upstairs on the same building's top floor was another small startup, 36Kr, which grew to become the first Chinese technology-focused online publication to list in the U.S. Two large, high-profile companies starting in the same housing community at the same time has given Jinqiu Gardens a somewhat mythical status within China's internet industry.

Is this a dance agency?

In picking the company's name, Yiming made an unorthodox choice—English name first and work back towards the Chinese. After much brainstorming, the team came up with "ByteDance," allegedly inspired by a famous Steve Jobs quote: "Technology alone is not enough. That it's

technology married with liberal arts, married with the humanities that yields us the result that makes our hearts sing.”⁴²

The logic being that “byte,” the unit of information in a computer, was technological sounding, and “dance” represented the liberal arts. From this, they devised the Chinese name, ⁴³ Zì jié tiàodòng (字节跳动), which literally translates as Byte Bounce, a change that was made in part due to concerns the company might be mistaken for a dance agency. If you think the name sounds a little strange in English, it arguably sounds stranger in Chinese.

Chinese internet startups created around this time tended to focus exclusively on the massive and abundant opportunities already present in the local domestic market with little thought given to overseas. Choosing an English name first provides a reliable indicator that the company really was thinking “Global from day one”—one of ByteDance’s somewhat clichéd company slogans.

Steve Jobs, whose quote inspired the name, had died just a few months before, sending shockwaves across China’s tech industry where he was widely revered. The “app economy” of internet services delivered through smartphone devices was Job’s brainchild and the infrastructure on top of which ByteDance would build its entire business.

Goofy Pics

During early 2012 Yiming had become utterly obsessed with thinking about data mining and information recommendation, researching all the major online content platforms from both Chinese and American markets. He had hit upon an important observation—many of the top-ranking non-game apps on Chinese app stores focused on light entertainment.

“We decided to crack the market through entertainment first. Amusement is a real demand,” said Huang He, the earliest mobile app developer at ByteDance who had previously worked with Yiming at 99Fang.

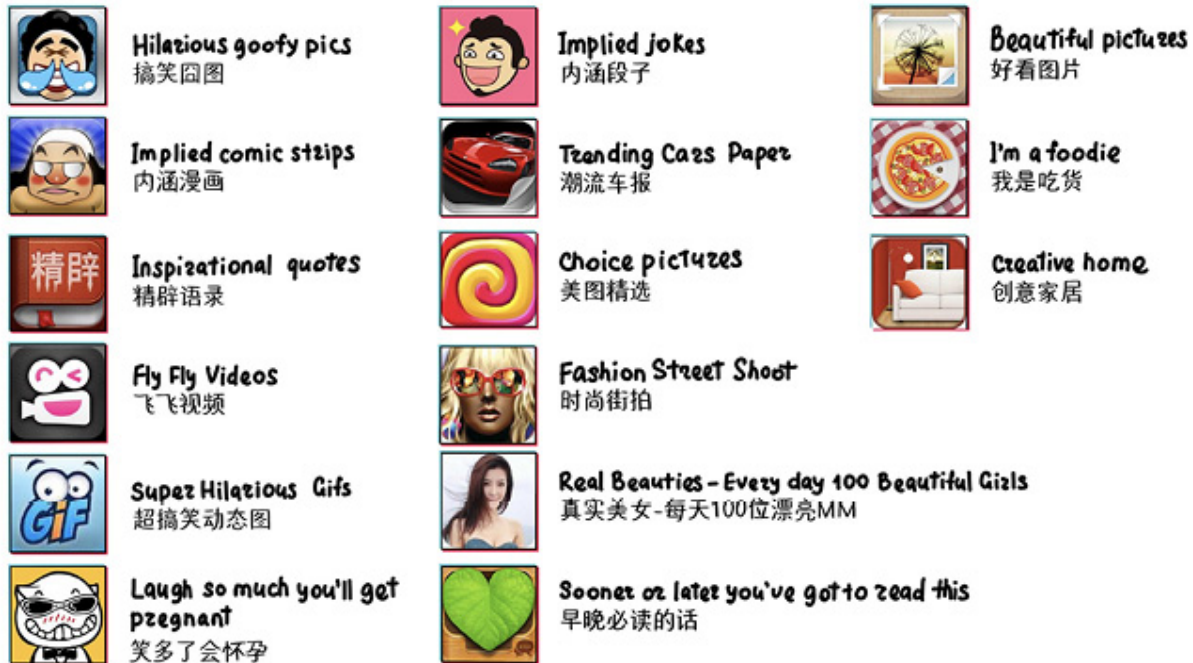
Even before ByteDance had officially registered as a business, the embryonic team cobbled together their first app entitled “Hilarious Goofy Pics,”⁴⁴ which served up an endless addictive feed of entertaining memes and silly pictures.



Above: 2012 screenshot of ByteDance's first app Hilarious Goofy Pics

A second app entitled "Implied Jokes" ⁴⁵ quickly followed with similar positioning focusing on internet memes. Implied Jokes turned out to be an instant hit and picked up millions of users within a few months. The difference in success between the two similar apps came down to a better name and optimization to save data.

In all, the nascent company released over a dozen apps for testing in the first half of 2012, experimenting with various themes and directions.



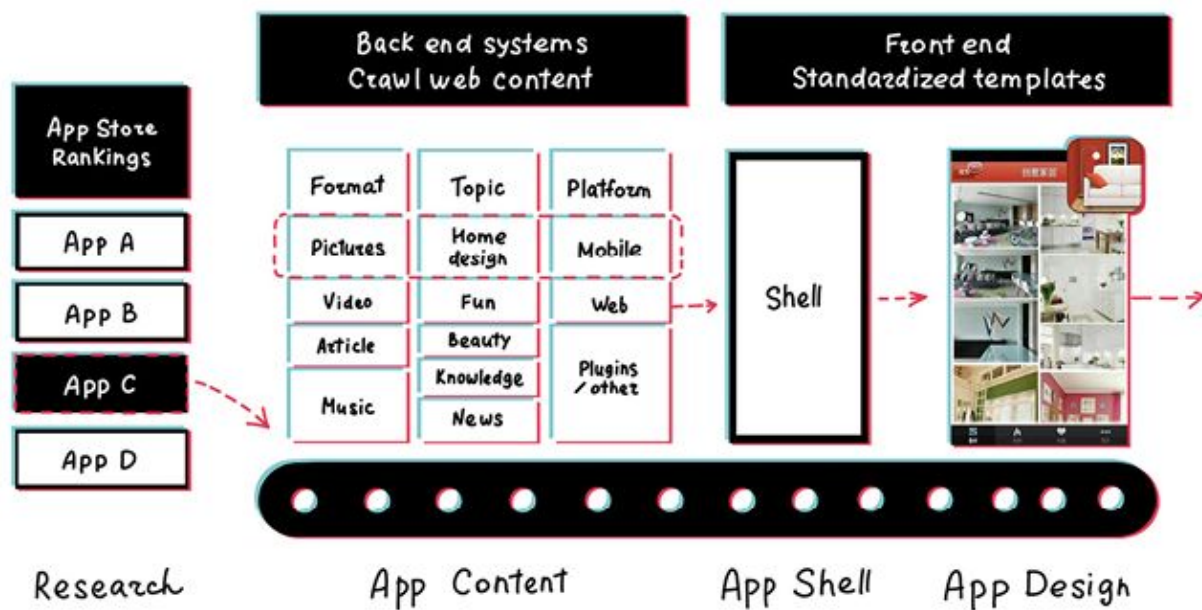
Above: ByteDance's earliest suite of apps released in 2012

To conceive names, the team researched top-ranking accounts on China's Twitter equivalent, Sina Weibo, and found that often it was vernacular names using straightforward language that did best. Plain descriptive titles such as "Beautiful pictures," "This evening's must-watch videos," or "Real Beauties—Every day 100 Beautiful Girls," while unimaginative, resonated well with many early mobile app users who could immediately understand what the app was for.⁴⁶

Previously at 99Fang, Yiming's team had built five real estate apps to test various positionings. Now at ByteDance, he further refined this strategy into one of rapid experimentation. Quickly push new ideas out, test multiple features, and let the market verify which has value - this became an enduring strategy of ByteDance.



Above: screenshots from several of the apps reveal their similar structure



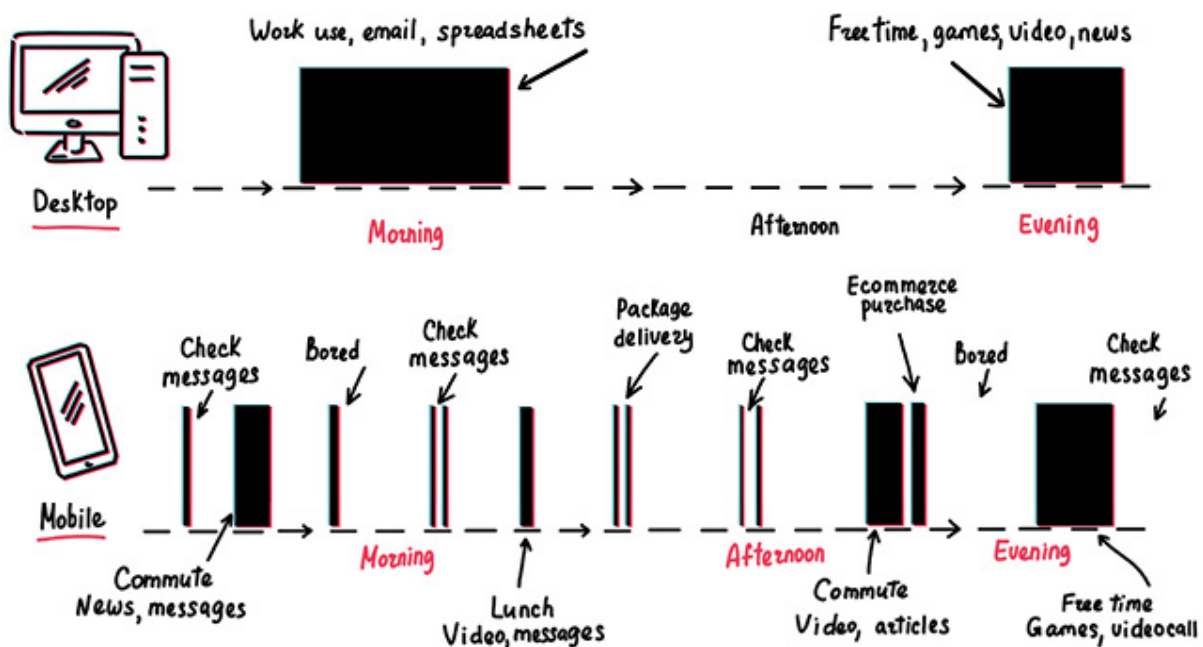
Above: the early factory production line system for generating apps fast

Given the considerable mobile development experience built up through 99Fang, creating these new apps for ByteDance was relatively easy for Yiming's team. "The cost of making an app was meager at that time. Design a (UI) frame, set a shell, add a filter to the content, and it popped out ready,"⁴⁷ explained developer Huang He.

That was just as well because one of the unfortunate consequences of building apps with names such as "Laugh so much you'll get pregnant" and

operating out of a retooled apartment office with the CEO wearing a T-Shirt and sandals to work was that recruitment suffered. The most talented engineers they did manage to hire through their own social networks left quickly. “There was no way to persuade high-level vice president candidates to join us. They all thought, ‘How can I possibly partake in something so lowbrow?’ There are even some engineers who won’t join them,”⁴⁸ reflected Yiming.

Fortunately for ByteDance, app development in China at the time was still rudimentary. Many competitor apps contained large packages of content bundled together with the app. This experience was somewhat like downloading a finished version of an e-book, a model that considerably bloated the app size. In comparison, ByteDance’s apps were light and small, just a few megabytes. Their content was continuously updated by connecting to their servers with the backend systems regularly crawling and classifying content from across the web.



Above: usage of mobile internet was highly fragmented - small pockets of time spread throughout the day. Desktop usage, in contrast, tended towards longer set blocks of time.

Consuming content such as articles, pictures, or videos on smartphones was a considerably different experience from the desktop PC internet. Yiming surmised there were three big pain points: the small screen,

fragmented time, and information overload. But as far as he saw, no one product in China had addressed these three pain points simultaneously.

They needed a flagship product, something more ambitious. “Although we tried many vertical fields at the time, we must do something big in the end,” explained Huang He.

Toutiao - Today’s Headlines

“Just as Zuckerberg founded Facebook to connect people and people, Travis (Kalanick) founded Uber to connect people and cars, Toutiao ⁴⁹ makes information and people more widely and efficiently matched.” ⁵⁰ was how Yiming later described the big picture vision for this new flagship product.

The team set about building a more ambitious app with a broad vision to aggregate and organize many content forms from across the internet. An app powered by big data and machine learning that would serve people with a tailored feed according to their personalized preferences and do away with human curators.

Up until this point, most Chinese news portals employed human editors and curated content manually following a model that hadn’t changed much since being popularized by Yahoo! News in the 1990s. Yiming believed that this wasn’t applicable anymore in the mobile age. As technology advanced, he felt humans would be entirely removed from the news curation process and replaced by automated systems leveraging big data and machine learning.

Jinri Toutiao was the title of the new app, a name which means “Today’s Headlines,” and which later became commonly abbreviated to just Toutiao or “Headlines.” The name won out from nearly a hundred candidates put forward by the team.

“‘Headline’ is the most important news. ‘Today’ adds a sense of timeliness. The most important news of the day. It is catchy, straightforward, and also creates a fear of missing out,” explained developer Huang He who came up with the winning name.

News apps were considered a lucrative business. The frequency of use was high as people opened news apps even if they only have a minute of idle time to catch the latest headlines. The retention was also high compared with many other popular categories such as mobile games, where titles

tended to have limited shelf lives. Staying informed is a basic human need; once a habit was built, a news app typically would not be deleted from thereon.



Above: the early Toutiao interface

It was relatively easy to add new extra features or categories of content to news apps and see them gain adoption. This contrasted with other tools like alarm clocks with high use and high retention but limited to one single use case. People tend to want their alarm clock just to be an alarm clock.

But to think of Toutiao just as a news app was somewhat misleading. It was positioned around news and was categorized in the app store together with other traditional news apps, but actually delivered all kinds of information, be it hard news, frivolous entertainment, or evergreen blog posts. “We weren’t really a news app. From the earliest beginning, we hoped to include all kinds of content... and become the information platform that understands you best,” explained Yiming.

During this time, the Chinese internet suffered from rampant piracy of content, with poor enforcement of intellectual property rights. Written content was exploited, freely scraped, and used by others, often to the point where it was difficult to find the original source. Faced with such an environment, it became an open secret that China’s online media environment was rife with “advertorials,” articles and stories paid for by businesses. How else could journalists sustainably make a living? Such

dynamics greatly strengthened the position of Toutiao as a broad aggregator of many online content forms. The app was essentially competing for online attention with all content distribution platforms, from Weibo and WeChat through to search giant Baidu and even web browser apps.

Toutiao was also winning over users with a superior experience. Even within the limited category of news apps, it outran others in product innovation, consistently iterating and being the first to introduce micro-optimizations that today are taken for granted. Unlike competitors, dragging down from the top of the feed caused the page to always refresh with new content. Knowing that network conditions were often adverse on people's commute to work, the app preloaded articles and displayed lower-definition pictures when necessary.

The team had also developed a cross-conversion-system to migrate users from their family of established apps over into Toutiao. Their existing apps acted essentially as self-owned acquisition channels through which they could funnel in users at no cost, particularly effective on early Android devices. ByteDance was extremely frugal in their first years with controlling costs. Throughout 2012 they spent one million yuan (\$158,000 at the time) on promotion and gained more than one million active users by the end of the year. The acquisition cost of a single activated user being less than 0.1 yuan (\$ 0.016).⁵¹ Building trashy meme apps like "Hilarious Goofy Pics" was simply a means to an end. These apps were a smart way to cheaply acquire users, which could later be converted over to the mothership platform Toutiao.

What does tech have to do with success?

ByteDance had yet to begin monetization, which meant Yiming was under pressure to secure investor funding. The company's earliest money was sourced from those who knew Yiming personally, such as Joan and several angel investors from the founder's network.

Toutiao had established itself as the core offering, so much so that media and employees dropped the name ByteDance and began simply referring to the company as Toutiao. Yiming went out to pursue Series B funding with a business plan centered entirely around their flagship news app.⁵² Yet investment was hard to come by.

Joan personally introduced Yiming to at least twenty of her venture capital friends. Not even one was optimistic about the company, and Yiming kept getting turned away. One left after talking for just 15 minutes and complained to Joan later: “It’s not my investment style to look at this kid.” Yiming was just too young-looking. He spoke quickly in a soft voice and gave off the air of being a gentle, mild-mannered software programmer rather than a tough business leader. He was the opposite of the brash and confident “Jack Ma style” entrepreneur that many Chinese investors favored. Only a few were willing to consider, but their quotations were low and below the expectations of Yiming. Their data and usage trends were decent. So what was the problem?

Many VC’s perceived Toutiao as a mere mobile version of a news portal website, of which there were many. Dominant desktop news portals NetEase and Sohu already claimed 200 million users each on their mobile apps, with other significant players such as Tencent and Phoenix ⁵³ also jostling for market share. The tail end of the market held many smaller niche players nipping at the better-known platforms’ heels. This was a mature “red ocean” market where the competition was fierce, and most of the spoils had seemingly been divided up by the big players. Most investors felt users were already extensively served.

No one questioned how ByteDance would generate revenue once they had gained a sizable user base; the Chinese tech world subscribed to the adage that “traffic is king,” and there always were ways to monetize a business once it had achieved scale. The key concern was ever getting to that scale in a market that had already been divided up.

Every serious investor in China by now was well aware of the rise of the mobile internet, but the full implications of the shift from desktop PCs to smartphones were still yet to become apparent. It was more than just a change in the device where people consumed information, but also a shift in the way information was distributed and consumed to fit the new medium. Yiming wanted to change how information was distributed—from human editor-driven to artificial intelligence-driven distribution based on big data and machine learning.

Joan recalled her frustration, “Everyone thought the technology was great, but how much does technology have to do with success? The existing

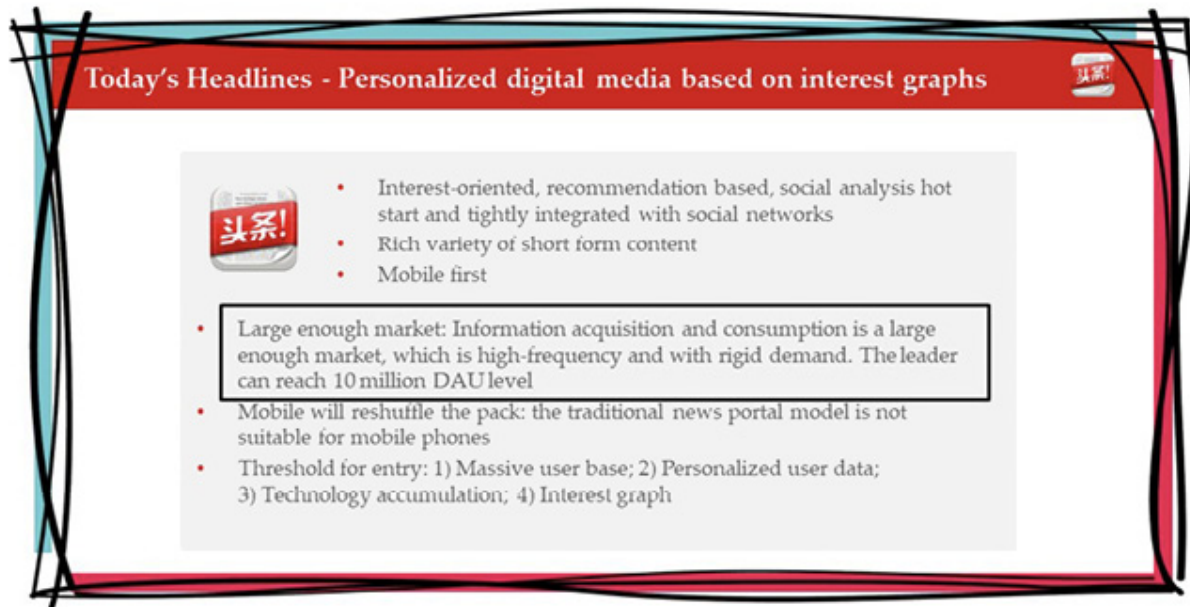
portals, without using this technology, also satisfactorily meet user needs. So they (investors) will question whether the value and benefit this technology brings to users is so great?” None of the Chinese venture investors grasped the true potential of the model or foresaw how powerful machine learning could be when applied to information distribution.

However, in defense of the investors who turned them down, it’s possible to argue that Yiming somewhat conservatively underplayed the potential opportunity. The business plan which they used to secure investment contained a revealing point (translated below).

“Information acquisition and consumption is a large enough market, which is high-frequency and with rigid demand. **The leader can reach 10 million DAU (daily active user) level.** ”

The venture capital model is to make many bets on early-stage companies hoping that some of those bets pay off big. The number of 10 million daily users quantifies the size of the opportunity being presented to investors. A big payoff requires a big market, yet Yiming understated his potential market for Toutiao by at least an order of magnitude. By September 2017, just four and a half years later, Toutiao had 120 million daily users. So instead of saying that Chinese venture capitalists failed to understand the potential of Toutiao, it’s perhaps better to say that Yiming over managed investor expectations. He was unable to present a grand enough vision of what his company could potentially become, something alluded to by Joan in her later article:

“I did not expect that for such a long period, so few investors would be optimistic about this product; at the same time, I did not expect that Toutiao (ByteDance) would become a super unicorn company.” ⁵⁴



Above: a translated section of the original business plan used to secure Series B investment round investment in 2013

The list of investors who turned down ByteDance's first few fundraising rounds, missing what would have likely ended up being the highest return deal of their careers, reads like a who's who of China venture capital.

Bob Xiaoping Xu, co-founder of ZhenFund, which ranked six years in a row as the top Chinese early-stage investment firm ⁵⁵ later admitted, "We passed on the angel round. I'll regret that forever." Sequoia Capital China, led by legendary venture capitalist Neil Shen, and praised for having "bought half of the Chinese internet," ⁵⁶ turned down ByteDance's Series A round. Something he later confessed was a colossal mistake, stoically reflecting, "This is part of the life of an investor... venture capital is a game of regret."

Prominent investors GGV Capital refused Yiming; the famed "unicorn hunter" Allen Zhu of GSR Ventures also passed. Billionaire internet entrepreneur and investor Zhou Hongyi later joked that he was the unluckiest of all "I invested early on, but along the way sold my shares at a meager price." ⁵⁷

Joan and Yiming were stuck; they had been turned down by some of the most famous and respected names in China's venture capital world and

needed to find someone to lead the Series B. The answer came from the other side of the world.

Russia to the rescue (via. San Francisco)

“Show up to YC at nine pm this Friday.”

This was the mysterious message sent out to all 43 startup teams taking part in the 2011 Y Combinator startup accelerator, Silicon Valley’s most prestigious program for new startups. The cryptic announcement from the program partners drove wild speculation amongst the founders as to what would happen. They weren’t told why they were supposed to be there, just that something important was happening. Some guessed a high-profile celebrity such as Steve Jobs ⁵⁸ might be speaking. Excitement grew as the time came close.

When it came round to Friday night, the room was full. All the entrepreneurs had gathered at the Y Combinator headquarters in Mountain View, row after row of young founders. A video conference call from Europe began. It was a middle-aged man, bald with thick eyebrows, speaking with a Russian accent. The big announcement came, “We are offering each of the 43 companies a \$150,000 investment.”

People were shocked. “The terms were insane! An uncapped convertible note ⁵⁹ with no discount. Effectively free cash,” ⁶⁰ exclaimed one of the batch’s founders, later recalling his reaction. 36 of the 43 startups had signed paperwork before the event was even over. ⁶¹ It was an unprecedented blanket investment based on the accelerator’s prestige alone and made without even seeing many of the startups. The brain behind it all was Russia’s most influential tech investor, billionaire entrepreneur, and physicist, Yuri Borisovich Milner.

Sitting in the audience that day was Matt Huang, a young ethnically Chinese entrepreneur from San Francisco. His startup, formerly a video aggregator, had pivoted into an advertising analysis platform called Hotspots.io. They gratefully snapped up Yuri’s offer without a second thought.

Fast forward a year, and after leaving Y Combinator, Matt’s startup had been acquired by Twitter, and he decided to take a break from the intense startup life to visit China for two weeks. Matt was curious to learn more

about the thriving tech scene and excited to scour around for potential angel investing opportunities. Through mutual connections, he was introduced to Joan from SIG, who put him in touch with Yiming. That was how Matt ended up in building 4 of Jinqiu Gardens, standing in one of the converted apartments that were ByteDance's offices.

Welcoming the visitor from afar, Yiming proceeded to patiently talk Matt through what they were doing. The dozen apps they had built, the news aggregator, and the personalized recommendation system. Matt was surprised. The quality of the technology they were using was comparable with what he'd seen in America. Yiming was technical, ambitious, and extremely clear thinking. He was easily as formidable an operator as any of the entrepreneurs Matt knew back in the U.S. The meeting lasted just 90 minutes. Still, Matt had seen enough to know he wanted to invest, something that Joan helped later facilitate with Matt staying in touch via email after returning to the states.

Later once all the local Chinese venture capitalists had turned down ByteDance, Joan came back to Matt to ask if he knew anyone in the U.S. who might be interested. One name immediately popped into Matt's mind—Yuri.

As it turned out, Yuri Miler's investment company DST (Digital Sky Technologies), already had offices in Beijing. It was one of the most prolific and successful foreign investors in China's internet industry, having backed an all-star lineup of Chinese companies, including Alibaba, JD.com, Meituan, Didi Chuxing, and Xiaomi.

DST sent partner Shou Zi Chew to meet with Yiming. He left that meeting optimistic, noting the app's growth trajectory, and judging Yiming to be a robust and capable founder with conviction in the company's direction. Yuri agreed to lead the Series B round, making an investment that netted him 7.2% of the company, which was reportedly valued at \$60 million.⁶² Today that stake, even after dilution, is worth multiple billions of dollars. Had the one hundred and fifty thousand dollars invested into each of the 43 Y Combinator startups failed to return a single dime, the move would still have been worthwhile, because that generosity led to the ByteDance deal.

As it later turned out, ByteDance's Series B was the last window of opportunity for many venture capitalists to bet on the soon-to-be unicorn. One year later, the company's valuation would skyrocket to half a billion dollars. Yiming had room to pick and choose the investors he wanted, including prestigious financial firms and other established internet companies that could also provide strategic resources such as access to data and users.

Despite many rumors, ByteDance avoided the fate of falling under the influence of one of the traditional big three giants of China's internet industry, Baidu, Alibaba, and Tencent, collectively known as the BAT. In particular, Alibaba and Tencent have built robust and broad ecosystems of internet services that command immense amounts of traffic and user data. Industry consensus holds that once a startup reaches a certain scale within the Chinese internet ecosystem, it must choose to accept investment from one of the BAT or risk being crushed by a competitor who has sided with one of the giants. The drawback is becoming a proxy of the corporate investor, which uses its portfolio companies to contain its rivals. ByteDance is the only Chinese internet company of its size that is still going at it alone. When a rumor had it that Tencent would invest in ByteDance, an employee grumbled to Yiming: "I didn't join ByteDance just to become a Tencent employee." "Neither did I" was Yiming's curt reply.⁶³

One well-established name that ByteDance did accept investment from was Weibo, which joined during the Series C round. Weibo had beaten Yiming's former employer Fanfou to win the crown title of "Twitter for China." That meant they could offer ByteDance a vast amount of data. Weibo opened up its APIs⁶⁴—the intermediaries that allow different apps to interact—allowing Toutiao to access Weibo's user data and activity, such as comments.

If a user logged into Toutiao with their Sina Weibo account, ByteDance would analyze their Weibo interests and reading preferences within seconds and use that insight to personalize the recommendations. It wasn't until later that Weibo came to realize ByteDance was a competitor. It eventually shut down the rival's access to its data and sold its shares in ByteDance.

You might ask how could one of the biggest names of the Chinese internet, Weibo, have failed to realize that ByteDance was its competitor? The

answer was much for the same reasons that none of the best minds in venture capital took them seriously.

Toutiao was defined as a news aggregator; thus, it was narrowly perceived as merely competing within the class of mobile news apps, a somewhat uninteresting and over-saturated category as far as most of the tech giants and investors were concerned.

“When the partners discussed, we felt the competition was too fierce. This small company had no chance,” explained Sequoia Capital’s Neil Shen.⁶⁵ The consensus held that the key to winning was not technology but other business factors, such as commanding a recognized brand and a loyal user base or enjoying the unfair advantage of receiving support in one of the BAT’s ecosystems. But it wasn’t just Chinese investors who were pessimistic; there was broad skepticism about the technology’s limits.

“News aggregation is a glory-less pursuit, where even the companies that do succeed have to grapple with the fact that no algorithm is ever going to be as good as the human brain at judging what news/content will appeal to said brain.”⁶⁶ was the opening to a distinctly negative article in the widely-read American tech blog TechCrunch assessing the chances of American startup Prismatic, which had a similar positioning to Toutiao. Today such viewpoints seem wildly outdated.

New digs

Gaining the backing of Yuri Milner marked a turning point. The company was able to move out of their converted apartment and into formal office spaces. Yingdu Mansions⁶⁷ was roughly 1km west of their old home along the main Zhichun road. This area had become home to multiple large internet companies that, along with Toutiao, became household names in China, including food delivery giant Meituan, run by Yiming’s friend Wang Xing, and phone manufacturer Xiaomi. A Walmart supermarket filled the first floor of their new building with ByteDance settled upstairs. The dress code was just as casual, and a company chef continued to prepare free meals for staff.



Above: A 2013 ByteDance job advert pictures Yiming at the front desk of the then-new offices at Yingdu mansions ⁶⁸*

Around this time, Yiming saw a slogan on a Beijing construction site that struck him as encapsulating this period.

Small place, big dream.

“I thought this was very suitable for us.”

³⁵ SIG led ByteDance’s Series A investment round, A+ round in December 2011 which included a \$1 million bridge loan and were heavily involved in Series B the year after. Source:

<https://www.cmtzz.cn/news/29846>

³⁶ <https://www.huxiu.com/article/268415.html>

³⁷ <https://www.huxiu.com/article/144262.html>

³⁸ <http://www.sig-china.com/index.php?category/home>

³⁹ <https://www.huxiu.com/article/144262.html>

⁴⁰ Image source: Baidu Maps street view

⁴¹ Image source: <https://v.douyin.com/3sYmyX/>

⁴² <https://www.youtube.com/watch?v=KII1MR-qNt8>

⁴³ Other Chinese names considered were: **zì jié wǔdòng** and **zì jié tiàoyuè** Byte Jump

⁴⁴ Chinese name: **gǎoxiào jiǒng tú**

⁴⁵ Chinese name: **nèihán duànzi**

⁴⁶ Given accusations later levied at the company, I wish to clarify I found no evidence of pornographic content on the app “Real Beauties—Every Day 100 Beautiful Girls.” However, there are indications that ByteDance was pushing close to the boundaries of what was legal in China. The app’s description contained a titillating promise:

“The highlight (of this app) is every day we contribute a bonus picture!

We won’t say too much about this matter, I believe you understand!”

<https://apptopia.com/google-play/app/com.ss.android.gallery.ppmg.google/intelligence>

47 <https://dy.163.com/article/EKI5CPM50511D84J.html>

48 <https://m.huxiu.com/article/141687.html>

49 Chinese name: Jīnrì tóu tiáo

50 <https://www.pingwest.com/a/61954>

51 <http://tech.sina.com.cn/i/2014-06-05/04399418360.shtml>

52 This early business plan of ByteDance is available for download here:

<https://www.slideshare.net/MatthewBrennan6/toutiao-2013-jan-series-b-funding-deck>

53 news.ifeng.com/

54 Unicorn companies are privately held startups valued over \$1 billion. Super unicorn or “Decacorn” are those with valuations exceeding \$10 billion.

55 <http://en.zhenfund.com/About>

56 <https://3g.163.com/v/video/VUSCJ8RK6.html>

57 <https://www.ixigua.com/i6797655980998918668/?>

58 Jobs sadly passed away in October of that year. Despite his deteriorating health, he was still appearing at events up until June.

59 A common early-stage startup financing method that avoids setting a valuation. Notes can be converted later into a specified number of shares of common stock or cash of equal value.

60 <https://www.quora.com/What-is-it-like-to-get-funded-by-Y-Combinator>

61 <https://techcrunch.com/2011/01/29/90-of-y-combinator-startups-have-already-accepted-the-150k-start-fund-offer/>

62 The investment was made through a fund called Apoletto, the investment vehicle of a foundation that supports Milner’s philanthropic endeavors. ByteDance’s valuation on the private secondary markets is reportedly in the \$100 billion range at the time of writing.

63 https://www.ixigua.com/pseries/6805466361402229262_6805715182820524556/?

In 2016 Tencent invested a small stake of 1.72% in ByteDance but has since sold its stake. Source:

<https://www.theinformation.com/articles/tencent-xiaomi-invested-in-tiktoks-parent-bytedance>

64 API: application program interface, a set of established rules that specifies how two software components should communicate

65 <https://www.ixigua.com/i6640043697771659783/?>

66 <https://techcrunch.com/2012/12/05/prismatic/>

67 <http://yingdudasha.cn/>

68 Image source: <https://m.weibo.cn/2745813247/3656157740605616>

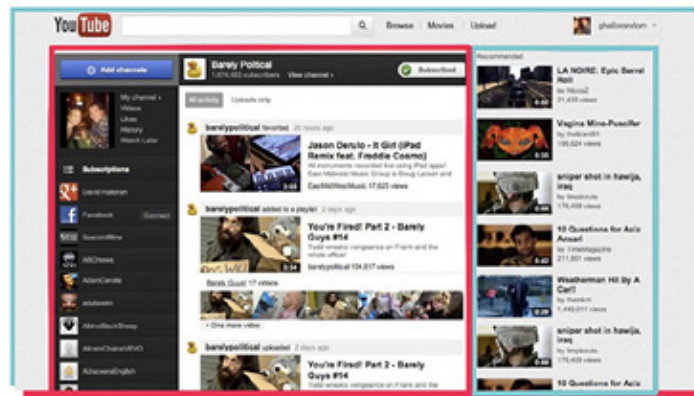
* “Real stuff” is my imperfect translation of gānhuò, which could also be translated as “the real McCoy” or “something of substance”

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CHAPTER 3

RECOMMENDATION, FROM YOUTUBE TO TIKTOK

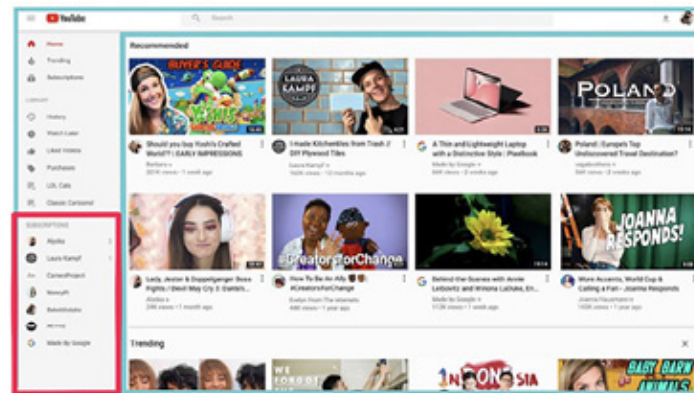
2011



53% of screen
real estate
Channel subscriptions

24% of screen
real estate
Recommendations

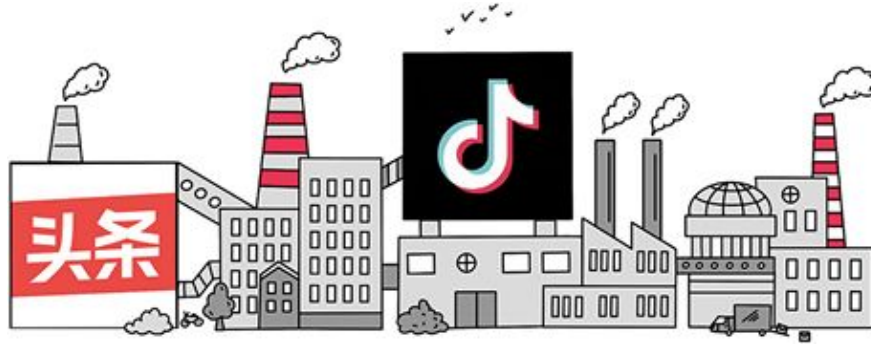
2019



7% of screen
real estate
Channel subscriptions

79% of screen
real estate
Recommendations

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Chapter Timeline

2009 – Netflix awards a \$1 million prize for an algorithm that increased the accuracy of their video recommendation by 10%

2011 – YouTube introduces machine learning algorithmic recommendation engine, Sibyl, with immediate impact

2012 Aug – ByteDance launches news aggregation app Toutiao

2012 Sep t – AlexNet breakthrough at the ImageNet challenge triggers a global explosion of interest in AI

2013 Mar – Facebook changes its newsfeed to a “personalized newspaper ”

2014 April – Instagram begins using an “explore ” tab of personalized content

2015 – Google Brain’s deep learning algorithms begin supercharging a wide variety of Google products, including YouTube recommendations

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IT WAS 2010, AND YOUTUBE HAD A BIG PROBLEM.

Despite being the third most visited website on the internet, “YouTube.com as a homepage was not driving a ton of engagement,” ⁶⁹ admitted John McFadden, technical lead for YouTube recommendations. Many people still thought of YouTube as a free and easy way to embed video content into other websites. The platform wasn’t viewed as a content destination, but rather as a series of one-off videos.

YouTube’s team tested a wide range of strategies to try and improve the site’s retention and keep people watching more videos for longer. In pursuit of this goal, they launched a new feature called “Leanback” as a way to queue up a series of videos to automatically play in sequence. They went so far as to outfit top content partners with professional camera equipment and even held livestreaming events. But the most significant change by far was betting on “YouTube Channels.” Channels was a way for users to easily subscribe to and watch collections of content from a single source, much like old-fashioned television channels.

So it wasn’t long before the homepage was redesigned around this new concept with a big blue “add channels” button as the primary call to action (see this chapter’s opening page). YouTube spent a whopping \$100 million on cutting deals with premium content creators, including celebrities like Madonna and Shaquille O’Neal, Hollywood production companies, and professional wrestling organization WWE. Their choice of partners reflected YouTube’s ultimate goal at the time; to transform itself into a TV-like entertainment destination.

However, a year later, the metrics showed that the average amount of time spent by users on YouTube remained flat, according to data from ComScore. ⁷⁰ Basically, none of the changes had much of an impact. It turned out that spending more on better content was not the right strategy because the problem was not that YouTube didn’t have great content. As YouTube’s engineering director Cristos Goodrow explained, “We believe that for every human being on earth, there’s 100 hours of YouTube that they would love to watch. And the content is already there. We have billions of videos.” ⁷¹

The problem was matching this vast amount of content to the right users. Encouraging people to subscribe to channels was one way of ensuring the content they liked reached them, but it still wasn't proving to be as effective as they had hoped. "We knew people were coming to YouTube when they knew what they were coming to look for," said technical lead McFadden. "We also wanted to serve the needs of people when they didn't necessarily know **what** they wanted to look for."

YouTube had long since added a sidebar of "Suggested videos," which were personalized recommendations based on the user's behavior history. But generating a useful personalized list wasn't easy. A passage from the 2010 technical paper "The YouTube Video Recommendation System", ⁷² neatly summarized the daunting technical challenge.

"Videos as they are uploaded by users often have no or very poor metadata.⁷³ Furthermore, videos on YouTube are mostly short form (under 10 minutes in length). User interactions are thus relatively short and noisy... [unlike] Netflix or Amazon where renting a movie or purchasing an item are very clear declarations of intent. In addition, many of the interesting videos on YouTube have a short life cycle going from upload to viral in the order of days requiring constant freshness of recommendation."

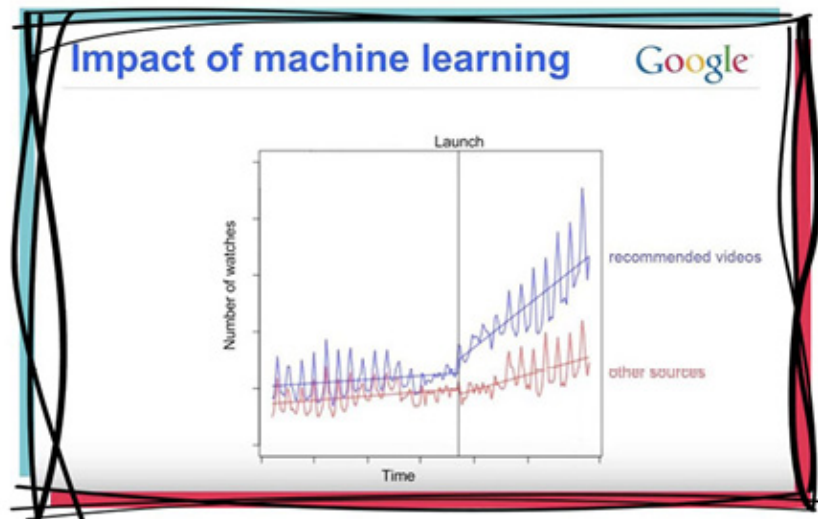
Uploaders often picked video titles that didn't describe the content accurately or selected thumbnails that failed to hint at the video's content. YouTube engineers shook their heads at one exceptional video, which failed to go viral largely because its title was the unsearchable "Check this out!" After much experimentation, the YouTube recommendation team found themselves relying on a version of a 12-year-old item-to-item collaborative filtering algorithm initially developed by Amazon back in 1998.⁷⁴

The beginnings of a revolution

2011 saw a breakthrough as Google started implementing a new machine learning system called Sibyl to make recommendations on YouTube.⁷⁵ The impact of Sibyl was immediate; by applying better technology to its recommendations, YouTube engineers discovered they had added a rocket-ship booster to the site's viewing numbers. The machine learning worked so well that soon, more people were choosing what to watch based on the

“recommended videos” list than any other way of picking videos, such as web searches or email referrals.

Google continued to iterate and further optimize the recommendation system, later switching from Sibyl to using Google Brain developed by the company’s now-famous moonshot laboratory group Google X, led by Stanford professor Andrew Ng. Google Brain leveraged groundbreaking new advances in deep learning. Whereas Sibyl’s impact had already been impressive, the results of Google Brain were nothing short of astounding. Over the three years of 2014 to 2017, the aggregate time spent watching videos on YouTube’s homepage grew twenty times. Recommendations drove over 70% of all time on YouTube.²⁶ There was increasing acknowledgment across social media that YouTube’s suggested videos had become eerie accurate at guessing what would interest you.

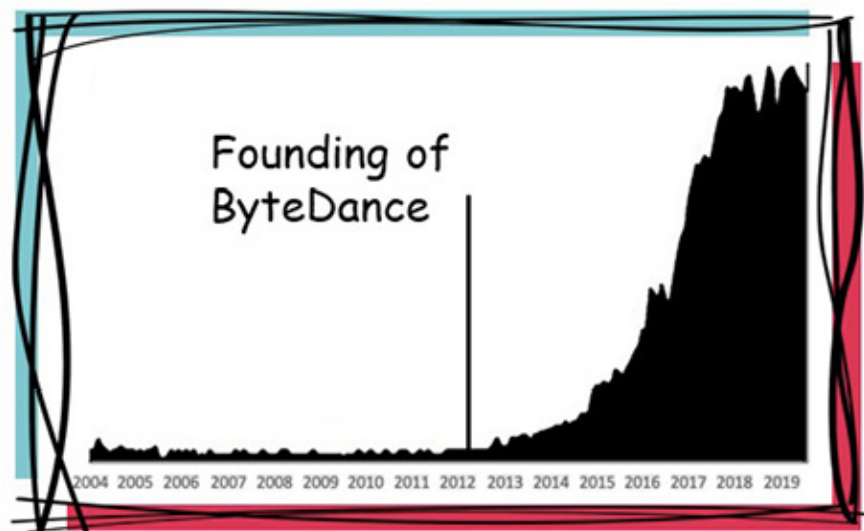


Above: The immediate impact made by YouTube’s use of machine learning algorithmic recommendation engine, Sibyl, in 2011.²⁷

Rapid progress in the field of AI, including the breakthrough known as “deep learning,” meant this method of content distribution would rapidly come of age with profound implications. For a company like ByteDance, the timing of these new advances could not have been better. They were in the early innings of a new era that would see the effectiveness and accuracy of algorithmic recommendation jump forward in leaps and bounds. YouTube was one of the first to benefit, but these advances would end up

supercharging all kinds of online experiences from Amazon's product recommendations to TikTok videos.

Many things need to fall into place for a new technology giant like ByteDance to arise. There needs to be a popular product that addresses a vast market. There must be a visionary founder or founders who are brave and smart enough to build a winning team. Just as important is luck. You must be in the right place at the right time to ride a massive wave of change. As legendary Chinese technology entrepreneur Xiaomi CEO Lei Jun judged, reflecting on lessons from his long career, "Seizing the opportune moment exceeds **far, far** beyond any other tactics." ⁷⁸



Above: Google Trends visualization for interest in the term "deep learning" from 2004 to 2019.

Many industry veterans in China recognized that personalized recommendation held great potential to improve the mobile experience. Yet Yiming was ahead of the pack as he not only sensed the opportunity but also acted decisively. The most critical factor driving ByteDance's success was Yiming's early clarity of vision that they must commit to going "all in" on recommendation and the timing of that decision. ByteDance got in on the ground floor for not one but two era-defining trends—the rise of smartphones and the rise of AI.

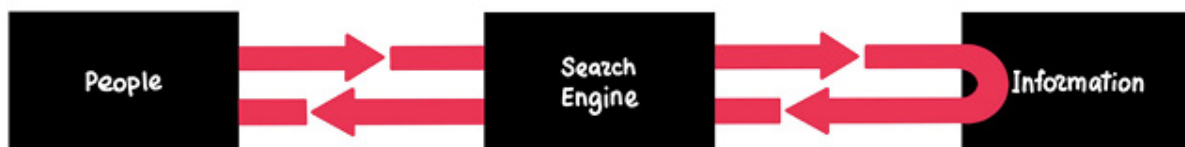
It started with a train ticket

Yiming recalls his earliest epiphany on the power of recommendation came as early as the late 2000s. The Chinese New Year was coming, and like hundreds of other migrants working in the big cities Yiming was searching online for a train ticket to his hometown. Train tickets are always scarce this time of the year, and securing one was typically a strenuous and frustrating task. People either had to constantly refresh online travel websites to check for availability or pay an inflated price to buy tickets from scalpers.

Yiming resolved to do neither; instead, he built a computer program from scratch that would automatically crawl ticketing sites and notify him when a spot became available. Creating the program over a single lunch break, it secured him a ticket in less than thirty minutes.

“It was a revelation. Search engines require people to find information. We are still using search here, but we can specify the condition for a trigger. When the condition is met, information will be delivered to people. Such is **the shift from people looking for information to information looking for people**,” said Yiming as he later reflected on the differences between search and recommendation.

Pull model: “people looking for information” ACTIVE



Push model: “information looking for people” PASSIVE



Above: contrasting push and pull models of “people looking for information” and “information looking for people.”

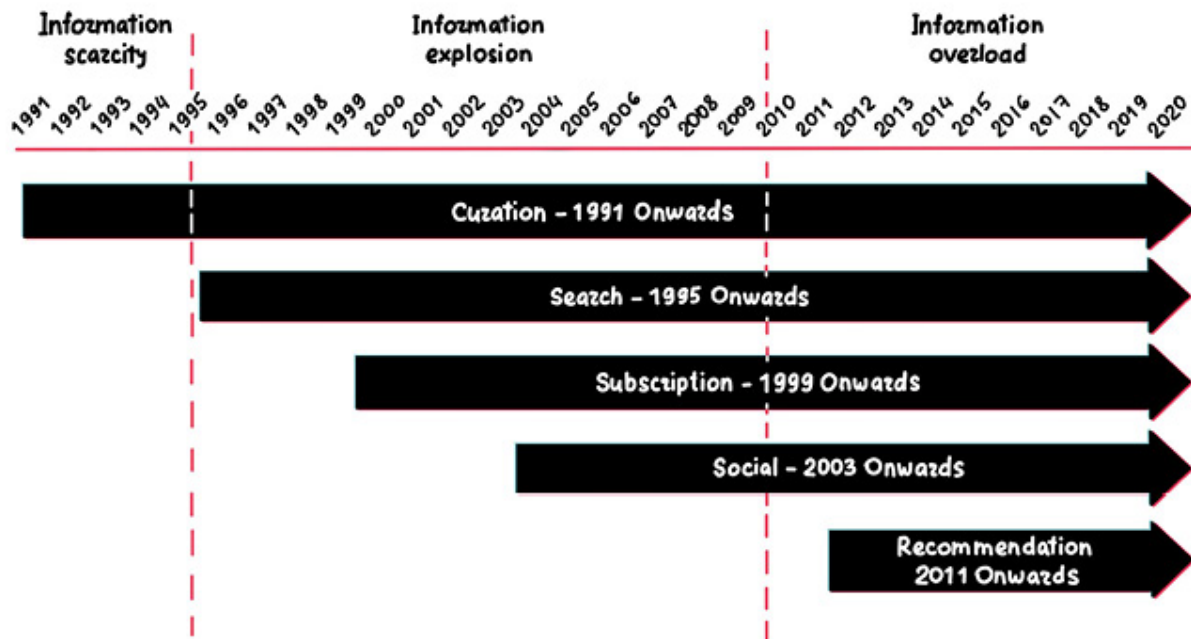
“That was my earliest dwelling on recommendation engines and the application thereof.” The idea was far from being a fully formed concept; however, the seed was planted in Yiming’s mind. Instead of something

proactively pulling information from the internet, it could be pushed or recommended to them based on their individual needs.

By late 2013 and with ByteDance already over a year old, Yiming was invited to give a presentation at a small industry conference⁷⁹ where he laid out a complete picture of his thoughts, which had progressed further.

“The internet period can be subdivided. The earliest period being portal websites, then search, then micro-blogging, then recommendation engines. These are different ways to send and receive information. We can see that each generational change in technology plays a vital role.”

Yiming laid out a framework⁸⁰ detailing the evolution of methods by which information was distributed in the internet age. The earliest method being “portal websites.”



Above: methods of internet empowered informational distribution.⁸¹

Human curation - portal websites, Yahoo! AOL

Portal websites are similar to pre-internet newspapers in that they are traditionally large centralized collections of content updated and organized by editorial staff. A key characteristic of portal sites is that human editors decide which content to display and give prominence to. This centralized human curation has its roots back in the internet’s earliest days when it was

possible to list all the major internet sites in a single manually curated directory.

In China, three major portal websites arose in the mid to late 1990s NetEase, Sina, and Sohu, which all mimicked the original portal site model of Yahoo!. All three companies went public on the Nasdaq in the first half of 2000 and became China's first generation of internet giants before the rise of Alibaba and Tencent. The portal site model remained remarkably resilient, and using human editors to select and curate the order of content was the way most of Toutiao's predecessors preferred to operate.

Curation by human editors is arguably not a true internet mode of content distribution but merely a continuation of the pre-internet forms of delivery such as newspapers or TV in that it shares the characteristics of being a one way broadcast with limited interaction or personalization.

Search engines – high intent

By the mid-1990's it had become abundantly clear that the web was now too large to index manually. The internet represented an explosion of information; suddenly, anyone could set up their own blog site and start publishing online. Search engines were an extremely effective use of technology to solve a pressing need—locating precise information across an unimaginably vast decentralized network.

The first popular search engines included names such as AltaVista and Yahoo!. Google arose later with better technology, which propelled them to dominate the category. Search is an extremely efficient and accurate way to match content to users with one major weakness—you must know the term you wish to search for. The user must expend mental effort to understand their need, type a relevant term into the search bar, and select the most suitable option from those presented. This makes search less ideal for news and entertainment, which hold a strong discovery element.

The search business turned out to be highly lucrative. Placing ads against search results has propelled Google and Baidu, which dominates China's internet search market, to become two of the most valuable giant internet companies in their respective markets.

Subscription - email and RSS

Most people associate email as a technology for work communication. Yet even in the 1990s, the first generation of digital marketers had worked out that email was also a powerful channel for distributing promotional content and maintaining a direct relationship with pre-existing customer bases. “Sign up for our newsletter” has been the primary website call to action for over two decades, with inboxes becoming increasingly cluttered ever since.

Around the turn of the millennium, the Rich Site Summary (RSS) standard was formed, which allowed people to subscribe to receive updates from websites directly using RSS feed reader software such as the popular Google Reader. Instead of checking 30 different sites for new content, everything could be managed centrally in one place. Much like “Web 2.0,” another forgotten term of the same era, RSS was all the rage at one point but has waned in popularity to the point that the term now needs to be explained to most people. Podcasts also use the RSS feed standard, and many podcast apps still favor the subscription model.

Subscription as a form of content distribution⁸² did not give rise to giant internet companies in the same way as search. This is partly due to being characterized by open standard formats not controlled by any one private organization, predominantly RSS and email. Another reason why subscription did not give rise to giant companies was that it was quickly superseded by another form of content distribution, which built upon and extended the basic concept further: What if we could subscribe to people?

Social - decentralization

Social networks need little introduction. They have proven to be immensely popular and effective at not just connecting and facilitating communication between groups of people, but also a potent method of distributing information. In a social network, you are essentially subscribing to individuals who are sharing content. This could be self-produced content, “here’s a picture of dad and me today,” but it’s also just as likely to be content from other sources “check out this travel article.”

This form of distribution is highly decentralized. The content displayed to the user is a function of the individual actions of their entire network. Everyone’s feed is personalized as a direct consequence of no two individual’s social networks ever being precisely the same.

There are many diverse social content distribution models, from the scrolling newsfeeds of Facebook and Twitter to large WhatsApp groups, which are a prevalent form of social content distribution in many emerging markets. Another model is exemplified by platforms such as Reddit or Digg, which distribute content based on upvoting. The point that unites all of these is that distribution of content is dictated by the users themselves, working within the constraints of the platform's rules.

Recommendation – Toutiao and TikTok

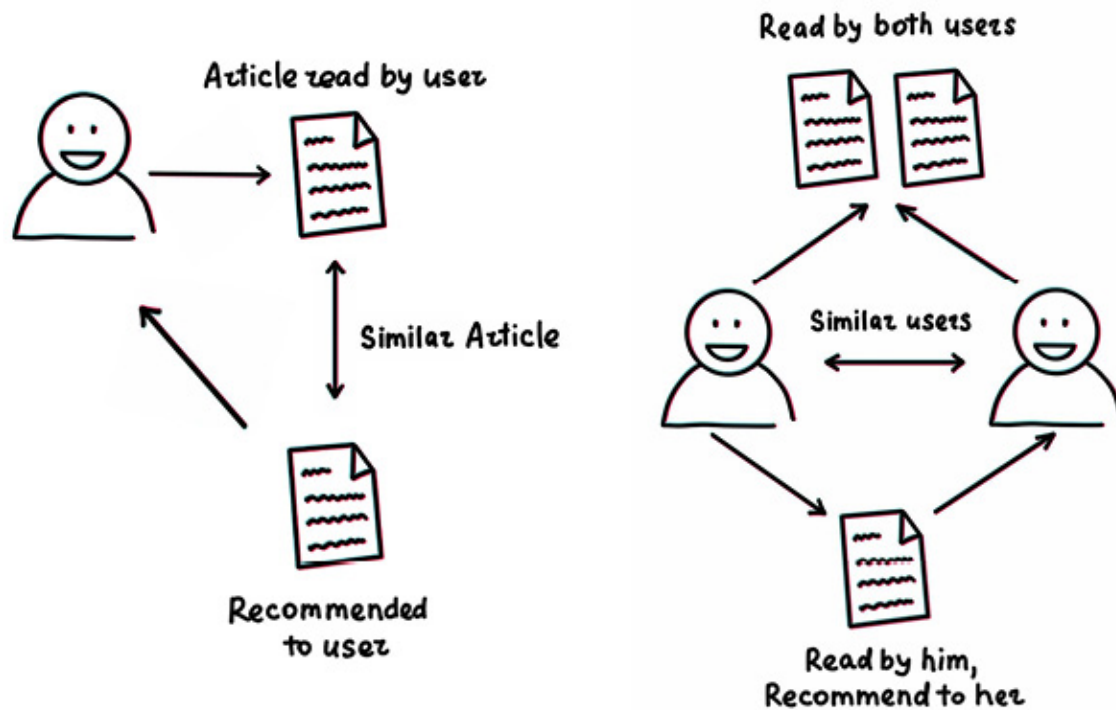
Recommendation is the most recent model to have come of age. When fully embraced, it can be highly accessible as recommendation removes the requirement to actively subscribe to channels or take specific actions such as add friends or tap 'like' buttons. Good recommendation has high technological requirements. With search engines, the user provides an unambiguous indication of intent by what they type into the search bar. Recommendation has no explicit declaration of intent; it must infer the user's preferences based solely on their previous behavior.

An early pioneer of online content recommendation was StumbleUpon,⁸³ founded in 2001, a recommendation driven browser toolbar extension. Netflix also famously awarded a \$1 million prize for an algorithm that increased their recommendation system's accuracy by 10% in 2009. The importance of online recommendation first became evident in e-commerce through prompts such as "Customers who bought this item also bought..."

⁸⁴

In general, recommendation systems rely on two key processes: "content-based filtering," and "collaborative filtering." The two concepts being relatively easy to grasp. A content-based filtering system will recommend content to users, similar to what they already like to consume. If the user enjoys watching videos with dogs and has been tagged as "dog lover," the system will recommend more dog videos.

A collaborative filtering system will base their recommendations on finding groups of users who enjoy similar content. Say Jane's and Tracey's interests are highly correlated. If Jane watches a video multiple times all the way through, a reliable indicator of interest, then the system will also recommend that video to Tracey.



Above Left: Content-based filtering, Above Right: Collaborative filtering.

Internet informational distribution

These various methods aren't mutually exclusive; it's possible to make use of them all. It's also rare for a platform to rely on one single approach to content distribution—what site or app doesn't use search in some form? However, most platforms lean towards one method being primary.

Platforms can evolve, with the mix of methods being relied upon shifting over time. An excellent example of this is YouTube, which went from relying heavily on channel subscriptions at one stage to firmly embracing recommendations. Screenshots on the opening page of this chapter visualize this change. Channel subscriptions are still a popular and accessible part of the YouTube experience, but recommendations have become the more prominent method to discover content.

	Passive – Low Control	Active – High Control
Human-driven	Social Users share content in a decentralized manner.	Subscription The user manually curates sources.
Machine-driven	Recommendation Preferences are inferred based on behavior.	Search The user indicates intent through a search term.

Someone with a voracious demand for information, such as an intellectual or a journalist, will likely have a preference for subscription and search. These methods hold the highest degree of accuracy and control. They also require the user to be more proactive and engaged, typing in search terms, and curating subscription lists. People with lower demand for information are more likely to hold a preference for social and recommendation. These methods are accessible and lend themselves well to news and light entertainment.

Active methods (subscription and search) are better for larger screen devices often used for serious work or study, where session times tend to be longer, and keyboards allow for accurate and fast input. Passive methods of content distribution are, in general, more suitable for the fragmented time and small screens of smartphones.

Before starting ByteDance, Yiming had already built deep practical experience with many methods of content distribution. Kuxun (2006-2008), a travel search engine, Fanfou (2008-2009), a Twitter-like social platform and 99Fang, a real estate portal that employed a combination of search with

basic recommendation. Yiming acknowledged the depth and breadth of his expertise in an interview.

“The efficient flow of information is the main theme of my entrepreneurship. I think that information transmission has a great impact on the benefits, cooperation, and cognition of human society... I care about information. Whether it’s a search engine’s keywords, or a social networking site that uses people as nodes or an interest engine that uses interests as a granularity, they are all information-based.”⁸⁵

Why RSS is doomed - Yiming’s article

In 2013, Google announced plans to shutter its popular RSS feed reader service Google Reader, after nearly a decade of operation. The move provoked an online outcry and intense backlash from a loyal base of users who found the service valuable.⁸⁶ Outraged fans formed an online petition to keep Google Reader operating, which gained over 50,000 signatures within a few hours.

Yiming himself was an early adopter of Google Reader, yet he was pessimistic about its future. Taking time out from his intense schedule as CEO, Yiming sat down to write a seminal article praising Google’s decision. Arguing that while the move drew outcry, Reader was never a mainstream product. Most of its loyal users worked in the media and internet industry, in which collecting and consuming information was crucial to their day-to-day work. Not only did they have to be self-disciplined, but they also needed to have exceptional information management skills, he argued.

“The subscription model is too demanding on the users’ end. They need to know ‘what I like and what I will subscribe to.’ Users will also be on the fence about whether they should subscribe to publications with a mix of interesting and uninteresting content.”⁸⁷ A better solution built for the masses, he believed, was Toutiao’s use of algorithms to recommend content based on past behavior. People’s behaviors change as screens get smaller, time becomes fragmented, content consumption becomes bite-sized, and access to information becomes bountiful, Yiming noted. Google cited the same factors for terminating its Reader service.

“As a culture, we have moved into a realm where the consumption of news is a near-constant process,” ⁸⁸ said Richard Gingras, Senior Director of News & Social Products at Google. “Users with smartphones and tablets are consuming news in bits and bites throughout the course of the day – replacing the old standard behaviors of news consumption over breakfast along with a leisurely read at the end of the day.” Google is looking at “pervasive means to surface news across [Google’s] products to address each user’s interest with the right information at the right time via the most appropriate means,” he added.

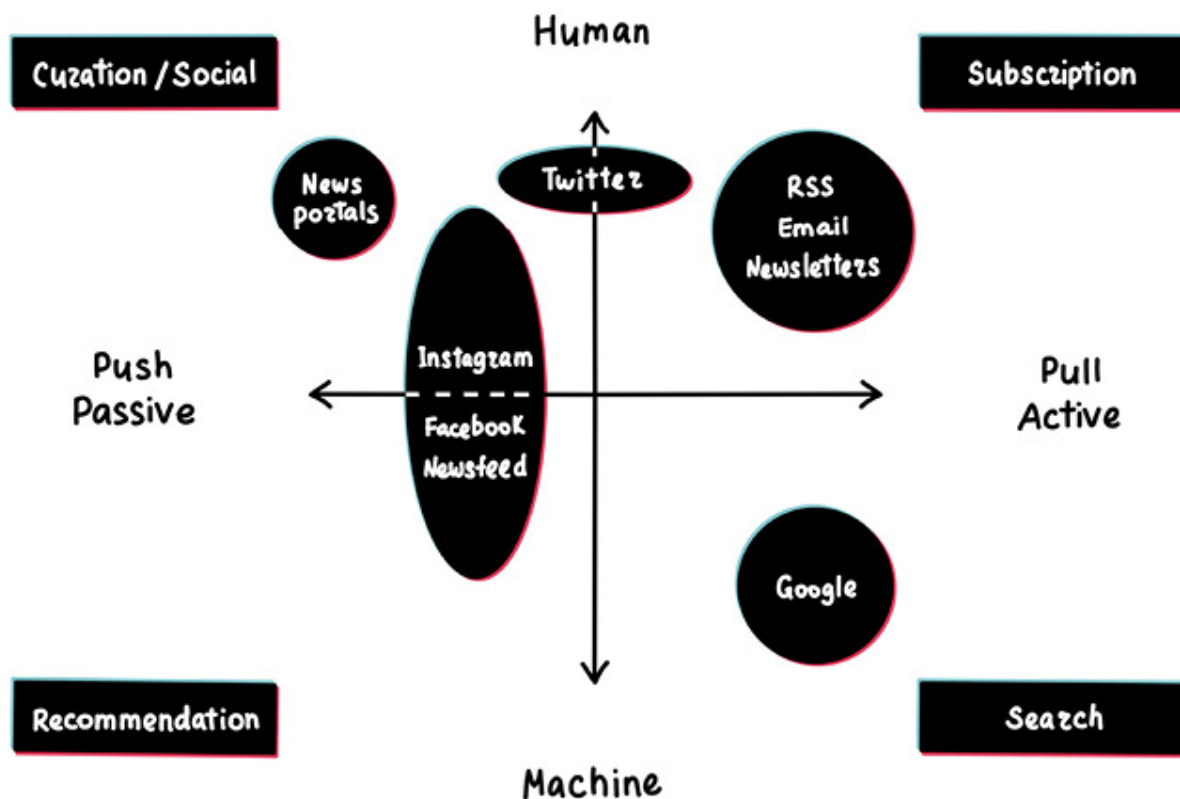
Much like Google’s new vision, Toutiao leveraged machine learning capabilities to anticipate user tastes and essentially give people what they want, when they want it. Rather than intense news consumption, Toutiao offered a more leisurely interest reading experience. Information was presented in an endless stream with a de-emphasis on content sources and a new focus on people’s interests.

The global landscape

Applying this framework of internet informational distribution to look at the global market shortly after the founding of ByteDance reveals companies like Facebook were already embracing recommendation.

“We want to give everyone in the world the best personalized newspaper in the world.” ⁸⁹ were the words of Mark Zuckerberg in 2013, announcing a significant set of changes to the Facebook newsfeed. The American giant saw the adoption of machine learning as central to remaining competitive.

The Facebook newsfeed was already adopting sophisticated technology to recommend an optimal mix of posts from friends and family, news media, adverts, and branded content. In 2014 Instagram’s popular “explore” tab was tweaked to display content precisely personalized for each user. YouTube leaned heavily into optimizing suggested videos as the most effective strategy for increasing engagement, starting from around 2011.



Above: the primary methods of distribution used by various American platforms around 2013

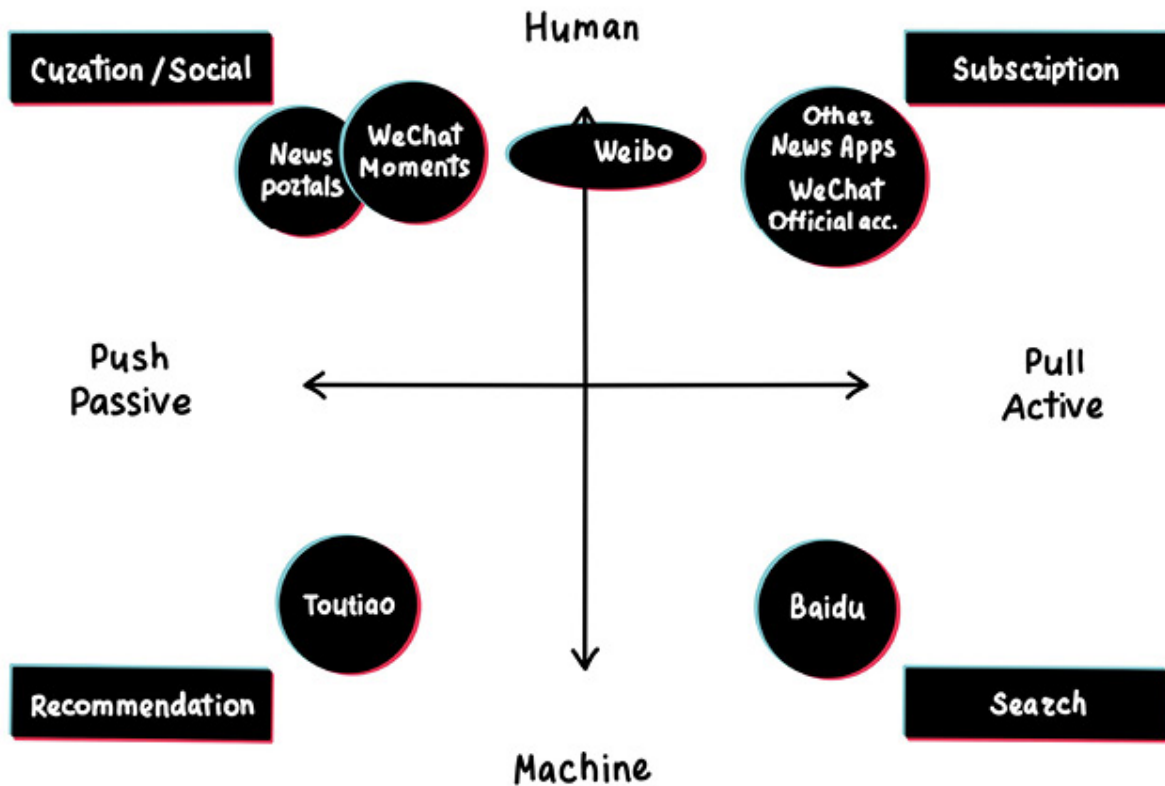
An opportunity ripe for the China market

Applying the same framework to analyze the competitive dynamics within the Chinese market shortly after the founding of ByteDance, we see the positioning of Toutiao was radically different from the other major platforms.

For search on both desktop and mobile, the clear market leader was Baidu. With a strong moat of technology and brand recognition. In the same way as Americans say, “Let me Google that,” Chinese also use Baidu as a verb. Baidu’s stranglehold on the lucrative search market would remain unchallenged.

The 800-pound gorilla of the Chinese mobile internet was “super-app” WeChat, which at the time embraced two ways of distributing media content, official accounts that followed a subscription-based model and a newsfeed entitled “moments.”

The newsfeed was, in many ways, the embodiment of WeChat founder Allen Zhang's philosophy. His stance at the time towards algorithmic recommendation could be described as, at best, suspicious, at worst, dismissive.⁹⁰ In his mind, WeChat's "moments" was meant to be a place for authentic communication between people. The feed was simply reverse chronologically organized posts from the user's contacts, even using filters on photos was removed as an option.



Above: the primary methods of distribution used by various Chinese platforms around 2013

By contrast, Sina Weibo, China's other giant gorilla of the mobile internet, was a media-focused organization. They had won their lucrative position in dominating microblogging, the so-called "Twitter of China," not through having the best technology or user experience but by getting a critical mass of big-name celebrities and media outlets to embrace the platform.

Weibo tagged users based on accounts they had subscribed to and used this as an indicator of their general interests guiding choices over which content to recommend. But it did not see improvement in this crude

recommendation technology as an essential business driver. The company was focused on better managing and extracting value from the influencers on their platforms and expanding their reach to the vast pool of users in lower-tier Chinese cities, a high growth market just opening up. Investing in better recommendations was low down the priority list.

Many market-leading news aggregation platforms still relied on human editorial staff to call the shots on which content would be given prominence. At the time, most smartphone news apps were clones of their PC counterparts. Subscribing to channels of content based on interests (e.g., finance, lifestyle, sport) was about the extent of personalization available.

There was a yawning gap in this competitive macro-environment, with ample white space for Toutiao to attack. Like all the venture capitalists who turned Yiming down for their round B investments, none of the big platforms took content recommendation seriously. Incredibly the largest player in the mobile space, WeChat, was suspicious of the technology.

Industry consensus held that even if recommendation engines did prove superior, the technology would be copied. It was not a sustainable way to win a long term meaningful share of an already crowded, competitive market.

ByteDance was about to prove them wrong.

69 <https://www.theverge.com/2017/8/30/16222850/youtube-google-brain-algorithm-video-recommendation-personalized-feed>

70 <https://www.sfgate.com/business/article/YouTube-s-effort-to-get-people-to-watch-longer-2352967.php>

71 <https://www.businessinsider.com.au/youtube-engineer-christos-goodrow-on-recommendation-engine-2015-7>

72 <https://dl.acm.org/doi/10.1145/1864708.1864770>

73 In this context, “lack of metadata” refers to videos uploaded without tagging with keywords, or accurate titles and descriptions

74 <https://glinden.blogspot.com/2011/02/youtube-uses-amazons-recommendation.html>

75 <https://www.datanami.com/2014/07/17/inside-sibyl-googles-massively-parallel-machine-learning-platform/>

76 <https://www.theverge.com/2017/8/30/16222850/youtube-google-brain-algorithm-video-recommendation-personalized-feed>

77 <https://youtu.be/3SaZ5UAQrQM?t=503> Image Credit: Tushar Chandra, Principal Engineer at Google Research and a co-lead for the Sibyl project

78 <https://www.bilibili.com/video/av49873394/>

79 <https://www.tmtpost.com/84589.html>

80 The theory presented over the next few pages expands upon and contextualizes the broad observations first presented by Yiming.

81 Years are chosen not by the first instance of usage, but when the method came of age—a choice which is admittedly open to debate in many cases.

82 Subscription as a form of payment did facilitate the rise of large internet companies such as Netflix and Spotify.

83 <https://en.wikipedia.org/wiki/StumbleUpon>

84 <http://www.cs.umd.edu/~samir/498/Amazon-Recommendations.pdf>

85 http://economy.gmw.cn/2018-03/23/content_28080924.htm

86 <https://bits.blogs.nytimes.com/2013/03/14/the-end-of-google-reader-sends-internet-into-an-uproar/>

87 <https://tech.qq.com/a/20130314/000123.htm>

88 <https://www.wired.com/2013/06/why-google-reader-got-the-ax/>

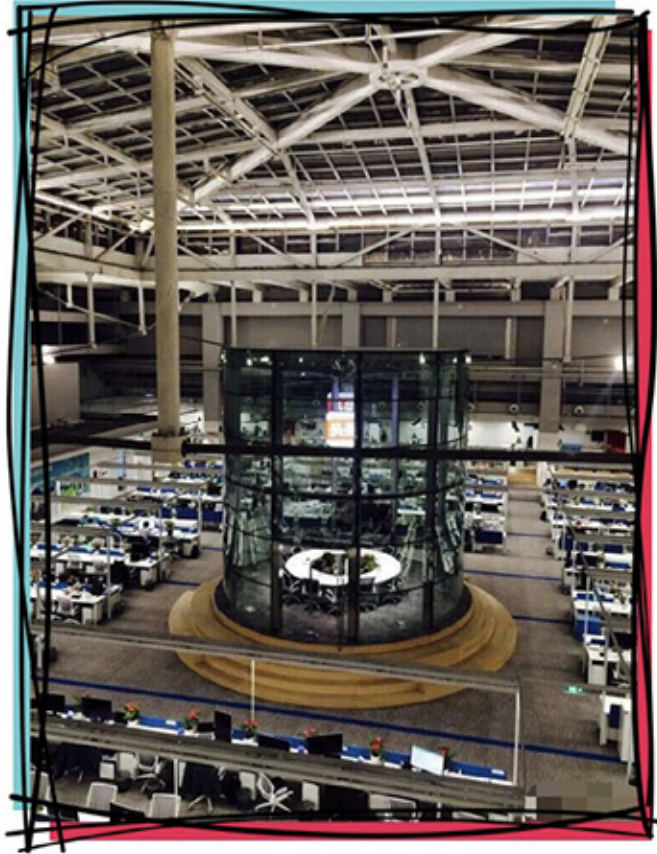
89 https://www.washingtonpost.com/business/technology/facebook-to-change-news-feed-to-a-personalized-newspaper/2013/03/07/b294f61e-8751-11e2-98a3-b3db6b9ac586_story.html

90 This stance has since softened considerably. Today WeChat has several recommendation content feeds.

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CHAPTER 4

IN CHINA, THE NEWS READS YOU

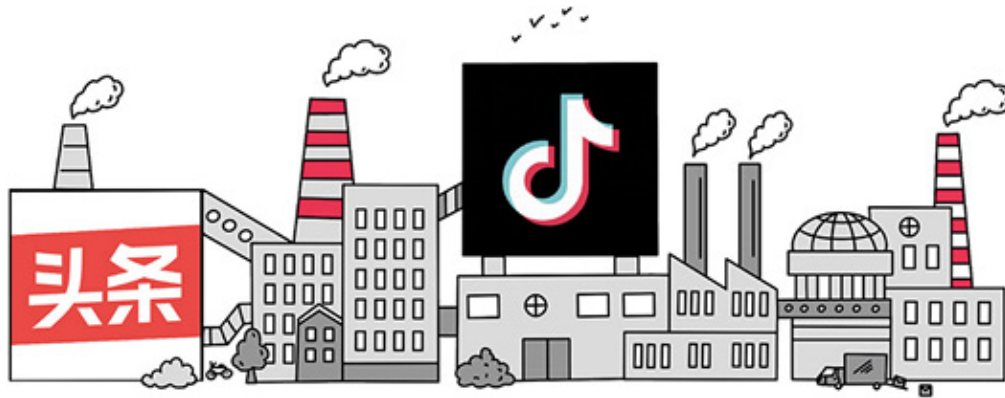


“We have built the largest machine-learning platform for content. That’s our weapon.”

Lei Li, ByteDance AI Labs

Image: the large central “fishbowl” glass meeting room, ByteDance’s Beijing head offices, former aviation museum AVIC Plaza.

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Chapter Timeline

2012 Sept – Toutiao 's personalized recommendation system goes live

2013 Aug – Zhang Lidong joins ByteDance to lead commercialization

2014 – Yang Zhenyuan joins ByteDance as VP of Technology

2015 Jan – Okina wa annual meeting

2016 Feb – Company moves to new offices at AVIC Plaza

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IN MID-2012, an email dropped into ByteDance's technical team's inboxes with the ominous title "Recommended Engine General Meeting." Yiming was determined to push forward on a topic that he saw to be critical to the company's future. The email continued: "To be an information platform, it is necessary to do a good job on the personalized recommendation engine. Do you want to start this thing now?"

Toutiao's early recommendation system, its so-called "personalization technology," was, at the time, rudimentary. Open the app, and the user would be bombarded with top-read articles to keep them immediately hooked. Next, it would mix in more targeted click-bait articles appealing only to specific demographics to test and determine who the reader was. The user clicking on the article with a big preview picture of a female car show model is probably male. The other user consistently reading moralistic feel good "chicken soup for the soul" articles is probably a senior citizen. Supplementing this guesswork was basic information such as the user's phone model, geographical location, and the time of day they opened the app.

This was a decent start, but it was far from where Yiming knew the company needed to be. He hoped to become the best in the industry and maximize this technology's potential to gain a sustainable advantage over others. Yet to get beyond their current level involved overcoming intimidating technical barriers. The small team had experience in search engines and mobile app development, but no one had deep expertise or capability to develop a state-of-the-art personalized recommendation engine. Many on the team expressed deep concern about the group's lack of technical ability to achieve Yiming's goal. Bringing in expensive outside experts wasn't realistic given they were a startup of tens working out of a converted apartment; besides hardly anyone in China actually had the expertise they were looking for.

Yiming was stubbornly insistent. As he saw it, they could choose to be slightly innovative and ride the mobile internet wave to moderate success, or they could go all in and shoot to achieve a fundamental breakthrough that would really create value. "We can't do recommendation now, but we can learn. After this meeting, I'll be the first to take action" was Yiming's promise to lead by example.

As part of the drive, Yiming got wind of an upcoming book, *Putting Into Practice Recommender Systems*.⁹¹ It was perfect for them and written by one of China's leading experts in machine learning at the time—Xiang Liang⁹². Yiming personally reached out to Xiang, who was then working as a researcher at video streaming site Hulu, to request a copy yet was rejected on account that the book was not yet published. Frustrated, Yiming decided to teach himself poring over what he could find online. Ironically, years later, Xiang ended up joining ByteDance, becoming one of the pivotal machine learning experts at the company's AI Labs. Yiming later described Xiang's refusal to send him a copy of his book as having severely delayed the progress of the company's recommendation capabilities, reflecting that "there were only a few companies that really made their minds up to make recommendation engines and many failed."⁹³

Sheer determination and self-study proved somewhat useful, but in the end, the significant breakthroughs for ByteDance came as a direct result of acquiring outside talent. Only by luring in experienced expertise from other organizations could Yiming hope to realize the vision of building a best in class recommendation engine. There was one company in particular that could provide much of that top-level talent—search giant Baidu.

Baidu, also based in Beijing, was one of China's largest internet companies; it held a strong reputation for hoarding top-level technical talent. Google exiting from the Chinese search market had removed Baidu's competitive pressure, allowing them to enjoy a dominant and relatively safe market position. In ByteDance's early years, Yiming considered Baidu his most important competitor. He was deeply afraid of its algorithmic talent and perceived it as the most capable company in China to move into the area that he hoped to dominate.

Fortunately for Yiming, Baidu was late to realize the importance of personalized recommendation. Baidu's search business was so unrivaled and profitable that it was blindsided by coming threats from ByteDance even as they were falling behind in the new area of mobile. Despite being touted in the media as "the Google of China," the comparison was unfair. Baidu did not have the luxury of controlling the dominant smartphone operating system, Android. It also didn't help that the company was perceived as suffering from faction infighting and bloated management.

There were constant rumors that that CEO founder Robin Li's wife was the one calling the shots on important company decisions.

The first technological breakthrough came in 2014 when ByteDance lured away Deputy Director of Search, Yang Zhenyuan, from Baidu, where he had been working previously for nine years. Yang was immediately given the title vice president of technology and set about orchestrating a major technical upgrade.



Above: In early 2014, the company was roughly 100 people in size.

Yang's joining opened the flood gates to many other Baidu engineers following him as ByteDance went all out to swipe the rich technical talent pool of Baidu with large pay packets and generous share options. By 2015-16, ByteDance started to break away and create a sustainable advantage in recommendation technology after capturing other big-name Baidu talents, including Chen Yuqiang and Zhu Wenjia⁹⁴. Zhu Wenjia later led the team responsible for developing the original recommendation systems used by Douyin and TikTok.

The company's technical expertise had reached such a level by 2016 that they could experiment with methods to algorithmically generate content. During that year's Olympics, a ByteDance-developed bot wrote original

news coverage, publishing stories on major events faster than traditional media and enjoying engagement levels comparable to articles produced by human writers.

Recommendation systems became ByteDance's core technology, underpinning everything from short videos on TikTok to articles in Toutiao through to comedy GIFs on their app Neihan duanzi (Insider jokes.)

Recommendation 101

In January 2018, ByteDance held a public meeting in Beijing to disclose how their algorithms work.⁹⁵ The goal was to allay specific criticisms from state media and internet watchdogs of spreading pornography coupled with general concern around its apparent lack of human oversight concerning content. At the event, the company's senior algorithm architect Cao Huanhuan went into detail around ByteDance's recommendation system's principles. His presentation informs much of what we cover below.⁹⁶

ByteDance's system centers around three profiles: the content profile, the user profile, and the environment profile. For the content profile, Cao gave the example of a written news article about an English Premier League football match between Liverpool and Manchester United. Keywords would be extracted from the article using natural language processing, in this case, "Liverpool Football Club," "Manchester United Football Club," "English Premier League," and names of several key players from the game such as "David de Gea."

Relevance values are then assigned to the keywords. In the example, "Manchester United Football Club" was 0.9835, and "David de Gea" was 0.9973, both very high as to be expected. The content profile also includes when the article was published, which helps the system calculate when it has become outdated and stop recommendation.

The user profile is built from various sources, including one's browsing history, search history, the type of device they are using, the device location, their age, gender, and behavioral traits. Users were divided into tens of thousands of latitudes based on social data and user behavior mining to build different profiles.

As you read posts recommended by the platform, it learns your preferences by tracking your behavior: what you choose to read, what you

opt to dismiss, how long you spend on a piece of content, which articles you comment on, and which stories you choose to share.

Finally, the environment profile is based on where the user consumed the content, for example, at work, at home, or during a commute on the subway, as people's preferences vary given the different situations. Other environmental traits include the weather and even the stability of the user's internet connection and which network they were on (e.g., Wi-Fi or China Mobile 4G.)

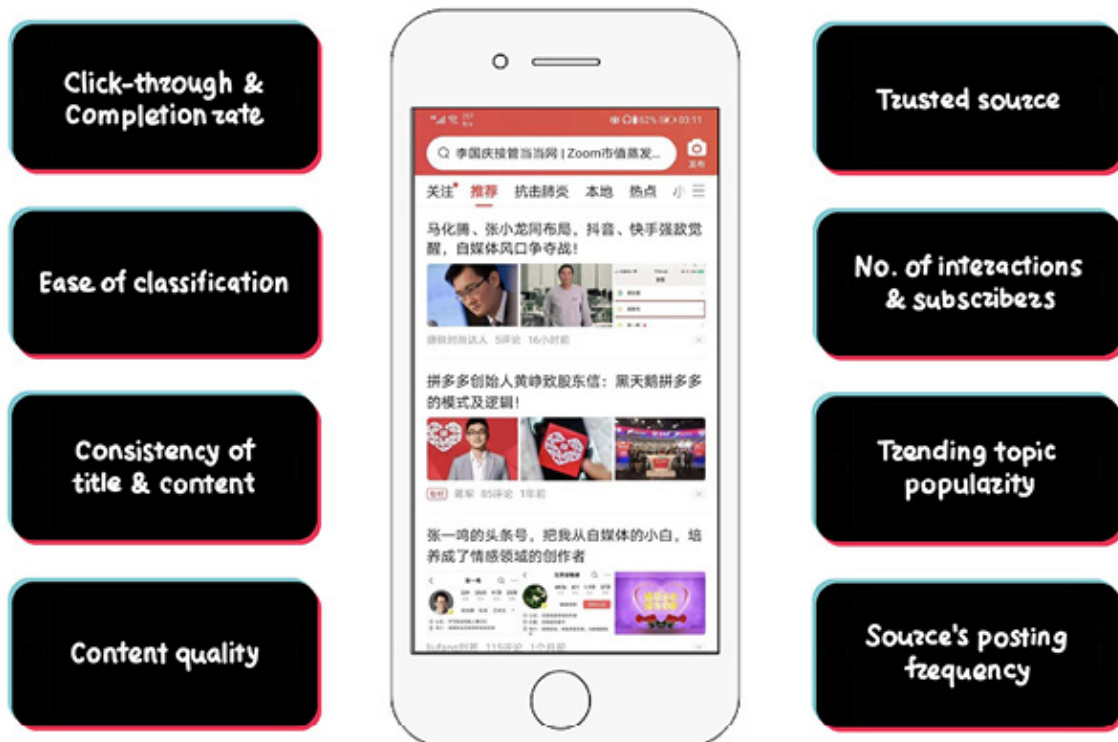
The system computes the strongest statistical match between the content profile, user profile, and environment profile that will optimize the percentage of articles read and the percentage of the articles finished (i.e., time spent).

This content distribution process involves allocating a “recommendation value” to each newly published story based on its quality and potential readership. The higher the value, the more suitable people the article will be distributed to. The story's recommendation value changes as users interact with it. Positive interactions such as likes, comments, and shares increase the recommendation value; negative actions such as dislikes and short reading times lower the value. The value also decreases over time as the content becomes out of date. For fast news cycle categories like sports or stock prices, a day or two could be enough for the value to see a significant decrease. For more evergreen categories, such as lifestyle or cooking, the process is slower.

Focusing on the first hundred articles to recommend to a new person was considered critical as retention was shown to drop significantly after this milestone.⁹⁷ Identifying and quantifying this “north star” key growth metric echoes how the early Facebook growth team famously focused on getting new users to add seven friends within ten days. According to former executive Chamath Palihapitiya, Facebook's early team “talked about nothing else” except this growth metric.⁹⁸ ByteDance's focus here led to high retention rates of over 45%,⁹⁹ more generally associated with mainstream social networks, and globally one of the highest average time spent on apps per user.

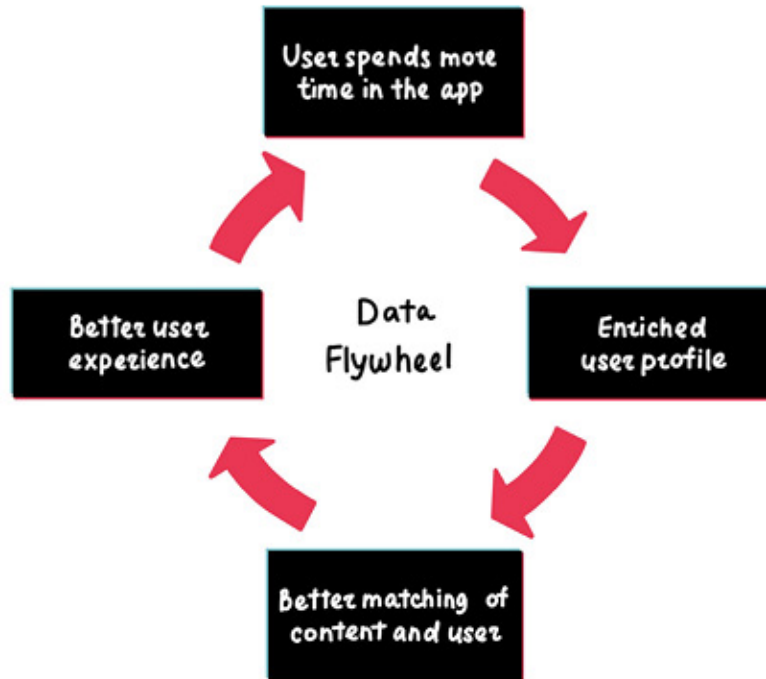
This core system for recommending written articles with Toutiao was later adapted and used for short videos with TikTok and Douyin. All these apps make use of the same ByteDance backend recommendation engine system. Videos are more challenging as they tend to be uploaded without keyword tagging or accurate titles and descriptions, making for an exciting computer vision challenge to work out what is actually in the video.

The beauty of relying on recommendations to improve engagement is that it creates a virtuous cycle of continual improvement over time, often referred to as a “data network effect.” The more time spent using the app, the more enriched becomes the user profile, which leads to more accurate content matches and better user experience. This naturally leads to more time being spent in the app, which further enriches the user profile and so on.

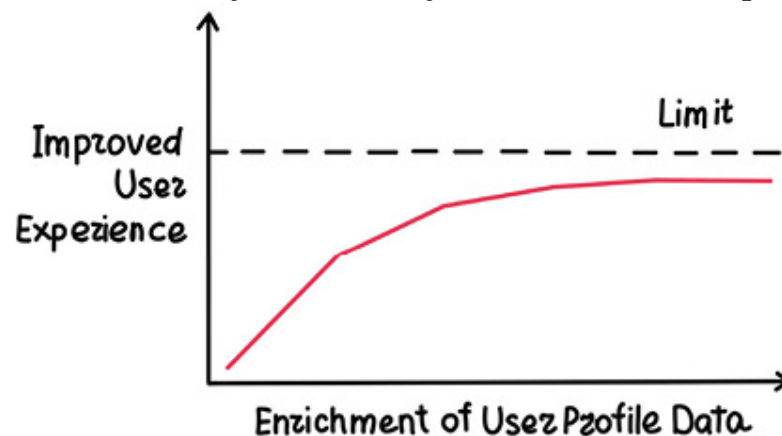


Above: Some of the factors influencing an article's recommendation value.

This virtuous cycle is powerful but doesn't continue indefinitely. The rate of improvement in user experience is initially fast but will asymptote over time as the user's profile is enriched further and further to the point where a fully accurate and detailed interest graph has been formed.



Above: The core data flywheel of ByteDance's content platforms.



Another limitation of their system was the necessary human element. Toutiao claimed to be a purely technology-driven company, doing away with human editors. Yet this characterization was somewhat misleading. The system still relied heavily on human labor, an army of staff conducting basic repetitive tasks such as tagging articles and manually reviewing content, which aided the machine learning. Reliably extracting key terms was critical for accurate recommendation, but technologies such as natural language processing could only get you so far.

Regardless, no matter how accurate their recommendation, merely having a better product than competitors was also not enough. To rapidly grow the

Toutiao user base and scale the company to unicorn valuations, the team had to master the darker arts of growth hacking.

Growth hacking - China style

The Shenzhen airport warehouse 3A was filled with phones, hundreds of thousands of them. Wall to wall, pallet after pallet, a seemingly endless stack of smartphones fresh from the factory production lines. Later that day, all of them would be loaded onto planes and shipped out across China's major cities, weaving their way through a byzantine system of provincial distributors, sub-distributors, and retail store networks before finally landing in the hands of a consumer.

A group of young men and women in grey overalls lined up for early morning duty. To a casual observer, these looked like a typical team of warehouse workers ready to load and unload cargo all day. Instead, the group had a very different task. "Right lads, you know the drill—five minutes for each batch of 12 phones, not a second more. Let's go!" called the group leader. They immediately set to work.

Another day's worth of repetitive tasks lay ahead, following the same sequence:

Use a special device to blow hot air on the phone box seal until the tape fell off. Carefully unbox the phone, making sure to keep everything in pristine condition. Hook the phone up to the machine, a thick plastic box, with a screen roughly the size of an iPad and a row of 12 USB ports.¹⁰⁰ Tap to select the correct options, then press "confirm." Wait. Once the machine had done its thing, unhook the phone and put it back in the original box exactly as you found it, and reseal the tape.

The whole process took no longer than five minutes and was repeated hour after hour, day after day. Just 86 of these machines running continuously over an eight-hour work shift would be enough to cover 100,000 low-end to mid-range priced Android devices. The goal was to batch pre-install over a dozen extra apps onto each phone, one of which was Toutiao.

Back in Beijing, Yiming and executive Zeng Qiang were pouring over the pre-installs spreadsheet as had become their daily routine. Numbers of total installations and activations from each distribution channel and manufacturer were laid out in neat rows. Multiple factors were broken

down and analyzed in detail: 30-day retention rates, device models, A/B tests, the coverage rates of China's myriad of cities and townships. A complex system had evolved to optimize the budget for what had become ByteDance's most effective way to growth hack new users and fast track their company – the grey market of cutting deals with distributors to pre-install apps onto phones after they've left the factory but before they reach the consumer.¹⁰¹



Above: A photo taken in Shenzhen of three app pre-installation machines in action.

Even by the Chinese internet industry standards, the market for pre-installing apps on smartphones was a chaotic, wild west. Yet there was always high demand because it remained a cost-effective channel to reach users of low-end or mid-range Android devices at scale. When ByteDance started allocating budget into app pre-installation, they paid roughly 0.4 yuan per install (\$ 0.06), which was above market rate at the time but still incredibly cheap given that over four years that price continually rose to more than 12 yuan (\$ 1.68).

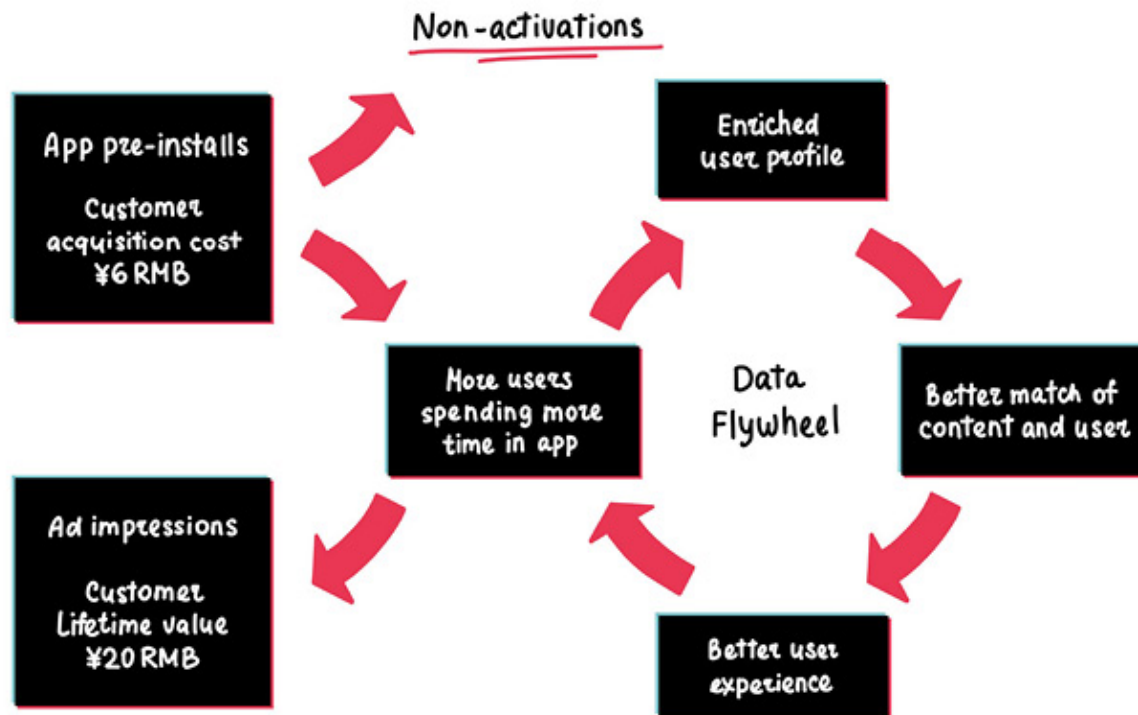
The app pre-installs practice worked because most consumers were either ignorant or indifferent to what software should be packaged on their phones. Buyers focused on the price, brand, and specs of the hardware; whatever software was on the device beyond the Android operating system simply didn't matter to most people. Many of the preinstalled apps were either deleted or perhaps used only once out of curiosity. Toutiao was as well-placed as any to convert those who did give their app a try. Reading news and other web content was a high-frequency activity and a stable need for most people. Toutiao's logo, a newspaper with a bright red banner and characters reading "Headlines," left little doubt about what the app was for. If they could get a user to try them out from a pre-install, they had a chance to get the data flywheel going, enrich their user profile, and start surfacing personalized content. If the user picked up the habit of using Toutiao, they typically didn't delete the app from thereon.

Few retailers questioned app pre-installs as the practice was rife, and, in an industry where the competition was cut-throat and margins thin, pre-installs were a welcome source of side revenue. As pre-installing apps became more lucrative, multiple levels of distributors and agents in the distribution chain embraced the practice. The manufacturer installed their set of apps, the first-level distribution agent added their batch, the second-level distribution agent added another round, even the retail stores themselves might add a few.¹⁰²

A national television news investigation highlighted the case of a new mobile phone bought with more than 60 pre-installed apps. With so many programs running and the phone's memory being only a few hundred megabytes, the device's performance was severely affected. Even worse, it was not uncommon for apps to be pre-installed using a method that involved reinstalling the phone's entire operating system. Apps loaded onto the phone in this manner were often given system root permissions, which prevented them from being uninstalled.¹⁰³

At any link in the chain of distributors, there was a chance that their installation process may result in removing some or all the previously pre-installed apps, which posed a severe risk to app developers like ByteDance. Any deal with an upstream manufacturer such as Huawei or Xiaomi to pre-install Toutiao on their phones may end up being negated by a downstream

distributor's reinstallation of the operating system with their own set of apps later. The solution the market settled on was for developers to pay only for apps that were activated. This put manufacturers that sat upstream at a disadvantage. They had little control over what happened with the distribution channels downstream and even less control over the user's decision to open and use the pre-installed apps.



Above: The app pre-install model (monetary values illustrative.)

Developers such as ByteDance found themselves going further down the chain to smaller distributors and local warehouses, finally even working with individual stores to ensure the apps weren't wiped and made it to the end consumer. Store managers in lower-tier Chinese cities could make handsome returns. A store's wholesale and retail sales combined could typically total 2,000 handsets a month. If each phone pre-installed 25 apps at an average price of 2 yuan, then the store manager's revenue would be 100,000 yuan (\$14,000) per month. As part of the buying process, store clerks would help set up the phone for the customer, insert the SIM card, set a password, sign the warranty, introduce some of the phone's more useful shortcuts, and take the opportunity to activate the pre-installed apps. In this way store managers could guarantee almost perfect activation rates as staff were activating the apps themselves as part of the sales process.

ByteDance chose to invest so heavily in pre-installs, and often paid above-market rates to secure the best partners that they ended up shifting the balance of power within the industry. Other app developers complained that ByteDance was driving up costs of pre-installs. In addition, Yiming was the first to accept billing based on numbers of installs rather than activation volume, a change that was highly favorable to manufacturers.

Other less orthodox but still effective methods of gaining installs involved hiring the services of “ground promotion” companies. Their typical tactics would require recruiting groups of female college students to stop people on the street and encourage them to install an app in return for a gift or a small cash reward. The methods didn’t work well with young people but unsurprisingly found favor with older men.

In the early years of Toutiao, ByteDance acquired tens of millions of users through pre-installation. This led to a situation where the company’s core user base was people using cheap Android devices, many of which were purchased with Toutiao already pre-installed. Over time, the content preferences of this group began to greatly affect the public’s perception of the company.

The app had gained a reputation for serving people a concoction of mindless, uncultured garbage. “We must face the fact that for 96% of people, their needs are so vulgar,” ¹⁰⁴ explained Gao Han, a senior UI designer, and ByteDance employee number twenty-two. He confirmed the app’s reputation was well-earned, “Toutiao is of course trashy. **It’s all click-bait**, all kinds of messy news. It’s indeed trashy. That’s right, I will admit it.”

Everyone knows that junk food is bad for you, but people still love to eat it. ByteDance vehemently denied actively pushing vulgar content ¹⁰⁵ yet was undeniably in the business of giving people what they want. It just so happened that what the masses of China wanted to feed into their brains every day was the mental equivalent of a big greasy cheeseburger—clickbait, celebrity gossip, and pictures of pretty girls. “You think that the whole of China comprises of social elites? The college education rate is only 4%,” continued Gao Han.

Such realities did not dissuade Yiming, who claimed that 60% of his daily information intake was obtained from his company's own products, predominantly Toutiao. During a media interview,¹⁰⁶ Yiming once took out his phone and talked a journalist through his Toutiao homepage. Scrolling through the feed, he praised the system's recommendation accuracy. Industry merger reports, stock market listings, news of executives switching companies—each piece of content he claimed accurately satisfied his personal preferences and held value to him.

Nestled between the industry news articles were two pictured headlines, “Basketball babes” and “Car show models,” precisely the type of low-brow content the app had become notorious for. “Were these also accurately calculated by the system?” the reporter hesitated to ask.

Time to make some money

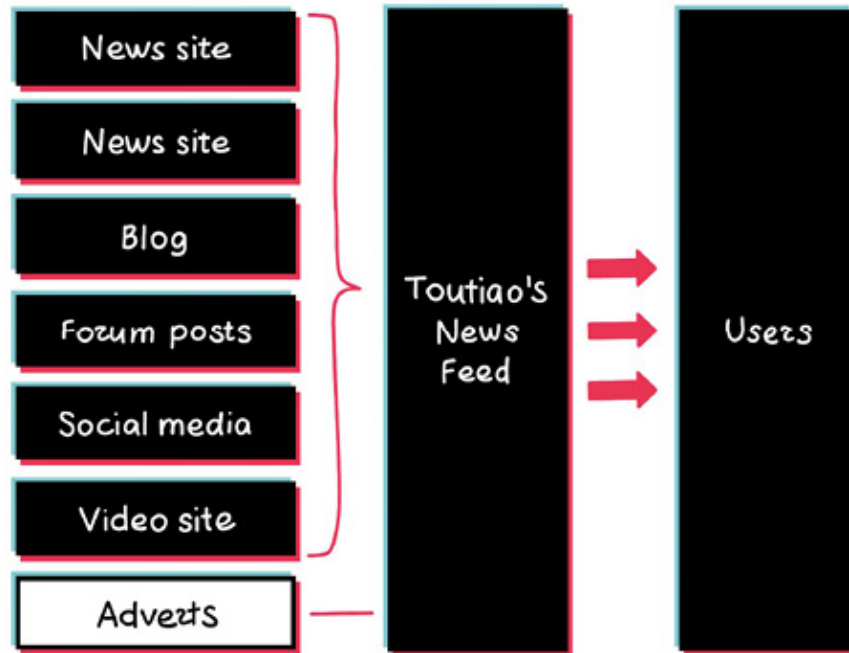
Growth hacking new users and feeding them click-bait was undeniably going well. The next challenge was to shift gears and focus on the subsequent stage of development – monetization. The obvious model was to serve ads in the newsfeed of Toutiao directly alongside other content.

The vision was simple: integrating personalized ads based on user data leveraging the same proprietary technology behind the app's content targeting. This model was scalable, had the potential to be automated, and in moderation, didn't annoy users too much.

The initial challenge ByteDance faced was simply the skepticism in the local market around the effectiveness of mobile ads. It seems incredible now, but many industry players at the time felt the phone screen was too small and not suitable for advertising. At that time, most mobile advertising was in formats such as banner or splash screen ads with low conversion rates and poor user experience. Newsfeed ads, as pioneered by Facebook, were far more established in the U.S. market than China.

During the desktop era, most Chinese brand advertisers did not trust online advertising. They were unwilling to switch from traditional media, which saw China fall behind the U.S. in the digital advertising space. During the mid-2010s, by far the most popular mobile newsfeed in China was WeChat's “Moments,” which actively shunned advertisements to protect the user experience. Parent company Tencent could afford to do so thanks to an

already proven business model reliant on businesses indirectly affiliated with WeChat, such as micropayments in games and content subscriptions for premium music and video.



Above: Toutiao aggregates multiple forms of content from across the internet and serves it to users in a single personalized newsfeed along with intermittent ads.

To lead monetization for ByteDance, Yiming knew he needed to find just the right person, someone highly capable and ambitious with deep experience and contacts in advertising. That person ended up being Zhang Lidong, who today holds the title Chairman of ByteDance China. Lidong serves as the legal representative ¹⁰⁷ for 36 different ByteDance branch companies, which demonstrates Yiming's absolute trust. He is arguably the second most important person in the company with a former vice president once claiming: "ByteDance has two irreplaceable people—one is Zhang Yiming, and the other is Zhang Lidong." ¹⁰⁸

Four years older than Yiming, Lidong was medium built with a broad nose and short black hair. He came from one of the poorest parts of Linfen, a city of over four million inhabitants in the bleak northern inland province of Shanxi. For many years during the 2000s, Linfen had the ill fortune to be named the most polluted place on the planet, ranking bottom of the World

Bank's air quality rankings.¹⁰⁹ Shanxi province holds roughly one-third of all the coal deposits in China.

Lidong had spent over eight years of his early career as a reporter. Ambition led him to the capital Beijing where he joined the economic news department of the state-run newspaper Beijing Times and eventually settled on covering the automotive industry.¹¹⁰ He rose rapidly through the ranks, becoming editor-in-chief and later attaining vice president level, gaining notoriety as one of the youngest general managers of advertising in the country. But Lidong's star was rising just as the newspaper business was experiencing a precipitous fall. Previously the Beijing Times had one of the largest distributions and healthiest businesses in China's newspaper industry; that was until the internet came along gutting readership—the impact was severe. It only took a few years for his former employer to suspend publication entirely on New Year's Day 2017.¹¹¹

Yiming wished to headhunt Lidong and invited him to come and talk at the company offices. As soon as he entered the meeting room, Yiming wrote the words “user volume, click rate, conversion rate, unit price, CPM, CPC” on a small whiteboard and put together a long list of complex and esoteric calculation formulas. The technical boss then spent the next few hours explaining the derivation process of these equations. Lidong later confessed that he didn't understand any of it, but that mattered little as Yiming's approach of using math to deduce the profit model of advertising had shocked him greatly.¹¹²

Shortly after settling in, Lidong began leading the company to take its first baby steps into advertising. The first client he found willing to test out ByteDance's advertising was a home appliance store operated by Chinese retail brand Gome, which was just 10 minutes' drive from their office in northern Beijing.

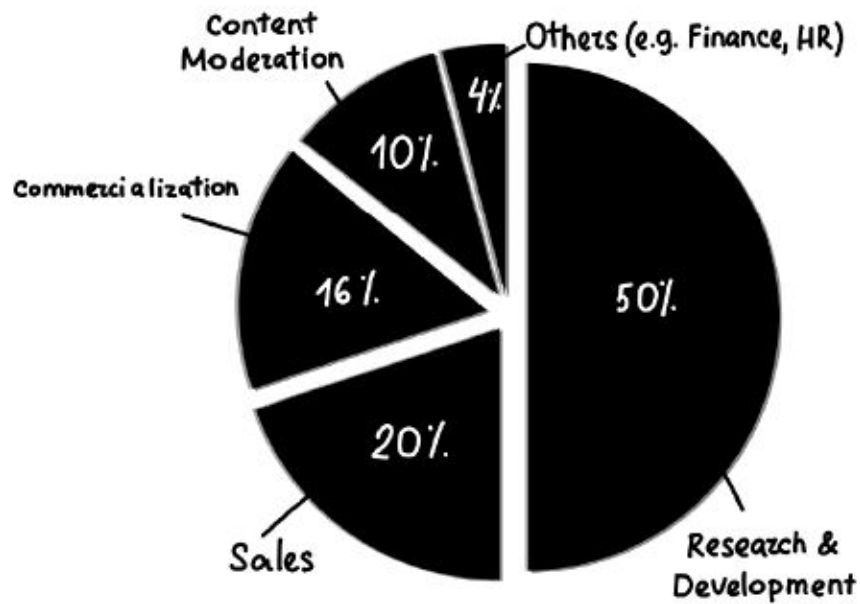
There was no back-end system built yet to support the insertion of adverts, so the test was directly hardcoded into the newsfeed. This very first advert was a coupon for an in-store promotion. After discovering the advert from the newsfeed, the person would visit the store and show the ad to staff during check-out to redeem a free bottle of cooking oil. Oil was a common promotional gift as Chinese households made extensive use of oil when stir-frying.

At first, only users within a three-kilometer store radius were exposed to the ad. The morning passed, and no one visited the store. Expanding the radius to ten kilometers resulted in more than a dozen people redeeming the vouchers. The team continued to expand the range; eventually, over a hundred people came until all the cooking oil had been redeemed. The team was excited about the small success with their first rudimentary location-based targeting advert. Yiming later said it reminded him of Steve Jobs' biography, in which the legendary entrepreneur recounted the broken-down old car he received from his father on his 17th birthday. "It's still a car," was Job's optimistic take.

"This is how I felt about our first ad," joked Yiming.

With Lidong in place, Toutiao's sales team grew from five people to several hundred within the space of two years. Lidong took many of the tried and tested techniques from the old newspaper advertising industry and applied them to the online ad business, going out and meeting clients face to face with ground promotion staff all across China. The first key accounts they secured came from Lidong's old work relationships in Beijing with car manufacturers.

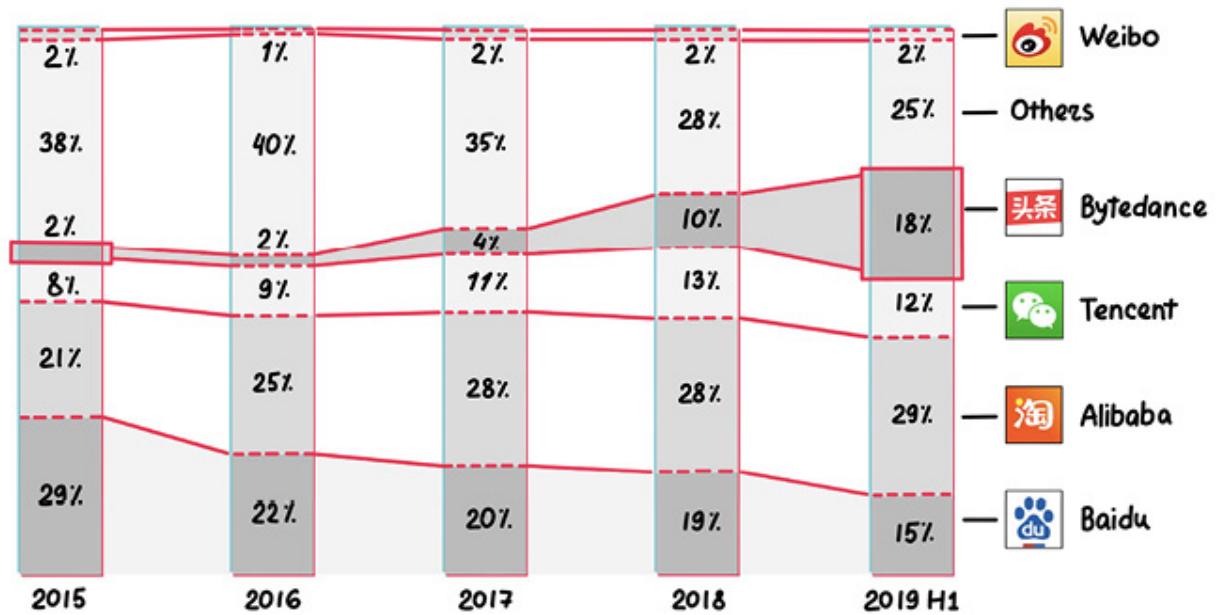
A sizable chunk of ByteDance's entire workforce consists of salespeople and staff focused on commercialization overseen by Lidong. His teams would sell ads and develop commercialization strategies for all ByteDance's apps, a model that reduced waste, overlap, and internal competition between teams.



Above: Recent estimates of ByteDance's China-based workforce. Sales and Commercialization teams combined account for 36% of headcount.

Reported ad revenues soared from 300 million yuan in 2014, and 1.5 billion yuan in 2015, to around 8 billion yuan in 2016, catapulting ByteDance into the advertising big league alongside China's established BAT giants.

Such rapid growth even beats out advertising heavyweights such as Google and Facebook when comparing the equivalent timeframes since each company's founding. ¹¹³



Above: ByteDance's share of China's online digital media market spend grew from just 2% in 2015 to become the second-largest player with 18% by the first half of 2019. During the same period, Baidu's market share roughly halved from 29% to 15%.¹¹⁴

TMD, the new BAT?

ByteDance was no longer a small fish; it had become an integral part of China's internet ecosystem. Yiming found himself being invited to China's most prestigious industry gatherings. Headcount swelled, and offices became overcrowded. Employees found themselves scattered across seven or eight different locations as space kept running out. Yiming's office was described as so small that "five people wouldn't be able to turn around in it."¹¹⁵ Eventually, they had to move.

Rather than sharing an office tower with others, the new head office building, AVIC Plaza, was entirely theirs and less than a kilometer west of their old offices, still along Zhichun road in Beijing. Décor began to somewhat resemble that of a modern Silicon Valley tech company, with walls decorated by stylish graffiti, a gym, a game room, and a well-stocked cafeteria. AVIC Plaza, the new head office, was characterized by its high ceilings and sizable open-plan central space—a vast former aviation museum, it had now become an "attention factory."

A new term, TMD, started appearing in the Chinese tech media to describe a group of rising internet stars, challengers to the old guard of Baidu, Alibaba, and Tencent.

T stood for Toutiao. Up until 2018, Chinese media always referred to ByteDance as “Toutiao.”

M stood for Meituan, the leading online food delivery and restaurant listing platform founded by Yiming’s friend Wang Xing.

D stood for Didi Chuxing, the Chinese ride-hailing equivalent of Uber.



Above: A group photo of leaders at the 2015 Wuzhen Summit, China’s most prestigious internet conference. Yiming’s position on the furthest left side at the back indicates he is arguably the least important person present. In contrast, Alibaba CEO Jack Ma, Tencent CEO Pony Ma, Baidu CEO Robin Li, and LinkedIn founder Reid Hoffman stand centrally and directly behind Chinese leader Xi Jinping.¹¹⁶

Yiming learned to take greater care of his appearance. Previously staff had noted, “some strands (of his hair) would stick out sometimes because of a bad sleeping posture.”¹¹⁷ At one point in his earlier career, inspired by Mark Zuckerberg’s well documented, efficient dress habits, he had bought 99 identical T-Shirts of the Chinese clothing brand Vanke and worn them one after another, for 99 days.¹¹⁸ Now he was clean-cut and better dressed, occasionally even wearing a suit in public.

Traveling to the United States to participate in an internet industry forum Yiming was pictured sat next to Microsoft's Bill Gates, looking at ease. Yiming's old college dorm roommate Rubo was stunned. "How did you get used to it?" he asked. "I haven't," retorted Yiming.

Yet, at the same time, it was clear that ByteDance was somewhat becoming a victim of their success. Chinese internet companies are not shy about cloning their competitors, and all the traditional Chinese internet giants were now eagerly copying every small innovation and product feature of Toutiao.

Everyone understood the power of personalization, which became a standard for all content platforms. This significantly weakened Toutiao's advantage. Scrolling feeds of news and entertainment content, a feature pioneered in China by Toutiao, started appearing in all kinds of content-based applications from search engines to browser apps. It was also becoming more evident that Toutiao's growth would inevitably slow considerably in a few years as it reached market saturation. The company needed to think about other ways to grow.

Video, the new frontier

When Yiming announced the company's 2015 annual meeting would be held on the small island of Okinawa, the staff were ecstatic. For a few days, they could swap the smog and cold streets of Beijing for the fresh air and milder climates of the romantic Japanese island nearly 2,000 kilometers to the southeast. Flying the entire staff off for a mini-holiday was practically unheard of in the work-obsessed "996" ¹¹⁹ culture of the Chinese tech industry. News of the Okinawa trip even made it into industry media, which judged "enjoying the lazy sunshine to really be the best reward." ¹²⁰



Above: A group of ByteDance staff poses for a picture on the island of Okinawa in 2015. ¹²¹

After the meeting, Yiming arranged for several managers to meet at an informal Japanese izakaya bar. Over the excitement of consuming specialty dishes and sampling local sake, the talk turned towards the company's future direction. Yiming brought up his idea, "perhaps it's time to try short videos."

The short video category of mobile apps was nothing new in China, even in 2015. The previous twelve months had seen many of the biggest industry names invest heavily in promoting their short video apps. Urban workers couldn't fail to miss that the subway stations were plastered with adverts for Tencent's Weishi and Weibo-invested Miaopai. Some early winners had emerged already. Local apps such as Meipai and Kuaishou were already well established with large active user bases. Many assessed that ByteDance had already missed the boat.

On the other hand, videos were already driving massive engagement for its news app. The previous year Toutiao had added professionally generated video content; typically videos of one to five minutes and viewing rates were booming. Additionally, they had rolled out several successful incentive programs to attract video creators.

The rise of short videos came at a time when mobile connectivity and infrastructure in China had improved significantly. 4G networks were now the standard, and free Wi-Fi at cafes and restaurants was ubiquitous. Mobile phone screens were becoming larger with improved resolution. The conditions were falling into place for mobile video content to thrive.

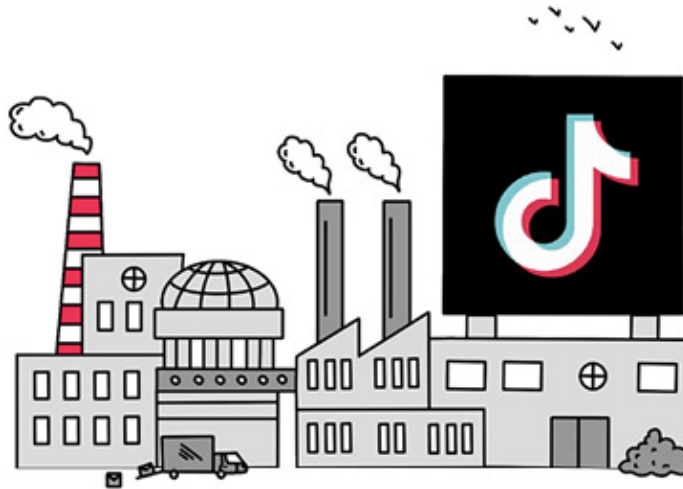
For ByteDance, this would be the beginning of a new era that would completely redefine the company.

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- 91 The book (Chinese only) can be found here
<https://github.com/jzmq/book/blob/master/novel/%E6%8E%A8%E8%8D%90%E7%B3%BB%E7%BB%9F%E5%AE%9E%E8%B7%B5.pdf>
- 92 Xiang's team came second in Netflix's 2009 one-million-dollar prize competition to improve their recommendation algorithm.
- 93 <http://www.nbd.com.cn/articles/2019-03-14/1310042.html>
- 94 Zhu Wenjia (Chinese: 朱文家)
- 95 <https://www.leiphone.com/news/201801/cEc03ORUAeiwynC.html>
- 96 Presentation slides with notes <https://cloud.tencent.com/developer/article/1052655>
- 97 <https://blog.ycombinator.com/the-hidden-forces-behind-toutiao-chinas-content-king/>
- 98 <https://youtu.be/raIUQP71SBU?t=1265>
- 99 <https://blog.ycombinator.com/the-hidden-forces-behind-toutiao-chinas-content-king/>
- 100 <http://www.wujimy.com/09/04/16/28503.html>
- 101 https://www.sohu.com/a/337627735_117091
- 102 <https://product.pconline.com.cn/itbk/bkxt/1507/6670604.html>
- 103 Author's note: I experienced this on a Samsung Android device purchased while resident in Chongqing during 2013. I hold fond memories of spending hours trying to remove a pre-installed version of Sina Weibo to no avail.
- 104 <https://zhuanlan.zhihu.com/p/53255283>
- 105 <http://tech.sina.com.cn/i/2016-12-14/doc-ifyxpipt1331463.shtml>
- 106 <https://www.pingwest.com/a/51495>
- 107 The legal representative holds legal responsibility to ensure the good standing of the company. They also act as a signatory for all company operational activities.
- 108 <http://tech.sina.com.cn/csj/2019-07-17/doc-ihytterm4337365.shtml>
- 109 <https://www.theguardian.com/environment/2007/mar/26/globalwarming.china>
- 110 <http://auto.sina.com.cn/news/2008-04-19/1529367261.shtml>
- 111 <https://zh.wikipedia.org/wiki/%E4%BA%AC%E5%8D%8E%E6%97%B6%E6%8A%A5>
- 112 <https://finance.sina.com.cn/chanjing/gnews/2020-03-14/doc-iimxyqwa0378441.shtml>
- 113 <https://blog.ycombinator.com/the-hidden-forces-behind-toutiao-chinas-content-king/>
- 114 Data source: Huacheng Securities
- 115 http://zqb.cyol.com/html/2015-10/16/nw.D110000zgqnb_20151016_1-08.htm
- 116 Image source: http://www.wuzhenwic.org/n_6821.htm
- 117 http://zqb.cyol.com/html/2015-10/16/nw.D110000zgqnb_20151016_1-08.htm
- 118 <http://tech.sina.com.cn/i/2016-11-28/doc-ifyyawmm3696549.shtml>
- 119 996 refers to the grueling yet common Chinese work schedule of working 9 am to 9 pm, six days a week
- 120 <https://www.leiphone.com/news/201502/rwfBdJFFBcjKWdoq.html>
- 121 Image source: <https://hihocoder.com/contest/challenge15>

PART TWO

THE FRONT END

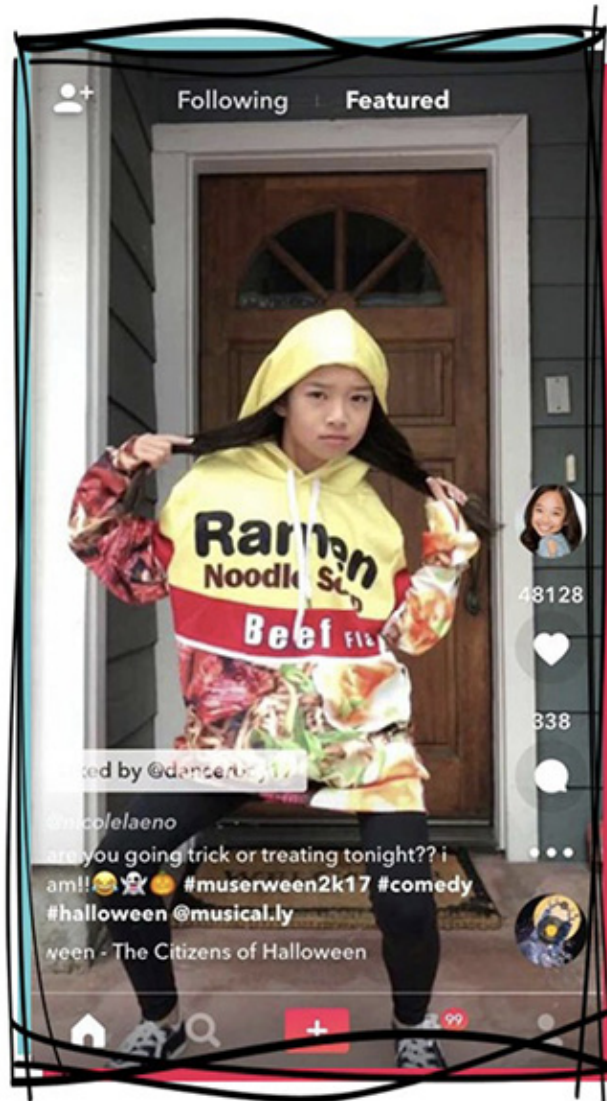
Full-Screen Short Video



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CHAPTER 5

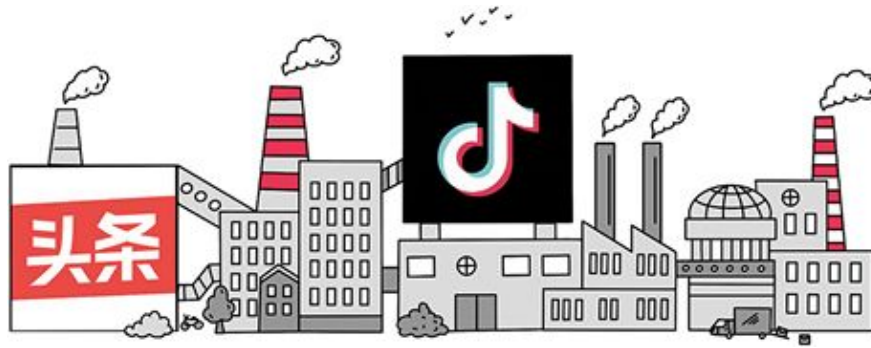
FROM PARIS TO SHANGHAI - MUSICAL.LY



“This is no question the youngest social network we’ve ever seen.”
Gary Vaynerchuk

Image: a 2017 screenshot of Musical.ly influencer Nicole Laeno, 12 years old at the time

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Chapter Timeline

2013 Jan – Vine, a popular six-second looping short video app, gains immediate popularity on launch

2013 June – Instagram incorporates 15-second video sharing

2013 July – A small team in Paris come up with the concept of full-screen vertical, music-driven short videos

2013 Oct – Mindie debuts on the Apple App Store

2014 Jan – Mindie announces seed funding and relocates to California

2014 April – The first version of Musical.ly is released

2015 Jan – Lip-sync app Dubsmash becomes a global sensation

2015 April – *Lip Sync Battle* premieres on Spike TV in the U.S.

2015 July – “Don’t judge me” challenge goes viral, powering Musical.ly to the top of the U.S. App Store

2016 May – Musical.ly secures Series C funding of \$133 million taking their total valuation to over half a billion

2016 Oct – Twitter announces it will discontinue Vine

OceanofPDF.com

ONE HOT SUMMER DAY in 2013, in a small basement near the *La Gar e de Lyon* train station in central Paris, four young men were typing away on their Macbook Pros. The group of friends were college students: Gregoire, Clément, Simon, and Stanislas studied together at the National School of Fine Art. Simon, the most technical of the group, went to the Epitech Engineering School. The small startup team had bonded over their shared passion for cinema and technology. Now they were trying their hand at iPhone app development.

Their first attempt, an app called “Ever,” allowed people to create collaborative story albums. Despite being beautifully designed, it was neither an original idea, nor was it incredibly useful. In the end, it proved to be a failure. They had poured months of hard work into it, but it was time to face the harsh reality—they needed to scrap the project and start over.

Looking for inspiration, the team started downloading and analyzing popular apps. Vine, a six-second video-sharing platform, had launched a few months earlier and became an overnight sensation. Vine’s demo version had created so much buzz that Twitter co-founder Jack Dorsey spent a reported \$30 million acquiring the company even before the app had officially launched. At the same time, Instagram had also just added the option for fifteen-second videos.

As the group kept playing around with Vine and other short video apps¹²², they saw that there was still plenty of room for improvement—both Vine and Instagram displayed videos in square frames. Given the vertical dimension of the iPhone screen, it was evident to them that the videos also needed to be upright, making use of the entire screen.

They built a test version to try out how the concept would look. With the video taking up the entire screen, it wasn’t a scrolling newsfeed anymore, so there was a question over how to switch to the next video. The team decided to experiment with using a swipe up action. It worked well. Going from one video to another held a sense of surprise and anticipation. People never knew what would be shown after another swipe.



- Scrolling newsfeed
- Square aspect ratio video
- Video covers roughly half the screen
- Account information, comments, likes displayed separately from video
- Home icon top left with drop-down menu

Above: The Vine interface and logo in 2013.

The camera on the early iPhones was rudimentary, but the team loved how Instagram had made use of filters to make normal iPhone photos look highly professional. Was there an equivalent for videos? Well, they soon found the answer—in music no less. **Adding music to videos produced the same results as adding filters to pictures**. It was a fast and easy way to instantly spice up videos by fusing professionally generated music with user-generated visuals.

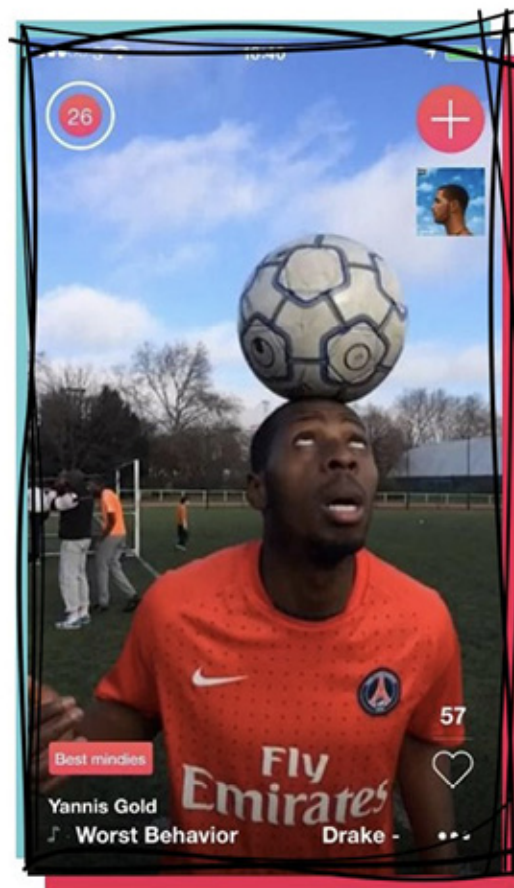
“Music is a catalyst for creativity,” Grégoire explained in a later interview. “It’s very complicated to shoot an interesting video, but if you put music on an awful video, it will make it interesting.”¹²³

At first, they opted for a design sequence where adding music was only possible after shooting the video. This was similar to Instagram—adding a filter only after choosing a picture. Later they realized they could switch this around. The music could be selected first and played loudly while users shot the video. Stanislas explained, “This transformed your camera into a karaoke engine.”

Even better, the team discovered that there was a developer API allowing them to access 30-second previews of all the music sold on iTunes. It was perfect. They could access a massive catalog of songs, including iTunes

exclusives. With popular tracks, the preview was cut perfectly to start right at the moment just leading into the song’s catchy hook. They couldn’t have asked for more.

By now, a prototype was coming together. Back on the basement computer, the team had a choice to make: If the video took up the entire screen, where did the menus and information about the video go? The answer was to overlay it on top and keep the information displayed to a minimum—the video title, the creator’s name, “like” and “share” buttons, an icon for the music track, and a large plus-sign button to create a new video.



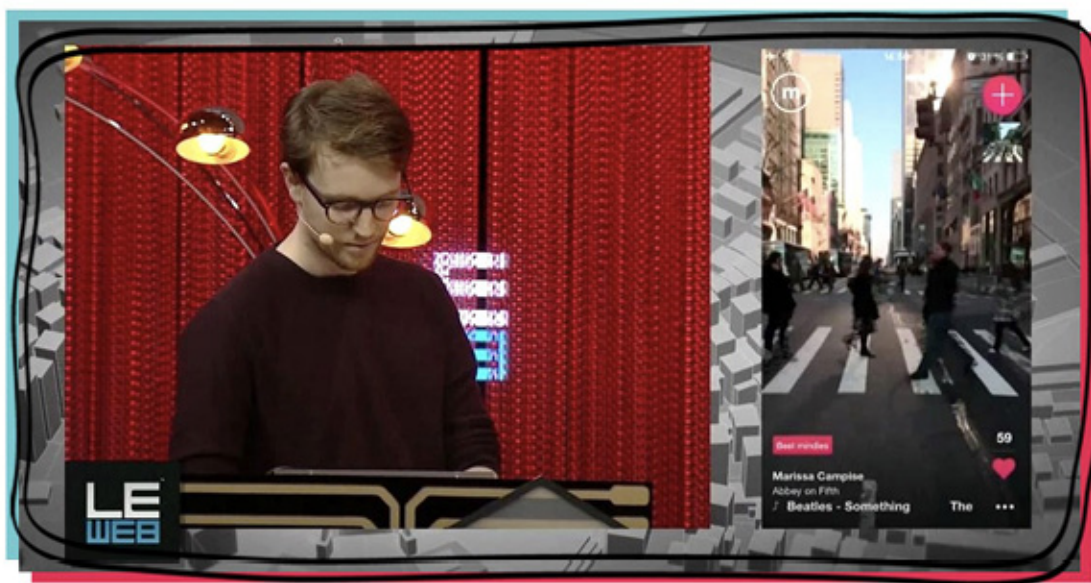
Above: a Mindie video taken January 2014, a soccer player wearing a Paris St. Germain football shirt balances a ball on his head to a clip of the song “Worst Behavior” by artist Drake.

Using what they’d created to take videos themselves, the young men realized music powered the experience more than the visuals—videos became the meme to accompany the track. The song itself would act as a

hashtag that facilitated music-driven content discovery. Stepping back to assess their work, the team agreed they'd struck gold. The app was refreshingly clean and minimalist—music-centric, vertical-screen short videos—a different experience from any other existing app on the market.

They then began searching around for the right name for their app. Clement came up with “M indie,” a combination of the words “mainstream and indie.” The video length was initially set to seven seconds, one more than the standard of short video market leader Vine, which they felt wasn't long enough for most music clips.¹²⁴

When M indie quietly launched on the App Store in October, it gained the attention of only a handful of tech industry media outlets.¹²⁵ But the few reviews it did earn were broadly positive: “A fun app that allows you to create short, shareable micro pop videos,”¹²⁶ one article stated.



Above: M indie co-founder Grégoire Henrion demonstrates the M indie app at LeWeb conference Paris in December 2013. The demonstration is instantly recognizable as the predecessor of TikTok.¹²⁷

The reception from early users was excellent, and the team had done enough that they could now go after serious funding. First courting investors locally in Europe, they soon flew across the Atlantic to New York and finally San Francisco. By January, they had sealed the deal on a seed funding round of \$1.2 million and were ready to pack their bags and leave Paris behind to relocate to beautiful, sunny California. Little did they know

their paths were set to entwine with another small group of entrepreneurs on the other side of the world, in China.

Alex Zhu @bullshitting

With long flowing hair, a light mustache, and a penchant for wearing scarfs, Alex Zhu looks more like an artist than a typical startup founder. Born in Anhui, traditionally considered a poor, landlocked province of China a few hundred miles west of Shanghai, Alex had recently emigrated to the United States to work for the German enterprise software giant SAP in Silicon Valley. His title was the glamorous sounding “Futurist of Education,” a job he later claimed was dull and easy. “Enterprise (software) is a good business, but it’s not sexy enough, and I want to be a sexy man,” Alex joked on stage at an event in his lightly accented English. ¹²⁸

Alex’s online profiles hinted at his underlying rebelliousness and eccentricity—listing his LinkedIn work location as Mars and opting for the Twitter handle “bullshitting.” A career of writing company blog posts and creating slide decks predicting the future trends of education left him unfulfilled. Out of frustration, in late 2012, Alex posted on Twitter:

“It’s time to un-enterprise myself and start doing something meaningful... Life is either an ADVENTURE or NOTHING.” ¹²⁹

Alex had become enamored with the idea of “Massive open online courses,” or MOOCs, as they were popularly referred to; it had become all the rage in education. Alex had thoroughly researched the area in his role as Futurist of Education and spotted one fatal flaw: MOOCs wouldn’t transform education as many hoped because more than 90% of the people enrolled in the courses never finished them.

Alex felt he had a solution. What he later described as a one-billion-dollar idea to transform education—make the lessons short and mobile-friendly. In his words, “a combination of Twitter and Coursera.” His vision was to create a vibrant peer-to-peer education platform that allowed people to create and consume mini-courses on all kinds of lifestyle topics such as cooking, yoga, or painting using nothing more than their phone.

Helping Alex clarify this vision was his friend, Louis Yang. Alex and Louis had both previously been directors at eBaoTech, ¹³⁰ a Shanghai company that provided services to insurance industry firms. eBao had

clients all over the world that sent Louis traveling extensively and eventually landing in California, where he reconnected with Alex. Louis was sold on Alex's idea and resigned from eBao to join him as a cofounder of the novel education app.¹³¹

Hailing from the southern Chinese province of Hunan and one year younger than Alex, Louis was youthful in appearance. He sported a wide, slightly chubby face with a distinctive wavy quiff of hair. Like Alex, Louis was an IT veteran, having worked at eBao for seven years. He also held a predisposition toward entrepreneurship. While in university studying thermal energy and power engineering, he built a student startup website called the "Sino Network" in his spare time.¹³² In 2005, he created an online customized T-Shirt service, but unfortunately, unlike today, the logistics in China at the time was underdeveloped and too expensive to make the idea viable.

Together Alex and Louis founded the company Cicada Education and raised a quarter of a million dollars from China Rock Capital Management Ventures¹³³ in an initial funding round. In 2013 they spent six months forming a team and building the prototype.

With the freshly built beta version of the app in hand, Alex excitedly jumped into creating his first lesson, picking an accessible light topic, "the history of coffee." It took him a painful two hours to piece together a lesson that was just three minutes long. Condensing useful and accessible knowledge into bite-sized lessons was far more challenging than he had imagined. Shortening the lessons didn't make research and curriculum development any easier. Even after two hours, Alex was utterly unsatisfied with the lesson he'd produced – a boring video composed of static images with a voice-over. With only 8% of the capital left from their initial seed investment, they felt that rather than return the funds to investors, it was better to pivot into a completely new direction.

"The day we released this application to the market, we realized it was never going to take off," Alex said. "It was doomed to be a failure."

The birth of Musical.ly

Alex was asked multiple times at events and in media interviews about what inspired him to create Musical.ly. His reply was always a variation of

the following:

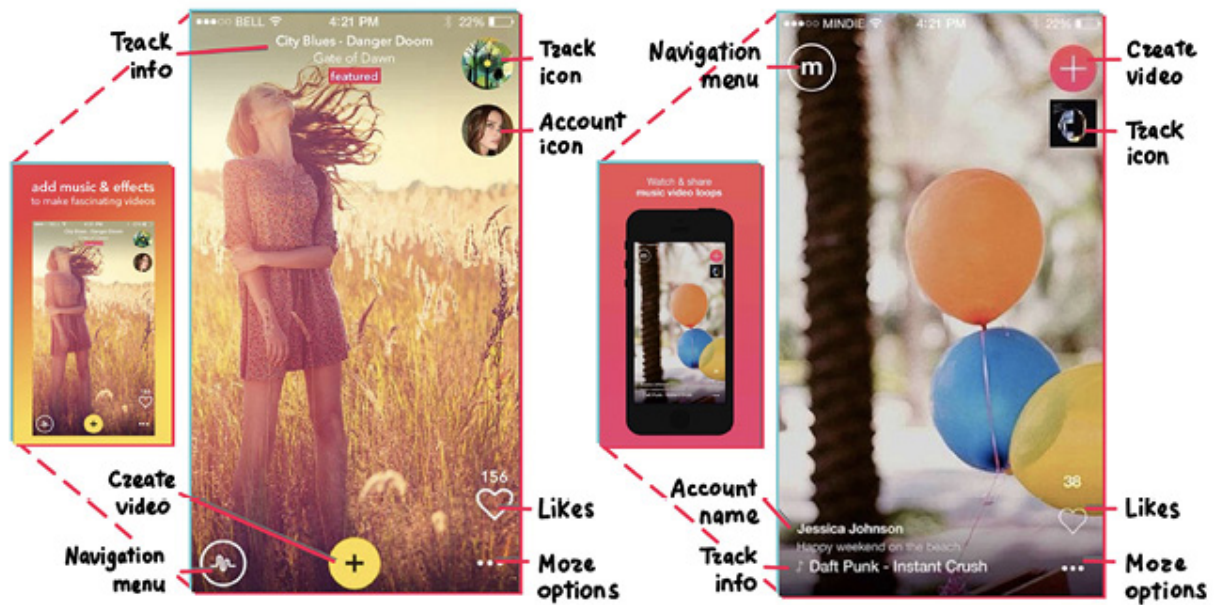
“I was on (the) Caltrain one day, from San Francisco to Mountain View, and it was crowded by teens. Observing their behavior, 50 percent were listening to music, and the other 50 percent were taking photos and videos with their speaker on top to add music. Teens are so passionate about social media, photos, videos, and music, that it made me think. Can we combine these three very powerful elements into one app and build a social network for music videos?” ¹³⁴

From there, the team was able to pivot and develop a prototype taking them in a completely new direction. Within 30 days, they had built an entirely new experience based on combining music and short videos—a social network tailored for young people. With an incredibly fast development time, Musical.ly 1.0 launched in April 2014.

The app opened to the song “Forever Young” playing to a looping background video of two young women playing on the beach. Below the homepage logo, the app’s tagline read: Musical.ly—instant music videos. ¹³⁵ This fairy-tale comeback story made for great fodder in media interviews, but it glossed over the reality of what the team had actually done—they had cloned Mindie.

It didn’t take long for the Mindie team to notice their new competitor. A search for “Mindie” on the App Store resulted in Musical.ly appearing right next to them. The two apps shared the same keywords. “At the beginning, we were really surprised... Everything was the same, even some of the App Store description, the logo color and gradients,” explained co-founder Stanislas.

The team realized they had made a mistake, leaving part of their code up on developer site Github which Musical.ly pounced on and used to speed up their development. Searching through user profiles, they found Alex Zhu’s account and saw he had been a highly active early adopter.



Above: a side-by-side comparison of the original Musical.ly user interface with that of Mindie. Images are taken from the app store descriptions of both apps at the time of Musical.ly’s initial release. Curiously, both pictures display the same clock time (4:21 PM) and battery life (22%).

Appropriating the idea of another more creative team was just the starting point for Musical.ly, but it was no guarantee of success. The situation of Alex and Louis’ startup remained precarious. They were still critically low on money with a team of less than ten, which made attracting engineering talent very tough.

Rather than focusing on one market, they chose to launch the app worldwide in multiple regions and languages, including China, where they opted to change the name to the unorthodox but memorable “Mamma Mia.” The logic was simple: “The first musical that Chinese people thought of was Mamma Mia,” ¹³⁶ explained Louis. But it didn’t help. Despite the name change, the app bombed locally.

“Instead of saying we chose the U.S. market, I would say the U.S. market chose us,” Louis explained in a later interview. “The U.S. is a music country—everyone there is so into music.” ¹³⁷ The early U.S. adopters were middle and high schoolers who, after finishing class, had plenty of time for leisure and entertainment. In contrast, Chinese teens typically had a grueling schedule of after-school tutoring and exam preparation homework.

If social media was about sharing one's life, Chinese teens didn't have much to share. "I have a cousin who studies 12 hours every day. How could he have the time to spend time on Musical.ly to do something creative?" noted Louis in the interview.

Even with some initial traction and reasonable user retention rates, raising money was difficult. More than 20 investors turned them down. Louis recalled investors telling him: "If you can give us just one example of a domestic Chinese team who's built up a social product abroad, we can consider investing." ¹³⁸ There wasn't one.

That meant the team had zero marketing budget, another reason why they had little hope in a hyper-competitive Chinese market where typical promotion strategies involved expensive influencer campaigns and large marketing teams.



Above: The early Musical.ly team (Louis Yang front center) pictured in their original offices at iCamp Startup Accelerator Shanghai ¹³⁹ in late 2014.

Instead, Musical.ly was simply relying on word of mouth and free growth hacking tricks. The iOS app store had begun heavily prioritizing search terms, which they took advantage of. The team stuffed a string of buzzwords next to the app's listed title to boost its search ranking. At one point, the app's name was as wordy as "Musical.ly—make awesome music videos with all kinds of effects for Instagram, Facebook Messenger."

The team put a noticeable link in the early version of the app that encouraged the first adopters to send feedback. Every day, they received emails with titles such as “The top three things I like about Musical.ly” or “The top three things I hate about Musical.ly.” In the space of a single month, they had received more than 200 emails from American users who helped them identify things to change.

To understand the sentiment of the users, Alex spent a great deal of time on the app and registered multiple fake accounts pretending to be a regular user.¹⁴⁰ He would comment on others’ videos and ask why they shared or created certain content, a user research tactic commonly deployed by Chinese tech bosses. They brought hundreds of their early loyal users over into WeChat messaging groups to have daily conversations. For every new feature, they shared mockups to gain immediate direct feedback. One of the first changes made based on these talks was to extend the length limit of the videos to 15 seconds to match the limit imposed by Instagram, the platform where teens wished to share their videos most.

#CreativityChallenge

In trying to encourage a sense of community amongst their early adopters, the team discovered that the most effective way was to regularly promote “challenges.” Challenges are essentially user-generated video memes.¹⁴¹ A challenge sets up a replicable, cookie-cutter structure that allows anyone to take part and make their own version. They could be anything from a simple set of dance moves to a goofy prank. Viral video memes were already blooming on other social media platforms before Musical.ly. Prominent examples included the “Harlem Shake,”¹⁴² which saw groups of people dancing together crazily, and the “ice bucket challenge”¹⁴³ where celebrities would challenge each other to pour ice-cold water on themselves to promote ALS awareness. The term “challenge” explicitly communicates the participatory and fun nature of the format.

On Musical.ly, users were encouraged to join trending challenges and make their own version. Challenges were a way of educating users and showcasing new ways to create videos. It also allowed the Musical.ly team to have greater control over the direction of content creation. The

challenges gave users a purpose to participate and create videos rather than just passively watch others.

“The key difference between Musical.ly and Vine is that on Musical.ly we lowered the barrier to content creation, so all the consumers are creators at the same time,” explained Alex. As he saw it, lowering the barrier to content creation was critical to Musical.ly’s success, where all content was produced by users.

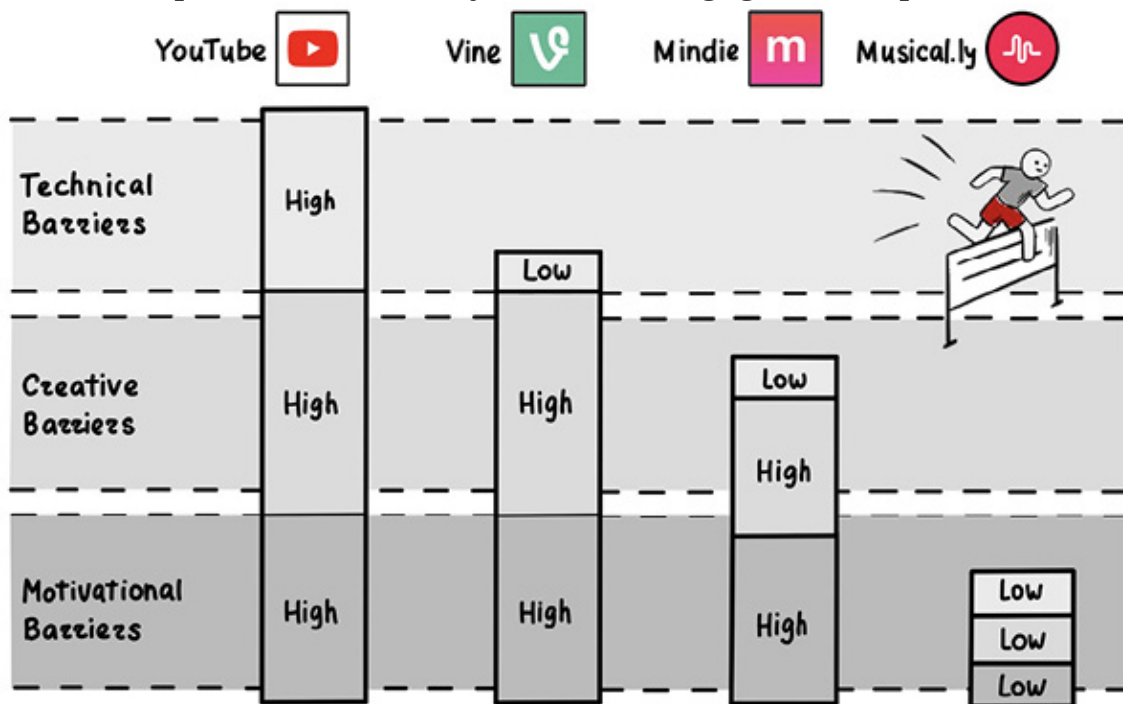
The most significant barrier to people creating videos wasn’t a technical one. All the new breed of short video mobile apps had easy-to-use mini-editing studios built in. Young users especially had no trouble working out how to add music, text and use the recording function. In particular, Vine’s recording feature was about as simple as one could imagine – point the camera and hold down a button to record for six seconds.

Shyness also wasn’t a problem as many young users loved recording themselves. The more substantial barrier was one of creativity and inspiration—coming up with a concept. Most users need to be inspired. Very few people had as much time, talent, and dedication as already famous Vine stars such as Zach King, notable for his mastery of digital editing techniques to create magic effects.

Music was one form of creative inspiration; on Mindie or Musical.ly, people could easily choose a favorite track to mime or dance to. But the Musical.ly team found that the promotion of daily challenges was a much more effective way to build a regular habit of content creation. Users didn’t need to think that much. All they had to do was follow the crowd and add their spin on an already familiar theme. Copying others wasn’t just okay; it was actively encouraged.

Challenges also helped combat the final most difficult barrier of all—**motivation**. There was a sense of immediacy. Users either chose to participate in the fun challenge while it was trending today or risk missing out. Participation also gave people a sense of being part of a wider community. Former Musical.ly vice president of operations, Zhou Bingjun, revealed in a 2016 presentation, that “on average, each (Musical.ly) user creates more than one piece of content per day.”¹⁴⁴

By the end of 2014, Musical.ly had established a core group of loyal users. Constant daily communication with hundreds of these Musical.ly fans through WeChat messaging groups meant that the team was close to their users despite being on the other side of the world. They were learning the nuances of American teen culture, and through the communication, fans were feeding them new ideas for challenges to promote or amplify. The team also continually monitored which videos gained traction on the platform and promoted ones they felt could engage and inspire others.



Above: the barriers to content creation across YouTube, Vine, Mindie, and Musical.ly.

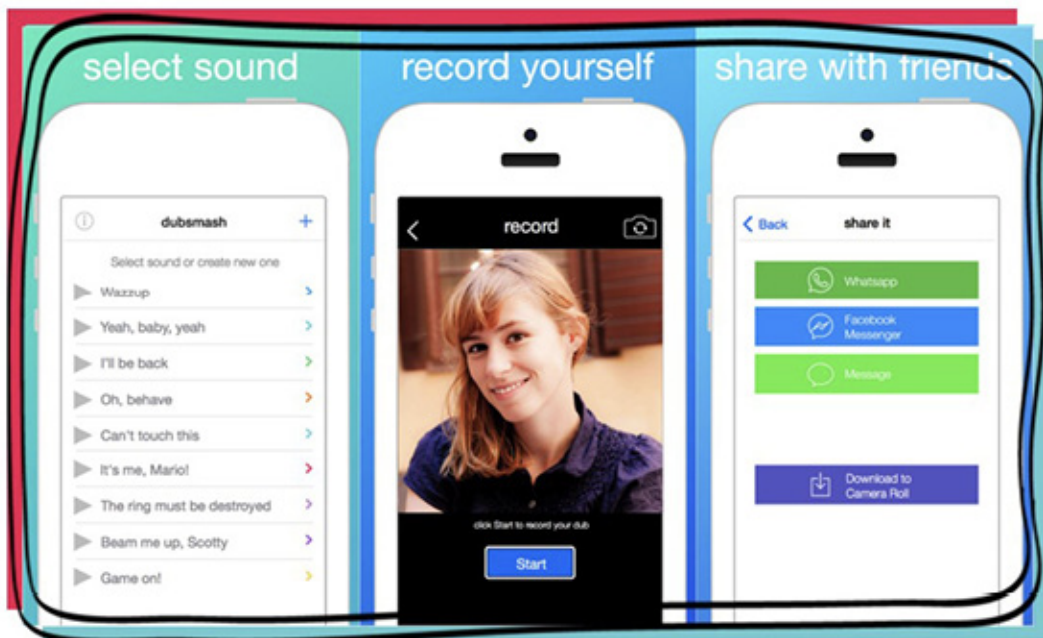
This hands-on daily work was time-consuming, manual in nature, and challenging to automate. American internet firms favored scalable, data and technology-driven growth methods. Yet Musical.ly was drawing from China’s standard playbook, where such low-tech tactics were commonplace and labeled under the term “operations.” ¹⁴⁵

It was a solid base, but they had failed so far to reach a tipping point that would allow them to scale. The founders were hesitant over whether to keep the business running, given the difficulty of securing funds, and their team was stripped down to a skeleton crew of just seven people. “Sometimes slow growth is worse than failing fast,” reflected Alex.

Dubsmash

“Lip-synching goes viral!” the smartly dressed news reporter exclaimed in serious-sounding tones while turning to face the studio camera. “That’s right, Jane, new app Dubsmash is taking the world by storm!” interjected her newsroom co-host. “The app has been downloaded more than 10 **million** times since it launched in October, now that’s **impressive** !”

In early 2015 news about Dubsmash was spreading like wildfire across global mainstream newspapers and television. The app was an overnight success built by three German engineers led by CEO Jonas Druppel. Dubsmash did just one thing—let users create ten-second lip-sync videos. People loved it.



Above: Screenshots of Dubsmash, taken from the app store description in 2015.

Dubsmash hit the number-one spot in the German app store just seven days after launch. Later, it achieved the top position in over 40 other regions beating out traditional leaders such as Facebook and YouTube.

What set Dubsmash apart was its precise positioning around a straightforward use case—it was a simple tool to make lip-syncs and nothing more. Dubsmash featured a substantial audio library of well-known

movie quotes and song clips to choose from, content that appealed to all personalities.

Unfortunately, Dubsmash faced the common challenge for small teams that unexpectedly met big success early on: their back-end infrastructure was utterly unprepared for the explosive growth. There was no user account system,¹⁴⁶ no login or registration, and no way to add friends or interact with other users. There was also no way to post within the app; videos were downloaded to the phone and shared through other social media platforms. This made it difficult to retain users, and the hype around Dubsmash faded.

The overnight success demonstrated the compelling use case for lip-syncing, and it was also an early indicator of what would happen next for Musical.ly.

Lip Sync Battle – the turning point

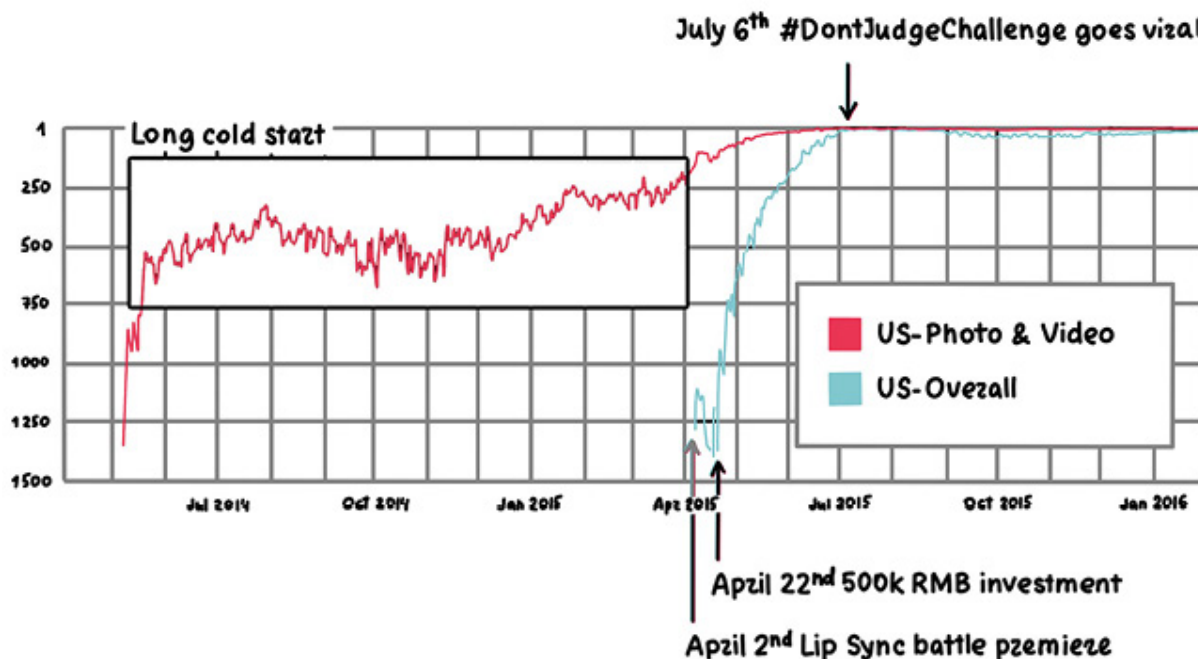
In early April 2015, the Musical.ly team in Shanghai noticed something unusual in its download numbers. Every Thursday night, there was an abnormal surge in installs. The Shanghai team set about trying to understand what caused this, conducting extensive research online and through its user feedback groups. Finally, the team hit upon the answer—*Lip Sync Battle*.

Lip Sync Battle was a new TV competition co-hosted by rapper LL Cool J on the now-defunct American television channel, Spike TV.¹⁴⁷ A spin-off from the popular *Jimmy Fallon Show* it was a success straight out the door. The first episode, which aired on April 2, was the highest-rated premiere in the channel's history. During and after the show, some of the audience would search for apps that let them record miming videos. Many stumbled upon Musical.ly, leading to the spike in downloads.

Inspired by the success of Dubsmash, “lip-sync” had been one of the keywords which the team had stuffed into the overly long app name and description. The team chose to double down on the lip sync use case as fast as possible, repositioning the app to make lip-sync far more prominent.

Short demonstration video clips were added to the onboarding process, so a new user could immediately grasp how to make lip-sync videos. The team curated a list of what it considered to be the platform's very best lip-sync content and made sure this was the very first batch of videos displayed to

impress new users. The app started sending notifications to new users, encouraging them to “post your first lip-sync video,” a move co-founder Alex felt “dramatically changed the retention.” Soon they had also added a duet feature enabling two users to collaborate on a video. In early April, Musical.ly’s iOS App Store ranking was around number 1,400; now every week, they were rapidly jumping up the charts by roughly 100 positions, and the best was yet to come.



Above: Musical.ly’s US Apple app store ranking early 2014 through to early 2016. ¹⁴⁸

#DontJudgeChallenge

On July 5th, 2015, Musical.ly’s operations team found some users were starting to post a new video type. The clips would begin with someone taking on the persona of an unsightly person, poorly dressed with unkempt hair, often with their face covered in makeup made to look like acne. After a few seconds, the videos would suddenly cut to the same person, almost unrecognizable from just a second before, posing in their best clothes with groomed hair, makeup, and favorable background lighting. The message of the videos was simple: don’t judge others by appearances.

The team joked together about this new meme. Some of the videos were surprisingly entertaining, showcasing the creativity of American teens.

Before leaving the Shanghai office at seven o'clock in the evening, Louis decided to mark a few of these videos as featured content, which would receive preferential traffic distribution. Thinking no more of the matter, he headed home as usual.

The next day, Louis awoke to find his actions from the evening before had triggered something entirely unexpected on the other side of the world. It was his birthday that day, and according to China's office culture, he could expect a cake and small gifts from the team. But nothing happened. Instead, when he walked into the office, he was greeted with an excited exclamation: "We're already number one! Don't you know?" a colleague shouted at him madly.¹⁴⁹

The meme had gone viral. Tens of thousands of videos with the hashtag #DontJudgeChallenge had flooded the platform, spilling out to Facebook, Instagram, Twitter, and other social media platforms. In one week, 400,000 different variations of the meme were created with an estimated combined view count of over one billion.¹⁵⁰

With newfound success and topping the American app store charts, funding became much easier. In the lead up to their breakthrough in 2015, Musical.ly gave away a sizable chunk¹⁵¹ of their company to Chinese mobile app developers Cheetah Mobile in return for five million yuan (\$700k), which allowed them to keep going during the long cold start period. After the breakthrough, they immediately worked on putting together a new round at a much larger valuation that injected 16.6 million dollars into the company.

Expanding rapidly, a new office was set up in the U.S. hosting a handful of marketing, business development, and content licensing specialists. Working out of a San Francisco WeWork office, they raided Alex's old company, SAP, for talent to form a small new team led by Alex Hofmann, under the title of president of north American operations. In addition to the founders, Hofmann would become the new face of Musical.ly at many media events and interviews. With funding secured, an expanded team, and a user base now measured in tens of millions, Musical.ly entered a new development stage.



Numa Numa

Viral lip sync
December 2004



The Harlem Shake

Viral video meme
February 2013

Above left: the internet's first viral lip-sync video "Numa Numa," ¹⁵² taken in 2004, pre-YouTube. Above right: viral video challenge meme, the Harlem Shake, 2013. ¹⁵³

Ariel

Ariel Rebecca Martin was effectively living out of a suitcase. A flood had nearly destroyed her southern Florida home, and now her entire family found themselves reduced to living out of a single room at her grandparents' house. Ariel was a green-eyed fourteen-year-old with long brown hair and boundless energy. Bored, she had invited a couple of her friends over, and together they were crashing on her grandparents' sofa.

Ariel swiped open her phone yet again to check Instagram when a video with the Musical.ly watermark caught her attention. Intrigued, she and her friends decided to download and check out the app together. Setting up their new profiles, her friend opted for the screen name "Queen," and another chose "Dancer." On a whim, Ariel went with "Baby" ¹⁵⁴ and started to think about which song to use for her first video. After working out how to use the recording effects, she busied herself, creating a sequence of hand gestures to accompany the lyrics. After an hour or so of practice, Ariel was ready to post her first video using one of her favorite songs—the Nicki Minaj hit "Stupid Hoe." ¹⁵⁵

Later, when Ariel found one of her first few videos had been featured on the app, she squealed with excitement. Being featured was the equivalent of the platform giving her an official seal of approval, and in the process, she

had gained thousands of new followers. Ariel immediately felt the urge to go back and change her profile name, regretting her random choice of “Baby.” After deliberation, she decided it was too late—her fans knew her as Baby Ariel now.

Over the coming summer months, she continued to post videos of her lip-syncing from the lilac-colored bedroom at her grandparents’ home. Since her entire family was working on repairing their house, she and her brother couldn’t go to the summer camp as planned. Instead, Ariel spent most of her free time playing with Musical.ly. Her videos continued to be featured, and her following was now already in the hundreds of thousands. Ariel found she had a talent for creating hand gestures to fit song lyrics, and her boundless energy and radiant smile made her a natural fit for short music videos. In a matter of months, 14-year-old Ariel had built the most famous account on Musical.ly to both her and her parent’s amazement.

The whole family quickly wised up to the opportunity. Ariel had struck digital gold. Her mother built her a website, and Ariel started schooling from home to keep up with a rigorous content creation schedule as she expanded out to other platforms. Fast forward five years, and today Ariel has 33 million TikTok followers, 9 million Instagram followers, 3 million YouTube subscribers, and over 1 million Twitter followers.¹⁵⁶ She’s released songs, been featured in movies, and was named by *Time Magazine* as one of the most influential people on the internet.

Ariel’s rise was a fairy tale and a dream come true for many young teens. Her success came so naturally and quickly that it almost felt preordained. In fact, this was not far away from the reality of what had actually happened. Ariel’s rise was part of a much larger deliberate plan, from which a small group of people, of which she was one, had benefited mightily. Musical.ly was building an attention economy.

Building an economy, the Musical.ly way

Being a platform where all the content is user generated required Alex and Louis to think carefully about nurturing an active community. They needed to ensure there was a stable group of committed creators regularly producing high-quality content. This would provide them with stickiness and longevity, hopefully avoiding the fate of being yet another flash in the

pan small social media platform. Alex compared the strategy they took to that of “nation-building.”

“You have to attract immigration, and in order to do so, you’ve got to make a small percentage of people rich first,” he said.

In his conceptualization, Musical.ly was like a newly discovered continent in need to entice new immigrants, much as the American colonies had been at one time. With few people, the new continent had a small gross domestic product. Distributing this pool of wealth evenly would result in everyone living miserably and failing to attract more immigrants. According to Alex, the solution was to deliberately foster an economy with a high degree of income inequality, which allocated most of the GDP to a small group of pioneers—in Musical.ly’s case, early users. Once this group got rich, the news would then spread to trigger a gold rush with others following in the footsteps of the first wave of immigrants, eager to test their fortunes in this new world.

The latecomers would also have a shot, or in Alex’s words, “let them live the American dream.” He likened this process to transitioning from a centralized, planned economy to a market-driven, decentralized one, nurturing a rising middle class.¹⁵⁷ The big stars could remain wealthy, but there needed to be ways for the talented but obscure, new creators to be discovered and rewarded.

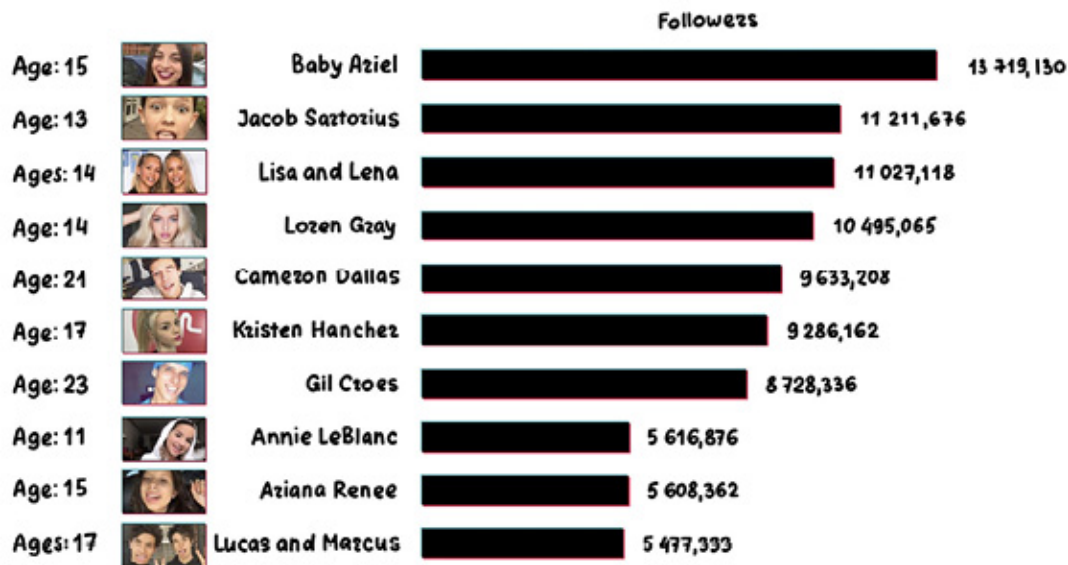
“Getting rich” was a metaphor for securing the heightened social status of being “Musical.ly famous.” The team had a god-like power to manipulate attention on the platform and ensure individual accounts received massive exposure. The content operations team could manually select at their discretion videos to be tagged as “featured.” Such videos would be actively promoted and were guaranteed high visibility. Under this system, just a few individuals working out of a nondescript co-working space in suburban Shanghai wielded considerable influence over American teens and the pre-teen culture by choosing which videos to feature, as illustrated by the #DontJudgeChallenge.

Using the metaphor of immigration between countries was Alex’s roundabout way of saying Musical.ly set the rules of the game. They drove vast traffic to specific individuals such as Baby Ariel, catapulting them

overnight to online influencer status. As the platform grew, this group became outright celebrities, in some part due to their creativity, persistence, and hard work, in most part due to the invisible hand of the Shanghai content operations team tilting the attention game heavily in their favor.

Younger than Snapchat

Once the platform had grown large enough to garner mainstream media attention, the first thing reporters picked up on was the users' age. "This is no question the youngest social network we've ever seen," exclaimed online marketing guru Gary Vaynerchuk in an article profiling the platform.¹⁵⁸ "Snapchat and Instagram skew a little bit young... but with Musical.ly, you're talking about first, second, third grade."

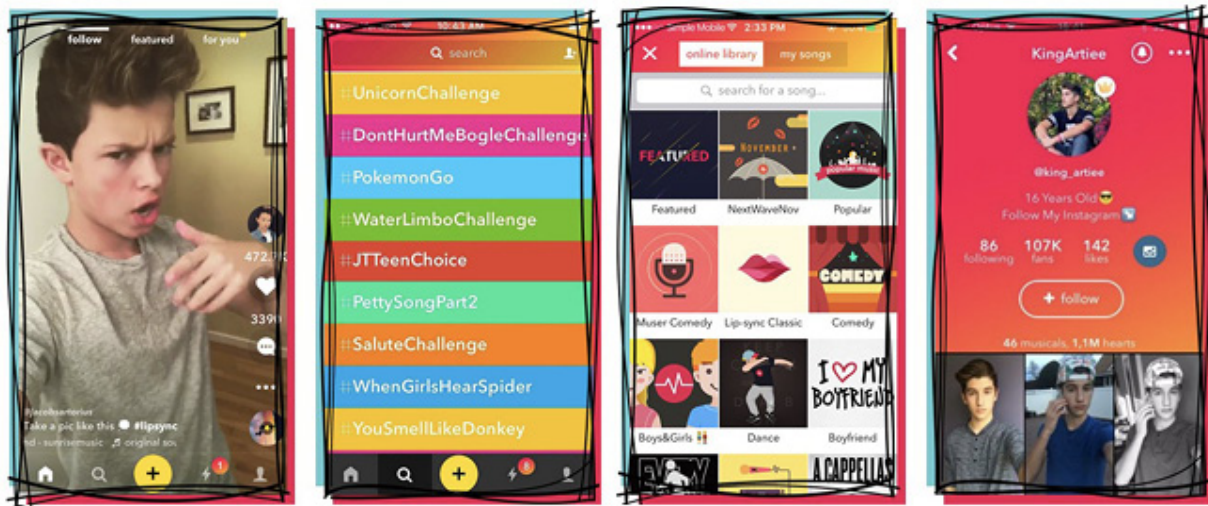


Above: A snapshot of the top 10 most popular Musical.ly accounts in late 2016 with numbers of followers and the ages of the creators at the time. Ages range from 11 to 23, with a medium average age of 15.¹⁵⁹

Musical.ly was especially appealing to young creators as more established mature platforms such as YouTube and Instagram had already grown into noisy and saturated ecosystems where gaining attention and attracting new followers was an uphill battle. Typically, teens had neither the photography skills nor the glamorous lifestyle necessary to be an Instagram influencer, nor did they have a witty writing style to build a Twitter account. But lip-sync and dancing were well-suited. Teens were creative, obsessive about sharing, and had plenty of free time after school. They were comfortable

shooting videos and held few inhibitions about using their phone's front-facing camera to record themselves. The Musical.ly team leaned into the youthful demographic with the app's design characterized by generous use of loud, garish colors.

Teens were a critical demographic for the music and entertainment industry. It wasn't long before artists started to test the waters. Singer Jason Derulo became the first major artist to adopt Musical.ly for sharing videos of himself dancing from his studio. Seeing his success in gaining teen attention, other celebrities were quick to grasp Musical.ly's marketing power. The floodgates opened with artists such as Selena Gomez, Lady Gaga, and Katy Perry, all jumping on the platform to connect with the young audience. Some began using it to drum up excitement for upcoming tracks by sharing snippets of unreleased songs.



Above: Screenshots of Musical.ly. A rainbow-colored list of trending hashtags includes #WhenGirlsHearSpider and #YouSmellLikeDonkey.

By 2016, Musical.ly's staff had grown to more than 50 with sizable operations and development teams in Shanghai. Alex claimed in media interviews that the company's server architecture was struggling to keep up with the demands of all their traffic.¹⁶⁰ In May of that year, the startup secured another round of funding, this time over \$100 million, valuing the company at over half a billion dollars.¹⁶¹ The Chinese tech industry lauded Musical.ly as kings of the overseas market. It was unheard of that a Chinese social app could gain so much success abroad. "Musical.ly proves that so-called "cultural differences" are actually paper tigers," were Louis' bullish

words to encourage other Chinese startups.¹⁶² By December, the app had 130 million registered users with an active monthly user base of 40 million, heavily weighted towards North America and Europe.¹⁶³

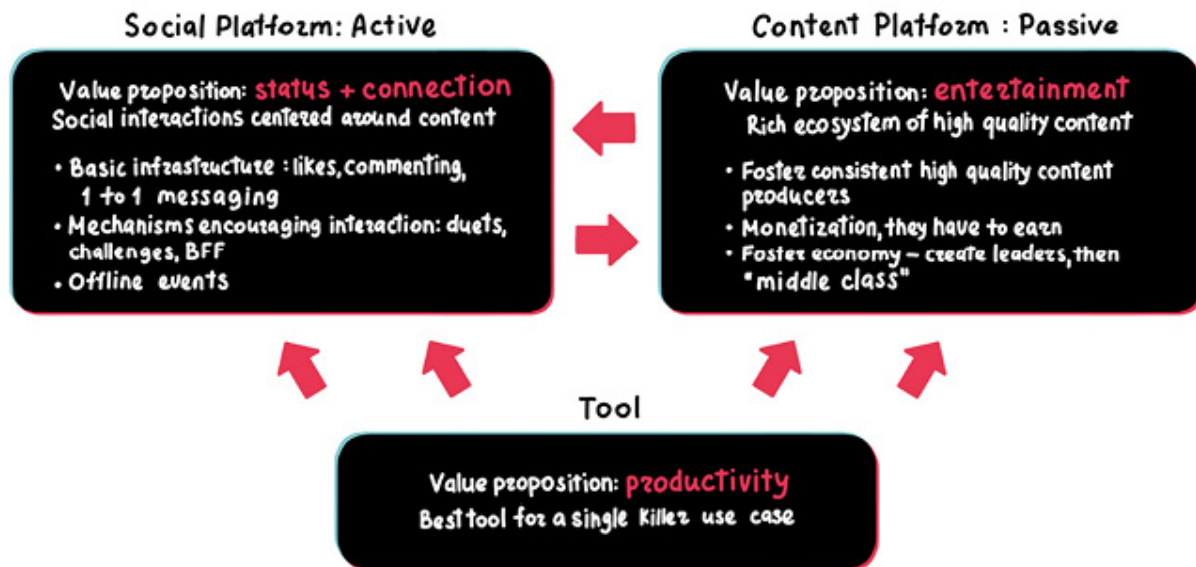
Analysis: from brush to canvas

“When you want to grow early on, you want to be a brush, meaning you have to be very specific; you have to solve a specific need very well... Later you want to be a canvas; you want all kinds of things to happen on this blank canvas.”¹⁶⁴ This was the metaphor Alex used to explain how he envisioned Musical.ly’s progression.

Through lip-syncing, Musical.ly gained popularity and found its initial product-market fit. The platform lowered the barrier to mobile content creation, allowing potentially everyone to become an entertainer. The initial value proposition was simple: The app was a tool that helped people make short lip-syncing videos after watching Lip-Sync Battle. Most people were looking to export their newly created videos to post on other platforms such as Instagram.

This type of early user retention strategy was not unique to Musical.ly. For example, YouTube began as a video hosting tool allowing people to embed videos on web pages for free, a revolutionary concept at the time. Instagram’s initial traction came from the “killer feature” of filters. People found that with a few simple taps, Instagram could make their photos look more professional.

Being a utility tool allowed Musical.ly to get its foot in the proverbial door and gave it the chance to leverage this initial traction to build something bigger. Once the tool helped attain a critical mass of content and creators, it could become a content platform, satisfying people’s desire for passive entertainment consumption—TV for mobile. Once a critical mass of users was attained, it could foster connections between those users, evolving into a social platform, satisfying people’s desire to connect and seek status. These values reinforce each other, as content becomes the starting point for interaction and status creation. Conversely, the social connections built through that app would keep users coming back to consume more content.



Above: An outline of the progression from tool to social and content platform.

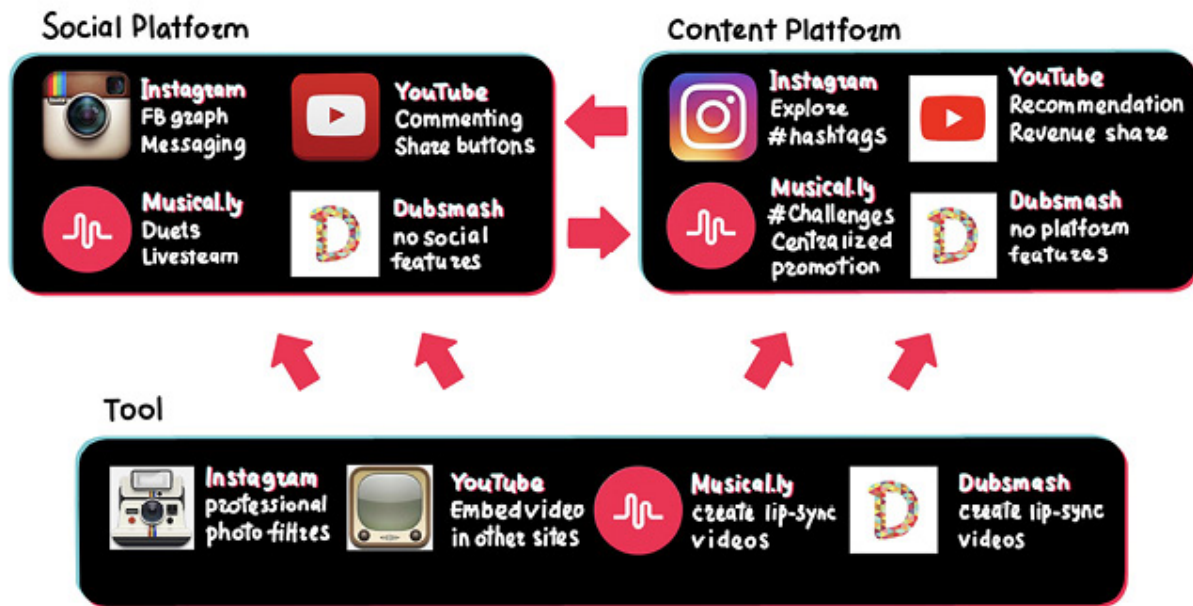
However, for many platforms, it's tough to achieve both goals equally well. For example, YouTube excelled as a repository of knowledge and entertainment, but it was relatively weak for making social connections. YouTube's limited social features such as commenting, were notorious havens for disrespectful, abusive trolling, and spamming.

In a behind the scenes video recorded during the first year of Musical.ly's operation, co-founder Louis, confidently looking into the camera, stated: "Our dream is to become the video version of Instagram."¹⁶⁵ Both Musical.ly co-founders commented publicly multiple times how they wished to make Musical.ly a social platform. "Social graph is the key!"¹⁶⁶ proclaimed Alex at one Shanghai event in 2016, following up later in the discussion with "We want the content to become more and more boring. As long as you can connect with your friends, your boring contents are attractive to your friends."

Musical.ly had initially done a great job of connecting and building community amongst teens. But by 2017, it hit a growth bottleneck and found itself struggling to break out of this demographic. It couldn't shake off the perception it was a platform just for teens.

The team had poured a lot of effort into building camera effects, live-streaming tools, and social features that fostered connections between users

and their favorite influencers, including “best fan forever” (BFF) and “Q&A” options, all of which proved popular. Teens and pre-teens loved sharing, taking part in challenges, and interacting with one another. This led the founders to believe the platform’s future direction must be that of a video-based social network somewhat similar to the positioning of Kuaishou, the leading short video app in China at the time.



Above: Examples of transitions from utility to social and content platforms.

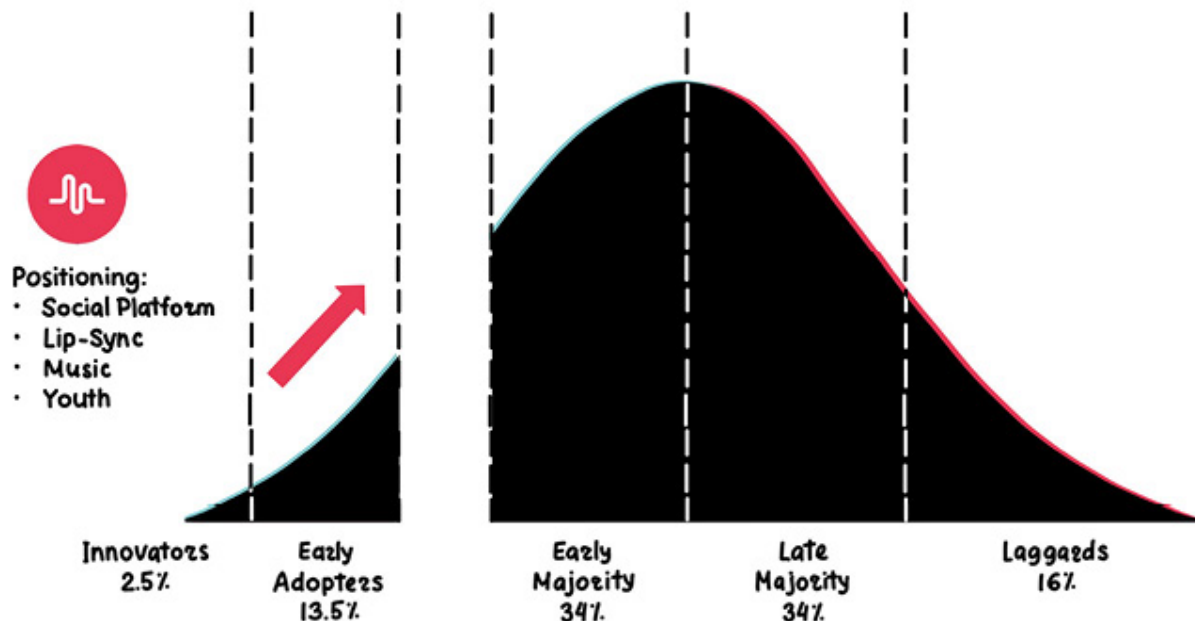
It was only later that Musical.ly came to the realization it had misunderstood its own product. The solution to growth did not lie in adding features or camera effects at all. The thing that needed to change most was the back-end architecture. “In doing user interviews and digging deeper into our data, we discovered Musical.ly was not a social network, it was a content platform,” revealed James Veraldi, an American working in Shanghai on product strategy with the Musical.ly team. **Musical.ly had spent years thinking it was the next Instagram; when in fact, it was the next YouTube.**

Musical.ly needed to lean more heavily towards the value of passive entertainment to grow beyond its already highly engaged early adopter teen user base and release its full potential.

Three major challenges confronted Musical.ly's further expansion: positioning, content diversification, and technology.

Positioning: The bright pink logo, the garish color scheme, the trending hashtags of #YouSmellLikeDonkey, and the featured videos of eleven-year-old influencers were all perfect for attracting pre-teens. Still, these elements were repellent to many other demographics. The user base skewing so young was also an issue as some creators over time were outgrowing the app and moving over to more prominent platforms like Instagram or YouTube. The last thing a status-obsessed teen wants to be is the oldest person in his or her peer group, still posting content on a social-media app for younger kids.

Musical.ly needed a redesign and a switch to more neutral positioning to appeal to an older crowd. Such a move was a significant gamble, however. A poor rebrand could quickly alienate people and set off an exodus of users from which it might never recover.



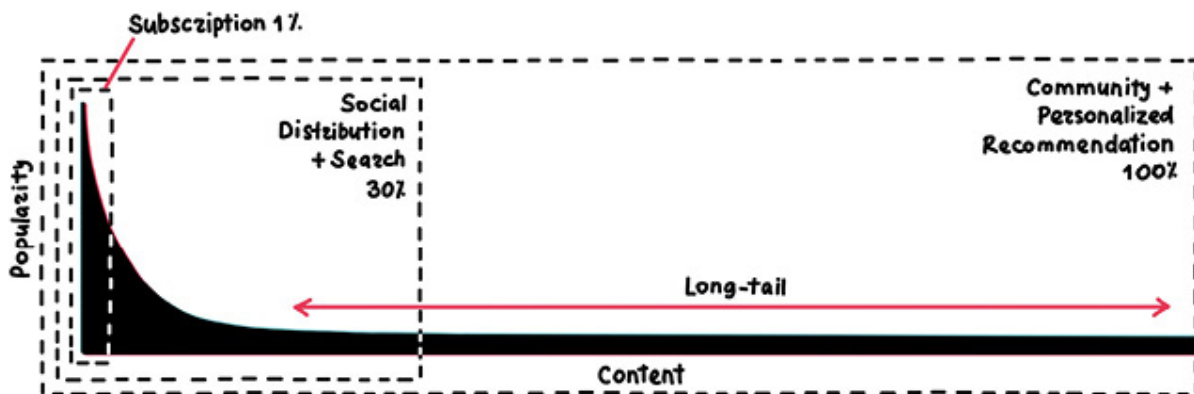
Above: Musical.ly had failed to “cross the chasm” and reach an older, more mainstream audience.

Content diversification: The founders knew they needed to broaden the content offerings. “Content diversification and use case diversification are **super, super, super** important to us. Musical.ly has to evolve into a more generic platform,” said Alex in a TechCrunch interview. ¹⁶⁷ Most of

Musical.ly's content only appealed to a young demographic. This was holding it back from transitioning into a comprehensive content platform with broad appeal, such as YouTube. The founders worked to encourage original video content in non-music related categories such as sports, fashion, and makeup.

Technology: The team experimented with two new options to display videos only from accounts the user already followed and only accounts in the same city as the user. Still, most people ignored these new options, opting for the default content feed. This default setting surfaced videos recommended by the system's algorithms, with the problem being that Musical.ly's technology wasn't as sophisticated as it would have liked.

In a short 2017 presentation at a Tencent Media event in Shanghai, Louis outlaid his thoughts regarding the use of technology to improve the platform's content distribution mechanism. His theory was highly complementary to the one presented earlier in Chapter 3 of this book.

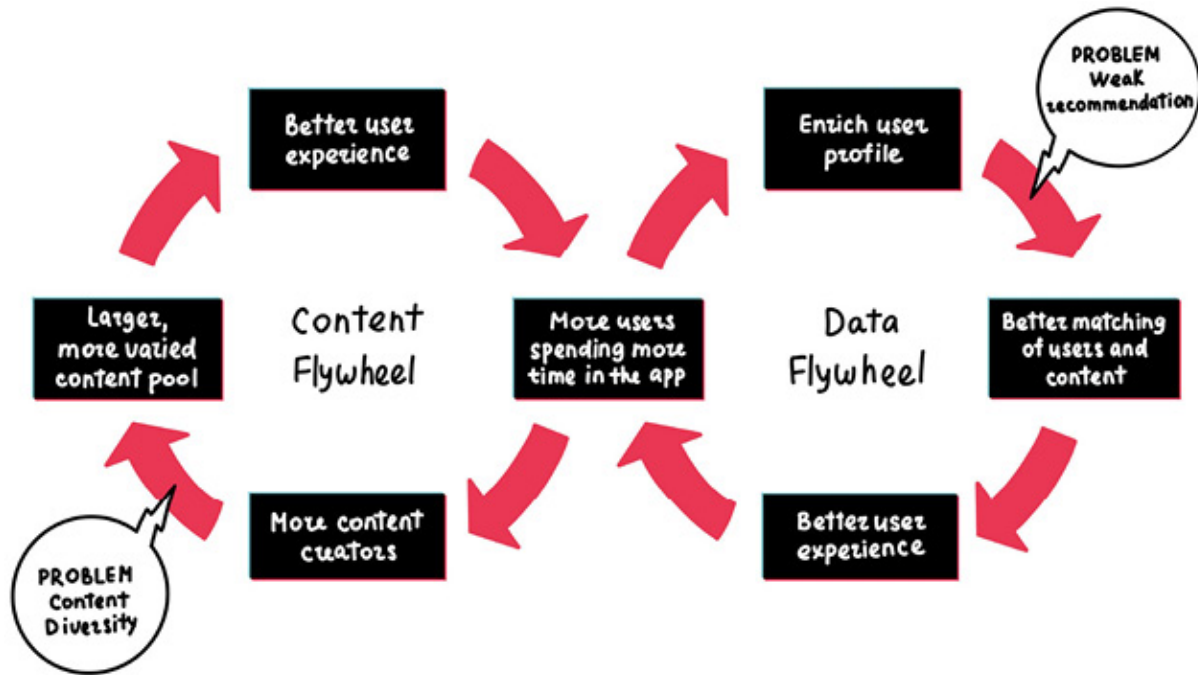


Above: Image adapted from Louis' presentation at the Tencent Media event in 2017.¹⁶⁸

Louis presented three models for the distribution of content. A traditional model that relied exclusively on a subscription mechanism which was viable because it only needed to ensure the top 1% of premium content reached people. Secondly, a social network model where the problem became how to distribute the top 30% most engaging content well. This was equivalent to the model used by Twitter. The final model was Louis' vision for the future of Musical.ly, one powered by personalized recommendation, where it was possible to match all of the platform's long tail niche content to an appropriate audience.¹⁶⁹ This vision was in line with comments from

his co-founder Alex, “The traffic model of Musical.ly has to become more and more personalized. So the parents’ experience will be completely different from the kids’ experience.”

Flywheels not flying



Above: Musical.ly should have benefited from two virtuous cycles but did not.

Better recommendation technology was needed to improve personalization, which would allow long-tail content creators to gain audiences and strengthen the diversity of content, enabling the platform to attract audiences beyond teens. Through the accumulation of both content and data, Musical.ly should have benefited from two virtuous cycles powering their growth, but as it turned out, both of these effects were greatly hampered.

More user behavior data wasn't leading to better recommendations because of limitations in Musical.ly's technology. More content wasn't leading to better user experience because the content lacked diversity. People who enjoyed dancing videos already had far more compelling content than they could ever watch; adding yet more dance videos didn't improve the user experience.

The startup was facing a “chicken and egg” problem. As users themselves generated all content, fewer older users meant there was little content suitable for them being produced. Why produce content if there’s no audience? This meant that when older users did join, they quickly left. The solution was to broaden the types of content offerings and slowly age up the user base.

“We realized while Instagram and Snapchat are social graph led, the fact that Musical.ly was content & creative led was an opportunity, NOT an obstacle. It would be mobile TV for the smartphone generation, and in order to age up, we needed to broaden the content offering and tell that story to the world,” explained product strategist James Veraldi in a later presentation.

Finally, back to Mindie

In an interview, Louis was asked to reflect on the differences in the competition between the U.S and China. His comments were exceedingly insightful.

“In the U.S., you have the possibility to market in an efficient way by creating word of mouth, because the competition pattern is different. There, if you are doing something innovative in an area, your competitor is going to try to create a differentiator, so both of you are competing with different characteristics.”

This is exactly how things had played out between them and the French entrepreneurs at Mindie. When Musical.ly began to gain success using video challenges to drive content creation, the Mindie team was reluctant to copy. Instead, the two apps diverged in focus with Musical.ly eventually moving into lip-syncing and Mindie opting to focus on social storytelling and working out a hack to integrate with Snapchat. The Mindie team was gracious in reflecting back on what had happened: “Musical.ly made a great work of optimizing and scaling what we initially created,” said co-founder Stanislas. Louis then turned to outline how he saw competition in his local market.

“Here in China, it’s a different way—if you do something right... they will follow, **they will do exactly the same thing** ... People here have a very different business logic, they think you can always get a market share very

quickly by powering in with money, and once you have that, you can kick the others out... people don't have that patience to grow up very gradually. Everything happening in China is so fast, so people lose their patience.” ¹⁷⁰

Here Louis was referring to a new, more challenging competition that had just begun. For by now, Musical.ly had already found themselves locked in a battle fighting a much larger and more fearsome opponent—ByteDance.

122 Notable video apps at this time included SocialCam, Viddy, and Mobli

123 <https://techcrunch.com/2013/10/17/mindie-is-an-immersive-music-and-video-jukebox-app-done-right/>

124 This was later extended to 10 seconds shortly after launch due to user feedback

125 Product Hunt founder Ryan Hoover and senior TechCrunch writer Romain Dillet were notable online proponents of Mindie

126 <https://thenextweb.com/apps/2013/10/17/mindie-like-vine-pop-music-soundtrack/>

127 Image source: <https://youtu.be/ibjbxRBMI30?t=175>

128 <https://youtu.be/wTyg2E44pBA?t=111>

129 <https://twitter.com/bullshitting/status/257573228887805952>

130 <https://www.ebaotech.com/>

131 <http://xiamag.com/41260.html>

132 Chinese name <http://xiamag.com/41260.html>

133 <https://crcmventures.com/>

134 <https://www.forbes.com/sites/mnewlands/2016/06/10/the-origin-and-future-of-americas-hottest-new-app-musical-ly/>

135 Using .ly had become a fashionable solution among American internet startups at the time to get around the problem that all the reasonable .com names had long been taken. (examples include: Bit.ly, Feedly, Strikingly, and Grammarly) No one seemed bothered that technically .ly is the internet country-code domain for the African nation of Libya.

136 <http://tech.163.com/14/1118/19/ABBTTNOK00094ODU.html>

137 <https://supchina.com/2017/09/13/can-pop-music-connect-teens-china-world-musical-ly-co-founder-louis-yang-wants-find/>

138 <https://36kr.com/p/5041108>

139 <http://www.icamp.ai/>

140 Many of these accounts can still be found on TikTok by searching “temporality”

141 The term meme was coined by Richard Dawkins in the 1976 book “The Selfish Gene” to explain how ideas replicate, mutate and evolve (memetics).

142 [https://en.wikipedia.org/wiki/Harlem_Shake_\(meme\)](https://en.wikipedia.org/wiki/Harlem_Shake_(meme))

143 https://en.wikipedia.org/wiki/Ice_Bucket_Challenge

144 <https://kknews.cc/media/vg4may.html>

145 Heavy reliance on “operations” is one of the Chinese internet’s defining characteristics with Alibaba being the most famous example of a heavily operations-driven organization. Cheaper labor is only part of the reason for the popularity and effectiveness of operations in China. The discipline itself is primarily a response to the wider online macro environment (e.g. walled garden ecosystems, KOL marketing, limited access to quality, scalable ad targeting, and ineffective SEO.)

146 Walkthrough of Dubsmash, April 2015 <https://youtu.be/xDDHkz18c-k?t=85>

147 <https://www.paramountnetwork.com/shows/lip-sync-battle>

148 Data source: App Annie

149 <https://zhuanlan.zhihu.com/p/27878425>

150 <https://www.youxituoluo.com/120223.html>

151 Cheetah Mobile held approximately 17.4% of Musical.ly equity as of December 31, 2016

152 <https://www.youtube.com/watch?v=KmtzQCSh6xk>

153 https://www.youtube.com/watch?v=8f7wj_RcqYk

154 <http://www.justjared.com/2017/08/03/baby-ariel-reveals-how-her-musical-ly-name-was-invented/>

155 <https://www.youtube.com/watch?v=LNwqJNi80Rc>

156 https://en.wikipedia.org/wiki/Baby_Ariel
157 <https://www.forbes.com/sites/mnewlands/2016/06/10/the-origin-and-future-of-americas-hottest-new-app-musical-ly/#7ee844285b07>
158 <https://cn.nytimes.com/technology/20160919/a-social-network-frequented-by-children-tests-the-limits-of-online-regulation/en-us/>
159 Data source: <https://youtu.be/zNGZCO7aISA?t=23>
160 <https://www.wired.co.uk/article/musically-lip-sync-app>
161 https://www.crunchbase.com/search/funding_rounds/field/organizations/last_funding_type/musical-ly
162 <http://www.ccidnet.com/2016/0623/10149446.shtml>
163 <https://youtu.be/ey15v81pwII?t=96>
164 <https://youtu.be/ey15v81pwII?t=265>
165 https://v.youku.com/v_show/id_XODU3NDEyNTcy
166 <https://youtu.be/E3aOxgyMUqk?t=188>
167 <https://youtu.be/ey15v81pwII?t=237>
168 Presentation: <https://v.qq.com/x/page/z002582m7e0.html>
169 https://www.sohu.com/a/205007924_247520
170 <https://supchina.com/2017/09/13/can-pop-music-connect-teens-china-world-musical-ly-co-founder-louis-yang-wants-find/>

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CHAPTER 6

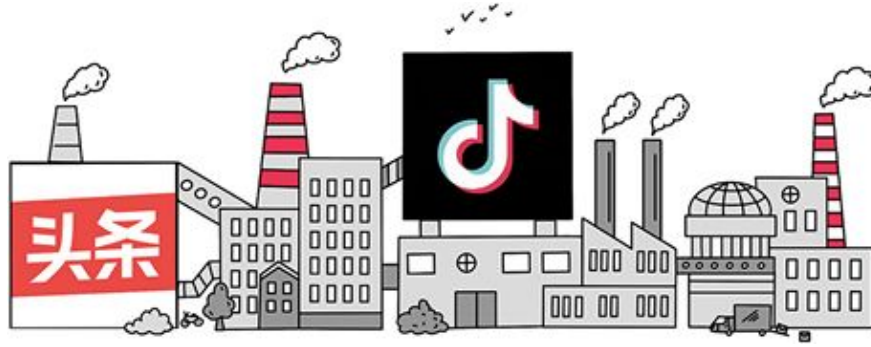
AWESOME.ME



*“Your product is too basic. You wanna get
on the highway with this broken up old car?”
Douyin early user feedback*

Image: The original Douyin team posed for a
picture together before the release of version 1.0

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Chapter Timeline

- 2013 – ByteDance hires “product guru ” Kelly Zhang
- 2015 July – Short video app Xiaokaxi u gains fleeting success
- 2016 Sept – ByteDance launches a clone of Musical.ly called A.me
- 2016 Dec – A.me is rebranded as Douyin
- 2017 Mar – First popular Douyin video
- 2017 Sept – Douyin’s recommendation system upgraded
- 2017 Oct – Engagement spikes over China’s Golden Week holiday
- 2018 Jan – Daily active users reach 30 million
- 2018 Feb – The “Karma’s a Bi*ch ” meme goes viral
- 2018 Nov – Daily active users reach 200 million

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YIMING STOOD ON THE STAGE, a tiny figure dressed in black. Behind him sat two massive LED screens at least 15 feet high, stretching out to either end of the large conference hall.¹⁷¹ Before him, a packed audience sat in darkness—hundreds of guests from content production houses and influencer management companies to journalists and influencers. Strobe lights reminiscent of a rock concert flashed dramatically across the ceiling.

Yiming was not naturally suited to public speaking. Staff used to lament his heavily accented Mandarin and emotionally reserved, soft-spoken deliveries.¹⁷² He had made considerable progress in the last two years. “With enough learning and perseverance, any skill could be mastered,” was always Yiming’s standard approach to his shortcomings. Wrapping up his opening presentation in true Steve Jobs fashion, he opted to finish on a high, with just one more thing: “In the next twelve months, at least one billion yuan will be used to subsidize short video creators.”

“1 Billion” lit up on the screen in bold red characters making Yiming look even smaller than before. This was the announcement that the audience had come for. It was another payday for video creators as platform companies like ByteDance burnt money in subsidies.

By 2015 the whole of China’s internet industry knew that engagement was shifting towards videos, and platforms were in desperate need to secure all kinds of video content. The problem was short video producers had yet to find a sustainable business model. They found it easy to get views but challenging to live a comfortable life. With numerous platforms jockeying for attention, one of the best ways to immediately attract content producers was simply to pay them. The strategy turned the field into a new battleground for subsidies. Billions of yuan were being thrown around to entice production studios and talented creators.

ByteDance entered the fray relatively late. There were dozens of Chinese apps that had already introduced new ways of consuming videos, including local adaptations of Vine and Dubsplash. Meipai, the short-video app under China’s largest photo-editing app Meitu, was well-loved by urban youngsters. Market leader Kuaishou had seized tens of millions of users in smaller Chinese cities. From slick, professional production houses and

fashion mega-influencers, through to rural farmers recording from the fields, China's short video market held something for everyone.

ByteDance's flagship app Toutiao was already featuring short videos, which were experiencing an explosion in usage. Over the first quarter of 2016, the average time an active user spent on the app per day was an impressive 53 minutes. This had grown to a formidable 76 minutes per day by the third quarter of the year. Over half of the growth came from short videos alone. Short video consumption within Toutiao during the first half of 2016 was growing at a monthly rate of 35%! ¹⁷³ There was simply no way the team could sit by anymore. They had to put serious effort into building dedicated short video platforms.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2011			 GIF Kuaishou 快手									
2012			 Miaopai 秒拍							 Youjiao 有料		
2013		 Xiaoying 小影						 Waishi 腾讯微视				
2014			 Xunlei yingyin 迅雷影音	 Meipai 美拍								
2015												
2016												

Above: Short video apps launched from 2011 to 2016 (non-exhaustive).

A three-pronged attack

Having committed to joining its competitors and entering the “red ocean” short video market, the next question for ByteDance was “How?” The answer was simple—copy the best. The company chose to attack the opportunity from multiple avenues and start by imitating what had already worked.

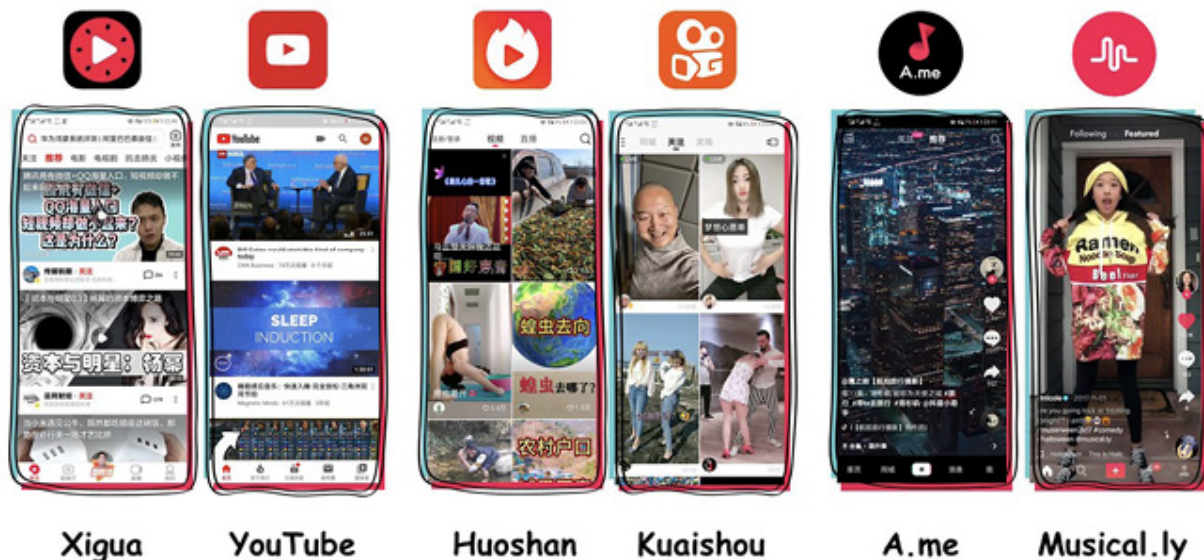
The three-pronged strategy consisted of “Xigua (Watermelon) Video,” ¹⁷⁴ which imitated YouTube, the clear global leader in online videos, and

“Huoshan (Volcano) Short Video,” which imitated Kuaishou, the clear leader in China’s short video market at the time. Finally, the wildcard in the mix was A.me, later known as Douyin, a clone of Musical.ly, the Western market leader of short videos for young people.

ByteDance rejected the idea of creating long-form video platforms similar to Netflix or Hulu. This market was already fiercely competitive in China, with three large players backed by the BAT giants burning money to take market share from each other. More importantly, the success of these platforms tended to be driven by securing exclusive high budget professional content. That didn’t play to the company’s strengths in algorithmic recommendation. Short-form video content, however, was a suitable match for the company’s already existing technological moat.

Of the three apps, the management team was the least optimistic about the chances of Musical.ly clone A.me, which was later renamed to Douyin. Musical.ly itself had already failed once to crack the China market, and growth was plateauing in international markets. The teen demographic which had embraced Musical.ly in America had little free time in China due to excessive homework and after school classes. Adding to the hesitation around A.me was the prominent case of lip-sync centric short video app Xiaokaxiu ¹⁷⁵. It had been a big initial hit in 2015 but saw its popularity fade rapidly, proving to be a flash in the pan.

Many felt this kind of entertainment-focused short video app wasn’t suitable for China, and A.me received relatively scant resources. It was basically a side bet. The far bigger budget went into promoting Huoshan Video as it followed a locally proven model and competed head-on with a thriving domestic competitor, Kuaishou.



Above: three apps comprised ByteDance's new short video strategy (Xigua, Huoshan, and A.me) and their original counterparts (YouTube, Kuaishou, and Musical.ly).

The responsibility to manage the Musical.ly clone would lie with Kelly Zhang, who oversaw user-generated content platforms, known in the industry simply as UGC. In her thirties with short hair and glasses, Kelly Zhang had been acquired by ByteDance a long with her startup—online picture community, Picture Bar¹⁷⁶—in 2013. She was an internet industry veteran and a serial entrepreneur having been a founding employee of two previous internet companies.¹⁷⁷

Kelly had a reputation for understanding how to cultivate communities, an essential process for UGC platforms. This discipline was quite different from the model of Toutiao, where users passively consumed but didn't create themselves. Kelly oversaw multiple ByteDance platforms, and the Musical.ly clone was just one project among many.

Amazing youth

Xiao'an was nervous—she was mid-way through her interview at ByteDance's head office. Finishing up at university, she was looking to enter the workforce with her first real job. She had interned one summer at ByteDance previously, and last week an old colleague from back then, Xiao'wei, had sent her a message: "I'm working on a new project. Would you like to take a look?" Now, here she was, sitting in the interview room.

She smiled somewhat awkwardly again, trying her best not to show her nervousness.

The interviewer leaned over to show her something on a phone. “The team you would be joining is working on a new app. This is a test version,” Xiao’an smiled politely and looked down at the handset. The video immediately jumped out at her from the screen. “What the hell is this? I’ve never seen **that** before,” she thought to herself. The video took up the entire screen and automatically looped. The “like” and “share” buttons were overlaid on top of the actual video itself. “This app is weird,” She thought to herself.

“What do you think?” asked the interviewer. “Oh, interesting,” was all she could politely muster.

Xiao’wei went on to arrange for her to meet some of the team members at a bustling barbeque restaurant off Zhichun Road, not far from the office. Summertime was when evening dining spilled out onto the street—the scene was loud and lively. People chinked glasses and toasted one another after a long day of work. Over cold bottled beer and juicy meat skewers, Xiao’an was introduced to the A.me team.

Zhang Yi was a product manager with tattooed arms who loved extreme sports. Every weekend, she would go out to the suburbs to ride motorbikes. Li Jian was a third-year university student. Xiao’wei had discovered him playing guitar on a live-streaming app. He had never worked in the internet industry before.

“Damn. This was a very young team. It’s going to be a lot of fun,” Xiao’an recalled her initial impressions of meeting her prospective colleagues. The team was inexperienced, but there was something that drew her to the group.

ByteDance by now had over 2,000 employees. The A.me team was less than ten staff working together in a small section of the head office’s second floor. It was operating somewhat like a small startup within the much larger organization. The app was officially registered under a separate company, Beijing Weibo Vision Technology Co., Ltd., controlled by Yiming’s old university roommate, Rubo.

No one in the team had much real experience making a full app from scratch. Around ten engineers were brought over and took a week to build the beta version of the app that Xiao'an had seen during her interview. It was a mess and full of bugs. Frustrations arose between the design and engineering teams. Aspects of the layout that the designers felt were blindingly apparent had to be explicitly marked up for engineering to build as they wished. 24-year-old designer Ji Ming recalled: "The things I worked on so hard to get out every day were so unsatisfactory. I really felt bad."

To improve user experience, the team invited students from nearby middle schools to chat at the company. They talked through what kind of apps they liked to use and why. Ji Ming found the students could only make judgments based on the existing apps in the market. It was difficult for them to imagine something they hadn't already experienced. This reminded the team of the famous Steve Jobs quote: "A lot of times, people don't know what they want until you show it to them."

A.me

By the end of September, the team was ready. With little fanfare and much indifference from within the industry, the first version of the app entered the market under the name "A.me—Music Short Videos." ¹⁷⁸

"The A.me app is a fashionable platform for sharing and shooting music short videos. Let the emotion of music take you away; show your confidence and personality. Reveal it whenever. Here every moment is you, and just you!" read the app's description. The "A" stood for "Awesome," and the app's logo was set as a red musical note on a distinctive black background.

The app was out there, but it didn't exactly set the world alight. The team received an automatically generated email every morning, listing the latest number of active users. After operating for some time, they decided to remove all the accounts created by internal staff to get a more accurate picture of the app's performance. The active user stats dropped by half the next day.

"It was so bleak," recalled Xiao'an. "I didn't know what to do every day...The features were so poor." She had no faith in the product itself and

felt embarrassed to track down small online influencers and ask them to post content on the app.

Among the early users invited to the platform was a little-known creator named “Xue Lao’shi,” a Chinese student studying in Canada. After trying out the app, he initially rejected the invitation to join. “Your product is too basic,” he complained. “You wanna get on the highway with this broken up old car?”



Above: App store showcase used to promote A.me version 1.0. ¹⁷⁹

Instead of gathering established influencers, the team banded together a group of around twenty official A.me endorsed “creators” who were enlisted from other platforms, including Musical.ly, to create videos. The goal was to have their content inspire and educate other early users to create videos.

Similar to Musical.ly in the U.S., some very young elementary school and junior middle students were attracted to use the app. Some of the first official challenges set up were targeted at this age group. But without the same culture and ecosystem as Musical.ly, these children were very limited in the kinds of content they could create.

The lack of people uploading interesting videos was a massive issue for the team. Yet without proving that the app could retain and grow their tiny user base, it would never receive proper support from the parent company. Heavily promoting the app in its current state would be like pouring water into a bucket with a big hole in the bottom.



Above: Twelve of the app's original seed batch of video creators wearing A.me branded T-Shirts.



Cute Puppet Dance
Big Snack Bag waiting
for you to grab

Together with Paris,
one song each
Top 3 get big
snack bags

Post 2000's, post 1990's
Pretend to be cute
competition

Above: some examples of in-app promotional banners for very early official challenges on A.me targeted to the female pre-teen demographic. ¹⁸⁰

Solution: treat them like royalty

The first thing the team did was to treat its existing small band of popular creators like royalty. The operations team would chat with them individually every day, earnestly listening to their ideas, and making them feel they were participating in the platform's growth and molding its direction. If a Beijing-based user encountered a problem that was not easy to explain online, they would be invited to chat over a free meal at ByteDance's company cafeteria.

Just like Musical.ly, the team made use of theme-based challenges to build communities and actively encouraged users to riff off each other's creations, building upon shared memes. Users were free to initiate their own challenges, which helped the team learn what kind of content people preferred and guide them in selecting their official challenges. Many of the first official challenges originated not from the team but ideas that came out when talking with the early video creators. They would reward the best video creators with gifts such as cameras, celebrity merchandise, or snacks, yet another way to make people feel special. Top content creators were ranked within the app on "Most Popular List," "Most Active List," or "Weekly Rookie List" – helping foster a sense of community.

The tactics A.me was using are part of what is termed "operations." Heavy reliance on operations to achieve growth is one of the defining characteristics of the Chinese internet. At Western tech companies, the role of user acquisition typically falls under the marketing, sales, or growth teams, which tend to systematically achieve user growth through highly scalable data and technology-driven methodology. In addition to these established techniques, Chinese companies also favor using manual, labor-intensive methods to promote and grow their platforms. Examples include paying for celebrity endorsements and media exposure, operating promotional accounts on other platforms, or running regular competitions and festival holiday promotions. Operations teams are typically active all day, maintaining relationships with outside stakeholders, including users, creators, and promotional partners.

Cheaper labor costs helped make the operations approach more viable, but the main driving factor was an online environment where infrastructure around digital marketing lagged behind Western equivalents. Established disciplines such as search engine optimization were, in large part, rendered ineffective within the Chinese “pay to play” walled garden ecosystems that had sprouted up.

A.me went as far as assigning account managers to individual creators who would do everything possible to please them, even buying them dinners and helping with school assignments or relationship issues. Next to the team’s workstation was a big box filled with all kinds of props for shooting videos, from wigs and glasses to funny placards. When an early user celebrated their birthday, the operations team would record an exclusive video for them. Around Christmas time, a team intern even went out his way applying for a credit card to order a Christmas tree on Amazon for Xue Lao’shi, the Canadian student who had initially refused to join.

Long cold start - time to quit?

The Chinese internet is a brutal battleground with little place for sentimentality. Underperforming products are unceremoniously shut down and forgotten every day. Douyin was basically operating as an internal startup team. For such projects, failure is often expected.

Given the poor performance, no one would have thought it strange had the team been shut down. “To abandon a project, mainly we look at the data,” described ByteDance executive Chen Lin, “but the leader also has to use their judgment.”¹⁸¹ A leader needs to judge **why** the data is bad.

Is the market segment smaller than they thought? Have they misjudged people’s needs? Or did they just execute poorly? Perhaps ByteDance doesn’t have the right company DNA to do something targeted at such young people? All these were legitimate doubts during these early months.

In the end, Yiming reasoned as such: “The logically correct thing is definitely right. And others have already verified (this path), our data is poor because we haven’t done a good job ourselves.”¹⁸²

Many of the issues leading to A.me’s early poor results were already on the way to being resolved. The app itself had been somewhat poorly constructed with many bugs and only basic features; this had mostly been

remedied. Its earliest positioning was confused, looking to appeal to pre-teens, teens and those in their early twenties. By renaming and rebranding the app they could focus on a more precise target—trendy urban youths.

The redesign

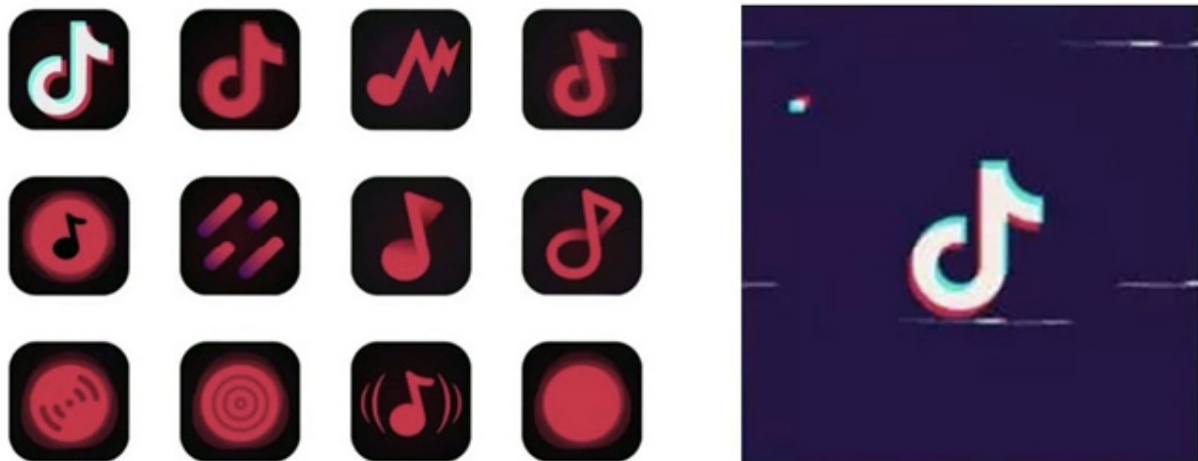
While A.me’s signature dark color pallet was popular, ¹⁸³ the name A.me wasn’t intuitive enough for Chinese people, and a rebrand was needed. Given music played a vital role in the app’s experience, the idea that both the creators and their audience would unwittingly move their bodies to the music inspired one team member to coin the name Douyin, literally “Shaky Beat” in English.

dǒu yīn

dǒu : Can be translated as “shake,” “shiver,” or “tremble.”

yīn: Can be understood as “sound,” “music,” or “musical note.”

“Douyin” was voted the favorite out of the hundreds of names considered. Next, it needed a logo.



Above: Original design concepts for the now-iconic logo used by Douyin and TikTok.

The young designer tasked with the redesign had his epiphany when attending a rock concert lit up by beams of swirling colorful lights with a surrounding dark stage. Inspired by the psychedelic visuals at the live event, he set about creating an image that would capture the show’s euphoric feel and decided to toy around with the musical note ♪ sign. He ran the icon through various filters and settled on using something known as a glitch

effect. The style is reminiscent of the static distortions of an old television suffering from a weak signal. The overall impression was one that perfectly conveyed the feeling of shaky movement. The musical note itself had been altered to form the shape of the letter “d,” signifying the first letter of the app’s name Douyin. The logo precisely matched what the team wanted the product to be. It was different, creative, and instantly recognizable.

Unfortunately, this future vision was still far away from the reality of where the app actually was. The numbers didn’t lie—the data throughout the first half-year of operation was unimpressive.

Nation building: the Douyin way

With its successful rebranding and a larger budget, Douyin was now repositioned upmarket as a trendy app for fashionable young urban elites. But to pull off such a transition, the team needed to solve their biggest problem head-on—the lack of quality young content producers.

The solution was art college students. The Douyin team went deep into art schools across the country to scout for good-looking students to be its users. Altogether, the team convinced hundreds of them to join, with the promise to help them get famous online. This proved highly effective. The influx of users helped build an original content pool and establish the app’s tone as cool and fashionable.

The operations team then started to manipulate the visibility of videos to encourage the type of trendy content they wished to cultivate. Consequently, videos that didn’t conform to the community’s tone and values would struggle to gain visibility.

Douyin mobilized its entire staff to lure in creators from competitors. They scouted for suitable talent across all major Chinese social media platforms and even overseas Chinese on Musical.ly, messaging them one by one. To speed up the process, they also began to strike deals with “multi-channel networks” (MCNs), a type of organization that emerged in YouTube’s early years to represent and professionally manage a collective of creators.

Simultaneously, the team was aggressively setting up accounts on other short video and social media platforms where they posted watermarked Douyin videos. As with Musical.ly, the watermark was the key. It was like a mini advert, for people intrigued by the videos would see the watermark

and search for the name on the app store. The unique user ID of the video creator was later added to the watermark next to the flashing Douyin logo. This small but critical change further encouraged people to share their videos on other platforms, as this could now lead people back to their accounts and grow their fan bases.

The operations team would continuously scan the platform for content that could be used for promotion on other platforms. By February, it saw the first hint that Douyin might have a chance at breaking out. A dance meme entitled “The Backrub Dance” ¹⁸⁴ that had originated on Douyin started to organically spread out onto many other platforms.

In March, another video caught the team’s attention. It was an uncanny impersonation of a famous comedian, Yue Yunpeng, similar in both look and style. The team repeatedly pinged the celebrity’s official social media accounts. After much persistence, they eventually attracted his attention, and the comedian shared the impersonation to his millions of followers. The video, watermarked with the glittering Douyin logo, received over eighty thousand likes and more than five thousand forwards. Douyin’s Baidu index, a local equivalent of Google trends, rose considerably the next day.

I’m not cool enough to use this

Douyin continued to double down on its brand positioning as an app for the coolest of the cool. It began a cinema advertising campaign in the early summer of 2017, a 30-second ad slot ¹⁸⁵ that was a fast-paced tour de force of wild shaking camera effects matched to a loud thundering electronic beat. One industry professional described their experience seeing the commercial as “dazzling,” adding “this product is too cool, so cool it’s not suitable for me.” ¹⁸⁶

Cinema ads were backed up by online promotional campaigns of short, playful, and interactive adverts featuring famous historical figures such as the Mona Lisa and Abraham Lincoln using Douyin. Amusing, original, and well-executed, these adverts went viral across Chinese social media garnering curiosity and building brand awareness.



Above: Douyin advertisements during 2017 feature the Mona Lisa wearing glasses and what appears to be Abraham Lincoln smoking a blunt.

That summer, Douyin clinched a sponsorship deal for a new talent show, *The Rap of China*.¹⁸⁷ The program was endorsed by several big-name celebrities and became a breakout hit with China's urban youth. Hip Hop culture became all the rage, in part because of the show. Douyin was the perfect tool for young Chinese to create their own rapping, breakdancing, beatboxing, and street dance videos. Female contestant VaVa, sometimes referred to as "China's Rihanna," commented: "All the people into Hip Hop are all on Douyin."¹⁸⁸

Somewhat echoing how *Lip Sync Battle* had been a game-changer for Musical.ly in the U.S. market, shows like "Hip Hop in China" and its rival program "This! Is Street Dance," pushed Douyin to the forefront of China's youth culture.

Knowing that young Chinese people were highly sensitive to the personal image they displayed online, a dedicated engineering team was set up to build best-in-class beautifying filters and special effects for Douyin. These lowered the barrier to content creation and gave confidence for users to shoot without makeup.

September saw ByteDance close a two-billion-dollar funding round valuing the company at 22 billion dollars. These inflows helped ensure a

steady stream of cash for Douyin's already lavish promotional efforts, which now entered the next stage—offline events.

The crowd goes wild for Yiming

Hundreds of fashionably dressed young people were arriving at 751 D.PARK, an expanse of industrial plants redeveloped into a hip culture venue in northeast Beijing. They were clad in baseball caps, brightly colored dresses, loose-fitting hip-hop style streetwear, and limited-edition sneakers. The site had been transformed into something akin to the stage of the talent competition *American Idol*, spanning two floors filled with strobe lighting, high volume music, and trendy backdrops. This was an exclusive party—three hundred top Douyin creators coming together to celebrate the app's one-year anniversary.

The online stars, billed as the “new generation of internet celebrities,”¹⁸⁹ weren't there to just socialize and enjoy themselves. Every influencer was aware of the unspoken competition to derive the best content from that night. They were all fighting to achieve a higher level of superstardom, and the medium of battle was short video.

The influencers who knew each other gathered in small groups as their assistants tirelessly captured fifteen-second videos of their carefully crafted skits. Loners roamed around the dance floor, absorbed in finding the ideal lighting for their lip-syncing selfie videos. Lesser-known influencers nervously approached more famous ones, proposing to record a dance together to potentially tap into their peers' following. Loud hip-hop music kept playing in the background as creators hurried to touch up the videos they had just shot. Once the editing was done, they uploaded their works and anxiously waited for the app's algorithms to judge who would grab more eyeballs.

Dance teams took to the stage to display their skills. The crowd bopped their heads back and forth as rappers attempted to impress with clever lyrics.¹⁹⁰ Later as the hosts were mid-way through giving out awards, a wave of noise erupted from the back of the crowd interrupting the proceedings.

It was Yiming. Dressed in a black baseball cap and grey T-Shirt and accompanied by Lidong. The audience went wild—the CEO had decided to

drop in unannounced! Immediately he was bombarded with requests to take pictures and videos. As those around him whooped and cried out wildly, the entrepreneur simply smiled and kept his hands calmly by his side, an awkward 34-year-old engineer type among the hyper fashionable, mostly teenage hip hop crowd.



Above: Yiming and Lidong appear at a Douyin promotional event marking the app's first anniversary in Sept 2017. ¹⁹¹

He already knew from looking at the data, but this was confirmation in the flesh—Douyin had built a robust community, with powerful momentum, and was on the verge of doing something special.

The breakout

October 1st marks the beginning of “Golden Week,” a seven day long official Chinese national holiday. Periods like these are big opportunities for China's internet industry. People's behaviors change for a week; many find more time for entertainment and to try new things.

Over October, Douyin's daily users doubled from seven to fourteen million; two months later, they reached 30 million. Over those three months, the 30-day retention rates jumped from eight to over 20%, the average time spent in the app soared from 20 to 40 mins. ¹⁹² It was as if some magic rocket fuel had suddenly been added, boosting every key metric. What had changed?

The answer was Zhu Wenjia. Zhu Wenjia, hired from Baidu in 2015, was widely considered to be one of the top three best people in the entire company when it came to algorithm technology.¹⁹³ He ran one of ByteDance's most capable engineering teams and had recently been assigned to work on Douyin. The team's work harnessing the full power of ByteDance's content recommendation backend led directly to the astounding October results.

The better the metrics, the more resources ByteDance placed behind the app as it now had good retention and was fast-tracked into becoming a strategically important product. Suddenly support was coming in from all over the company—people, money, user traffic, celebrity endorsements, brand collaborations, and most importantly, full integration and optimization of ByteDance's powerful recommendation engine. Chinese stars with massive fan bases such as Yang Mi, Lu Han, Kris Wu, and Angelababy opened accounts, joining in publicity campaigns, and a nationwide “Douyin Party” event roadshow was planned. Douyin had become the hottest upcoming app in China.

ByteDance ramped up the investment in all three short video products, including Douyin. People, resources, and advertising budget were all raised, leading an industry insider to comment later: “The sudden rise of Douyin wasn't without good cause. Yiming threw more money at this than anyone and dared to hunt down and grab the best people.”¹⁹⁴

Commercialization began with the first three brand ad campaigns paid for by Airbnb, Harbin Beer, and Chevrolet. Douyin's advertising business would soon make rapid progress. ByteDance already had hundreds of sales and marketing staff who would shortly be able to add Douyin's advertisement inventory to their sales targets.

Yiming revealed in a later interview that the company had made it compulsory for everyone on the management team to make their own Douyin videos with goals to gain a certain number of likes or suffer forfeits such as doing push-ups. It wasn't good enough to just look at charts and data; management needed to understand short videos from a creator's perspective also. Yiming had watched Douyin videos for a long time but creating his own was “a big step for me,”¹⁹⁵ he admitted.



Above: Yiming’s personal Douyin account (3277469). Seventeen videos at the time of writing, including clips from his global travels. ¹⁹⁶

“Oh well... karma’s a bi*ch”

The video opened to a young woman yawning, dressed in pajamas with messy morning hair. Wearing glasses and with no signs of makeup, she casually lip-syncs the line, “Oh well... Karma’s a bi*ch” and throws a silk scarf into the air. Suddenly loud background music explosively begins. In an instant, she transforms into a glamorous alluring fashion model, almost unrecognizable from a second before. A new meme had taken hold of Douyin. ¹⁹⁷

“Karma’s a bi*ch” was a new version of the original “Don’t judge me” challenge that had propelled Musical.ly to top the U.S. app store three years earlier. The meme was another breakthrough for Douyin; people loved watching the shocking transformations. Compilations of the meme’s videos started popping up online. In particular, the makeup skills of some women left many men in disbelief. “Karma’s a bitch” left an impact on mainstream culture and gained widespread recognition and publicity, even making waves out into English language global media. ¹⁹⁸

Douyin was also increasingly hyper-charging the popularity of catchy pop songs with strong hooks. In late 2017, a track known as the “Ci-li-ci-li song” exploded on Douyin. The song’s catchy energy was undeniably infectious. Yet, it was the novel set of dance moves that had become associated with the track’s hook that turned the music into a meme and dramatically amplified its success.¹⁹⁹

The track had actually been released back in 2013 by Romanian reggae and dancehall artist Matteo, under the name “Panama.” Four years after its debut, the song’s unexpected and explosive spike in popularity led the singer to hastily organize an Asia tour to capitalize on his track’s sudden fame. A YouTube video ²⁰⁰ shows him meeting Chinese fans at the Hangzhou airport who demonstrate their moves to him in the arrivals hall. With the dance having been created entirely in China, the bewildered artist finds himself in the awkward situation of not knowing how to follow the moves to the song for which he is famous.

Perhaps the most reliable indicator of the platform’s increasing influence on society was how the name, Douyin, had started to enter everyday colloquial vernacular, becoming synonymous with short video. The meaning of “Let’s shoot a Douyin!” needed no explanation.

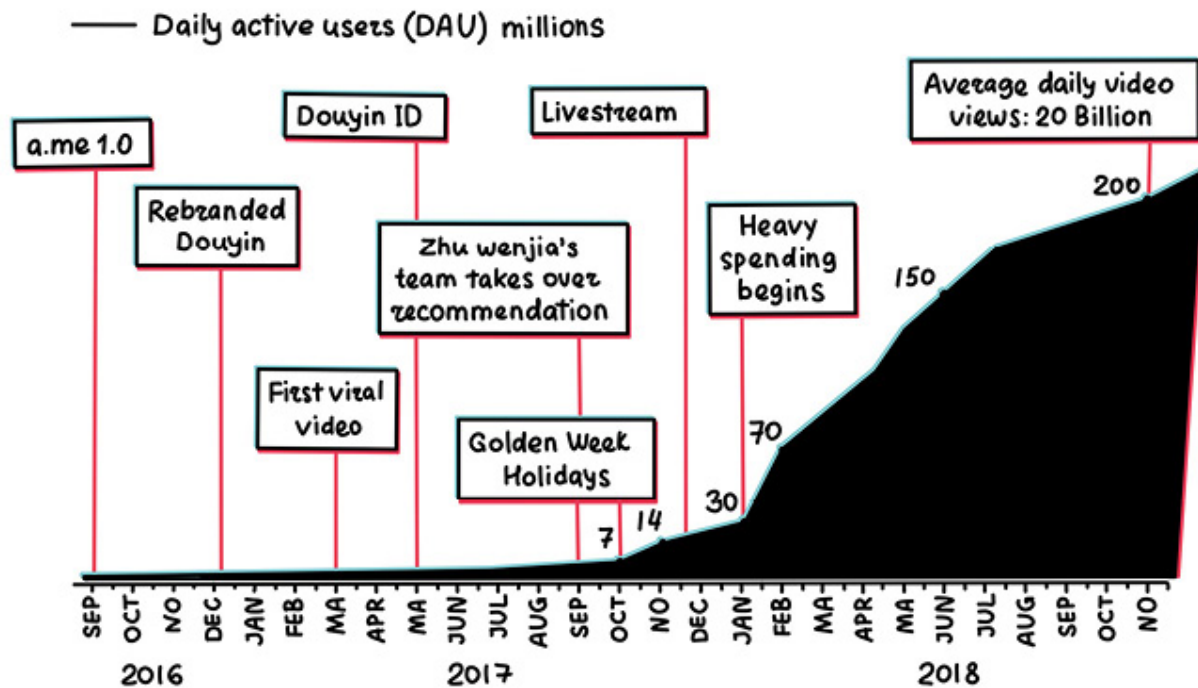
Make it rain

ByteDance knew they now had a winning formula. Retention was good, word of mouth was excellent, a large, vibrant community of video creators had been fostered. The recommendation engine was doing its job of surfacing the best content. Douyin’s fire was already burning bright; now, it was time to pour gasoline on things and **spend, spend, spend**.

The holiday week of Chinese New Year is another unique annual opportunity for app promotions. Hundreds of millions travel home to be reunited with their families and find themselves with free time to relax. An entertainment app like Douyin was the perfect way to pass the time; word of mouth spread naturally between family members.

To step up its efforts further, Douyin directly gave out money to users by running a Chinese New Year “lucky money” campaign. Users could collect small cash amounts in special videos by tapping on the “red packet” icons—a digital manifestation of cash-filled envelopes people give to each other

during the holiday. ByteDance also went all out, spending wildly, buying adverts and promotions across major online channels to acquire users, spending about 4 million yuan a day ²⁰¹ (over half a million dollars). The combination of all these effects sent Douyin to the top of the Chinese app store charts. Various reports stated Douyin's daily users jumped from around 40 to 70 million over the February to March period covering Chinese New Year, with some of the top accounts seeing their follower numbers quadruple.



Above: A chart mapping the progress of Douyin, from zero to 200 million daily active users, during the first two years of operation.

The success simply emboldened ByteDance to double down on Douyin, upping the daily promotion budget to 20 million yuan (\$2.8 million) to buy traffic on every available channel that would take their money. By April, daily users had exceeded 100 million.

With a sizable user base, Douyin began attracting attention from digital marketers, who saw it along with WeChat and Weibo as the “must-have” social media platforms for brands. The “Big-two” was becoming the “Big-three.” ²⁰² Through Toutiao’s enormous media and advertising platform, the company already had relationships and extensive contacts with all of China’s major professional media, marketing agencies, and brands. The

push came from ByteDance to encourage everyone to start creating videos and allocate budget towards Douyin.

Douyin continued to grow well for the rest of 2018, breaking out into the mainstream and gaining popularity across society.

Why Douyin? Drivers of success

Douyin started as a Musical.ly clone—vertical full-screen 15 second short videos, with a swipe up motion and music-driven discovery. The core experience hadn't changed since Mindie's original version was built in a Paris basement, back in 2013. Yet Musical.ly and Douyin had achieved drastically different results, which begs the question: Which factors drove Douyin to succeed where Musical.ly had failed?

a. Infrastructure

Firstly, we need to acknowledge that regardless of anything ByteDance did or didn't do, simply by launching three years later than Musical.ly, Douyin already enjoyed more favorable conditions for success. By 2017 fast, affordable, stable, and ubiquitous 4G internet had become widely available across China. A video-first mobile app like Douyin could only gain mainstream adoption once the appropriate network infrastructure was widely in place. Cheap, fast video uploading and downloading meant that videos could be created and consumed on the go instantly. By 2017 data package plans had reached a level of affordability such that people were willing to stream video using network data during their daily commute on the subway or while they queued up at supermarket checkouts, something unthinkable just a few years earlier.

Additionally, other supporting technologies had also matured to a level where they could immensely enhance the experience. In a presentation ²⁰³ discussing the rise of Douyin, Kelly Zhang drew attention to four factors—full-screen high definition, music, special effects filters, and personalized recommendations. Smartphone screens had overall gotten much larger with higher definition, greatly improving the video watching experience. Face recognition and augmented reality effects had become commonplace, which allowed

for more engaging, fun special effects and filters. Image recognition and computer vision had made very considerable advances, greatly reducing the need for manual audits of inappropriate content and allowing for the classification of videos that lacked meta-data. Most relevant of all were the advances made in big data and recommendation technology in which ByteDance specialized, which leads us nicely into the next reason.

b. Support from the mothership

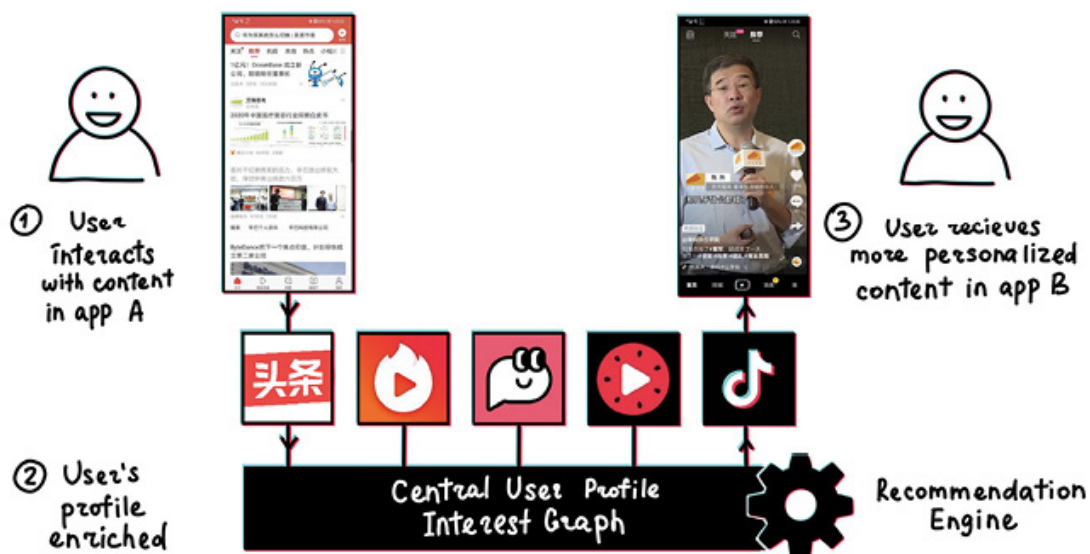
On the surface, Douyin's relationship with ByteDance during early development was similar to "Instagram and Facebook" or "WeChat and Tencent," i.e., a small, agile startup in a much larger established organization. Under no pressure to monetize early and spared the distractions of negotiating new investment rounds, they could simply focus on growth and building the best product. Remaining independent from their parent company, while also benefiting greatly from the access to technical expertise, money, and infrastructural systems, most startups could only dream of.

Yet, due to the unique way ByteDance is organized, the support received by Douyin was undoubtedly more significant than the well-known case studies of Instagram or WeChat. Douyin (and later TikTok) had a founding team, but no actual real "founders" in the traditional sense. That's because unlike most big western social media platforms, Douyin's success wasn't born from the vision of an individual; instead, it arose from a systematic process of experimentation within the organization.

The app came into being as part of ByteDance's original three-pronged short video strategy when the company chose to build their own versions of three successful and already proven models—YouTube, Kuaishou, and Musical.ly. All three would plug into the company's existing technology stack and big data – the most critical elements being the recommendation engine and existing user profile interest graphs.

ByteDance's "powerful weapon," ²⁰⁴ to quote the head of its AI Lab, was its content recommendation engine and a pre-existing database of

millions of user-profiles and interest graphs. ByteDance's core content apps shared the same backend technology stack and user data. That meant, for example, an article read and liked in one ByteDance app could directly affect the content recommendations made to them on another app.



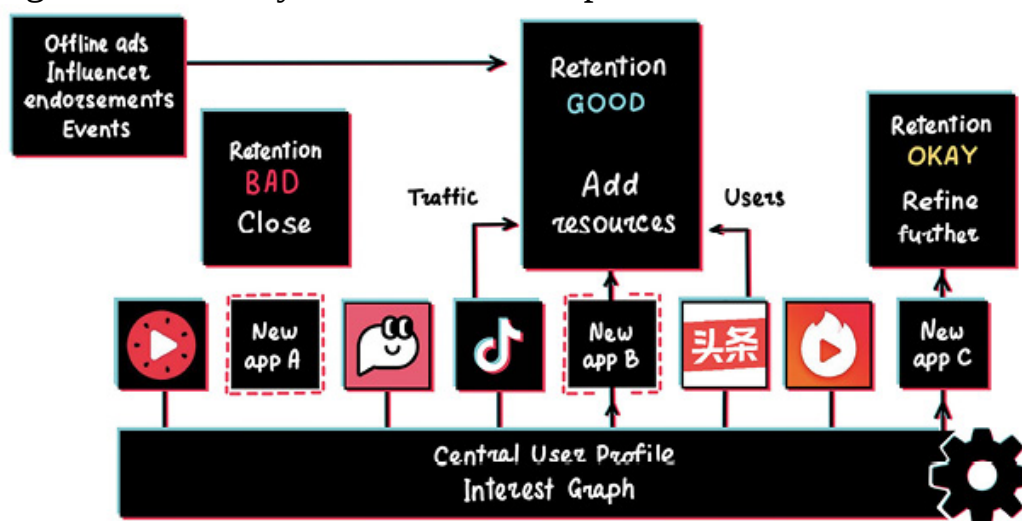
Above: The process in which a user's interaction with content in-app A leads to more personalized content shown to them in app B.

"Every one of us has an interest graph in the backend that ordinary users can't see. For example, the celebrities I am interested in, the ten companies I care about the most," ²⁰⁵ explained Yiming in one interview.

Similar to newsfeeds, short-form videos were a perfect match for this process. The user would typically swipe or tap the screen multiple times per minute, and each interaction revealed a little more about their preferences and could be used to further enrich their interest graph. In contrast, long-form videos provided much less data because people could watch a 45-minute episode of a drama series without touching the screen once.

Even as a late mover, ByteDance's executives felt they must make serious efforts to break into the short video market. Video was not only clearly the future trend, but particularly with short-form videos, ByteDance was the Chinese company best placed to reap the benefits.

Once its short video platforms got through their cold start period and gained initial traction, ByteDance simply needed to assess which are doing best and allocate resources and support appropriately. This “app factory” model is key to understanding ByteDance’s success. Pure content-based platforms are susceptible to the problem of losing popularity and going out of fashion. People often delete entertainment apps once they become bored with them. They may forget to reinstall the app when they upgrade their device with little consequence. Constantly releasing and testing new products is an effective way to mitigate this risk. ByteDance was set up to reinvent itself continually.



Above: each new app benefits from the shared technology stack, and resources are allocated based on performance.

The practice of experimentation in building multiple apps to test which gain traction can be traced back to Yiming’s previous startup, 99Fang, which created five separate apps, all focusing on real estate. Over many years Yiming and his team had greatly refined this process. Resources could be allocated based on clear metrics; if the data for one app showed strong engagement and retention, further resources would be assigned to boost its performance.

After a long cold start, Douyin’s quantitative data eventually revealed it to be promising. Budget, engineering talent, traffic, celebrity endorsements, and attention from higher management all naturally congregated towards an effective channel of growth. Another aspect of this model was boldness. If something had momentum—as Douyin

clearly did—then Yiming was not hesitant to authorize huge budgets to accelerate the process. That was especially the case with short video, as a late mover Yiming knew that time was against them.

c. The power of recommendation

Previously Musical.ly had wasted much time and effort mistakenly believing their app was first and foremost a social network. With Douyin, the company culture and technology stack was aligned with the actual true value of this format—a content platform. In a content-based community, content is more critical than people. Douyin was a recreation of television entertainment for the mobile age, not a new video first Facebook.

ByteDance’s system was powerful in its ability to identify quality videos even from small accounts and distribute them out to a broad audience. A major appeal of Douyin, and later TikTok, was that they gave ordinary people opportunities to get famous. A farmer who made videos from her little shed in the most far-flung region of China held the potential to become an overnight success on Douyin by virtue of her talent. As one prominent American venture investor said later of TikTok:

“It’s like a digitization of American Idol or America’s Got Talent, which is you’ve got these people who have aspirations to show their talents... whether it’s extreme sports videos, or comedy, or singing, or music... even if they don’t have a following today, if they make something great and they put it on the platform, there’s a chance that they’ll get discovered.” ²⁰⁶

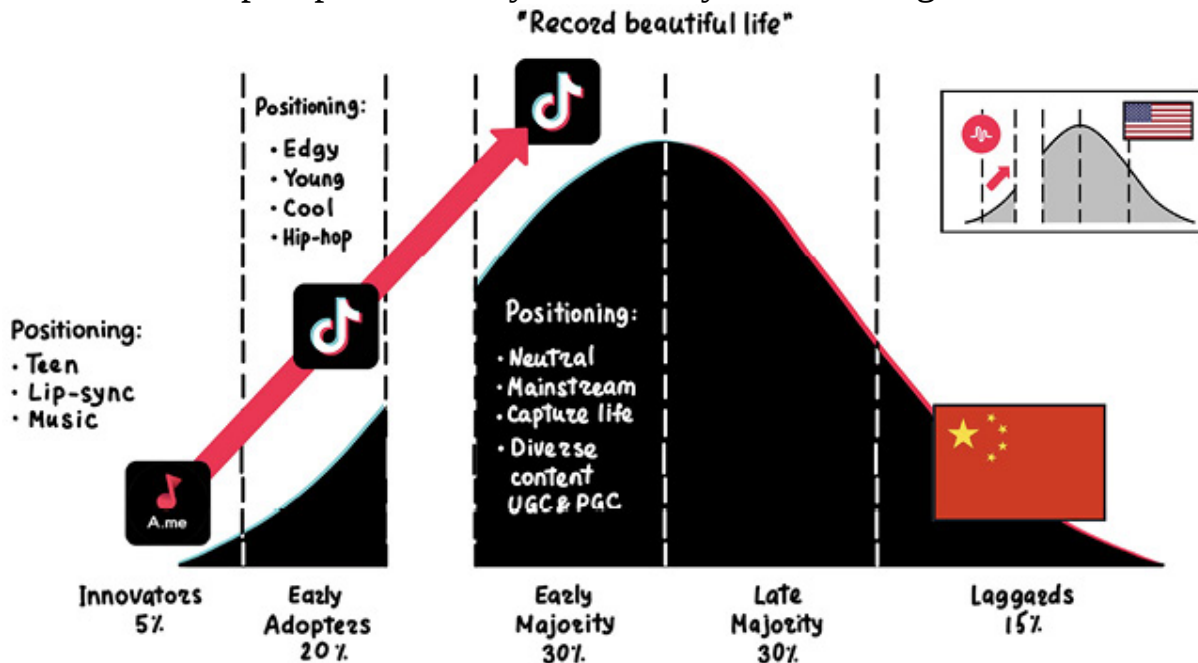
The upside was everyone felt they had a chance. The downside was an unpredictable and volatile allocation of traffic. It wasn’t unusual to see so-called “influencer” accounts with two or three colossal hit videos amongst an extensive back catalog of videos with unremarkable, mediocre engagement.

d. Positioning – “Record beautiful life”

2016, 2017, and 2018 saw three vastly different positionings for Douyin. The app and its content ecosystem evolved so quickly that the A.me of late

2016 was unrecognizable from the Douyin of early 2018.

At the very beginning, A.me's positioning mimicked Musical.ly, targeting female pre-teens through to those in their early twenties. By summer 2017, Douyin was an app for the coolest of the cool—young trendsetting art students and hip-hop idols. Early 2018 saw yet another significant shift.



Above: As Douyin broke through to different sections of the market, its positioning changed. Something Musical.ly struggled to do in Western markets.

ByteDance put in place a deliberate and systematic strategy to broaden the platform's content extending into all kinds of mid-tail and long-tail content niches. Travel, food, fashion, sports, games, pets, every major category had a rich and diverse body of content to satisfy all tastes. Promotions shifted from signing big trendy celebrities that appealed mostly to young people into rapidly accelerating content diversity and building a platform more accessible to the mass market. The app's original slogan had been "Let worship start here—devoted to the new generation of music short video communities."²⁰⁷ In early March, this was updated to the simpler, more neutral "Record beautiful life."

"Imagine that a full-screen video turns your phone into a window. You can see a vast world through this window. Douyin is a projection of this colorful world."²⁰⁸ explained Yiming during a speech to company staff.

Positioning around music had helped immensely with attracting the young early adopters' group. Musical.ly, Douyin, and TikTok are all robust music discovery platforms, something which goes back to the earliest insights of the Mindie team—**adding music to videos was the same as adding filters to pictures**. But trending hot new songs will not help as much with engaging middle-aged office workers or seniors. Being positioned as a music platform can confuse consumers as to the actual value of what you are offering, which is not music but general entertainment.

This shift in positioning was necessary to make the app more accessible, but it was also a reaction to a crackdown from Beijing's government. Early 2018 saw ByteDance fall foul of the online regulators and suffer an intense regulatory backlash, with accusations that they had failed to regulate the content on their platform correctly. Tens of thousands of videos and accounts were removed, and amongst other measures, an anti-addiction system was put in place.

This didn't stop online marketers flooding the platform. Consensus snowballed among marketers that Douyin was an opportunity that must be grabbed. Herd mentality kicked in—get in now and build a following while the competition is still low.

The Douyin cash cow

Douyin proved to be very conducive to advertising, using a targeting system that leveraged years of experience from running ads on Toutiao. Full-screen autoplay video ads could be inserted at regular intervals between standard videos with only a small notification, marked “advert,” appearing at the bottom of the screen next to the videos' text description. Marketers quickly worked out that ads shot in raw, less polished manner could easily be mistaken for regular user-generated videos and trick people who otherwise would immediately skip anything they thought was a promotion.

Douyin was similar to Facebook or Instagram in one crucial regard. Influencers, brands, businesses, and ordinary people voluntarily spent their own time and money, creating very high-quality videos and then uploaded those videos to the platform for free. Contrast this with long-form professionally generated video platforms such as Netflix, Disney+, Hulu, or

Tencent Video. Such platforms spend vast sums of money to either license or produce high-quality content themselves, typically charging a subscription fee to reap back these substantial upfront costs.

ByteDance knew they needed to build infrastructure to help these creators monetize their hard-earned account followings. The easier it was to make money on Douyin, the more effort people would take to create high-quality content and build active communities. Livestream tipping was a tried and tested model prevalent on Kuaishou; Douyin successfully integrated this model also into Douyin. Arguably a more direct way to monetize attention was e-commerce. Once accounts hit a certain number of followers, creators could add e-commerce links to their videos to buy items, which opened immense opportunities. Douyin's algorithmic distribution model led to popular videos being rewarded with vast amounts of traffic. Combining this with e-commerce meant videos of novel products that lent themselves well to being demonstrated in a short few second video could drive very significant sales; top categories included clothing, cosmetics, plus food and beverage products. During an online shopping festival held in December 2018, sales from the top 50 Douyin accounts reached over 100 million yuan (\$14 million), driving 1.2 million transactions in a single day.²⁰⁹

Anguished peers – Tencent

How did competitors react to the unexpected breakout success of Douyin? Answering this question provides an informative counterexample to our later analysis of Facebook's reaction to TikTok. The local rival we will look at is Tencent.

Owner of famed “super-app” WeChat, Tencent apps account for roughly half the time Chinese spend on their smartphones. The Shenzhen headquartered internet leviathan dominates China's online social and lucrative gaming industries. In short, any casual observer would expect Tencent to end up dominating short videos in China.

Tencent entered the short video market early with a clone of Vine called Weishi in September 2013. By coincidence, the office of Weishi was just one block away from ByteDance in northwest Beijing. Initially, Weishi ²¹⁰ pursued the path of inviting many celebrities to the platform. CEO Pony Ma recorded videos on his own account to further attract media attention. The

app gained initial traction and after heavy promotions over Chinese New Year, claimed 45 million daily users.²¹¹ Yet with only basic features, no clear direction, and a lack of internal company support, it soon faded away, sharing a fate similar to that of Vine itself.

Weishi videos were eight seconds, a tad longer than Vine's six-second limit but still very compact. Normal users found it hard to complete a story arc in just eight seconds. There were no filters or beautifying special effects to make it easy for people to create. Compounding this effect, the early Android and Symbian smartphones that dominated the Chinese market at the time had low-resolution screens and poor cameras.

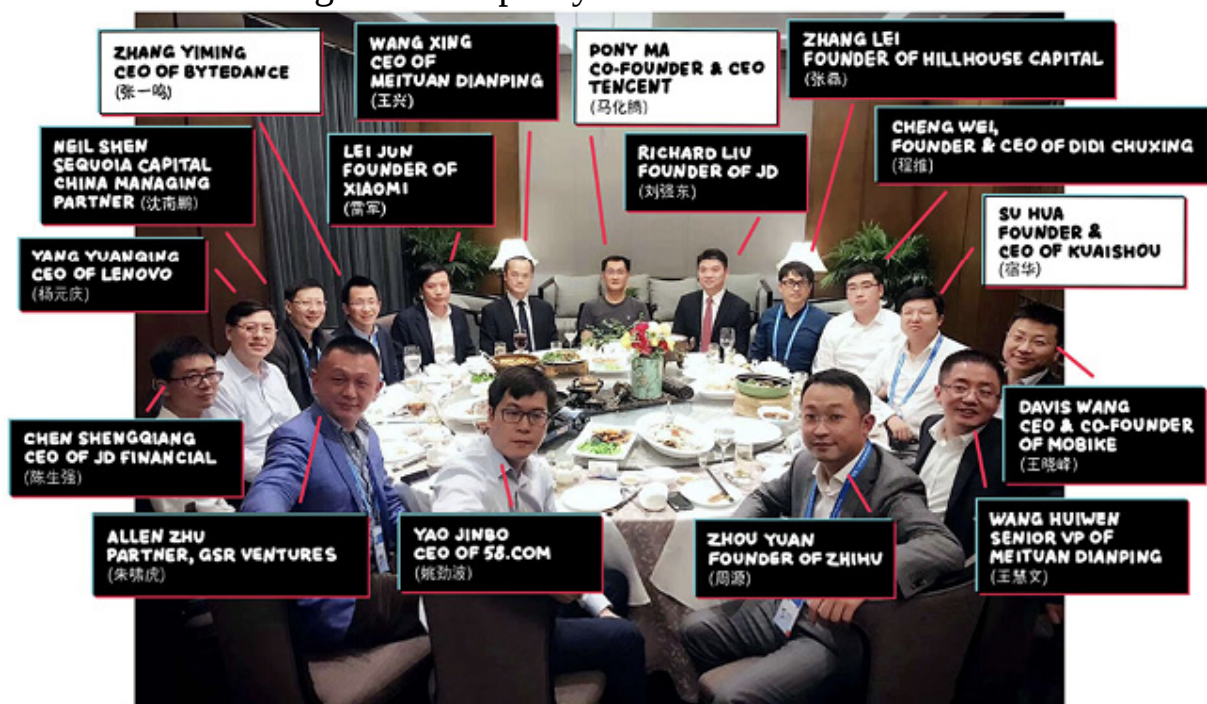
Short video was in an exploratory phase at the time, and the industry hadn't worked out its true value. Everyone was muddling through, and there were few clear monetization opportunities. No one expected short video would grow to be such a large market. Long-form video equivalents to Netflix, Hulu, or YouTube were considered much more critical with established commercialization models such as premium subscriptions and pre-roll adverts.

In summary, Tencent had acted upon the trend of short video much earlier than ByteDance. Within just eight months of Vine becoming popular in the U.S. market, Tencent already had released its local counterpart. Yet this failed to translate into any kind of early mover advantage. By March 2015, the general manager of Weishi had left the company, and the app stopped updating.

Fast forward to early 2017, and Tencent was on a high. Super-app WeChat's payments business was booming. Tencent Video was leading the Netflix style long-form video market, while the gaming department's smash hit "Honor of Kings" experienced explosive growth. Tencent had hit a home run; the game was a money printing machine with a monthly turnover exceeding \$400 million (3 billion yuan).

Rather than compete across every internet category itself, Tencent developed a strategy to invest in allies, backing best in class companies with a minority stake and lightly integrating their services into the broader Tencent ecosystem.

The company announced it would invest \$350 million into Kuaishou, the market-leading short video app at the time, pushing its valuation to \$2.5 billion. The deal was widely judged to be a smart move—another critical ally in the booming sector of short videos and another savvy Tencent investment, which would lead to a handsome return. It followed a pattern of similar moves Tencent had taken in e-commerce, search, food delivery, and ride-hailing. One month after the Kuaishou investment, Weishi announced that it had officially closed. At the exact time, Douyin was coming out of its long cold-start period, Tencent was formally bowing out of the short video market and choosing to back a proxy.



Above: Yiming attends a famous gathering of many of the most influential Chinese internet industry CEOs in 2017. The seating reveals the pecking order of importance with Tencent CEO Pony Ma holding the best position. Alibaba's Jack Ma is notably absent, as is any female.²¹²

The speed at which Douyin came out of left field to become a breakout success took the entire industry by surprise. By the Chinese New Year of 2018, the emergence of Douyin had become clear to all. Tencent scrambled to react. It doubled down on Kuaishou, increasing its stake as the lead investor of a new \$1 billion funding round. But Kuaishou's core user base was predominantly in rural areas of China, far removed from Douyin's

stronghold of young urbanites with greater purchasing power. Tencent felt threatened. In early 2018, it decided to re-enter the short video field—Weishi would be resurrected. ²¹³

Weishi 2.0 - operation kill Douyin

In early 2018, Tencent quickly organized a team to revive Weishi, rebranding and revamping the entire platform to a structure similar to Douyin. Internally Weishi was branded as a “strategic product,” a title bestowed by higher-ups to only a handful of critically important platforms. The title signaled to the entire organization that Weishi was now a big deal. In just a few months, the Weishi team expanded to 400 people.

Tencent’s approach to taking out rivals was consistent—clone the competitor’s product and leverage its immense online services empire to drive vast numbers of downloads. At a Weishi team strategy meeting, a simple direct “north-star” goal was set—match Douyin in terms of user numbers, retention, and time spent in the app.



Above left: A 2013 advert for the original Weishi reveals its substantial similarity to Vine, even the color pallet was similar. Above right: A screenshot of the updated 2018 Weishi shows its striking resemblance to Douyin.

Tencent completely blocked ByteDance from purchasing any form of online advertisement on their platforms. Additionally, Weishi signed a handful of young celebrities²¹⁴ to post videos exclusively on its platform. April 2018 saw Tencent announce a massive three-billion-yuan subsidy program for Weishi content creators. This move created much discussion online and made public Tencent's ambition to rival Douyin.

"Download Weishi" pop-ups and notifications started appearing across all kinds of Tencent platforms, sometimes in quite spammy and annoying ways. Even hyper-conservative WeChat added a promotion for Weishi when users posted videos.²¹⁵

After half a year, Weishi reached four million daily users. "With Tencent's resources, this speed can only be regarded as a complete mess,"²¹⁶ was the harsh judgment of an executive at a Tencent invested company. The retention of Douyin was around 80%, whereas Weishi was only 43%. Weishi's usage time was just a quarter that of Douyin.

What went wrong?

The first thing that needs to be stated upfront in dissecting Tencent's failure is that no one, including ByteDance, foresaw how important Douyin would become. Weishi 1.0 entered the market too early, and Weishi 2.0 joined too late, both missing the window of opportunity. ByteDance executive Chen Lin later judged Tencent's choice to shut down Weishi 1.0 as a "huge mistake."²¹⁷ It took a little over half a year for Douyin to go from an insignificant blip to explosive growth. This gave competitors like Tencent almost no time to react.

Along with most others, Tencent misjudged the market opportunity and missed the best time to face off with Douyin. Weishi 2.0 was entering into a market where Douyin already occupied a critical mass of mindshare and had become synonymous with short video.

The advantage of such a situation was familiarity with Douyin meant that people immediately understood how to use Weishi. Such a shallow learning curve lowered barriers to switching. The disadvantage was a lack of differentiation; early on, even Tencent staff struggled to pin-point what was unique about Weishi. Once people discovered the app was just a bland clone, they simply gave up and moved back to Douyin.

In 2017 Douyin had worked out a precise positioning—“young music community.” This was its entry point into the market. The team had created a strong identity and built a vibrant community of content creators before transitioning into a more generic and mainstream positioning along with the growth of the users. Being a latecomer under high pressure to grow fast meant Weishi skipped this phase entirely and went directly for the mass market. Consequently, it struggled to build a true community or a distinctive identity.

To rapidly build a large content pool, Weishi fast-tracked the whole process, paying specialized professionals, agencies, and studios. This type of professionally generated content was less able to inspire individual participation. Mimicking others through shared memes and challenges was the key driver to motivate users to record their own videos for both Musical.ly and Douyin. This lack of user-generated content created a vicious cycle; fewer user-generated meme videos to mimic meant people posted less, which meant there were even fewer videos to emulate, and so on.

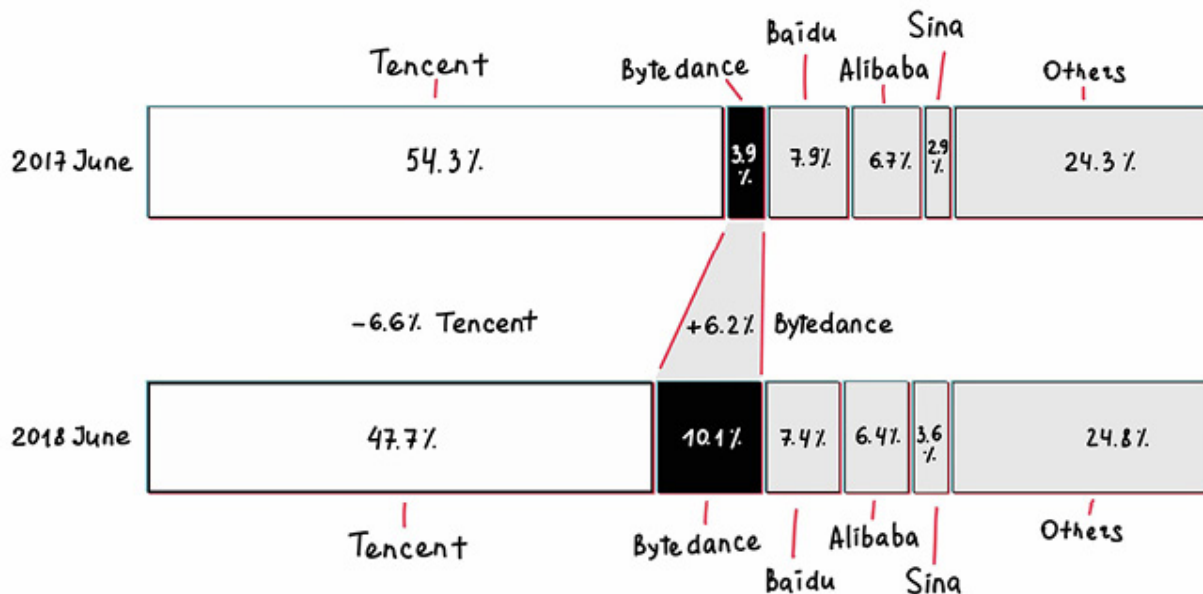
One distinct advantage Tencent held over ByteDance was its dominant position in social networking. Logging into Weishi using a WeChat account meant people could connect easily with friends and family who also used the app. This kind of “social graph” integration worked well for many of Tencent’s top gaming titles, but it was much less effective for short video.

Douyin’s success relied heavily on algorithmic recommendation, which removed the need for social relations. While the social graph held some potential for improving those recommendations, it also brought with it negative aspects as adding family, acquaintances, or co-workers can inhibit the sense of freedom. Getting things right was a delicate balance. Users want to be comfortable about what they upload and who they follow—this often requires a degree of anonymity and disconnection from their daily contacts. Meanwhile, the content needs to be relatable and organic enough to feel a sense of belonging and participation.

A major threat

By mid-2018, it was clear to the Chinese internet industry that ByteDance was now a threat to Tencent. WeChat met the need for communication,

while Douyin fulfilled a need for entertainment. On the surface, these are entirely different, but they played very similar roles for their parent companies—sucking up attention and acting as distribution channels for other services. Traffic is king.



Above: From June 2017 to June 2018, time spent on ByteDance apps as a percentage of all time spent on mobile in China grew 6.2% while Tencent's share fell over the same period by 6.6%. ²¹⁸

At the time, ByteDance did not yet constitute a severe threat to Tencent in the lucrative mobile games market, but it was easy to imagine a world in which they did. Tencent only needed to look back at its own past. During the PC era, the company had moved gradually from social into games, step by step, overtaking the previous market leaders. Dominating social had gifted the company a sustainable unfair advantage over competitors—low-cost distribution channels. ByteDance now enjoyed the same low-cost distribution advantage. Given gaming was one of China's most lucrative online services, it would surely only be a matter of time before ByteDance started to operate its own games and begin to endanger Tencent's core business.

Tencent had burned a lot of cash and placed considerable time and effort into making Weishi. Even when it was abundantly clear the app had little chance to become a viable competitor to Douyin, Tencent continued to

persevere. “If you don’t compete, you may lose more.” was the judgment of one analyst.²¹⁹

171 <https://cn.technode.com/post/2016-09-20/toutiaohao/>

172 <https://www.pingwest.com/a/51495>

173 <https://36kr.com/p/5053185>

174 Watermelon is a clumsy four-syllable mouthful in English but a more concise two characters in Chinese - **Xīguā** . The Chinese character for melon holds connotations of cuteness. The general point being the name sounds much better in Chinese than English.

175 <https://www.xiaokaxiu.com>

176 <https://www.36kr.com/p/204254>

177 Digital Red and Qianzhi Unlimited, where she had spent 3 and 4 years, respectively

178 The original website amemv.com now redirects to TikTok.

179 Image source: <http://www.pc6.com/az/357603.html>

180 Image sources: <https://v.douyin.com/JdY39Xp/>

181 <https://youtu.be/cva6C8G-GAc?t=21597>

182 <https://zhuanlan.zhihu.com/p/91711796>

183 Douyin's distinctive dark interface was inspired by Meipai (), an already established trendy short video app for Chinese youths. <http://www.downxia.com/zixun/11926.html>

184 <https://v.douyin.com/JewUd4v/>

185 <https://v.qq.com/x/page/v0509dplj3n.html>

186 <https://mp.weixin.qq.com/s/kc-10P4vIJX01oj5ptsjJQ>

187 https://www.iqiyi.com/a_19rrgxwtfh.html

188 <https://new.qq.com/omn/20180515/20180515A1H2ZP.html>

189 <http://news.ifensi.com/article-14-4374160-1.html>

190 <https://www.ixigua.com/i6467320259031335437/>

191 Image source: http://science.china.com.cn/2017-09/03/content_39110253.htm

192 <https://mp.weixin.qq.com/s?biz=MjM5MDczODM3Mw==&mid=2653030018&idx=1&sn=fb673ae1cefb132b330245fbc3dc958&>

193 <https://36kr.com/p/1723823521793>

194 <https://zhuanlan.zhihu.com/p/91711796>

195 <https://hans.vc/toutiao/>

196 <https://v.douyin.com/31Ur7P>

197 <https://www.youtube.com/watch?v=YHUN1UofcYM>

198 <https://www.theverge.com/tldr/2018/1/26/16937712/karma-is-a-bitch-riverdale-kreayshawn-meme>

199 https://youtu.be/U49yH_F031U?t=6

200 <https://youtu.be/kFVevz3HFMI?t=51>

201 <https://36kr.com/p/5136013>

202 The popularized phrase among marketers was “2 Micros, 1 Shake” **Shuāng wēi yī dǒu**

203 <https://www.youtube.com/watch?v=aqQtY-wG9Dg>

204 <https://www.technologyreview.com/2017/01/26/154363/the-insanely-popular-chinese-news-app-that-youve-never-heard-of/>

205 http://ly.fjsen.com/2015-09/25/content_16680876_all.htm

206 <https://thisten.co/njlwa/8eZbR0bz89GTOHnoYA4HFOtu1xiX68cH5IFFJYP>

207 <https://new.qq.com/omn/20180515/20180515A1H2ZP.htm>

208 <https://m.pedaily.cn/news/441247>

209 <https://walkthechat.com/simple-guide-to-douyin-store-case-studies-and-how-to-create-one/>

210 Chinese: **wēi shì**, a literal translation would be “Micro Video”

211 <https://www.pingwest.com/a/106343>

212 Image source:

213 <https://weishi.qq.com/>

214 Most notably Huang Zitao , Zhang Yunlei and Song Zuer

215 <https://www.ifanr.com/minapp/1101125>

216 <http://m.caijing.com.cn/api/show?contentid=4608050>

217 <http://finance.sina.com.cn/chanjing/gsnews/2018-05-06/doc-ihacuuvt8132798.shtml>

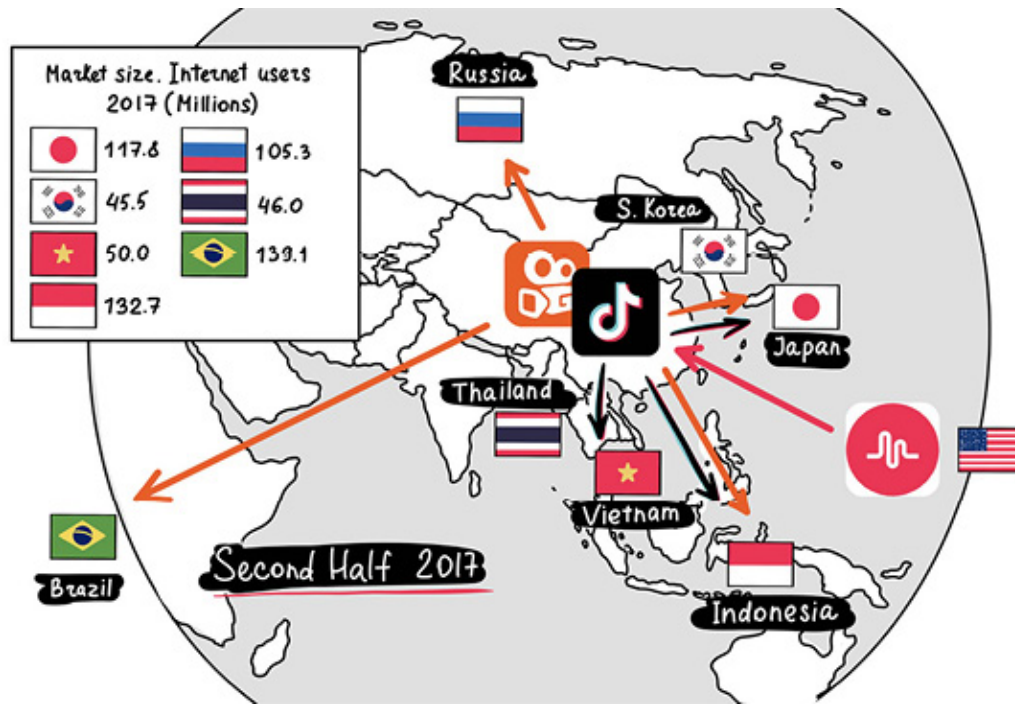
218 Data source: QuestMobile

219 <https://mp.weixin.qq.com/s/kc-10P4vIJX01oj5ptsjJQ>

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CHAPTER 7

GOING GLOBAL WITH TIKTOK

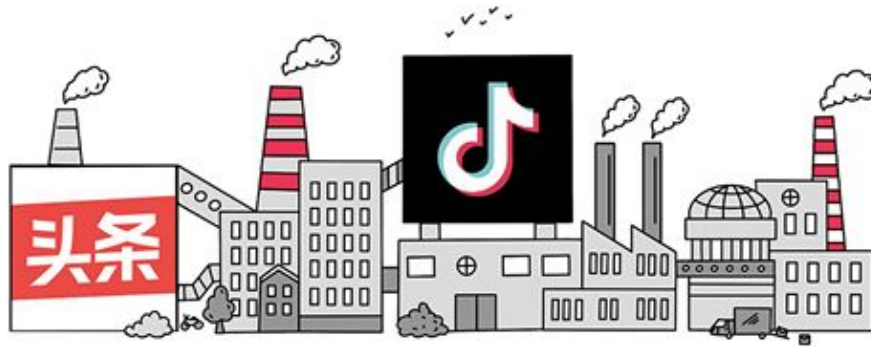


“If I were to do a startup again, I wouldn’t choose to do short video, because you really need to spend a lot of money.”

Kelly Zhang, ByteDance China CEO

Image: Douyin and Kuaishou start expanding into selected international markets under the names TikTok and Kwai, respectively. Musical.ly reenters the China market at the same time.

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Chapter Timeline

2017 Feb – ByteDance acquires Flipagram

2017 May – First version of TikTok launches on the Google Play store

2017 June – Musical.ly re-enters China under the name Muse

2017 Nov – TikTok briefly ranks no. 1 on the App Store in Japan

2017 Nov – ByteDance acquires Musical.ly for \$800 million

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“GOING GLOBAL IS A MUST” was Yiming’s direct message to Chinese staff.

The reasoning underpinning his absolute insistence was undeniable—four-fifths of the world’s internet users lay outside China’s borders. In the internet world, the fixed costs of product development were high, but the marginal cost of serving an incremental new user was often close to zero. Restricting itself to only one market, even the world’s largest internet market, would ensure it was all but impossible for ByteDance to truly compete with the Googles and Facebooks of this world. Yiming’s longheld dream of “going global” had been part of the organization since founding, and in preparation, he had persistently studied English for several years, reaching professional working proficiency.

“Google is a company without borders,” stated Yiming at one event. “I hope ByteDance will be as border-less as Google.” ²²⁰

As part of its globalization plan, ByteDance acquired Los Angeles-based Flipagram for \$50 million in February 2017. The app helped people gather pictures taken on their phone and turn them into slideshows with music. ²²¹ At the time, Flipagram claimed 36 million active users. To oversee the company’s integration into ByteDance, Yiming brought a small group over from Beijing to the offices in LA. “The team was embarrassed, because very few of them had been abroad, and they all expressed a lack of self-confidence,” ²²² recalled Yiming in a later speech. The concerns were many, but, in the end, Yiming concluded, “the actual operation was not as difficult as imagined.” ²²³

Yiming had a simple strategy for evolving ByteDance to the next level – hire or acquire the absolute best people and infuse their knowledge into the organization. To improve the company’s nascent recommendation engine, Yiming relentlessly poached top-level experts from Baidu. To start monetization, he headhunted one of the rising stars in traditional media advertising, Zhang Lidong. Similarly, Flipagram, along with other early deals such as news aggregators Dailyhunt in India and BABE in Indonesia, allowed Yiming access to vital local business know-how and expertise. ByteDance could pick the brains of experienced founders and quickly accelerate its understanding of local market nuances.

In China, the short video market had already become what venture capitalists called a highly contested “red sea.” The competition had intensified as the format became more popular. Every mid to large-sized internet player in China wanted to carve out their small slice of the pie, prompting many to look abroad for growth opportunities. There were some encouraging success stories.

At the time, the best examples of domestic Chinese developers going abroad, aside from the notable exception of Musical.ly, were utility apps that held universal cross-cultural appeal and did not require localization. Examples included video editing apps such as VivaVideo and VideoShow, which had achieved impressive numbers with minimal investment. By the end of 2016, VideoShow claimed 100 million registered users worldwide, and monthly active users exceeding 11 million with zero marketing spend.

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By the second half of 2017, Douyin had proved itself in China. Sophisticated technologies, including video analysis, AR filters, and ByteDance’s proprietary recommendation engine, had laid the foundations for rapid growth. In theory, at least, these technologies transcended geographical boundaries, but the team needed to verify this assumption. Was Douyin’s success in China truly translatable into other markets? Could they build a systematic playbook to enter other countries regardless of culture and localized content preferences? We were about to find out, as ByteDance announced in August 2017, they would plow hundreds of millions of dollars into helping Douyin go international.

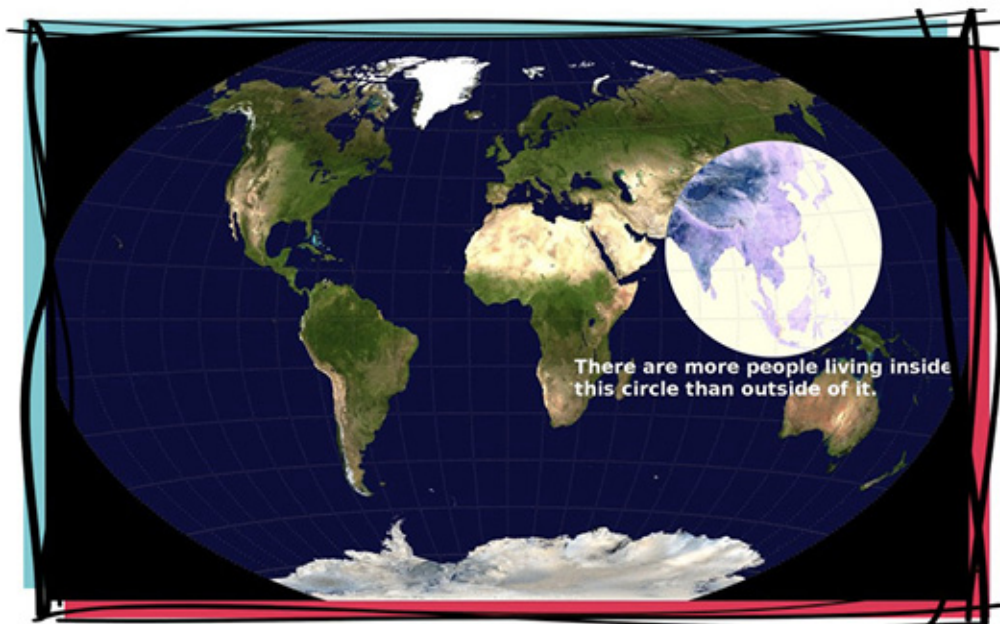
A new identity: TikTok

China’s local internet holds so many unique characteristics ²²⁵ tailored to local habits and penchants that it is common practice for popular Chinese apps to create a separate “international version.” Douyin was no different. When it came to deciding what to change, one of the first decisions was to change the name but keep with the distinctive black visual identity and logo.

“TikTok, the sound of a ticking clock, represents the short nature of the video platform.” ²²⁶ was the official explanation behind the new name. It was easy to pronounce in many major languages while also alluding to the

central importance of audio and drawing attention to the short-form nature of the content. Unlike its predecessor Musical.ly, the name drew no direct connection to music, lip-sync or dance.

In the earliest months, TikTok used “awesome me” as its tagline and the original A.me website amemv.com ²²⁷ before purchasing tiktok.com. The slogan ‘Make every second count’ was later added for Western markets as a message to content creators, reinforcing the short nature of the platform’s content.



Above: Asia holds more than half the world’s population. ²²⁸

When Chinese internet companies begin internationalization, they typically first look towards their backyard, Asia, home to over half the world’s population. Many Asian countries share broad similarities with China, with mobile-first users who skipped the 1990’s and 2000’s Web 1.0 and 2.0 desktop internet eras. In general, Asian countries exhibit highly active social media use and strong demand for online entertainment, characteristics that boded well for TikTok’s expansion.

TikTok saw it necessary to tailor their approach to each country leveraging localized promotion channels and native-language influencer ecosystems. Some markets would be easier than others, yet one of them stood out as being the toughest nut to crack—Japan.

Japan

TikTok's first offices outside China were hidden deep among the bustling streets, endless stores, and vibrant nightlife of Shibuya ward, Tokyo, a mecca for Japan's young fashionistas. "I'm sorry it's a little small here," ²²⁹ an embarrassed employee apologized to a visiting journalist. On the sixth floor, within a shared office space, lay the local operations staff for TikTok Japan. The earliest team had fewer than five people, yet the room was so small they were unable to all work inside at the same time. ²³⁰ Stacks of materials piled up underneath the desks. The words "work hard, have fun, make history" had been cut out in white paper and pasted on to the wall.

Exceedingly few Chinese internet services thrived in the inhospitable Japanese environment, making the market somewhat of a litmus test. "If something can be accepted by Japanese, then basically Southeast Asian users and other countries across Asia can all accept it," ²³¹ explained a veteran Chinese software executive.

Japanese are known to be self-conscious and value privacy. Many prefer to remain anonymous online and are reluctant to use their real name or show their face on publicly accessible social media accounts such as Twitter or Instagram. TikTok expected users to not only show their faces but film themselves.

Another challenge was Japan's tight labor pool. Talented young Japanese held a strong preference to work in large corporations or state agencies. An unknown new entry from China, like TikTok, would find it almost impossible to recruit top-level local talent. Given such, TikTok resorted to hiring Chinese staff with in-depth knowledge and understanding of Japanese society. Several had previously worked in Japan for other Chinese tech brands, including WeChat Pay and Shenzhen-based drone manufacturer DJI.

Lastly, the Japanese had a reputation for being cautious of internet products from rival east Asian economies. A prominent example was Japan's dominant messaging app, Line, which did its absolute best to downplay the Korean roots of its parent company Naver.

The early strategy for entering Japan followed a similar path of Douyin in China during its first low-key "cold start" period. Douyin already had a

large pool of quality videos that could easily be batch imported over into TikTok and used as a foundation, but this was not enough. One lesson learned from Douyin had been that user-generated content apps first need to cultivate a committed group of local, high-quality seed creators that define the tone of the community and can generate memes for others to mimic. That took time. Rushing in and spending heavily on ads without any kind of community would be counterproductive.

The Tokyo team placed much effort into identifying and approaching suitable online influencers for the new platform. This group could create high-quality content, and build awareness, in addition to a part of their existing follower base converting over to the new platform. There were two kinds of influencers: celebrity stars and niche area KOLs (Key Opinion Leaders). Celebrities had broader audiences, usually measuring in the millions, while KOLs in niche areas, such as cooking or dance, possessed smaller but loyal and engaged follower bases.

The big problem was the gatekeepers—talent management agencies that controlled access to celebrities and the best KOLs. For TikTok, these organizations were an impenetrable fortress; no one knew TikTok, which meant no agency would take them seriously.

At last, a breakthrough came with female celebrity Kinoshita Yukina.²³² Once the operations team discovered she had become a user, they immediately contacted her representative office. Kinoshita enjoyed using TikTok very much and was open to collaboration, but her agency expressed strong reservations. “It took around six or seven rounds of discussions to seal the deal finally. The star studios in Japan are particularly prudent, so we need to talk to them time and again to familiarize them with our product and show our sincerity for cooperation,”²³³ explained the then director of TikTok Japan.

Having painstakingly secured their first star, the road to convincing others became less bumpy. Kyary Pamyu Pamyu, a singer with five million Twitter followers, girl band “E-girls” and famous YouTube blogger “Ficher’s”²³⁴ were some of the earliest big names to endorse TikTok officially.

Additionally, the operations team ran promotional accounts on other platforms. TikTok Japan's Twitter account was registered in May 2017, making it probably the very first TikTok promotional account.²³⁵ Videos posted reveal a similar content style to early Douyin, dance and lip-sync for young people.

Some early adopters described discovering TikTok from Twitter and embracing the app because of its advanced video editing suite, filters, and special effects. Neither YouTube nor Instagram offered such diverse options making TikTok a useful tool to create videos to upload on other platforms. Each video's watermark acted as a mini advert, helping drive downloads of TikTok. "TikTok comments mostly discussed how the video was filmed, and the shooting techniques, comments on Twitter will be different,"²³⁶ noted early Japanese TikTok influencer Kotachumu, comments that reveal the strong video production focus of TikTok's earliest users.

A key reason why TikTok was able to do surprisingly well in Japan was a lack of similar products in the market. Ultimately, TikTok found it was competing with itself. Both the local short video competitors such as MixChannel²³⁷ and the famous Silicon Valley offerings of Facebook, Snapchat, and YouTube failed to offer anything similar.

To address a widespread cultural aversion to individualism, the operations team emphasized challenges that allowed people to participate together in groups²³⁸ and filters that could be used to make faces less recognizable, reducing self-consciousness and allaying concerns over physical appearance. Much operational expertise had been built up from Douyin that could be transferred over to TikTok Japan. This included a proven back catalog of highly engaging challenges that would generate online buzz, luring in more local stars and celebrities.

As mentioned, TikTok's early staff were Chinese with in-depth local knowledge, but soon it switched to hiring locals exclusively. As the platform's reputation grew bit by bit, hiring became more manageable, and gradually, decision-making power was transferred away from Beijing to the Japanese branch. Later ByteDance became more confident and began offline promotion with adverts for TikTok filling the Tokyo Metro.

To get around a ban on product placements on Japanese TV networks, the team cleverly worked out it was easier to feed producers interesting, report-worthy stories. Television reporting on TikTok anecdotes began to grow, becoming “almost a daily occurrence” by early June 2018, according to TikTok Japan staff.

Globalize products; localize content

A similar story to Japan was playing out in other Asian countries as ByteDance tested the waters and refined their systematic playbook. TikTok’s early markets were Asian, including Vietnam, Korea, Thailand, and Indonesia. At the same time, its Chinese competitor Kuaishou was also making its first moves outside of China, opting to focus its efforts on Korea together with several large-population emerging markets such as Russia, Brazil, and Indonesia.

The indications were pointing towards TikTok having the potential to work anywhere. At its core, the app simply served up personalized videos based on user profile tagging. This basic concept was able to create a compelling experience across all cultures. In one interview, Yiming described the ByteDance approach as to “globalize products and localize content.”

Standardized elements – universal across all markets

- **Branding** : the TikTok name, logo, and distinctive visual identity
- **UX, UI**: The core features and design, product logic
- **Technology**: recommendation, search, classification, facial recognition

Localized elements – tailored to specific geographies and languages

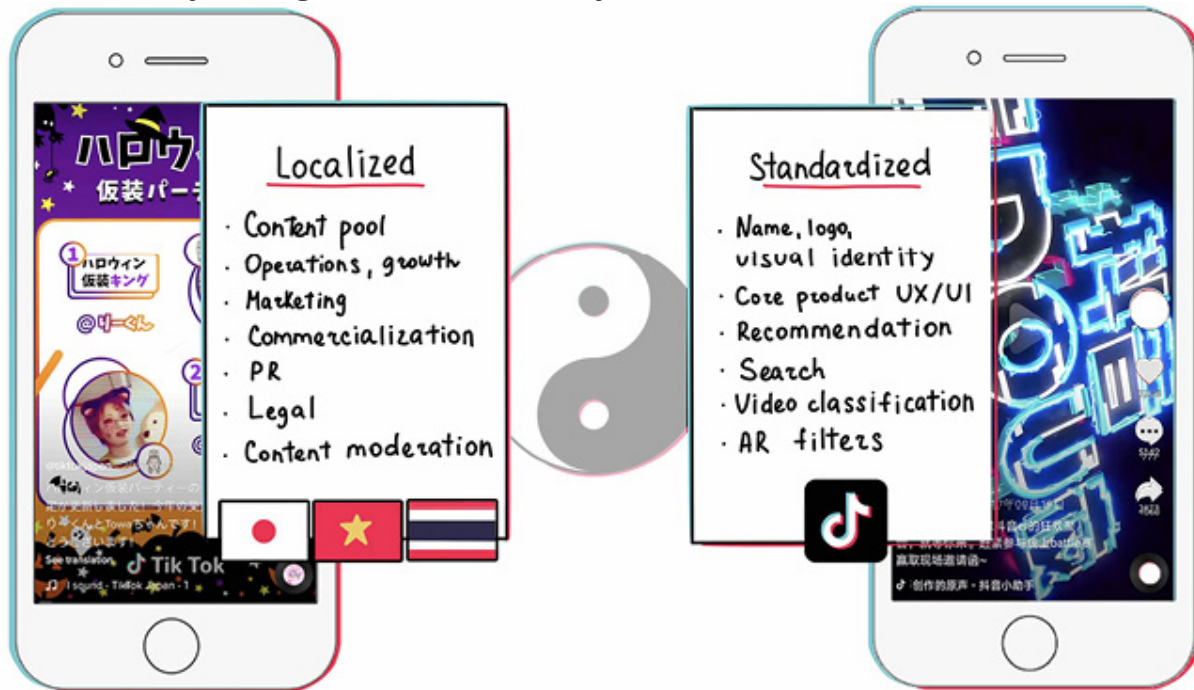
- **Content**: the pool of recommended videos
- **Operations**: marketing, promotion, and growth

Ancillary services could also be localized once the user base reached scale. These included

Commercialization: ad sales, business development

Others: government relations, legal and content moderation

Central to this system was the concept of regionalized content pools based on geography, culture, and language.²³⁹ The core TikTok experience was the “For You” feed, which was localized for each market. Japanese users would not be recommended content from Indonesian accounts and vice-versa. Each country or region was essentially an isolated island.²⁴⁰



Above: Localized and standardized fuse together creating a tailored experience for each market.

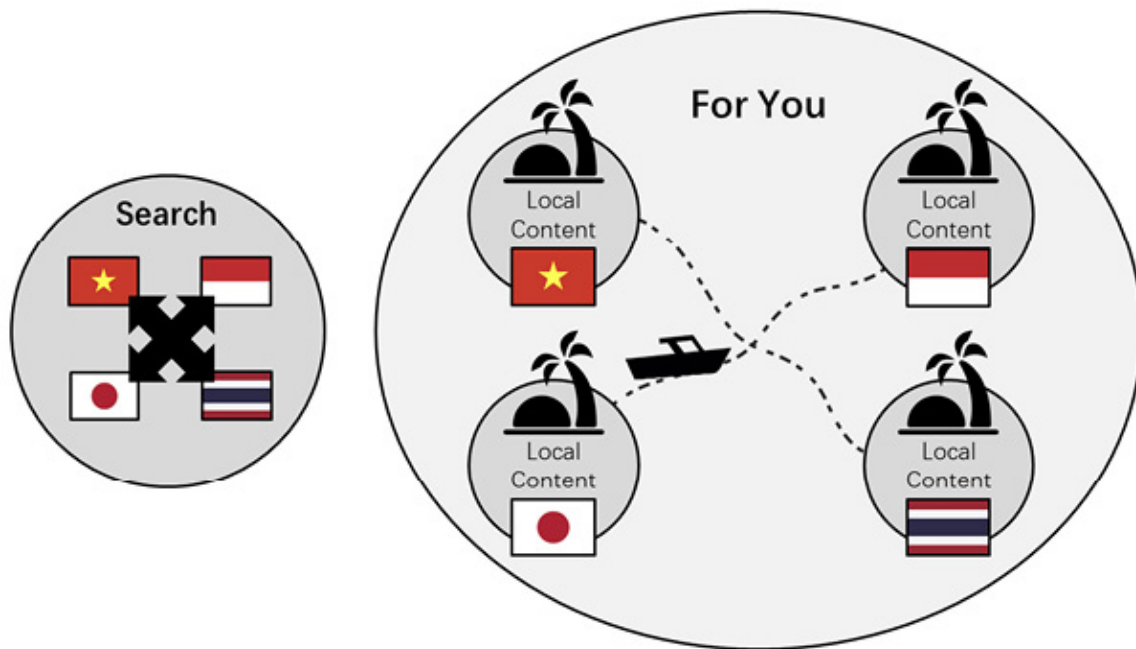
For a Japanese user to find and view videos from a friend in Indonesia, they would have to use the search function, which allowed users to see videos from any global TikTok account or hashtag they wished to seek out. Yet only a small proportion of traffic came from people using search; by comparison, the “For You” feed was the app’s core experience and where almost everyone spent most of their time.

Occasionally, content would be imported between countries to diversify categories and expose users to different video styles. People needed to be shown the possibilities and educated to create new forms of content.

Given how Douyin and TikTok were separate networks, there was much scope for beneficial exchanges between the operations teams. When something became popular in China, the teams would judge if it was

suitable to introduce into other markets. Vice-versa, something trending on TikTok, may be brought into Douyin.

Videos imported from Douyin and other regions could be used to kickstart a new region's content pool. These were essential “teaching materials” to guide and inspire. The videos had been verified as enjoyable and reproducible, allowing users to imitate them easily.



Above: Search was global, while each region drew from their isolated local pool of videos to display a personalized ‘For You’ feed.

Musical.ly comes home

In 2015 the China version of Musical.ly under the unorthodox but memorable name “Mama Mia” had utterly flopped. “We only went to America because we found no opportunity in our own market.” explained co-founder Louis Yang. Initially, they had concluded that Asian people were too reserved and not good at expressing themselves freely through video.

Two years later, the founders finally felt it was time to take another shot. Back in 2015, the Musical.ly team was less than ten people; now, they employed around a hundred people worldwide with extensive operations and engineering teams based out of Shanghai. They had gained invaluable

experience, coupled with the financial backing of prestigious venture capital firms. Even so, “coming home” to China wasn’t easy.

Musical.ly broke all the standard promotional rules. They didn’t create adverts. They didn’t subsidize content creators, and they didn’t pay celebrities. Since its founding, Musical.ly had spent almost nothing on marketing. Instead, they relied on word of mouth and leaned heavily into fostering strong user communities. Groups of highly motivated super fans ran the company’s various social media accounts on other platforms and organized offline meetups, all without pay. Some users loved Musical.ly so much they just wanted to help.

That kind of organic growth strategy was simply not viable in China, where the internet market was a relentless, unforgiving, dog-eat-dog world. The competition was brutal, the pace of change was rapid, and the cost of acquiring users had long been far higher than in overseas Western markets.

To make it in China, Musical.ly needed to take a different approach and be willing to spend. The founders decided to base the China operations team in Beijing, away from their well-established Shanghai headquarter offices. A new team with a more competitive culture.

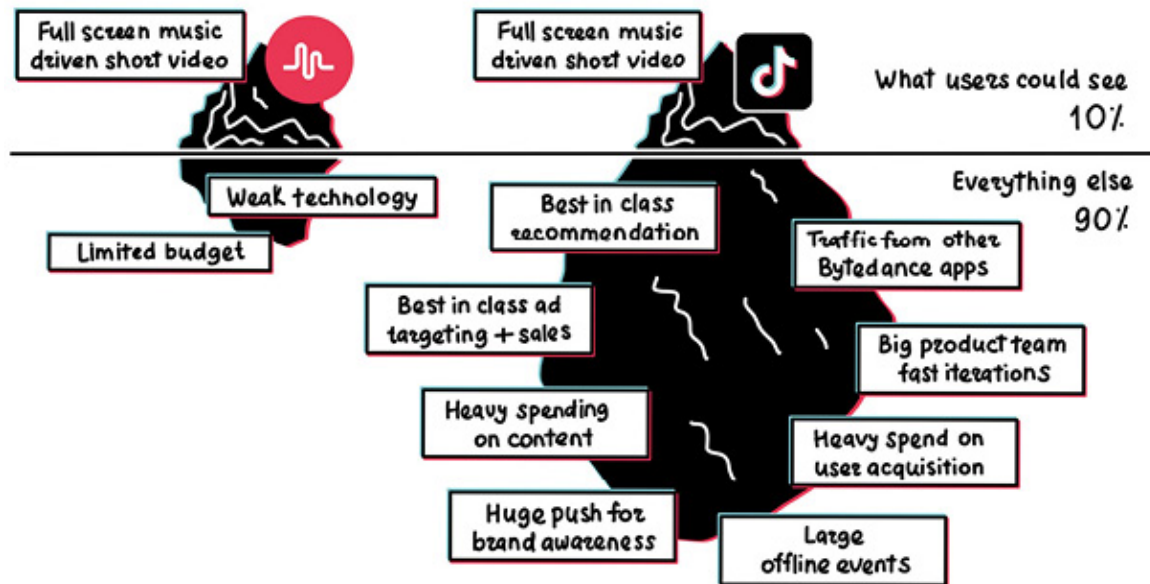
Early June 2017 saw Musical.ly officially re-enter China. Ditching the old “Mamma Mia” name, they rebranded under the identity “Muse,” looking to leverage their already established reputation in the American and European markets. It did little to help.

They were now a latecomer in a previously fragmented short video sector that was rapidly consolidating around the largest companies with the most resources. Co-founder Louis reflected: “I regret not (re)entering China earlier. Now the streets are shaking, making me extremely uncomfortable.”²⁴¹ Louis was referring to Douyin, the Chinese character “dou,” meaning “to shake.”

Muse, at its China peak period, garnered hundreds of thousands of daily users. Douyin’s daily users by the second half of 2017 were measured in the multiple millions and proliferating rapidly.

On the surface, Muse and Douyin were remarkably similar. Fifteen second full-screen short videos, swipe-up motion, music-centered content discovery, etc. But it was everything that users couldn’t directly see that

was leading to a vast difference in results. Douyin's success relied heavily on its parent company's resources, from technology to promotion and user acquisition. Muse was completely outgunned, their promotion budget was minimal, and their technological capabilities weak in comparison. They also suffered from limited brand awareness and no point of differentiation.



Above: Musical.ly and Douyin/TikTok seemed similar at first glance, but this was only the tip of the iceberg.

Founder Alex Zhu bluntly compared how he saw competition between the U.S. and Chinese markets. “In the early days of Musical.ly, our competitors were all American companies, and we hardly felt any competitive pressure. But since last year, we began to compete with Chinese companies, including ByteDance, whether in market operation or the speed of product iteration, there’s an order of magnitude difference with American companies.” ²⁴²

Musical.ly struggles to make money

Failing to compete well in China was just one of a long list of warning signs that all was not well at Musical.ly. The biggest issue was how to make money. Musical.ly had created an ecosystem of influencers and brought exposure to many brands, but the platform itself was having a tough time capturing any of the value it created.

In the U.S. and European markets, advertising was the default model for entertainment and social applications from YouTube to Twitter and

Facebook. But adverts were rarely seen on Musical.ly. There was no backend infrastructure to automate the purchasing of adverts by advertisers. Instead, the process was handled manually through sales teams. Yet Musical.ly had a limited sales network. In its largest market, the U.S., six out of nine advertising agencies surveyed by a reporter, were unaware that Musical.ly even had an advertisement offering.²⁴³

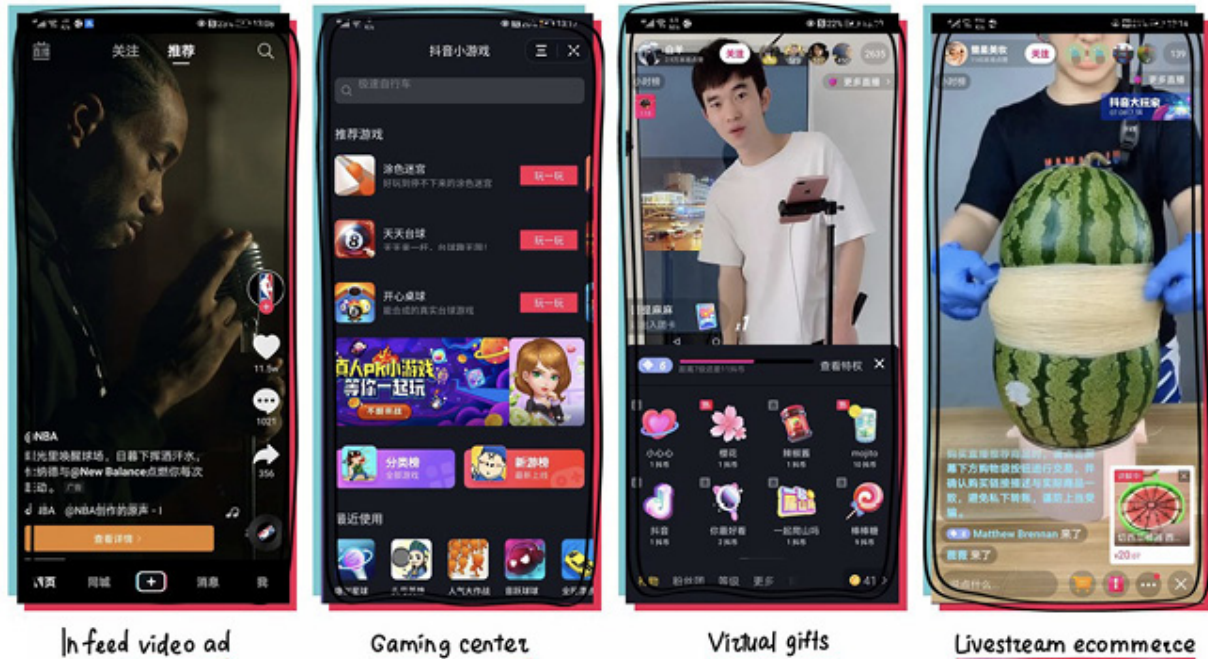
Advertisers viewed Musical.ly mainly as a young experimental platform that was still on the way to figuring out what its advertising deal should be. What unique value could Musical.ly provide that advertisers could not already get with the well-established and sophisticated advertising platforms run by Google and Facebook? The brands that did consider advertising on Musical.ly complained that the fees were too high. Adverts reportedly ranged from \$75k to \$300k for a single day, with more extensive packages for \$2.5 million.²⁴⁴ For an emerging platform with unverified advertising effectiveness, Musical.ly had priced itself out of the market.

Douyin – a license to print money

Musical.ly's monetization woes stood in stark contrast to the stellar performance of Douyin. Despite similarities in app design, their financial performance lied at different ends of the spectrum. By 2019 ByteDance's revenue was estimated to be 120-140 billion yuan, approximately \$17-20 billion, with Douyin accounting for \$10-12 billion of that total, roughly 60%.

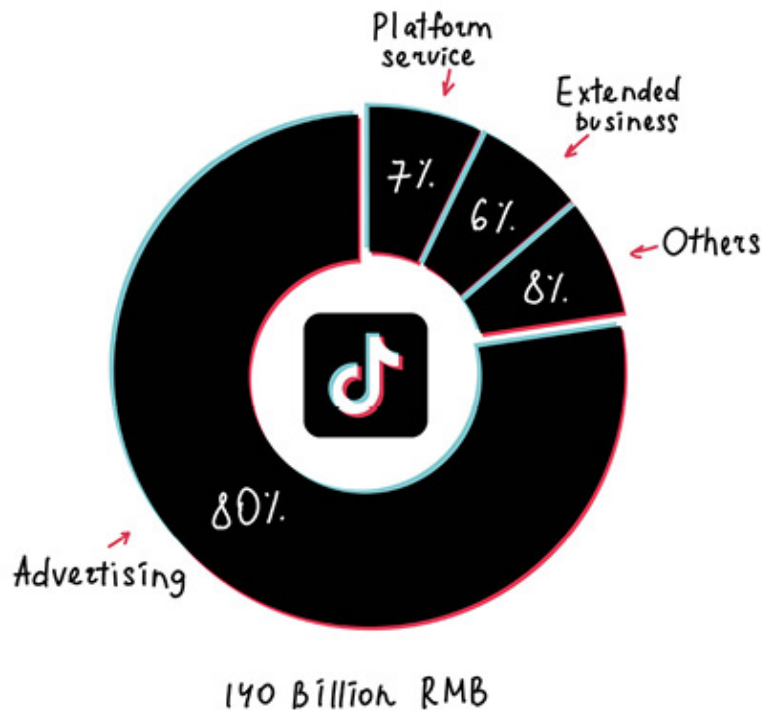
ByteDance's commercialization teams had brainstormed a multitude of ways to capture value, converting eyeballs into dollars and squeezing every last cent out of Douyin. Roughly 80% of Douyin's revenue was attributed to advertising, including multiple formats, such as sponsored brand challenges and full-screen adverts that played immediately upon opening the app.

Among the various formats, the lion's share of revenue came from the "in-feed video ads." These adverts took over the entire screen and played automatically, appearing like a regular TikTok video with just a tiny advert label located at the bottom of the screen next to the video's description text.



Above: Screenshot examples of Douyin's various monetization models, including in-feed video ads, games, virtual gifting, and e-commerce.

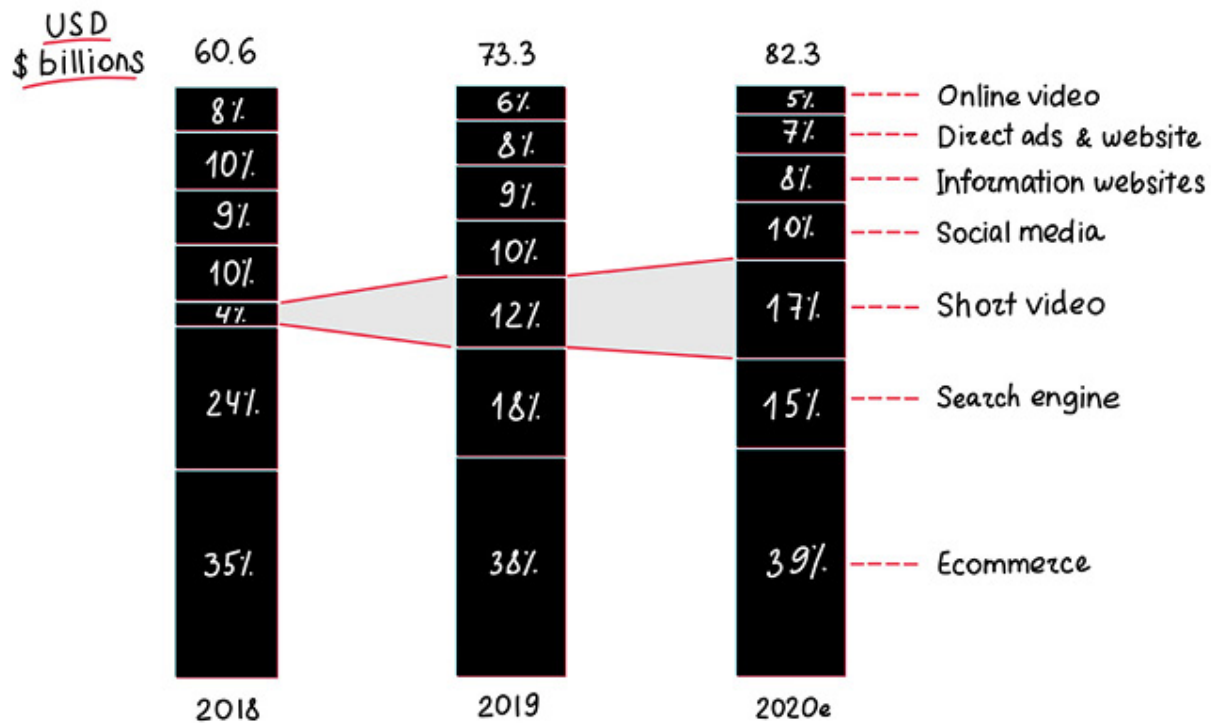
Marketers quickly worked out that if they made their promotion look like a typical user generated video rather than a professional, slick, polished advert, they could easily trick ad-adverse users into watching the first few seconds, enough time to get their message across. Accordingly, the format proved popular and effective.



Above: the estimated 2019 revenue breakdown of Douyin.²⁴⁵

Another monetization method is platform commissions from “Star-Chart,”²⁴⁶ ByteDance’s official influencer data management platform. Brands that wish to work with influencers on Douyin must run their campaigns through Star-Chart or risk having the promotional video removed without notice. The platform takes a cut of all fees paid by brands to influencers. There are also revenues from live-stream show tipping and live e-commerce.

“Extended businesses” included revenues generated through games, paid knowledge, and e-commerce. “Others” included revenues generated through blue tick account verification fees and “DOU+,” a system through which any user or creator can pay the platform to boost a video’s visibility.



Above: Internet advertising in China by category. Short video became China's second-largest advertising category behind e-commerce and exceeding search, a shift that was primarily led by Douyin.²⁴⁷

Toutiao was a slick, well-oiled advertising machine with large sales teams, sophisticated infrastructure, and established reputation among advertisers—all of which were leveraged into Douyin. The closest equivalent in Western markets was how Instagram benefited from parent company Facebook's pre-existing advert expertise and infrastructure.

Short video as a category has seen its share of total China online advertising spend rise rapidly in recent years, tripling from 4% in 2018 to 12% in 2019 and projected to rise further to 17% for 2020. During the same period, search engine advertising fell from 24% to 15%.

Time to sell?

By late 2017 threats to Musical.ly were coming in from all sides. In China, Douyin was taking off like a rocket ship while Muse had failed to make a meaningful impact. Targeting teenagers and pre-teens led Musical.ly to encounter growth bottlenecks. As teen influencers became older, they began to outgrow the app's young audience. Musical.ly was bleeding influencers

and their fans to larger, more established platforms such as Instagram and YouTube.

Time and energy were invested trying to diversify their offerings. Musical.ly built and tested a small suite of complementary apps —“Pingpong” and “Squad” for video messaging, with “Live.ly” for live-streaming,²⁴⁸ all of which fizzled out after failing to gain traction. There was an increasing number of indications that Musical.ly’s growth and engagement had peaked.²⁴⁹ TikTok, however, was already making encouraging progress across various Asian markets, including notoriously tough Japan.

Fortunately, Musical.ly was still in a favorable position to sell their company to a larger organization. There was no shortage of suitors; even as growth stalled the company remained an attractive acquisition target. Western markets are potentially lucrative but difficult to enter. Musical.ly had built a meaningful brand that resonated strongly with American and European teens. ByteDance, Facebook, Tencent, Snapchat,²⁵⁰ and Kuaishou, had all expressed interest at various times and engaged in talks with the founders.

Sometimes referred to as the “Berkshire Hathaway of tech,” Tencent was a prolific investor in other internet companies. Pulling out of talks with Musical.ly and having missed the opportunity to acquire WhatsApp in 2014, Tencent instead opted to gain a foothold in Western social networking via a \$2 billion stake in Snapchat.

Kevin Systrom, the then CEO of Instagram, had met in person with Musical.ly’s founders in Shanghai and later persuaded Mark Zuckerberg to consider a deal. Serious yet inconclusive talks were held at Facebook’s Menlo Park headquarters. Several senior Facebook executives opened Musical.ly accounts and tested the platform, including Zuckerberg, who often interacted with the company’s founders through the platform.²⁵¹ Media later reported that Facebook had “ultimately walked away out of concern about the app’s young user base and Chinese ownership, according to a person familiar with the matter.”²⁵² Both were risks that Facebook would already have been well aware of before going into the talks. In reality, there was another unreported complication—Fu Sheng.

Fu Sheng was an angel investor in Musical.ly. He was also CEO of Cheetah Mobile, a Beijing -based publicly listed mobile app development company with hundreds of millions of users around the world. ²⁵³ As part of their globalization drive, Cheetah Mobile had invested five million yuan (\$700k) in Musical.ly at a time when few believed the company had many prospects “everyone felt that there was no technical barrier to doing 15-second videos with songs,” ²⁵⁴ stated the then general manager of Cheetah Mobile’s investments. As a savvy businessman and industry veteran, the terms of his investment had empowered him with the right to veto any future acquisition deal. With several of the world’s largest and most deep-pocketed internet companies circling around Musical.ly, Fu Sheng knew precisely how to play the game and make sure he got his pound of flesh.

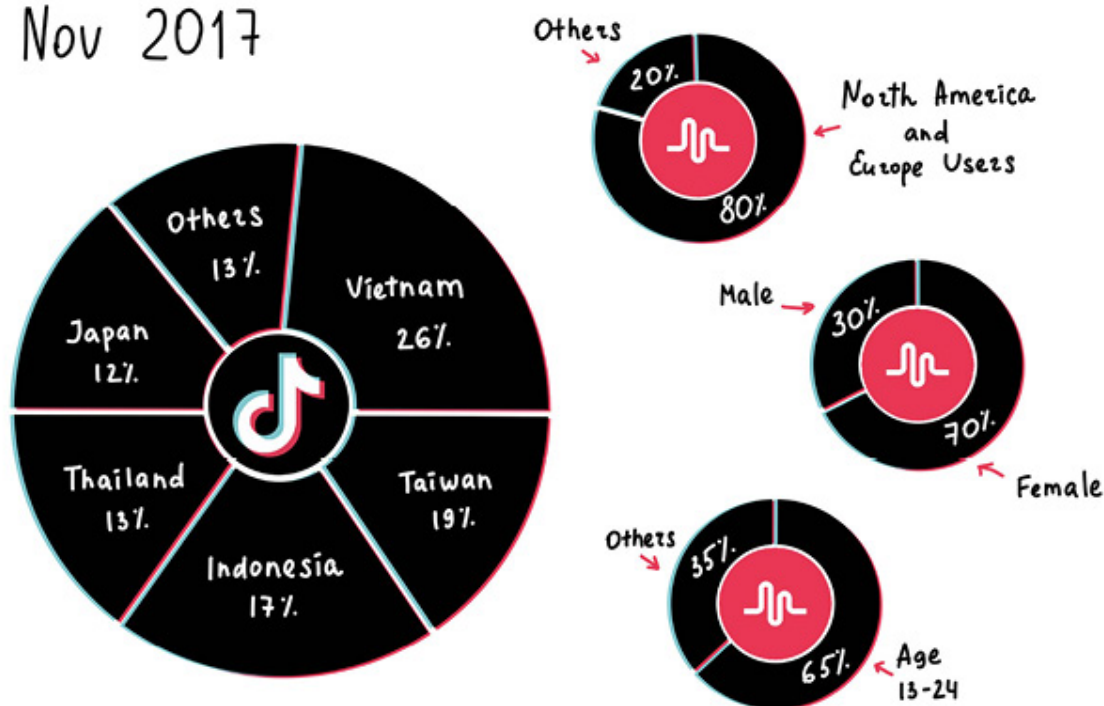
Yiming gets the deal done

With a willingness to spend big and long-harbored ambitions to be a global company, fast-growing ByteDance was near the top of the short-list of companies vying to acquire Musical.ly.

Yiming and the Musical.ly co-founders had been acquaintances for several years. Still, discussions only began seriously in Spring 2017 ²⁵⁵ with ByteDance executive and former Uber China spokesperson Liu Zhen overseeing the acquisition talks. “For us to get a deal done, Yiming has to figure out a way to work with Fu Sheng... it was **not** an easy deal.” ²⁵⁶ revealed Musical.ly board member Hans Tung.

To meet Fu Sheng’s exacting demands, the final agreement involved bundling together three separate transactions. Yiming committed to buying Cheetah Mobile’s news aggregation app News Republic for \$86 million and invested \$50 million into Cheetah Mobile’s live streaming platform Live.me. For Musical.ly and its investors, the price tag of \$800 million was a satisfactory return.

Nov 2017



Above: TikTok and Musical.ly data for November 2017.²⁵⁷ TikTok's user base was spread across various East and Southeast Asian markets. Musical.ly's user base skewed heavily towards Western teenage girls.

For ByteDance, the acquisition achieved several goals at once. Firstly, it was a cost-effective way to acquire a broad base of Western market users—Musical.ly had minimal overlap with TikTok's Asian user base.

Just as importantly, it prevented larger companies from acquiring and revamping Musical.ly as a shortcut to competing with TikTok, blocking off the most likely competitive countermeasure ahead of time. Lastly, as with Flipagram, it allowed ByteDance to learn what they could from Musical.ly's years of operation experience and apply those insights across the organization.

ByteDance announced the deal to the world on November 10th, 2017; they now owned Musical.ly. It was their largest acquisition to date. Two months earlier, ByteDance had received a new round of investment led by American growth equity firm General Atlantic doubling the previous \$11 billion valuation taking the company to \$22 billion. A year later, that

number would more than triple to \$75 billion. The company's bets on short video were paying off at a scale few had felt possible.

220 <https://hans.vc/toutiao/>

221 <https://youtu.be/vDXvJfXe3hw?t=23>

222 <http://finance.sina.com.cn/chanjing/gsnews/2019-03-14/doc-ihsxncvh2481985.shtml?>

223 <https://www.tmtpost.com/3324980.html>

224 <https://new.qq.com/omn/20171205/20171205A0XBIO.html>

225 To name a few: usage habits, cultural expectations, licensing and content regulations, payment system and app store SKU integrations and social sharing platforms.

226 <https://newsroom.tiktok.com/en-us/musical-ly-and>

227 https://twitter.com/tiktok_japan/status/884317356704399360

228 Map created by Reddit user Valeriepieris

229 <https://www.jiemian.com/article/2241255.html>

230 <https://36kr.com/p/1722597179393>

231 <https://www.huxiu.com/article/332665.html>

232 https://en.wikipedia.org/wiki/Yukina_Kinoshita

233 <https://36kr.com/p/1722597179393>

234 https://www.youtube.com/watch?v=qrM_5qNhW-8&t=2s

235 [https://twitter.com/search?q=\(from%3Atiktok_japan\)%20until%3A2017-09-01%20since%3A2017-04-01&src=typed_query](https://twitter.com/search?q=(from%3Atiktok_japan)%20until%3A2017-09-01%20since%3A2017-04-01&src=typed_query)

236 https://www.sohu.com/a/235450902_403354

237 <https://mixch.tv/>

238 <https://36kr.com/p/1722597179393>

239 For example, native English-speaking countries such as the U.S., Australia and Canada were grouped together within one region, despite their disparate geographies.

240 This strict “region lock” system has since been loosened as the TikTok content ecosystem has matured. Today it is possible for accounts from one region to become visible in other regions, especially those that have been whitelisted. Yet it is still the case that an account opened in one region, such as Egypt, will find it difficult to gain exposure in another region, such as Brazil.

241 <http://www.yuanzhonghe.com/view.php?id=303>

242 <https://it.sohu.com/20180124/n529099946.shtml>

243 <https://digiday.com/marketing/musical-ly-starts-selling-ads/>

244 <https://www.musicbusinessworldwide.com/vine-and-musical-ly-transformed-the-music-industry-then-they-disappeared-will-history-repeat-itself/>

245 Data source: EqualOcean Analysis

246 <https://star.toutiao.com/>

247 Source: SCMP reporting and analysis, QuestMobile, Qianzhan Industry Research Institute, iResearch, IDC

248 Interestingly the web domain live.ly was never purchased by Musical.ly. The disgruntled domain owner set up a prominent promotional message and link for rival live-streaming app Periscope.

249 <https://musicindustryblog.wordpress.com/2017/11/10/musical-ly-sells-for-800-million-but-peaked-by-being-too-silicon-valley/>

250 <https://www.zhihu.com/question/67915440>

251 <http://tech.sina.com.cn/roll/2017-11-19/doc-ifynwnty4928120.shtml>

252 <https://www.bloomberg.com/news/articles/2019-11-13/why-facebook-passed-on-buying-the-predecessor-to-tiktok>

253 Cheetah Mobile has since fallen from grace with its apps being identified as malicious by Google and removed from the Play store in February 2020.

254 <http://tech.sina.com.cn/roll/2017-11-19/doc-ifynwnty4928120.shtml>

255 <https://hans.vc/ByteDance-musical-ly-merger/>

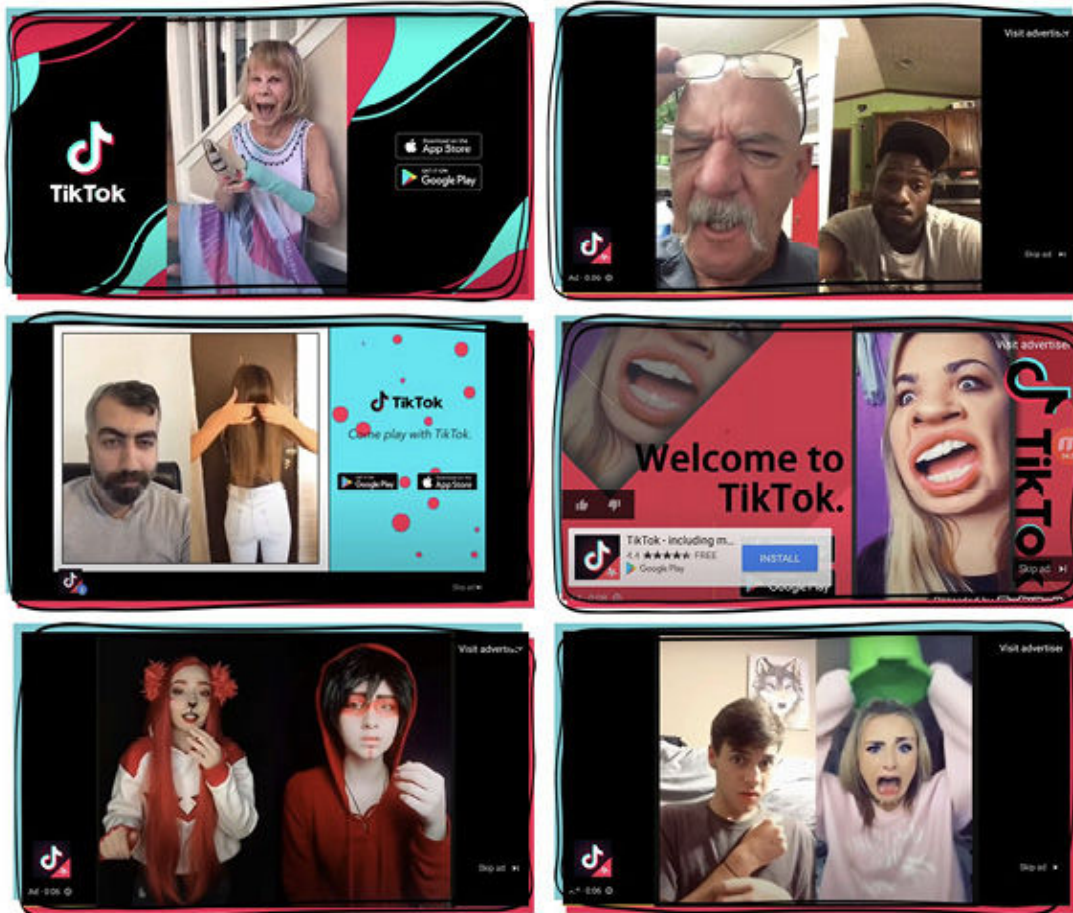
256 <https://youtu.be/YsPeT2oHQLY?t=2098>

257 Data source: Cheetah Global Labs, Musical.ly

[OceanofPDF.com](https://oceanofpdf.com)

CHAPTER 8

CRINGE!!!



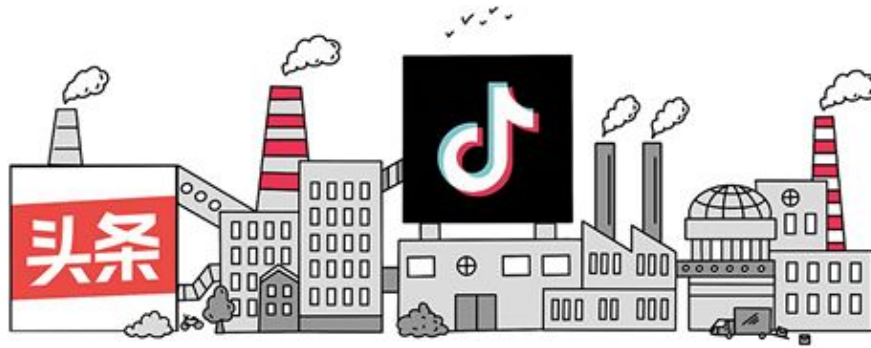
“Musical.ly disappeared, and we were all so happy.

*And then... **TikTok, TikTok, TikTok !** ”*

PewDiePie, the world’s most popular YouTuber

Images: examples of paid adverts for TikTok run on YouTube in late 2018

[OceanofPDF.com](https://oceanofpdf.com)



Chapter Timeline

2018 April – TikTok becomes the most downloaded non-game app in the world for the first quarter of 2018

2018 Aug – Musical.ly and TikTok merge. Making TikTok available in the United States for the first time

2019 Jan – The first advert appears on TikTok in the U.S., a splash screen ad for GrubHub

2019 Feb – ByteDance hires TikTok’s first General Manager for the U.S., former YouTube executive Vanessa Pappas

2019 April – “Old Town Road” begins a record-breaking run of 19 weeks as the No. 1 single on the Billboard 100

2019 June – ByteDance hires former Facebook VP of Global Partnerships, Blake Chandlee, to join TikTok

2019 Sept – The National Football League announces a multi-year content partnership with TikTok, an inflection point for brand adoption

2020 May – ByteDance hires former Disney executive Kevin Mayer as the CEO of TikTok

YOUR WORDS CAN BE REGARDED AS SLANDER,” replied the wealthiest man in China, ²⁵⁸ Pony Ma. The publicly shy introvert CEO of Tencent had become uncharacteristically inflamed, lashing out online in defense of his company. His barbed reply was directed towards Yiming, as the two entrepreneurs’ companies had become fierce rivals.

Before Ma made the fiery comment, Yiming had posted two pictures online—a TikTok logo and a rankings chart showing TikTok as the world’s most downloaded app in the first quarter of 2018. Above, he wrote just three words in English “celebrate small success.” Underneath Yiming called out Tencent for its copycat tactics in countering the growth of Douyin—blocking the app from Tencent’s social networks while simultaneously promoting its own virtually identical clone.



Yiming: WeChat's blocking and Weishi's plagiarism cannot stop Douyin's march forward

Pony Ma: Your words can be regarded as slander

Yiming: The former is not appropriate for discussion, and the latter has been notarized. I don't want to have a war of words. I just couldn't help but make a complaint and was criticized by our public relations. I will send the materials to you individually.

Pony Ma: There are too many things to notarize with you

Above left: Yiming’s online post made in May 2018 with translation (above right)

Openly boasting about TikTok’s success while criticizing a rival for what were arguably standard China industry practices had pushed Pony Ma over

the edge. Two of China's most mild-mannered, introverted, and reserved leaders were now involved in a public spat. Screenshots of the exchange rapidly went viral on China's internet, attracting widespread discussion and speculation.

TikTok succeeding where WeChat had failed was a bitter pill for Tencent to swallow. Despite investing hundreds of millions of dollars into advertising and promotions in 2013 and 2014, Tencent's flagship super-app WeChat had failed in its ambition to become a global hit. Today, outside of mainland China, WeChat is used predominantly by members of the Chinese diaspora and those doing business with China. ByteDance, on the other hand, had verified its hypothesis—TikTok did indeed hold the potential for widespread adoption across the globe, and the app was making rapid progress.

China had a universe of internet services parallel to that of the West, targeting mainly users within China. TikTok was the first China-made consumer internet product that had built a genuinely far-reaching global footprint. The main reason for this was its strong first-mover advantage. As ByteDance had discovered in early markets such as Japan, TikTok was essentially competing with itself. The experience that the app provided was simply different from other social media and social networks.

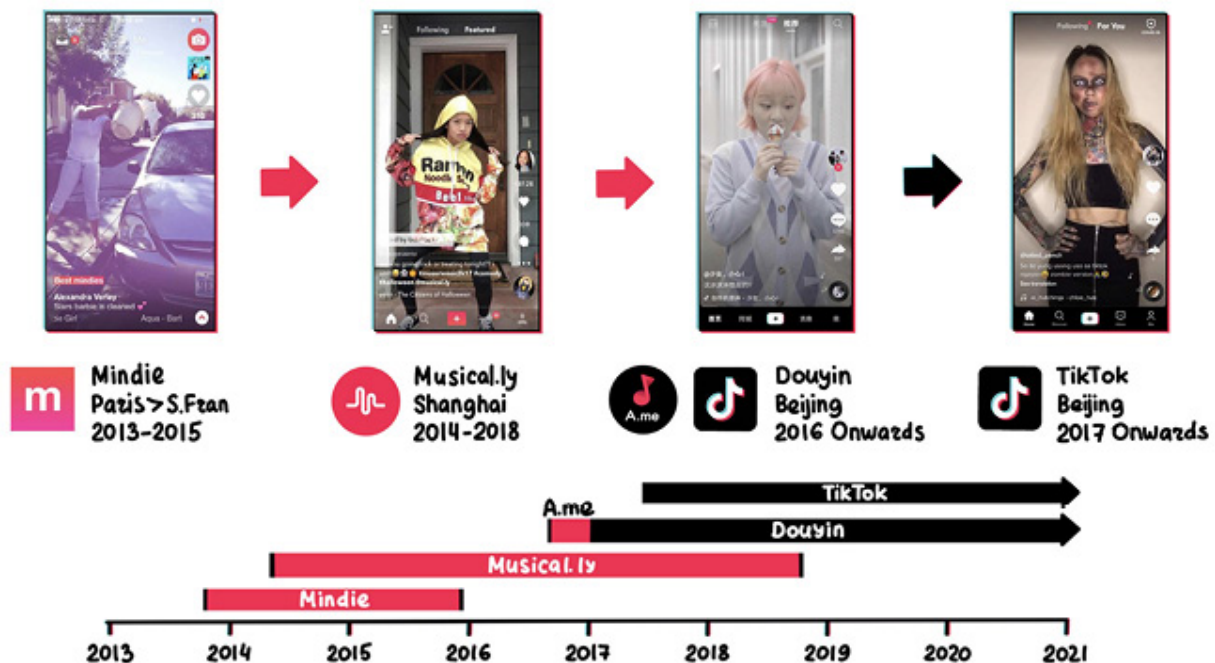
Behind TikTok's rise was a broader global trend. From e-commerce to payments and mobile gaming, China was now leading the world in many consumer internet services. By 2018 it was widely understood in China that short video was the “killer app” for the 4G mobile era. The local internet industry had dissected and poured over every detail of what had made Douyin a success. TikTok-style feeds began popping up across all kinds of apps as players scrambled to integrate short videos into their offerings. Yet awareness around the power of the format did not exist to the same extent in the West.

The era of “copy to China,” a period when Chinese internet firms eagerly pounced on new trends from Silicon Valley and made Chinese clones was coming to an end. With short videos, at least, the tables had turned. It would soon be time for Silicon Valley to play the game of “copy from China.”

Bring back, Musical.ly!

August 2nd, 2018 was a major milestone in the history of TikTok – the date when Musical.ly and TikTok merged. Overnight Musical.ly users found the app had morphed into “TikTok - including musical.ly” with their old videos and accounts now migrated over into a new app.

In the space of roughly one year, TikTok had grown from zero to claiming 500 million monthly users, ²⁵⁹ mostly in Asia. By comparison, Musical.ly, which had been in operation for five years, harbored just 100 million monthly users. At the time of the Musical.ly acquisition, ByteDance had implied that the two platforms would be kept independent. Yet just nine months later, the Musical.ly brand was discarded as the app found itself being swallowed up by the advance of the much larger and more successful TikTok.



Above: TikTok, the international version of Douyin, began as a clone of Musical.ly, which started life as a clone of Mindie. Making TikTok arguably the clone of a clone of a clone. ²⁶⁰

One could be forgiven for believing not that much had fundamentally changed. While the logo and interface design had been revamped, the core experience of swiping through short 15 second videos had stayed pretty much the same as before. Yet, loyal Musical.ly users loathed multiple key changes. ²⁶¹ There was widespread outcry online with petitions to “bring back Musical.ly,” with many users leaving TikTok. ²⁶² “Sometimes, you

have to let go of previously valuable things and even embrace frustrating a cohort of users in order to transition into a bigger opportunity,” explained Musical.ly product strategy manager James Veraldi.

Several popular social elements were removed, including Musical.ly’s leaderboard, a feature that displayed the most popular videos posted each day by country. The feature gave a sense of community to the young users, and its removal was unpopular. The leaderboard centralized attention towards the most popular video categories, which for Musical.ly was teenage dance, meme, and lip-sync videos. This didn’t fit with TikTok’s goal to age up the user base and encourage diversification of content categories.

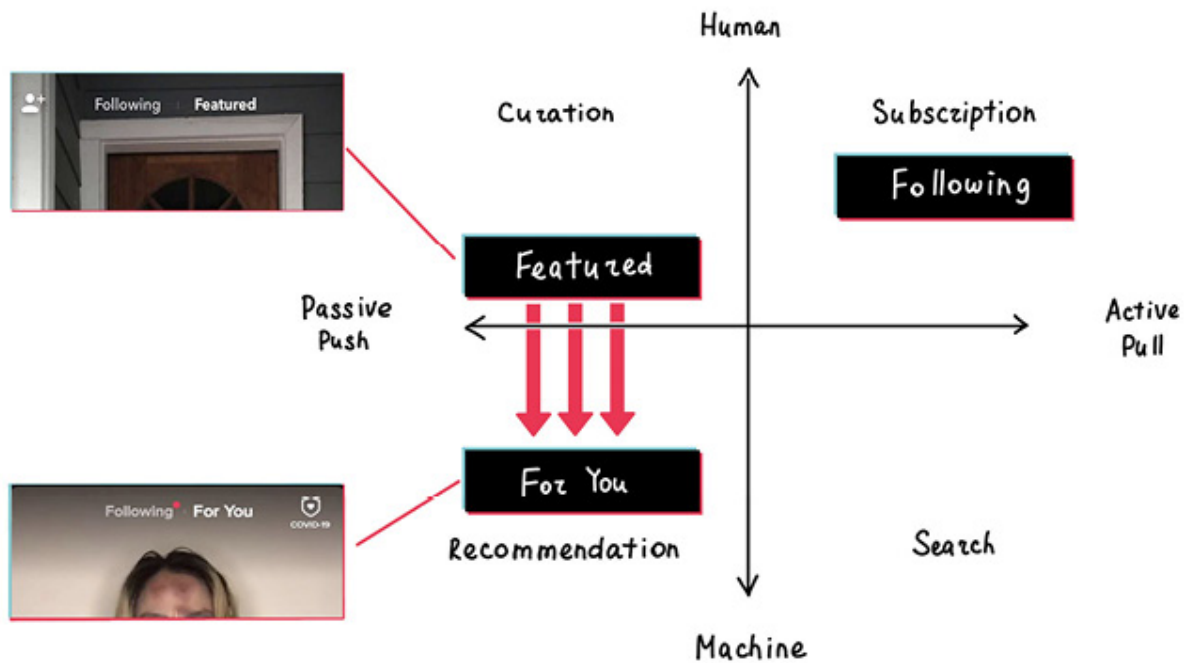
The signup process was significantly streamlined, removing the need to register or log in with an existing account. New users only needed to select from a list of interests (e.g., animals, comedy, art) and start watching videos within seconds.²⁶³ The app would use a “shadow profile” based on the device ID, which allowed for personalization of content even for those without registered accounts.

Asking someone to create an account before they even knew if they liked the app was putting the horse before the cart. The new process allowed people the freedom to experience TikTok without committing.

Video sharing was aggressively encouraged. Once a clip had looped several times over, indicating that the viewer found the video interesting, an attention-grabbing “share” icon would flash. Videos shared to other platforms now contained the flashing TikTok watermark. The logo was difficult to ignore as it alternated from one corner of the video to the other, continually vibrating.

The single most significant change was switching over to the same backend infrastructure used by Douyin. For users, this change was expressed as the main “Featured” feed being replaced by the new “For You.” The two titles accurately summarized the difference, “For You” was entirely personalized using sophisticated machine learning technology. “Featured” was Musical.ly’s old system, which used a combination of less advanced recommendation mixed with videos manually selected by the content team.

Time spent in the app was said to have doubled after switching the backend over to Douyin's system. "The content we wanted was in the app, but it was hidden by our initial architecture. Once we changed this, content diversity followed in droves," explained James Veraldi, former head of product strategy at Musical.ly. "I asked some (ByteDance) friends who had some data on the before and after. The step-change in the graph was anything but subtle," ²⁶⁴ revealed former Hulu and Amazon employee Eugene Wei.



Above: The switch from "Featured" into "For you" was a move from a feed that relied on a hybrid mix of curation and recommendation into one driven entirely by recommendation.

This single change was transformational, an effect which echoed that of YouTube's introduction of the Sibyl machine learning backend system in 2011, "the content is already there. We have billions of videos. " explained YouTube head engineer Cristos Goodrow at the time. Use of machine learning to classify and recommend videos had been the key to unlocking the potential of both platform's vast pools of content.

TikTok cringe compilation - part 14

"Why are **moms** using TikTok? Why is **anyone** using TikTok?" ²⁶⁵ shouted the world's most popular YouTuber towards the camera. Swedish gamer

PewDiePie was recording his second of fifteen “TikTok Cringe Compilation” videos after the first had proved to be a hit. Each episode was ten minutes of him reacting to painfully embarrassing TikTok videos.

TikTok hadn’t paid anything to PewDiePie. The A-list global internet mega-celebrity was creating video after video about TikTok because his audience loved it. This should have been the kind of authentic influencer promotion that online marketers dreamed of. Every video was essentially a free ten-minute advert for TikTok distributed out to a loyal 80 million follower base. Yet, at the same time, PewDiePie wasn’t exactly endorsing the app.

TikTok was bizarre. An endless stream of people posting weird content with almost a total lack of self-awareness. Mindless comedy skits, lip-sync, and just outright wacky oddball creations. The kids making these videos could be forgiven; they were just kids. But the adults posting on the app came off simply as creepy and weird. Countless numbers of TikTok cringe compilations started appearing on YouTube, many with millions of views. Criticism of the app became widespread, with the shaming of TikTok users becoming a regular occurrence on Twitter and Reddit.²⁶⁶

In China, Douyin had first garnered attention as a popular app for urban youths, associating itself with art students and fashionable hip-hop lovers. Yet in America, it was the absolute opposite. TikTok had entered the public consciousness as a cringe app for losers and misfits. What was going on?

The answer was ByteDance’s truly massive advertising campaign across major Western social media platforms such as YouTube, Instagram, and Snapchat. The advertising campaign’s budget was reported by the Wall Street Journal to be over \$1 billion in 2018.²⁶⁷ ByteDance became Facebook’s biggest Chinese customer as it grew TikTok’s footprint with app-install ads.²⁶⁸ Many Americans suddenly found TikTok ads were everywhere they looked online.

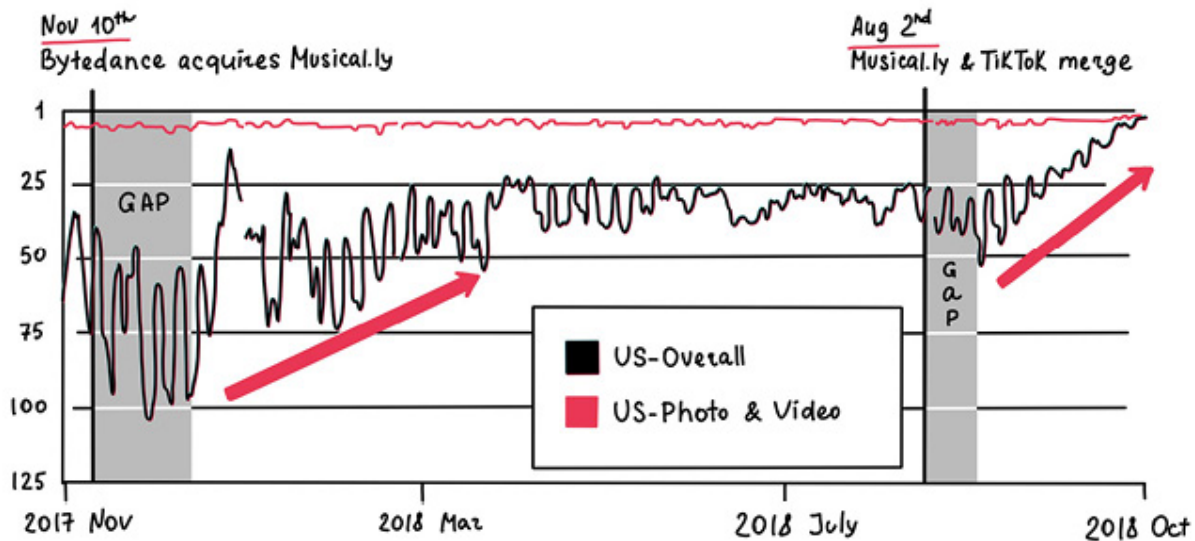
The company also spent heavily on traditional billboard ads and outdoor advertising. They ran an expensive TV advert right after the New Year’s Eve ball dropped in New York City’s Time Square. Adverts for TikTok popped up at famous landmarks around the world from the Burj Khalifa in Dubai to the London underground through to the Las Vegas strip.



Above: ByteDance product guru Kelly Zhang posts pictures online of TikTok advertising in December 2018. Left: New York Times Square, middle: the Burj Khalifa, Dubai, top right: Planet Hollywood, the Las Vegas Strip, bottom right: Piccadilly Circus, London.²⁶⁹

The initial warm reception towards TikTok across various Asian markets was highly encouraging – it seemed Douyin’s success really could be replicated globally. Yet the more successful TikTok became in Asia, the more attention it attracted from competitors; all major internet companies had advanced systems in place to keep track of new trends and changes in mobile usage habits. ByteDance had to move fast to grab the window of opportunity to leverage its advantage. In general, Western internet companies look down upon directly cloning competitors. Even so, if an established giant like Google or Facebook chose to promote a similar product to TikTok vigorously, it could significantly hamper their progress.

Facebook’s success in cloning rival Snapchat’s video feature “stories” demonstrated a fate that may easily befall TikTok. This meant speed was of the essence, and the most effective way to scale up fast with a combination of massive spending on online app install ads matched with building brand awareness through offline ads.



Above: U.S. App store download rankings for Musical.ly (later TikTok) from October 2017 – 2018. Shortly after merging Musical.ly with TikTok, the app store ranking improved considerably, boosted massively through aggressive spending on ads.

These TikTok ads are disturbing!

Usually, when a company wants to spend big on online advertising and introduce a brand to a new market, they will work with a creative agency. Expensive consultants will be hired, veteran advertising professionals with years of industry experience will create smart concepts. The process will involve carefully crafted brand messaging, extensive Gen Z focus groups, professional actors in expensive recording studios, crews of video editors, and graphic designers to ensure everything is perfect.

Yiming was never one to keep to conventions. When buying his first apartment in Beijing, rather than consulting with real estate agents, discussing with family or personally visiting housing compounds, he found a short-cut. Yiming crawled the web for data, scraped everything into spreadsheets, and crunched the numbers in a single evening.

When it came to advertising TikTok and newly acquired Musical.ly, ByteDance found a similar shortcut, but the strategy was somewhat unorthodox—it would simply use videos from the app itself. The platform's terms of service gave it the right to do so.²⁷⁰

After manually identifying and removing potentially inappropriate content, the company implemented a systematic process to experiment with various videos.²⁷¹ The adverts didn't actually say anything about what TikTok was or why anyone would want to use it; they simply needed to pique people's interest. The goal was simple—find the clips that got the most people to click on a big blue “install” button.

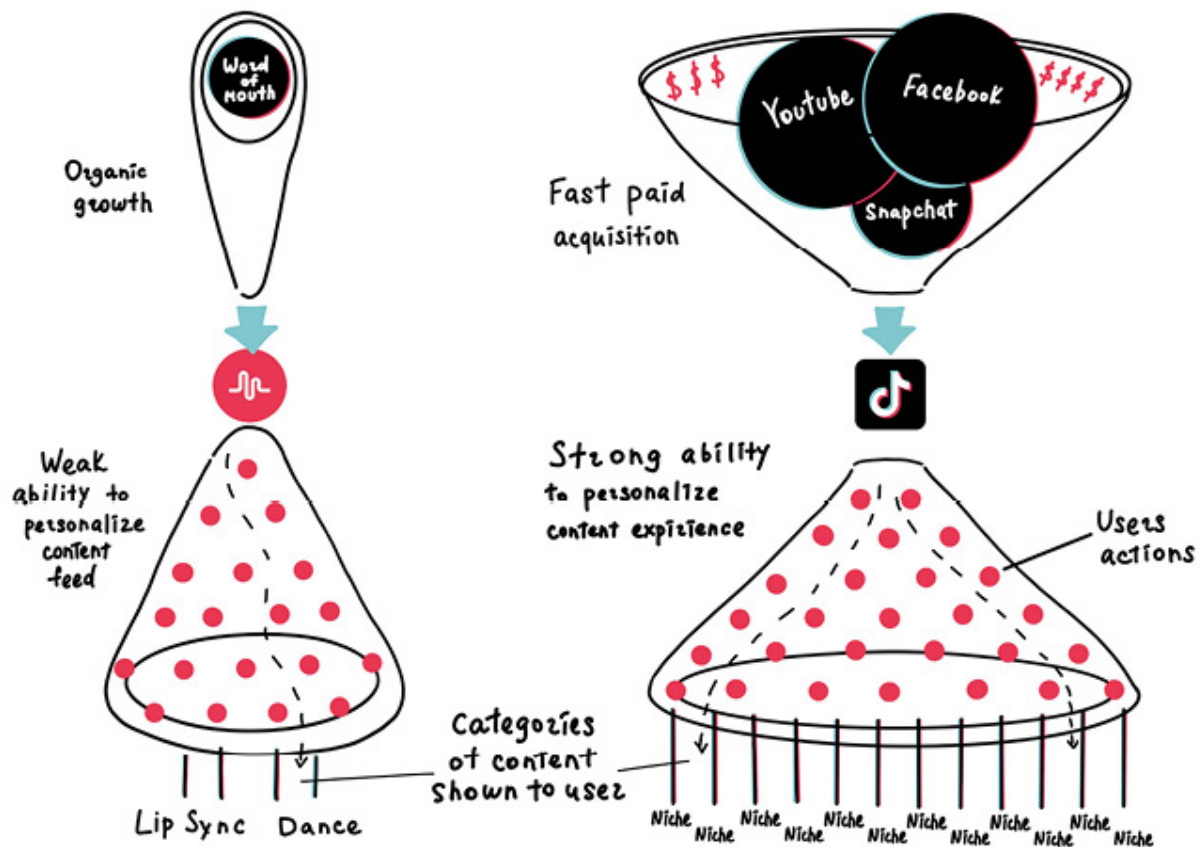
This ad buying process was run from Beijing by the company's experienced growth hacker teams. There was just one issue—the teams had a laser-like focus on conversion metrics but little understanding of the actual video content. Whatever converted best would be used more regardless of what the actual video showed. It turned out that wacky, outlandish, downright weird videos worked really well at getting people to click big blue “install” buttons.

Many of these weird ads were attracting social misfits. When these people started using TikTok, they, in turn, made strange videos that would attract more social misfits and so on.

TikTok's video classification systems were highly sophisticated and able to accurately identify and classify all kinds of subculture content—automatically. The system was also able to tag users more effectively based on their actions and precisely match them with content in a way that Musical.ly had never been able to do.

A prominent example were “Furries,” a stigmatized and misunderstood community of people who derive enjoyment from dressing up as animal characters in large fursuits.²⁷² Furries were big early adopters of TikTok in the U.S.²⁷³ Many built significant followings as the colorful cartoon-like animal costumes proved attractive to the app's large pre-teen user base, bringing the sub-culture to a new audience.

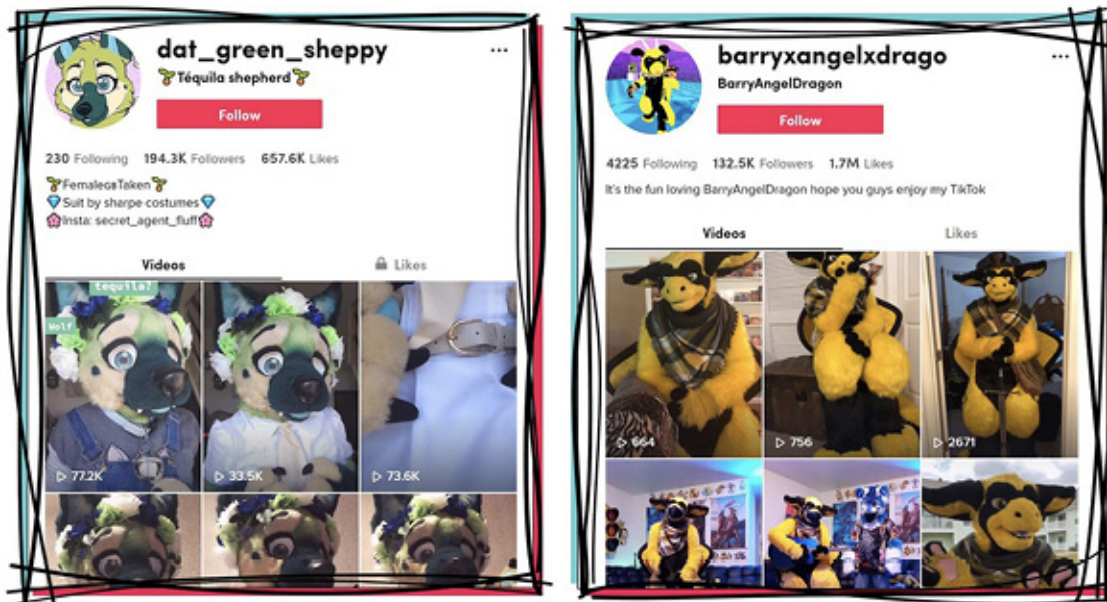
Other notable early TikTok adopter communities included cosplayers and gamers. The animosity between these groups led to the “Furries Vs. Gamers War”²⁷⁴ meme, a lighthearted imaginary conflict which saw gamers pretending to have been kidnapped by furries and roleplaying acts of espionage, feigning to have infiltrated the ranks of the furries.



Above: TikTok was acquiring new users at a much faster rate than Musical.ly. It was then accurately and efficiently matching those users with niche content based on their personal preferences in a way that Musical.ly never could.

TikTok contained a “duet” feature, which allows two videos to appear side by side, splitting the screen. Duet had previously been restricted in Musical.ly, but now users could respond to any video by recording one of their own. With many weird niche subcultures like furies on the platform, “duet” became popular, quickly transforming into a bullying and harassment tool. As a countermeasure, settings were later added, allowing users to disable duets.

Since merging Musical.ly with TikTok in August 2018, the platform was moving in a vastly different direction – not everyone was happy. “TikTok’s early (unintentional) positioning in the states basically was cringe.” explained an early TikTok employee who wished to remain anonymous. The app had an awful image problem. It was widely perceived as being only for misfits and kids making lip-syncing videos.



Above: examples of TikTok furry accounts. Adults dressed as large anthropomorphic animal-like characters.

“I haven’t seen one piece of content on there made by an adult that’s normal and good. To be a grown adult doing a cute karaoke video on an app and trying to make it go viral is odd behavior.” was the brutal assessment of Instagram influencer Jack Wagner, interviewed in one of the earliest American media articles covering TikTok.²⁷⁵

The colossal spend on adverts was effective at getting downloads, but they were also ruining the reputation of the platform, leading the then small U.S. based TikTok team to express concerns to the China head offices. In China, Douyin had never had such a problem. The seed group of early adopters had been carefully selected, and the app had built an outstanding brand image with carefully crafted glitzy cinema adverts, savvy viral marketing campaigns, and sponsorships of hit talent shows.

“If you look at history, a lot of inventions first started with a toy, with things that seem to be irrelevant, but have the potential to become something much bigger.”²⁷⁶ postured Musical.ly co-founder Alex Zhu in an interview, echoing an observation previously made by many industry practitioners.²⁷⁷ TikTok’s early reputation for wacky cringe videos had made it seem like a toy and hard to take seriously. The situation had echoes of the initial characterizations of Snapchat being written off as an app solely for college students’ “sexting” each other with disappearing pictures. Widely

criticized and with retention rates in the U.S. rumored to be as low as 10%,²⁷⁸ TikTok was not seen as a threat to anyone but itself.



Above: anti-TikTok online memes, from late 2018 and early 2019.²⁷⁹

Yet those writing the platform off failed to anticipate how fast TikTok would change. The algorithmic nature of the platform's content distribution makes it easy to "tilt the table" in favor of specific content types. ByteDance could reduce exposure to monotonous videos of teenagers lip-synching and dancing and instead highlight the growing variety of new content categories such as magic tricks, street comedy, sports, or arts and crafts.

Due to the massive influx of users driven by the big-spending on ads, creators found it easy to grow large fan bases quickly due to the imbalance between the supply of and demand for good content. Instagram, YouTube, and others were saturated with people competing for attention. TikTok was wide open and began to attract its own batch of content creators and online marketers—eventually, those looking to gain attention online will always follow the numbers. The dynamic was similar to the metaphor Alex Zhu had used years earlier with Musical.ly – to encourage immigration to your new country, "some people need to get rich first."

Celebrities start talking

“There’s a really cool app that I’ve been getting into lately called TikTok, do you guys know that?” ²⁸⁰ asked Jimmy Fallon, host of the popular NBC American late-night talk show *The Tonight Show*. The studio audience was silent. Evidently few people, if any, knew what he was talking about. “If you don’t have it, download it.” he continued, pulling out an iPhone to explain further.

Fallon was the first big-name American celebrity to endorse TikTok. ²⁸¹ The segment on his show went out to millions of viewers and was so glowingly positive that many assumed it must have been a paid for, although this was later denied. ²⁸² When other popular TV program, *The Ellen Show*, also ran a highly positive feature of TikTok, the eponymous host joked, “If TikTok wants to send me some money to sponsor it. I’ll play it again.” ²⁸³

ByteDance was never afraid to shell out big bucks to celebrity stars in return for endorsements. Yet in many cases TikTok didn’t actually need to sponsor—the app simply promoted itself. Just as they had found in Japan, TV shows and media would naturally pick up on the app’s goofy going-ons. Highlight reels of TikTok videos were extraordinarily engaging and always seemed to get the audience reacting. The result was continued free exposure.

Old Town Road

Montero Hill, an unemployed 19-year-old with an empty bank account, was sleeping on the floor of his sister’s house. He had dropped out of college to pursue his dream of becoming a famous rap artist, recording tracks in the closet of his grandma’s house and releasing the music on audio distribution platform SoundCloud under his artist title “Lil Nas X.” Atlanta native Montero came from a humble background; his parents had divorced when he was six, leaving him to be raised by his mother and grandmother in a run-down public housing unit.

The one thing Montero had going for him—he was a skilled and tenacious online marketer—having mastered the art of producing viral Twitter posts. ²⁸⁴ He spent several hours each day online promoting himself and his songs. Yet despite these efforts, his dreams of fame and fortune had failed to

materialize. The material lack of success contributed significantly to his perpetual state of anxiety, daily headaches, and insomnia.²⁸⁵

Searching YouTube one day in late 2018, Montero discovered a catchy banjo sample²⁸⁶ from a young producer in the Netherlands, also working out of his bedroom. Straight away, he sensed the track was exceptional and purchased the rights to the music for \$30. Montero set about creating a unique blend of country music-inspired lyrics to accompany the backing track. The result was a novel combination of disparate music styles – a heavy bass sound juxtaposed with strumming banjos and Montero’s playful lyrics “Cowboy hat from Gucci, Wrangler on my booty.” He named the track “Old Town Road” and termed its genre as “country trap.” Taking advantage of a special deal at a local recording studio, he spent just \$20 to rush out a recording in less than an hour.

Montero began resolutely promoting the new track online. He created cowboy-themed Twitter memes that tapped into the resurgence of interest in cowboy culture driven in part by “Red Dead Redemption 2,” the most popular game of the past year. The outstandingly catchy track had become a moderate success, yet after two months of relentless effort, interest in the song began to wane.

Everything changed on February 16th, when the track was picked up by a small TikTok influencer, Michael Pelchat, who created a video of himself dressed as a cowboy dancing to the song.²⁸⁷ A meme quickly formed around the track with people transforming themselves into dancing cowboys at the exact moment the beat of the groove kicked in.²⁸⁸ Interest around the song exploded with millions of people using it as the background sound to make short videos. “It went crazy!” exclaimed Pelchat, “Everyone was dressed like a cowboy for like three weeks.”²⁸⁹

TikTok’s meme virality superpowered the song’s popularity. Interest across America snowballed so quickly that radio station networks resorted to using an MP3 audio file ripped from YouTube to get the track on the air.²⁹⁰ TikTok had propelled Old Town Road to being a phenomenon, *the* hottest song in America at a time when Montero hadn’t even signed with a record label.

“Old Town Road” ended up as one of the most successful songs of all time, winning multiple music awards. The track became the longest-running number one single in Billboard Hot 100’s 61-year history, ranking 19 weeks at the top. Montero was signed by Columbia Records and later recorded a version with country music legend Billy Ray Cyrus. Reflecting on his fairytale success story, Montero was in no doubt as to the cause, “TikTok helped me change my life.” ²⁹¹

The fairytale story speaks to the critical importance of content distribution. Even an exceptionally savvy marketer like Montero found distribution on saturated social media platforms such as Twitter, Instagram, and Facebook an uphill struggle. In a world of abundant music where the barriers to creating professional quality tracks were so low—Montero had spent just \$50—it follows that merely creating an exceptional quality song is no guarantee of success. Instead, “cutting through the noise” is often the deciding factor. One wonders how many other “Old Town Roads” are out there waiting to be discovered?

The power of memes

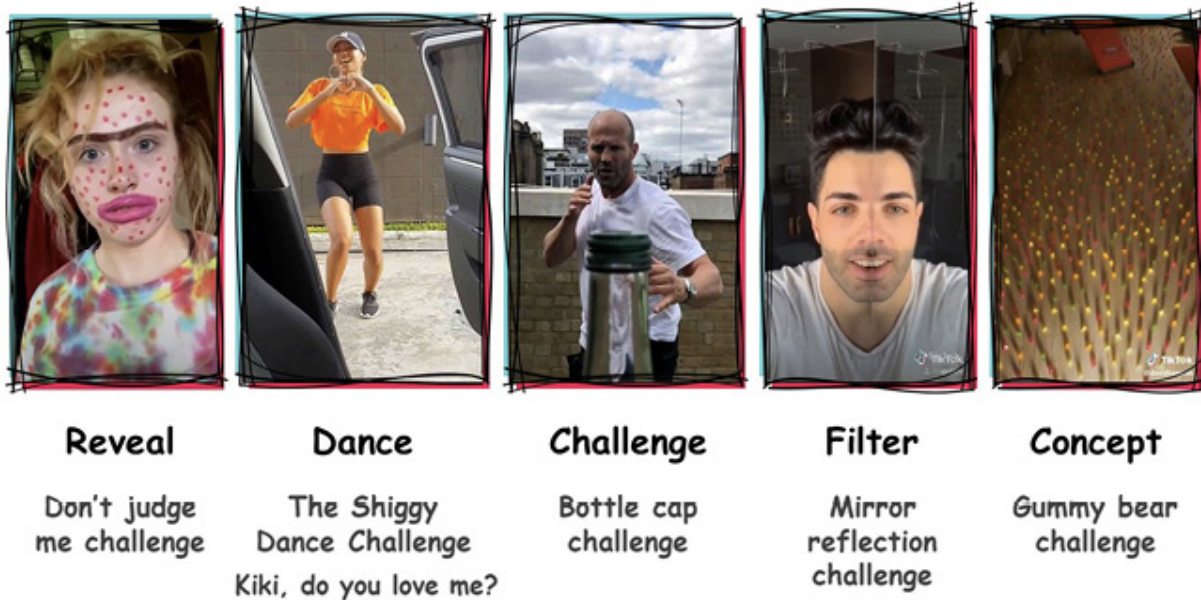
Old Town Road was the best illustration yet of the exceptional power of music-driven, user-generated video memes. Memes drastically lowered the creative and motivational barriers to content creation, providing a cookie-cutter structure allowing anyone to take part. These memes were referred to commonly as “challenges,” a term that explicitly communicated their participatory nature.

Memes were undeniably a critical driver of TikTok’s success. The videos typically came across as frivolous fun and resistant to logical analysis. Yet, with enough exposure, formulas begin to emerge. Similar to how all storytelling could be reduced down to seven basic plotlines, ²⁹² video memes could also be categorized into a fixed number of genres.

Reveal memes involve a short set up, followed by a dramatic transformation or reveal following the structure of the accompanying song. The set up happens during the song’s introduction, with the reveal beginning at the exact moment the song’s main riff or hook kicks in,

amplifying the dramatic effect—a mini-story sequence compressed into 15 seconds.

Several of the all-time most popular video memes are reveal memes, such as the “Don’t Judge Me Challenge” that originally powered Musical.ly to top the U.S. app download charts in 2015. Other prominent examples include the iconic “Karma’s a Bitch” (Douyin, 2017) and “The Harlem Shake” (YouTube, 2013).



Above: Examples of the various video meme formats.

Dance memes involve mimicking a set sequence of novel dance moves or hand gestures that accompany the lyrics or beat of a song. The portrait ratio of vertical video lends itself particularly well to capturing an individual dancing. This category is widely accessible and popular for teens recording themselves in bedrooms with the ultra-low barrier to take part being merely to move your body or mouth lyrics. Musical.ly’s first set of popular teen influencers included many, such as Baby Ariel, who mastered the art of creating hand gestures to accompany song lyrics.

Challenge memes involve completing a difficult, unpleasant, or skillful task. An early example of this format was the “Ice bucket challenge” (2014), which saw celebrities record themselves pouring buckets of iced water over their heads. The famous “Bottlecaps challenge” (2019) required twisting the cap off a bottle with a kick. In China, the “A4 Waist Challenge”

(2016) saw women prove their slenderness by covering the thinnest portion of their waists with nothing but a piece of A4 paper.

Filter memes are based around the use of a specific special effect. ByteDance quickly realized that users would adopt innovative and fun AR video filters to create memes. A newly released filter could completely solve the creativity problem of making a video enjoyable for others to watch. On TikTok, the filter would, much like a hashtag, act as a channel for content discovery, meaning those who adopted popular filters early had the chance to gain high exposure and many new followers.

An instructive example was the popular “Mirror reflection challenge” ²⁹³ (2020). The filter simply mirrored the left half of the screen onto the right side. Users quickly experimented and learned to align their faces to create a multitude of exciting effects and reveals.

Concept memes are precisely that – a concept. One that is novel but sufficiently replicable by others who, in turn, may add their own twist. An early non-video example was the craze for lying face down in public places, termed “Planking” ²⁹⁴ (2011). The “Mannequin challenge” ²⁹⁵ (2016) was another notable example. Those being recorded would remain frozen in action, like [mannequins](#) in a shop window, while a camera panned over them. The overall effect was as if time had suddenly stopped.

A noteworthy concept meme was the “Gummy Bear Challenge” ²⁹⁶ (2019), started by Czech TikTok user David Kasprak. The iconic chorus of the Adele song “Someone like you” plays as the camera slowly pans over hundreds of small gummy bears giving the impression they are the ones singing the song. Legions of bears wailing in unison “Never mind, I’ll find... someone like youuuuu”—their collective sorrow is palpable. The video is a mesmerizing 15-second artwork, all the more notable for its lack of a human protagonist.

Popular video memes straddled the “goldilocks zone” of difficulty. Too easy for others to replicate meant they quickly became tedious, too complicated for others to copy, and they could not spread. Prior exposure to a meme facilitated the mental processing of new variants as it provided a familiar structure into which the new information could be integrated. Music was the most potent trigger for making these mental connections.

The human brain is naturally attuned to detecting patterns in its environment. The second Old Town Road started playing on a new TikTok video, the viewer connected what they were seeing with all the other Old Town Road videos they had experienced. It immediately established familiarity and set up expectations of what would unfold.

Old Town Road was a particularly potent combination—an accessible dance meme, matched with a “reveal” of people switching to cowboy-themed clothes at the instant the song’s chorus began.

Professionally generated music videos had become established with the rise of television channel MTV in the 1980s. Smartphones and memes were helping to form a new category of “user-generated music videos.” Some artists even began to write lyrics to new songs with short video-friendly hand gestures in mind.

Struggling competitors

In the summer of 2019, Mark Zuckerberg stood at the front of the auditorium at Facebook headquarters in Menlo Park, facing an assembled group of company staff. He was midway through fielding questions as part of an employee open Q&A session, a company tradition that allowed for a direct communication line between front line staff and the CEO.

An engineer in the audience raised his hand. “Are we concerned about TikTok’s growing cultural clout among teens and Gen Z, and what is our plan of attack?”²⁹⁷

“TikTok is doing well,” was Zuckerberg’s unequivocal answer. Picking out India and young Americans as two strong TikTok user groups, he went on to outline the company’s plan to deal with the new competitor—Lasso, a TikTok clone. Facebook would first use Lasso to target markets such as Mexico, where TikTok did not already have a significant presence before competing head-on in markets where it did.

Fast forward four months to November, and Lasso had failed to make a meaningful impact. The app had fewer than half a million downloads, mostly in Mexico. While many videos on TikTok had hundreds of thousands of likes, “the videos in Lasso’s nearly identical feed typically have a few dozen,”²⁹⁸ noted one highly critical New York Times article.

Experimenting further, another clone of TikTok “Reels,” this time built into Instagram, began testing in the Brazilian market in late 2019.

By early 2020 Facebook COO Sheryl Sandberg was publicly admitting that the company was concerned about TikTok. “Of course, we worry about it,” ²⁹⁹ she said, after noting that her children loved using the app, later adding, “They’ve gotten to bigger numbers faster than we ever did.”

A pattern that had begun in China was now repeating itself in the West. When ByteDance came up with a breakthrough product, the internet giant best placed to compete with it failed to identify the threat until it was too late.

In 2012, Yiming saw the opportunity to aggregate content via a mobile newsfeed and use recommendations to personalize the experience. ByteDance was just 30 people working out of an apartment. China’s second most valuable internet company at the time, search giant Baidu, was far better positioned to build this service, given the high technical requirements. Yet Baidu’s leadership failed to identify the importance of the opportunity.

Similarly, in 2017 Tencent gave up competing directly in short video, shutting down its own service Weishi and opting to take a minority stake in the then market leader Kuaishou. As the usage of Douyin exploded, Tencent scrambled to re-enter the competitive field yet ultimately found it was too late—Douyin had already run away with the market.

Facebook was the latest incarnation of this pattern. Their misstep somewhat echoed that of Tencent’s. Early on, it had correctly identified Musical.ly’s potential and, at one point, had seriously looked to acquire the app outright. Yet later, it had severely underestimated TikTok.

Tencent could arguably be forgiven, given the rise of Douyin in China was so swift and took the entire industry by surprise. Facebook had no such excuse. It knew exactly what had happened in China with Douyin. ByteDance was spending unusually large sums of money on Facebook ads acquiring users for the international version of an app that had taken China by storm in a way that far exceeded the success of Musical.ly in Western markets. Whether through hubris or the distractions of the ‘Cambridge

Analytica' scandal, Facebook had a much longer window of opportunity to react, yet it still dropped the ball.

Facebook's attempts to replicate TikTok were timid in comparison with Tencent's efforts to copy Douyin in China. Tencent quickly mobilized a team of several hundred, enticed video creators with hundreds of millions of dollars' worth of subsidies and ran massive promotions across its family of existing products. In contrast, Facebook conservatively tested across a few selected developing countries, refraining from making any big moves.

In China, once Tencent realized the severe threat that Douyin posed to several of its core business lines, it took action to cease all ByteDance advertising across its services. By contrast, Facebook and Google, despite knowing that TikTok was now their rival, continued to allow the Chinese company to advertise across their platforms, spending heavily to convert users over to TikTok.

ByteDance's position was still far from desirable. It needed to get to scale fast. By far, the most effective way to do so in the American market was, as one former Facebook employee summarized, to "pay its future competitors billions of dollars in ads, giving up critical data on their targeting and successful installs along the way."³⁰⁰ Yet this undesirable situation was at least far preferable to China, where rivals simply flat out refused to take ByteDance's money.

Snapchat founder Evan Spiegel took a somewhat different view of TikTok. Despite concerns from staff and analysts citing the two app's overlapping target user groups of young people, the chief executive argued that the companies didn't compete. "I think at a high level, looking at TikTok, we definitely consider them a friend," was Spiegel's assessment given on an earnings call. Perhaps a more appropriate phrasing would be, "The enemy of my enemy is my friend."

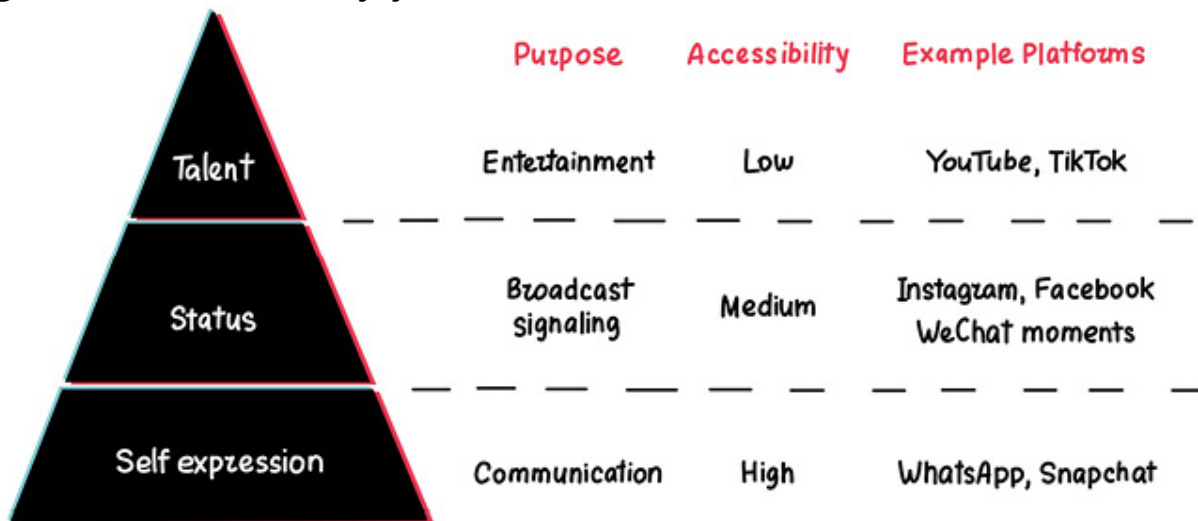
On a panel discussion at a German design conference in early 2020, ³⁰¹ Spiegel elaborated further how he saw TikTok, introducing a model to conceptualize the value of TikTok's content and how it differed from social network apps such as his own Snapchat.

Spiegel conceptualized a three-level pyramid of communication technology, with the base being composed of self-expression and

communication. He classified these as universal behaviors that anyone would feel comfortable doing. On top of this was the next layer of “status,” positing that “social media in its original construct is really about status, showing people that you are cool, getting likes and comments.” He reasoned that “status” was less accessible, holding a narrower base of appeal and a limited frequency of engagement given “people only do things that are cool once a week or once a month and not every day.”

The final pyramid layer was termed “talent.” Talent based content was more interesting than status-based content. It consisted of people creating media to entertain others. Producing this content took time and creativity. Fewer people were willing to learn a new dance or patiently shoot a music video.

TikTok is primarily an entertainment platform. The ‘social’ aspect is relatively weak to the point of being almost non-existent for most users beyond reading, writing, and liking strangers’ comments, similar to how most use YouTube. Spiegel proposed that entertainment content had the potential to draw people away from more influencer-based status content, given that it is more enjoyable and fun.



Above: Spiegel’s pyramid

This pyramid helped illustrate how Spiegel reasoned TikTok was not a threat. The core value of the two platforms was very different. TikTok provides people with entertainment from strangers, while Snapchat connects friends.

TikTok could be viewed as the true smartphone era successor to television. The simple swipe-up motion to load the next video was psychologically similar to flicking between TV channels with a remote control. Not knowing exactly what is coming next provides an addictive element of anticipation.

TikTok worked without any need to register an account, subscribe to channels, add friends or expend mental effort choosing which piece of content to consume. Just like television, one simply turns it on. Accessible and intuitive, TikTok was a place for people to zone out and relax.

Where's the moat?

Such easy onboarding and lack of reliance on a traditional social graph left many people struggling to pin down precisely what was TikTok's "moat." What was stopping its deep-pocketed, established competitors from moving in and eroding its market share?

It turned out that American companies competing with TikTok would hit exactly the same barriers that Chinese companies had met in competing with Douyin. Building a basic TikTok clone and gaining a small market share was easy. Creating a superior version of TikTok, once it had become established in a market, was exceedingly difficult.

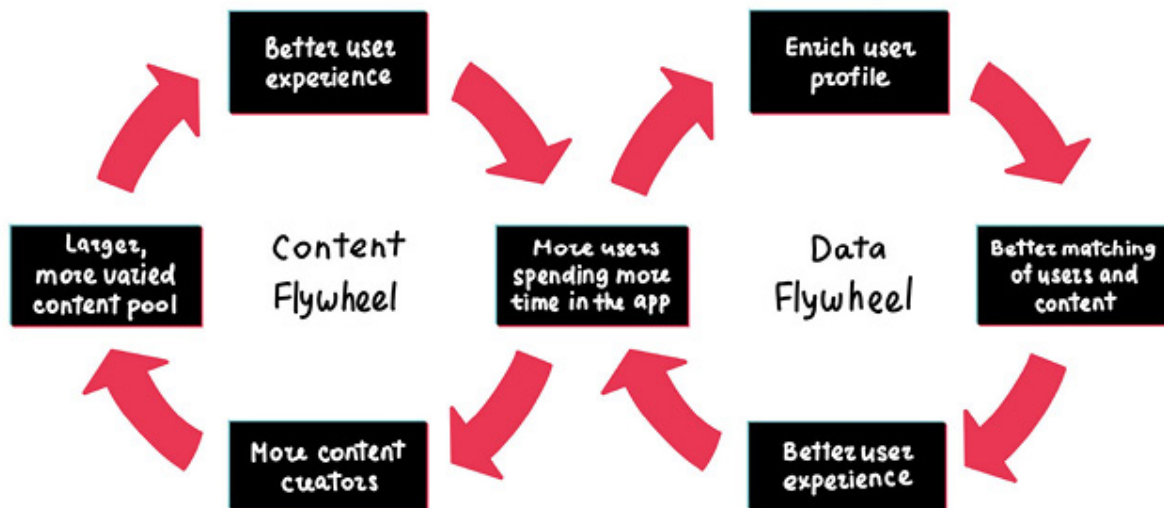
Only the largest internet companies had the resources to realistically match TikTok's advantage in automated video classification and content recommendation systems. The technology allowed for superior matching of long-tail content with users, which further amplified the already existing advantage in content. The more time users spent in TikTok, the more detailed their user profile interest graph became. Put simply—the more you used TikTok, the more personalized it became. This dynamic ensured the early experience of using any clone would always be inferior.

Other technical barriers that competitors would struggle to match included the app's sophisticated use of computer vision to automate the classification and tagging of videos. It was also continually releasing novel and creative augmented reality video filters.

The TikTok brand had already become synonymous with music-centric short video. This brand cachet was powerful as TikTok had reached a critical mass of mindshare among the general public with the name entering

the common vernacular. Saying “Let’s shoot a TikTok,” didn’t need to be explained. The more users TikTok attracted, the more buzz there was about it. There were always those who seek to be different, yet most people follow the crowd.

Arguably the most potent defense was the rich ecosystem of skilled video creators. These people invested time and creativity into making unique content for the platform that was timely and meaningful to their niche audiences. TikTok had the most diverse and vibrant ecosystem of short video content creators. Building that community took time and could not be easily recreated at scale. Potential new competitors to TikTok would risk running into the same issues as Tencent’s Weishi—people would try out the alternative app. Yet, once they discovered it had inferior content, they would simply give up and move back to TikTok.



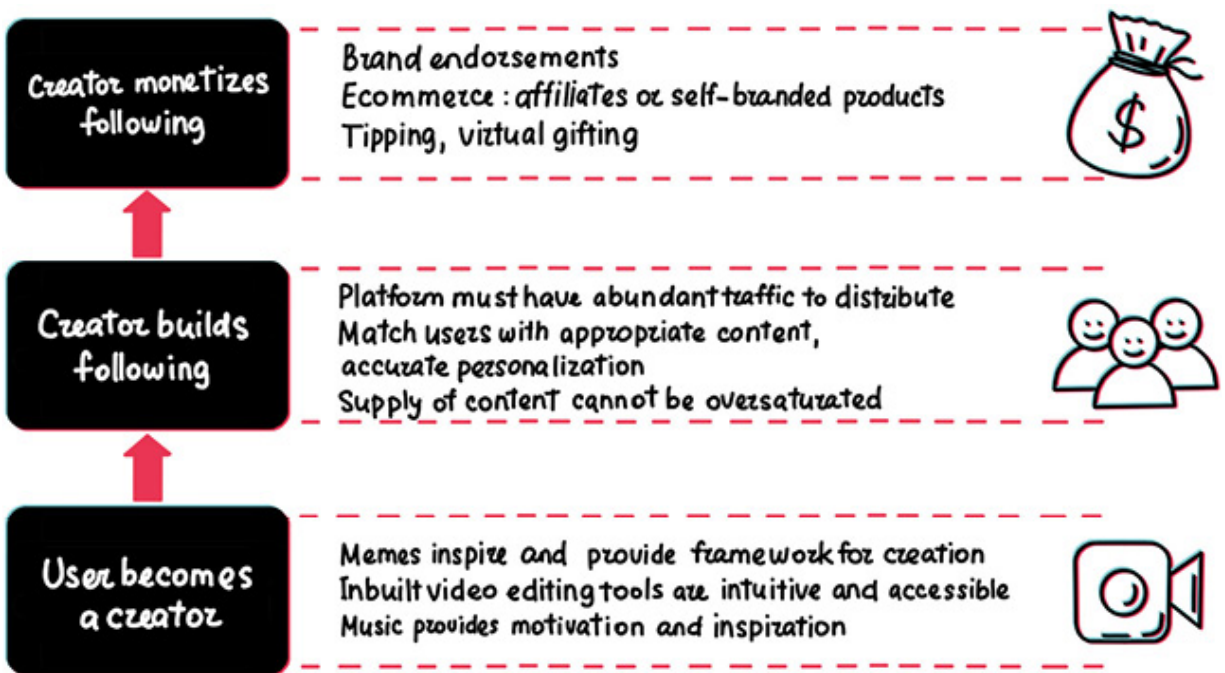
Above: TikTok’s dual virtuous flywheels

Fostering a healthy ecosystem of creators required three things. Firstly, users need to convert into becoming creators. Secondly, creators need to be able to find their audience and build a following. Lastly, they need a way to monetize that following either directly or indirectly.

TikTok had considered all three steps. The art of converting users into creators was a matter of heavily amplifying the use of memes as a framework to inspire and motivate. Videos could be migrated between regions to be used as “educational materials,” showing users what was

possible. Backing that up was best in class, intuitive to use video editing tools and a constant pipeline of new fun filters.

Due to the high new user growth through paid ads, building a following on TikTok was easy. Multiple stories appeared online of people rapidly gaining large followings with what could only be generously described as “very average content.” This dynamic was by no means unique to TikTok; the rapid rise of a new content platform will naturally spawn a new crop of influencers who grow along with the platform due to the temporary imbalance between supply and demand of content. Well established older platforms such as Instagram suffer from the opposite dynamic, as they were saturated with accounts producing high-quality work competing for attention.



Above: The three-step process of converting users into long term quality creators.

Monetization was the most challenging step. How does one extract value from such massive amounts of attention? For most people, regularly creating high-quality video content was a full-time job. Without a sustainable way to make a living from their account, creators would become frustrated and move their following over to other platforms, or simply stop creating.

Many influencers found indirect ways to work with brands to do endorsements and soft ad promotions. To facilitate this further, TikTok set up a marketplace matchmaking brands with creators³⁰² and began to test monetization features that had proved effective in China, including embedding store links within videos so people could buy what they saw.

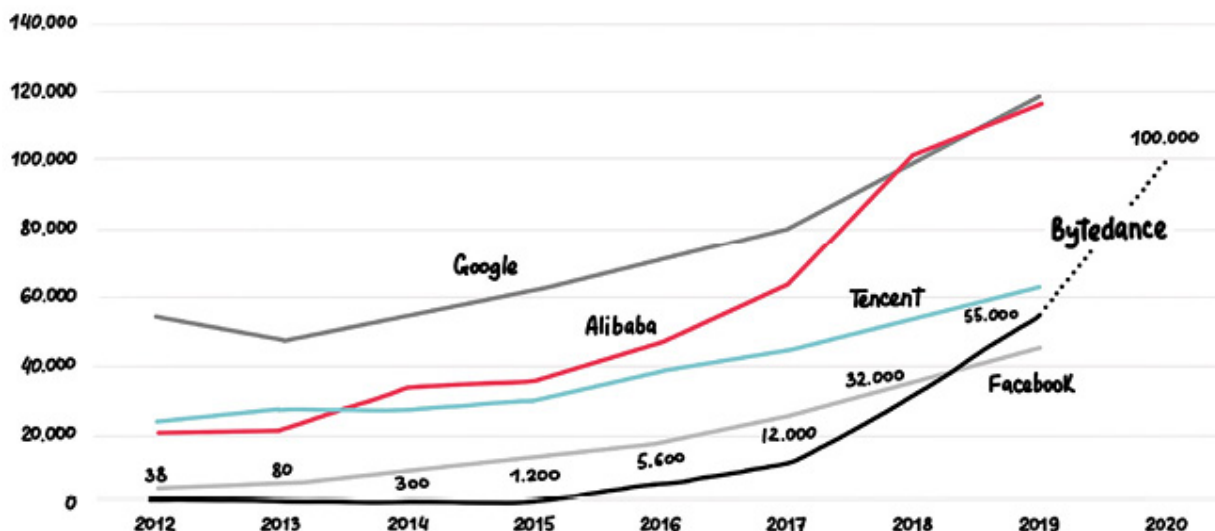
TikTok also adapted a tried and tested China strategy, setting up a “Creators fund” to subsidize influencers who met specific criteria. In 2020, they announced a fund starting with \$200 million, growing to \$1 billion in the U.S. over three years.³⁰³

To manage this fund and the many other initiatives, TikTok’s U.S. team needed to expand considerably. Onboarding so many employees so quickly was perhaps the biggest challenge of all.

Building TikTok America

Yiming was now spending most of his time away from China, fast-tracking all aspects of the international expansion and head-hunting the global business leadership team. The China operations were to be run by his trusted generals, Kelly, and Lidong.

In a letter³⁰⁴ to staff marking the company’s eighth anniversary, Yiming publicly disclosed the company’s 2020 target to reach a hundred thousand employees. This goal would see the company headcount leap ahead of Facebook and Tencent, with the lion’s share of new hires being outside China.



Above: The number of ByteDance employees from founding through to 2020 in comparison with other well-known internet service companies.³⁰⁵

When hiring for critical positions, Yiming operated from a tried and tested playbook—aggressively pursue the best people and offer them generous compensation. The first significant American hire was Vanessa Pappas in February 2019.³⁰⁶ Vanessa had spent seven years at YouTube, working her way up to become global head of creative insights and specialized in growth tactics for influencers and celebrities. She became general manager for TikTok in the U.S.

Four months later, Facebook vice president of twelve years, Blake Chandlee, jumped ship over to TikTok. When asked why he joined, Chandlee replied, “the metrics on TikTok were astounding.”³⁰⁷ Erich Andersen, a senior Microsoft executive of 25 years, joined as “global general counsel,” ex-U.S. Air Force combat security specialist Roland Cloutier became “global chief security officer.”

By late 2019 ByteDance had already begun to set up shop on Facebook’s doorstep. Moving into an office space previously occupied by messaging app WhatsApp, ByteDance wasted no time poaching Facebook staff offering salaries as much as 20% higher.³⁰⁸ News broke in May 2020 that TikTok had rented 232,000 square feet of prime office space at Times Square, New York.³⁰⁹

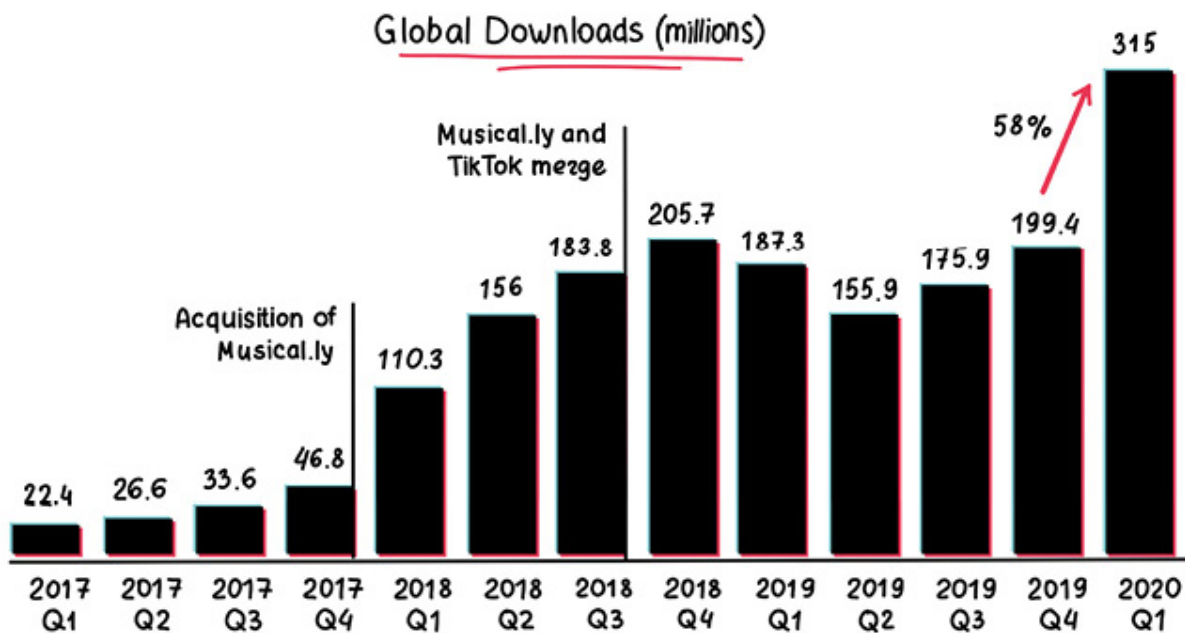
TikTok was taking America by storm, ByteDance had built China’s first genuinely global breakout hit internet product, far exceeding the international forays of rivals Tencent and Alibaba. Silicon Valley’s fearsome giants had proved themselves slow in responding to the new upstart allowing ByteDance free rein to dominate the short video category.

TikTok’s revenue in 2019 paled in comparison to its China operations. Yet, it was the growth engine of the future, the product that would power the company forward through the next stage of development beyond China.

Straddling both China and the West would put the company in a uniquely powerful position. By following the advertising revenue model already developed by Douyin, the revenue potential for TikTok was enormous. It was not only on course to become a cash-cow, but would also act as a gateway for other services. ByteDance would be able to partially allocate

advertising inventory to its own suite of products, accurately targeting and acquiring new users across the globe with speed and minimal cost without having to hand over money and data to the online advertising duopoly gatekeepers of Facebook and Google.

As the coronavirus pandemic spread across the world in early 2020, people found themselves quarantined inside their homes for weeks, leading into months. While airlines, hotels, and restaurants went bankrupt, demand for online entertainment skyrocketed, causing downloads of TikTok to hit all-time highs. The app was the perfect fix for those desperate for distraction, locked inside suffering from the cruel combination of stress and boredom. Downloads and user engagement spiked accordingly, particularly in America. By October 2019, TikTok's monthly users in the U.S. stood at just under 40 million, eight months later that number had grown to more than 91 million.³¹⁰



Above: TikTok global quarterly downloads.³¹¹

A strong team of industry veterans had already been assembled, but Yiming had higher ambitions still. In May 2020, news broke of his latest catch – Walt Disney executive Kevin Mayer was joining ByteDance. Mayer's hire sent shock waves through the American business community at a time when awareness around TikTok's parent company was still low. It was a marquee signing of a top-level business leader from one of America's

most respected companies. Mayer's title was CEO of TikTok and COO of ByteDance, with China reporting directly to Yiming.

The rise of TikTok had caught America's tech industry by surprise. Long essays³¹² and deep-dive articles dissecting the app's background and novel innovations started to appear with frequency.³¹³ By mid-2020, TikTok had become impossible to ignore, the app had been downloaded a shocking 2 billion times³¹⁴ and was indisputably the world's hottest platform.

It was almost too good to be true.

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259 <https://www.reuters.com/article/us-bytedance-musically/chinas-bytedance-scrubs-musical-ly-brand-in-favor-of-tiktok-idUSKBN1KN0BW>

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Musical.ly June 2016: <https://youtu.be/z4haZtTAToI?t=21>

Douyin May 2018: <https://youtu.be/4EcMiZfaK8Y?t=47>

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261 <https://www.youtube.com/watch?v=D7zE1NKWiu4>

262 <https://www.change.org/p/everybody-change-tik-tok-back-to-musical-ly?>

263 <https://www.zackhargett.com/tiktok/>

264 <https://www.eugenewei.com/blog/2020/8/3/tiktok-and-the-sorting-hat>

265 <https://youtu.be/9MXIiQuvT0A?t=229>

266 <https://www.buzzfeednews.com/article/krishrach/tiktok-twitter-thread-weird>

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269 Image source: <https://www.toutiao.com/a1621444337593351/>

270 Multiple TikTok users were surprised to find themselves featured in adverts for the app <https://adage.com/article/digital/tiktok-users-are-surprised-find-themselves-ads-app/2204996>

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274 <https://www.youtube.com/watch?v=ZQxMW23uTJU>

275 <https://www.theatlantic.com/technology/archive/2018/10/what-tiktok-is-cringe-and-thats-fine/573871/>

276 <https://www.spiegel.de/international/business/as-a-chinese-company-we-never-get-the-benefit-of-the-doubt-a-e1e415f6-8f87-41e9-91ae-08cfa90583b3>

277 <https://cdixon.org/2010/01/03/the-next-big-thing-will-start-out-looking-like-a-toy>

278 <https://www.theinformation.com/articles/chinas-bytedance-plans-slack-rival-even-as-losses-mount?shared=6e04505df99b45f2>

279 Image source: <https://medium.com/@NateyBakes/tiktoks-growing-pains-in-the-west-attack-of-the-memes-b96e26593649>

280 <https://www.youtube.com/watch?v=Lq7CCoCO6j4>

281 Coincidentally, the show that had propelled Musical.ly to the top of the app store in 2015, Lip Sync Battle, was a spin-off from The Jimmy Fallon Show.

282 <https://variety.com/2018/digital/news/tiktok-jimmy-fallon-musically-app-downloads-1203032629/>

283 <https://www.youtube.com/watch?v=jq25KtRjarw&feature=youtu.be&t=106>

284 <https://www.buzzfeed.com/pablovaldivia/lil-nas-x-tweets>

285 <https://twitter.com/LilNasX/status/1099455087670423553?>

286 The original track sample was from the song “34 Ghosts IV” by 9 Inch Nails

287 <https://www.tiktok.com/@nicemichael/video/6658388605418867974?refer=embed>

288 <https://www.youtube.com/watch?v=LxwpKKK3P4s&feature=youtu.be>

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EPILOGUE



“We’re looking at TikTok. We may be banning TikTok. We may be doing some other things.”
- Donald J. Trump³¹⁵

³¹⁵ July 2020, the Trump re-election campaign runs anti-TikTok adverts on Facebook
[https://www.facebook.com/ads/library/?active_status=all&ad_type=political_and_issue_ads&country=ALL&impression_search_field=has_impressions_lifetime&q=tiktok&view_all_page_id=153080620724&sort_data\[direction\]=desc&sort_data\[mode\]=relevancy_monthly_grouped](https://www.facebook.com/ads/library/?active_status=all&ad_type=political_and_issue_ads&country=ALL&impression_search_field=has_impressions_lifetime&q=tiktok&view_all_page_id=153080620724&sort_data[direction]=desc&sort_data[mode]=relevancy_monthly_grouped)

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WHAT A DIFFERENCE A YEAR MAKES . The world in which I began writing this book and the world we find ourselves in today seem so very distant.

The COVID-19 pandemic has turned our lives upside down. Working from home, travel bans, and mask-wearing became a part of daily life. A military border clash between India and China led to the deaths of 20 Indian soldiers. 59 Chinese apps, including TikTok found themselves subsequently removed from Indian app stores. India was TikTok's largest market accounting for roughly one-third of the app's users globally.

Donald Trump, I imagine, had never even heard of TikTok when I started writing in the summer of 2019. It's hard to imagine how the app's fortunes in the U.S. could have become more dramatic in the past few months as TikTok finds itself entangled in global geopolitical rivalries and U.S. election campaigns. As of this moment, September 2020, ByteDance is taking legal action against the president after being forced to break off and sell part of TikTok, with bidders including Microsoft and Oracle vying to acquire the valuable asset. The fate of TikTok in the U.S. lies in the balance.

Rather than make speculations that will surely become dated within days, I've decided to draw a line underneath this work and stop here. At the beginning of this book, I outlined the goal of this work as bringing value to the broader discussion and understanding around TikTok, ByteDance, the rise of short video, and Chinese internet firms in general. Sixty thousand words later, and I feel we've only scratched the surface.

Some tough choices were made in choosing what to cut, and there are many aspects I regret not having time to flesh out and explore in more detail: Yiming's obsession with "building a company as a product," the profound impact of TikTok on India's rural society before its ban, the media backlash against Toutiao over content copyright in 2014, ByteDance's forays into enterprise software, gaming, and ed-tech, to name a few.

One thing I'm sure of, this won't be the last book to be written about this company. The story of ByteDance and TikTok is to be continued....

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PEOPLE

People mentioned in this book listed in alphabetical order of anglicized names. Chinese is followed by pinyin romanization in bracket s.

ALEX HOFMANN, former Musical.ly president of North America, left after the ByteDance acquisition

ALEX ZHU (z hūjùn), co-founder of Musical.ly, later CEO of TikTok

ALLEN ZHANG (zhāngxiǎolóng), founder of WeChat and Foxmail, senior executive VP at Tencent

ARIEL REBECCA MARTIN (BABY ARIEL), social media influencer, most followed Musical.ly account 2016-2018

BLAKE CHANDLEE, joined TikTok in June 2019 as VP of global business solutions, having spent twelve years at Facebook as VP of global partnerships

CAO HUANHUAN (cáohuānhuān), senior algorithm architect, former principal research engineer at ByteDance, Beijing

CHEN HUA (chén huá), former Kuxun co-founder and CEO

CHEN LIN (chén lín), ByteDance executive, previously CEO of Toutiao, now responsible for innovative products and education. Employee number 12

CHEN YUQIANG (chényǔqiáng) , former Baidu engineer, played an instrumental role in improving Toutiao's algorithmic recommendation capabilities

CLEMENT RAFFENOUX, co-founder and chief product officer of Mindie

ERICH ANDERSEN, former chief intellectual property IP consultant of Microsoft, joined ByteDance as global general counsel in 2020

FU SHENG (fù shèng), an early investor in Musical.ly and CEO of Cheetah Mobile. Held veto power over Musical.ly acquisition deals.

GAO HAN (gāohán), ByteDance employee no. 22, senior UI designer

GREGOIRE HENRION, French serial entrepreneur, co-founder, and CEO of Mindie, later co-founder and CEO of YOLO

GU WENDONG (gǔwéndòng), ByteDance VP of finance, former deputy general manager of the CreditEase big data innovation center

HONG DINGKUN (hóngdìngkūn), ByteDance VP of technology

HUANG HE (huánghé), ByteDance's earliest developer, responsible for building their first apps

HUA WEI (huá wēi), senior ByteDance executive, with a focus on strategic investments and HR. Led several early company investments

JAMES VERALDI, former head of product strategy at Musical.ly

JASON DERULO, the first established music artist to embrace Musical.ly and exclusively premiere a music video on the platform

JIMMY FALLON, the first mainstream American celebrity to publicly endorse TikTok. "Lip-Sync Battle," the show that helped propel Musical.ly to popularity in 2015, was a spin-off from *The Jimmy Fallon Show*

JOAN WANG (wáng qióng), a trusted friend of Yiming and ByteDance's most important investor, managing director of SIG China

JONAS DRUPPEL, German co-founder and CEO of Dubsmash

KELLY ZHANG (zhāng nán), CEO of ByteDance, China. Previously CEO of Douyin, also briefly responsible for TikTok

KEVIN MAYER, a former high-ranking executive at Disney, joined as COO of ByteDance and CEO of TikTok in May 2020, resigned in August of the same year.

KINOSHITA YUKINA, the first well known Japanese celebrity to endorse TikTok

LIANG RUBO (liáng rǔbō), Yiming's close friend and roommate at Nankai university, ByteDance employee no. three, held various senior technical roles, later shifting to focus on HR

LIL NAS X (MONTERO LAMAR HILL), American songwriter. Creator of "Old Town Road," the longest-running no. 1 single in Billboard Hot 100 history and propelled to popularity through TikTok.

LIU JUN (liú jùn), an early investor in ByteDance

LIU XINHUA (liúxīnhuá), former president of ByteDance International, later chief growth officer of Kuaishou, currently an investment partner at Gaorong Capital

LIU ZHEN (liǔ zhēn), former head of strategy for Uber China, later responsible for international strategy at ByteDance. Niece of Lenovo founder Liu Chuanzhi

LOUIS YANG (yáng lù yù), co-founder of Musical.ly

MADRAC ZHU (zhùzì nán), former vice president at Musical.ly, currently responsible for monetization strategy at ByteDance

MATT HUANG (huánggòngyǔ), an angel investor in ByteDance through Starling Ventures. Co-founder of crypto-asset investment firm Paradigm, previously, partner at Sequoia Capital

NEIL SHEN (chénnán péng), founding and managing partner of Sequoia China, an investor in ByteDance

NIKHIL GANDHI, head of TikTok India until the app's ban. Previously VP at Walt Disney, India

PONY MA (mǎhuàténg), co-founder and CEO of Tencent

ROBIN LI (lǐ yànhóng), co-founder and CEO of Chinese search engine Baidu

ROLAND CLOUTIER, cybersecurity expert previously employed by the U.S. Air Force and Department of Defense, joined ByteDance in 2020

SHOU ZI CHEW (zhōu shòu zī), a former partner at the Beijing offices of Yuri Milner's investment firm DST, consulted for Yuri's Series B investment in ByteDance

SIMON CORSIN, co-founder and chief technical officer of Mindie

STANISLAS COPPIN, co-founder and chief growth officer of Mindie

SU HUA (sù huá), CEO of Kuaishou, China's second most popular short video app

TIM GONG (gōng tǐng), leads SIG China's private equity and venture capital activities

VANESSA PAPPAS, TikTok's first general manager for the U.S. market, former YouTube executive, joined 2019 Feb

WANG XIAOWEI (wángxiǎowèi), the first product manager of Douyin

WANG XING (wáng xìng), serial internet entrepreneur, friend of Zhang Yiming, founder of Xiaonei, Fanfou, and Meituan

WEI HAIJUN (wéi hǎijūn), former general manager of Cheetah Mobile investments, led Cheetah Mobile's Series A investment into Musical.ly

WU SHI CHUN (wúshìchūn), Kuxun co-founder and COO

XIANG LIANG (xiàng liàng), former Hulu employee who currently works as a researcher at ByteDance AI Labs, famously refused Yiming an early copy of his book *Putting into Practice Recommender Systems*

XIE XIN (xiè xīn), VP for enterprise efficiency, previously worked at Kuxun with Yiming

YAN SHOU (yán shòu), ByteDance senior executive, focus on strategy, investment, and gaming

YANG ZHENYUAN (yángzhènyuán), key ByteDance executive technical staff. VP of the recommendation algorithm business, joined 2014, having previously worked at Baidu for nine years

YURI MILNER, Russian investor, entrepreneur, and philanthropist, early ByteDance investor

ZACH KING, a popular visual illusion video creator, currently operates one of the most popular accounts on TikTok

ZHANG HANPING (zhānghànpíng), father of Zhang Yiming

ZHANG LIDONG (zhānglídōng), chairman of ByteDance China, joined 2013. A former journalist and vice president at Beijing Times newspaper

ZHANG YI (zhāng yī), The second product manager of Douyin

ZHANG YIMING (zhāng yí míng), CEO and founder of ByteDance

ZHOU BINGJUN (zhōubǐngjùn), former Musical.ly vice president of operations

ZHOU HONGYI (zhōuhóngyī), co-founder, Chairman, and CEO of the Internet security company Qihoo 360

ZHOU ZIJING (zhōuzi jīng), angel investor in ByteDance

ZHU WENJIA (zhūwén jiā), joined ByteDance from Baidu in 2015, responsible for implementing Douyin's recommendation system, later

CEO of Toutiao

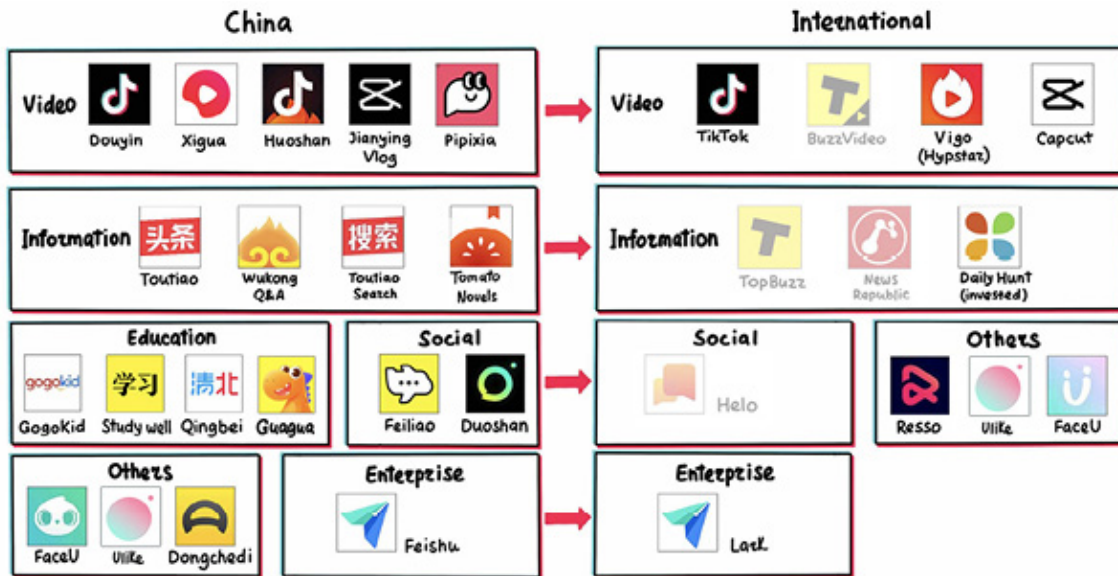
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BYTEDANCE APPS

There are a **lot** of ByteDance apps!

Below is a list of the most important titles as a reference resource to accompany the book. This list is not exhaustive.



Operational (China)

N.B. Light version apps (jisuban) not listed

Logo	Name	Description	Year	Official website
	Toutiao 今日头条	News aggregator	2012	toutiao.com
	Douyin 抖音	Short video	2016	douyin.com

	Xigua video 西瓜小视频	Short video	2017	ixigua.com
	Douyin Huoshan edition 抖音 火山版	Short video (CN)	2016	huoshanzhibo.com
	CapCut 剪影	Video editing	2019	lv.ulikecam.com
	Duoshan 多闪	Messaging	2019	duoshanapp.com
	PipiXia 皮皮虾	Memes & joke videos	2018	h5.pipix.com
	Feishu 飞书	Enterprise productivity	2019	feishu.cn
	Gogo Kid	K-12 education	2018	gogokid.com.cn
	Dong Chedi 懂车帝	Automotive	2017	dcdapp.com
	Wukong Q&A 悟空问答	Question & answer platform	2017	wukong.com

	FaceU 激萌	Beauty selfie camera	2018	faceu.com
	Feiliao 飞聊	Interest-based social	2019	feiliao.com
	Toutiao Search	Search engine	2019	toutiao.com/search
	QingBei Net School 清北网校	K-12 education	2019	qingbei.com
	Ulike 轻颜相机	Beauty selfie camera	2018	m.ulikecam.com
	Tomato Novels 番茄小说	Reading	2019	None found
	Guagua Dragon English 瓜瓜龙英语	K-12 language study	2019	ggl.cn
	Study well 好好学习	Education	2018	None found

Operational (International)

Logo	Name	Description	Year	Notes
	TikTok	Short video	2017	tiktok.com
	CapCut	Video editing	2020	faceueditor.com
	Lark	Enterprise productivity	2019	larksuite.com
	DailyHunt	Indian news aggregator	2016	dailyhunt.in
	Face U	Beauty selfie camera	2018	faceu.com
	Resso	Indian music streaming app	2020	resso.app
	Ulike	Beauty selfie camera	2019	None found
	BABE	Local news aggregator	2016	babe.co.id

Discontinued

Logo	Name	Description	Operation	Notes
	Implied jokes 内涵段子	Meme aggregator	2012 - 2018	Shut down due to government criticism
	Musical.ly	Short video	2014 - 2018	Merged into TikTok
	Flipagram	Short video	2017 - 2020	Merged into Virgo video
	News Republic	News aggregator	2017 - 2020	news-republic.com
	Helo	Indian market	2018 - 2020	helo-app.com
	TopBuzz	News aggregator	2015 - 2020	topbuzz.com
	Vigo video (Hypstar)	Short video	2017 - 2020	vigovideo.net

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2019-11-23

https://blog.csdn.net/sinat_26811377/article/details/103217551

2018-05-03

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